

3. Detailed Audit Controls

3.1. Information Technology (IT) Security Guidelines

3.1.1. Implementation of Information Security

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Security Policy							
3.1.1.1	An information security policy document has been formulated that includes physical, personnel, procedural and technical controls and was submitted to the Controller before commencement of operation.	1. Obtain the softcopy/hardcopy of the information security policy 2. Check the following 1. Controller received the policy approved by CA management before commencing operations 2. Policy contains physical, personnel, procedural and technical controls 3. information is classified depending upon its sensitivity for the CA as per IT Act-2000 4. both confidential and non-confidential information addressed in the policy 5. CA has prepared detailed manuals for performing its activities 6. CA ensures security and risk management controls are implemented 7. policy has been approved by CA management. Verify by checking approval date. 8. policy has been communicated to all employees via mail or training	Mandatory	IT CA Rules 19.2			

		session 9. Obtain name of personnel responsible for ensuring that security procedures implementation and verify roles and responsibilities defined 3. verify roles and responsibilities defined 4. Verify the all recent changes done to the policy since last audit 5. For sample recent changes done to the policy validate details were submitted to Controller within two weeks						
3.1.1.2	CA shall ensure the security policy clearly addresses security of both confidential and non-confidential information. The Certifying Authority shall prepare detailed manuals for performing all its activities and shall scrupulously adhere to them.	1. Obtain the softcopy/hardcopy of the information security policy 2. Check the following 1. Controller received the policy approved by CA management before commencing operations 2. Policy contains physical, personnel, procedural and technical controls 3. information is classified depending upon its sensitivity for the CA as per IT Act-2000 4. both confidential and non-confidential information addressed in the policy 5. CA has prepared detailed manuals for performing its activities 6. CA ensures security and risk	Mandatory	WebTrust 3.9.5, IT Regulations 3				

		management controls are implemented 7. policy has been approved by CA management. Verify by checking approval date. 8. policy has been communicated to all employees via mail or training session 9. Obtain name of personnel responsible for ensuring that security procedures implementation and verify roles and responsibilities defined 3. verify roles and responsibilities defined 4. Verify the all recent changes done to the policy since last audit 5. For sample recent changes done to the policy validate details were submitted to Controller within two weeks						
3.1.1.3	The Certifying Authority shall as approved, in respect of security and risk management controls continuously ensure that security policies and safeguards are in place. Such controls include personnel security and incident handling measures to prevent fraud and security breaches. The security policy shall be approved by management,	1. Obtain the softcopy/hardcopy of the information security policy 2. Check the following 1. Controller received the policy approved by CA management before commencing operations 2. Policy contains physical, personnel, procedural and technical controls 3. information is classified depending upon its sensitivity for the CA as per IT Act-2000 4. both	Mandatory	IT Regulations 3, WebTrust 3.1.1				

		confidential and non-confidential information addressed in the policy 5. CA has prepared detailed manuals for performing its activities 6. CA ensures security and risk management controls are implemented 7. policy has been approved by CA management. Verify by checking approval date. 8. policy has been communicated to all employees via mail or training session 9. Obtain name of personnel responsible for ensuring that security procedures implementation and verify roles and responsibilities defined 3. verify roles and responsibilities defined 4. Verify the all recent changes done to the policy since last audit 5. For sample recent changes done to the policy validate details were submitted to Controller within two weeks						
3.1.1.4	The Certifying Authority shall as approved, in respect of security and risk management controls continuously ensure that security policies and safeguards are in place. Such controls include personnel security and incident handling measures to prevent fraud and	1. Obtain the softcopy/hardcopy of the information security policy 2. Check the following 1. Controller received the policy approved by CA management before commencing	Mandatory	IT Regulations 3, WebTrust 3.1.1				

	security breaches. The security policy shall be approved by management, published and communicated to all employees.	operations 2. Policy contains physical, personnel, procedural and technical controls 3. information is classified depending upon its sensitivity for the CA as per IT Act-2000 4. both confidential and non-confidential information addressed in the policy 5. CA has prepared detailed manuals for performing its activities 6. CA ensures security and risk management controls are implemented 7. policy has been approved by CA management. Verify by checking approval date. 8. policy has been communicated to all employees via mail or training session 9. Obtain name of personnel responsible for ensuring that security procedures implementation and verify roles and responsibilities defined 3. verify roles and responsibilities defined 4. Verify the all recent changes done to the policy since last audit 5. For sample recent changes done to the policy validate details were submitted to Controller within two weeks						
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3.1.1.5	An authorized personnel shall be responsible for ensuring that security procedures within their area of responsibility are carried out correctly	1. Obtain the softcopy/hardcopy of the information security policy 2. Check the following 1. Controller received the policy approved by CA management before commencing operations 2. Policy contains physical, personnel, procedural and technical controls 3. information is classified depending upon its sensitivity for the CA as per IT Act-2000 4. both confidential and non-confidential information addressed in the policy 5. CA has prepared detailed manuals for performing its activities 6. CA ensures security and risk management controls are implemented 7. policy has been approved by CA management. Verify by checking approval date. 8. policy has been communicated to all employees via mail or training session 9. Obtain name of personnel responsible for ensuring that security procedures implementation and verify roles and responsibilities	Mandatory	WebTrust 3.9.8				
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		defined 3. verify roles and responsibilities defined 4. Verify the all recent changes done to the policy since last audit 5. For sample recent changes done to the policy validate details were submitted to Controller within two weeks						
3.1.1.6	A review process shall be designed for maintaining the security policy. Review dates and responsibilities should be recorded.	1. Obtain the softcopy/hardcopy of the information security policy 2. Check the following 1. Controller received the policy approved by CA management before commencing operations 2. Policy contains physical, personnel, procedural and technical controls 3. information is classified depending upon its sensitivity for the CA as per IT Act-2000 4. both confidential and non-confidential information addressed in the policy 5. CA has prepared detailed manuals for performing its activities 6. CA ensures security and risk management controls are implemented 7. policy has been approved by CA management. Verify by checking approval date. 8. policy has been	Mandatory	WebTrust 3.1.3				

		communicated to all employees via mail or training session 9. Obtain name of personnel responsible for ensuring that security procedures implementation and verify roles and responsibilities defined 3. verify roles and responsibilities defined 4. Verify the all recent changes done to the policy since last audit 5. For sample recent changes done to the policy validate details were submitted to Controller within two weeks						
3.1.1.7	Any change made by the Certifying Authority in the security policy shall be submitted by it within two weeks to the Controller	1. Obtain the softcopy/hardcopy of the information security policy 2. Check the following 1. Controller received the policy approved by CA management before commencing operations 2. Policy contains physical, personnel, procedural and technical controls 3. information is classified depending upon its sensitivity for the CA as per IT Act-2000 4. both confidential and non-confidential information addressed in the policy 5. CA has prepared detailed manuals for performing its	Mandatory	IT CA Rules 19				

		activities 6. CA ensures security and risk management controls are implemented 7. policy has been approved by CA management. Verify by checking approval date. 8. policy has been communicated to all employees via mail or training session 9. Obtain name of personnel responsible for ensuring that security procedures implementation and verify roles and responsibilities defined 3. verify roles and responsibilities defined 4. Verify the all recent changes done to the policy since last audit 5. For sample recent changes done to the policy validate details were submitted to Controller within two weeks						
3.1.1.8	Test	1. Obtain the softcopy/hardcopy of the information security policy 2. Check the following 1. Controller received the policy approved by CA management before commencing operations 2. Policy contains physical, personnel, procedural and technical controls 3. information is classified depending upon its	Optional	Test				

		sensitivity for the CA as per IT Act-2000 4. both confidential and non-confidential information addressed in the policy 5. CA has prepared detailed manuals for performing its activities 6. CA ensures security and risk management controls are implemented 7. policy has been approved by CA management. Verify by checking approval date. 8. policy has been communicated to all employees via mail or training session 9. Obtain name of personnel responsible for ensuring that security procedures implementation and verify roles and responsibilities defined 3. verify roles and responsibilities defined 4. Verify the all recent changes done to the policy since last audit 5. For sample recent changes done to the policy validate details were submitted to Controller within two weeks						
3.1.1.9	Test2	1. Obtain the softcopy/hardcopy of the information security policy 2. Check the following 1. Controller received the policy approved by	Optional	Hello				

		CA management before commencing operations 2. Policy contains physical, personnel, procedural and technical controls 3. information is classified depending upon its sensitivity for the CA as per IT Act-2000 4. both confidential and non-confidential information addressed in the policy 5. CA has prepared detailed manuals for performing its activities 6. CA ensures security and risk management controls are implemented 7. policy has been approved by CA management. Verify by checking approval date. 8. policy has been communicated to all employees via mail or training session 9. Obtain name of personnel responsible for ensuring that security procedures implementation and verify roles and responsibilities defined 3. verify roles and responsibilities defined 4. Verify the all recent changes done to the policy since last audit 5. For sample recent changes done to the policy validate details						
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		were submitted to Controller within two weeks						
Self-Assessment								
3.1.1.10	A documented security self-assessment plan shall be prepared by the CA	1. Obtain the self-assessment plan developed by the CA 2. Verify on a sample basis the for last two self-assessments conducted 3. Collect report of the self-assessment plan 4. Check the following: a. Findings of self-assessment were shared with the management b. Action was taken to address the finding	Mandatory	IT CA Rules 2.e				
3.1.1.11	Self-Assessment shall be performed on a periodic basis and the findings shall be reported to management and discussed for closure	1. Obtain the self-assessment plan developed by the CA 2. Verify on a sample basis the for last two self-assessments conducted 3. Collect report of the self-assessment plan 4. Check the following: a. Findings of self-assessment were shared with the management b. Action was taken to address the finding	Mandatory	IT CA Rules 2.e				
Risk Assessment								
3.1.1.12	The CA's security program shall include an annual Risk Assessment	1. Obtain copy of annual Risk Assessment plan 2. Check the following: a. Threats are identified and documented as part of risk assessment b. Standard operating procedures are present for combating threats. c. controls are in place to prevent misuse, alteration and destruction of sensitive information d. adequate cyber security tools are implemented by the CA such as DLP, Web Gateway, Email Gateway, Proxy Servers, and Firewalls, SSL/TLS, IDS/IPS, SAST, DAST, etc. e. a vulnerability assessment by the above mentioned tools are carried out at a pre decided interval. Report any scheduled VA left out. f. security plan has been developed g. updates and patches are applied in timely manner	Mandatory	IT CA Rules SCHEDULE-II 17.3 CA Browser Forum 5				

		3. Check the security plan covers the following safeguards: a. administrative, b. organizational, c. technical, and d. physical 4. Verify formal change management procedures have been developed						
3.1.1.13	A list of foreseeable internal and external threats that could result in unauthorized access, disclosure, misuse, alteration, or destruction of any critical information shall be maintained.	1. Obtain copy of annual Risk Assessment plan 2. Check the following: a. Threats are identified and documented as part of risk assessment b. Standard operating procedures are present for combating threats. c. controls are in place to prevent misuse, alteration and destruction of sensitive information d. adequate cyber security tools are implemented by the CA such as DLP, Web Gateway, Email Gateway, Proxy Servers, and Firewalls, SSL/TLS, IDS/IPS, SAST, DAST, etc. e. a vulnerability assessment by the above mentioned tools are carried out at a pre decided interval. Report any scheduled VA left out. f. security plan has been developed g. updates and patches are applied in timely manner 3. Check the security plan covers the following safeguards: a. administrative, b. organizational, c. technical, and d. physical 4. Verify formal change management procedures have been developed	Mandatory	CA Browser Forum 5				
3.1.1.14	The sufficiency of the policies, procedures, information systems, technology, and other arrangements that the CA has in place to counter cyber threats shall be verified.	1. Obtain copy of annual Risk Assessment plan 2. Check the following: a. Threats are identified and documented as part of risk assessment b. Standard operating procedures are present for combating threats. c. controls are in place to prevent misuse, alteration and destruction of sensitive information d. adequate cyber security tools are implemented by the CA such as DLP, Web Gateway,	Mandatory	CA Browser Forum 5				

		Email Gateway, Proxy Servers, and Firewalls, SSL/TLS, IDS/IPS, SAST, DAST, etc. e. a vulnerability assessment by the above mentioned tools are carried out at a pre decided interval. Report any scheduled VA left out. f. security plan has been developed g. updates and patches are applied in timely manner 3. Check the security plan covers the following safeguards: a. administrative, b. organizational, c. technical, and d. physical 4. Verify formal change management procedures have been developed						
3.1.1.15	A security plan shall be developed covering the following: • Security procedures to manage the risks • administrative, organizational, technical, and physical safeguards • cost of implementing the specific measures • security breach plan in case of any incidents	1. Obtain copy of annual Risk Assessment plan 2. Check the following: a. Threats are identified and documented as part of risk assessment b. Standard operating procedures are present for combating threats. c. controls are in place to prevent misuse, alteration and destruction of sensitive information d. adequate cyber security tools are implemented by the CA such as DLP, Web Gateway, Email Gateway, Proxy Servers, and Firewalls, SSL/TLS, IDS/IPS, SAST, DAST, etc. e. a vulnerability assessment by the above mentioned tools are carried out at a pre decided interval. Report any scheduled VA left out. f. security plan has been developed g. updates and patches are applied in timely manner 3. Check the security plan covers the following safeguards: a. administrative, b. organizational, c. technical, and d. physical 4. Verify formal change management procedures have been developed	Mandatory	CA Browser Forum 5				
3.1.1.16	Hypervisors, operating system, database, and network device patches and updates	1. Obtain copy of annual Risk Assessment plan 2. Check the following: a. Threats are identified and	Mandatory	WebTrust 3.6.18				

	shall be applied in a timely manner when deemed necessary based on a risk assessment and follow formal change management procedures	documented as part of risk assessment b. Standard operating procedures are present for combating threats. c. controls are in place to prevent misuse, alteration and destruction of sensitive information d. adequate cyber security tools are implemented by the CA such as DLP, Web Gateway, Email Gateway, Proxy Servers, and Firewalls, SSL/TLS, IDS/IPS, SAST, DAST, etc. e. a vulnerability assessment by the above mentioned tools are carried out at a pre decided interval. Report any scheduled VA left out. f. security plan has been developed g. updates and patches are applied in timely manner 3. Check the security plan covers the following safeguards: a. administrative, b. organizational, c. technical, and d. physical 4. Verify formal change management procedures have been developed							
Training									
3.1.1.17	The CA shall ensure all personnel performing duties with respect to operation of a CA,ESP, CSP shall receive comprehensive training covering the following: 1. CA/ESP/CSP/RA security principles and mechanisms 2. All PKI software versions in use on the CA system 3. All PKI duties they are expected to perform 4. Disaster recovery and business continuity procedures	1. Obtain the training plan and material for CA employees 2. Check the following: a. All relevant topics are covered in the training material b. Records of training are maintained by CA (e.g. Attendance sheet) c. Refresher training is conducted d. Training is given personnel in case of implementation of every new tool by the CA e. Personnel are trained with the update business continuity policy and procedures at regular intervals f. Training requirements are reviewed at least once a year 3. Obtain list of personnel entrusted with validation specialist duties 4. Check the following a. Validation specialist are trained to maintain required skill level b. All Validation Specialists have passed the examination provided by the CA 5. Obtain list of personnel	Mandatory	X.509 Policy 5.3.3, IT CA Rules SCHEDULE-III.15					

		responsible for trusted roles 6. Check if the changes made in CA, ESP, CSP operations were communicated with individuals responsible for trusted roles 7. Verify only trusted personals are performing verification at all levels						
3.1.1.18	Refresher training must be conducted as and when required, and the Certifying Authority must review these requirements at least once a year. Records of all trainings shall be maintained and reviewed periodically	1. Obtain the training plan and material for CA employees 2. Check the following: a. All relevant topics are covered in the training material b. Records of training are maintained by CA (e.g. Attendance sheet) c. Refresher training is conducted d. Training is given personnel in case of implementation of every new tool by the CA e. Personnel are trained with the update business continuity policy and procedures at regular intervals f. Training requirements are reviewed at least once a year 3. Obtain list of personnel entrusted with validation specialist duties 4. Check the following a. Validation specialist are trained to maintain required skill level b. All Validation Specialists have passed the examination provided by the CA 5. Obtain list of personnel responsible for trusted roles 6. Check if the changes made in CA, ESP, CSP operations were communicated with individuals responsible for trusted roles 7. Verify only trusted personals are performing verification at all levels	Mandatory	IT CA Rules SCHEDULE III 16 CA Browser Forum 5.3.3				
3.1.1.19	CA shall ensure personnel entrusted with Validation Specialist duties maintain a skill level that enables them to perform such duties satisfactoril	1. Obtain the training plan and material for CA employees 2. Check the following: a. All relevant topics are covered in the training material b. Records of training are maintained by CA (e.g. Attendance sheet) c. Refresher training is conducted d. Training is given personnel in case of implementation of every new tool by the CA e. Personnel are trained with the update business continuity policy and procedures at regular intervals f. Training requirements are	Mandatory	CA Browser Forum 5.3.3				

		reviewed at least once a year 3. Obtain list of personnel entrusted with validation specialist duties 4. Check the following a. Validation specialist are trained to maintain required skill level b. All Validation Specialists have passed the examination provided by the CA 5. Obtain list of personnel responsible for trusted roles 6. Check if the changes made in CA, ESP, CSP operations were communicated with individuals responsible for trusted roles 7. Verify only trusted personals are performing verification at all levels						
3.1.1.20	All Validation Specialists shall pass an examination provided by the CA on the information verification requirements	1. Obtain the training plan and material for CA employees 2. Check the following: a. All relevant topics are covered in the training material b. Records of training are maintained by CA (e.g. Attendance sheet) c. Refresher training is conducted d. Training is given personnel in case of implementation of every new tool by the CA e. Personnel are trained with the update business continuity policy and procedures at regular intervals f. Training requirements are reviewed at least once a year 3. Obtain list of personnel entrusted with validation specialist duties 4. Check the following a. Validation specialist are trained to maintain required skill level b. All Validation Specialists have passed the examination provided by the CA 5. Obtain list of personnel responsible for trusted roles 6. Check if the changes made in CA, ESP, CSP operations were communicated with individuals responsible for trusted roles 7. Verify only trusted personals are performing verification at all levels	Mandatory	CA Browser Forum 5.3.3				
3.1.1.21	All individuals responsible for trusted roles shall be made aware of any significant changes in the CA, ESP CSP, or	1. Obtain the training plan and material for CA employees 2. Check the following: a. All relevant topics are covered in the training material b. Records of training are	Mandatory	X.509 Policy 5.3.4 CA Browser Forum 5.3.4				

	RA operations, as applicable.	maintained by CA (e.g. Attendance sheet) c. Refresher training is conducted d. Training is given personnel in case of implementation of every new tool by the CA e. Personnel are trained with the update business continuity policy and procedures at regular intervals f. Training requirements are reviewed at least once a year 3. Obtain list of personnel entrusted with validation specialist duties 4. Check the following a. Validation specialist are trained to maintain required skill level b. All Validation Specialists have passed the examination provided by the CA 5. Obtain list of personnel responsible for trusted roles 6. Check if the changes made in CA, ESP, CSP operations were communicated with individuals responsible for trusted roles 7. Verify only trusted personals are performing verification at all levels						
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3.1.2. Information Classification

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Information Classification							
3.1.2.1	Documented processes shall be created for classification, declassification, labeling, storage, access, destruction and reproduction of classified data.	1. Obtain copy of the process for handling classified data 2. Check the following a. procedure exists for classifying data based on its sensitivity for the CA b. procedures exist for destruction, access and reproduction of data c. Information is labeled in accordance with CA's information classification documented procedures	Mandatory	IT CA Rules 19.1, WebTrust			
3.1.2.2	CA shall ensure information labeling and handling are	1. Obtain copy of the process for	Mandatory	WebTrust 3.2.4			

	performed in accordance with the CA's information classification documented procedures basis following categories: • Confidential • Restricted • Internal • Unclassified • Top Secret • Secret • Confidentiality • Restricted • Unclassified	handling classified data 2. Check the following a. procedure exists for classifying data based on its sensitivity for the CA b. procedures exist for destruction, access and reproduction of data c. Information is labeled in accordance with CA's information classification documented procedures						
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3.1.3. Information Management

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
System Administration							
3.1.3.1	The CA shall designate a trained "System Administrators" who will ensure that the protective security measures of the system are functional and who will maintain the security posture of CA's organization	1. Obtain the name of system administrator appointed by the CA 2. Check the following: a. Roles and responsibilities of system administrator have been documented. b. a designated System Security Administrator has been appointed for System Administrator with roles and responsibilities documented c. proper training was provided to Security Administrator d. Security Administrator is solely responsible for changes done to information 3. Obtain the password management policy 4. Validate the process implemented in case of forgetting password or changeover to another person 5. Verify if the instance of usage of administrator's passwords is	Mandatory	IT CA Rules SCHEDULE-II 5.1			

		documented securely 6. Obtain the Active Directory of CA employees 7. For sample user accounts verify the System Administrator promptly disabled access if the user is identified as having left the Data Centre, changed assignments, or is no longer requiring system access 8. Validate privileged access is given to users only on a need-to-know and need-to-do basis containing proper business justification and also only after the authorization is documented. 9. Obtain sample of security violations and verify the process followed for recording, investigating and management reviews done. 10. Conduct walkthrough to verify the PKI servers are monitored continuously 11. Obtain list of users who have access to the system 12. Verify only authorized personnel have access and no generic user is enabled or active						
3.1.3.2	The System Administrator may have a designated System Security Administrator who will assume security responsibilities and provide physical, logical and procedural safeguards for information	1. Obtain the name of system administrator appointed by the CA 2. Check the following: a. Roles and responsibilities of system administrator have been documented. b. a designated System Security Administrator has been appointed for System Administrator with roles and responsibilities documented c. proper training was provided to Security	Mandatory	IT CA Rules SCHEDULE-II 5.1				

		Administrator d. Security Administrator is solely responsible for changes done to information 3. Obtain the password management policy 4. Validate the process implemented in case of forgetting password or changeover to another person 5. Verify if the instance of usage of administrator's passwords is documented securely 6. Obtain the Active Directory of CA employees 7. For sample user accounts verify the System Administrator promptly disabled access if the user is identified as having left the Data Centre, changed assignments, or is no longer requiring system access 8. Validate privileged access is given to users only on a need-to-know and need-to-do basis containing proper business justification and also only after the authorization is documented. 9. Obtain sample of security violations and verify the process followed for recording, investigating and management reviews done. 10. Conduct walkthrough to verify the PKI servers are monitored continuously 11. Obtain list of users who have access to the system 12. Verify only authorized personnel have access and no generic user is enabled or active						
3.1.3.3	CA shall ensure System	1. Obtain the name of	Mandatory	IT CA Rules SCHEDULE-II 5.1				

	Security Administrator is properly trained before assigning the system security responsibilities	system administrator appointed by the CA 2. Check the following: a. Roles and responsibilities of system administrator have been documented. b. a designated System Security Administrator has been appointed for System Administrator with roles and responsibilities documented c. proper training was provided to Security Administrator d. Security Administrator is solely responsible for changes done to information 3. Obtain the password management policy 4. Validate the process implemented in case of forgetting password or changeover to another person 5. Verify if the instance of usage of administrator's passwords is documented securely 6. Obtain the Active Directory of CA employees 7. For sample user accounts verify the System Administrator promptly disabled access if the user is identified as having left the Data Centre, changed assignments, or is no longer requiring system access 8. Validate privileged access is given to users only on a need-to-know and need-to-do basis containing proper business justification and also only after the authorization is documented. 9. Obtain sample of security violations and verify the process followed for recording, investigating and						
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		management reviews done. 10. Conduct walkthrough to verify the PKI servers are monitored continuously 11. Obtain list of users who have access to the system 12. Verify only authorized personnel have access and no generic user is enabled or active						
3.1.3.4	System Administrator shall solely be responsible for creating, classifying, and retrieving, modifying, deleting or archiving information.	1. Obtain the name of system administrator appointed by the CA 2. Check the following: a. Roles and responsibilities of system administrator have been documented. b. a designated System Security Administrator has been appointed for System Administrator with roles and responsibilities documented c. proper training was provided to Security Administrator d. Security Administrator is solely responsible for changes done to information 3. Obtain the password management policy 4. Validate the process implemented in case of forgetting password or changeover to another person 5. Verify if the instance of usage of administrator's passwords is documented securely 6. Obtain the Active Directory of CA employees 7. For sample user accounts verify the System Administrator promptly disabled access if the user is identified as having left the Data Centre, changed assignments,	Mandatory	IT CA Rules SCHEDULE-II 5.1				

		or is no longer requiring system access 8. Validate privileged access is given to users only on a need-to-know and need-to-do basis containing proper business justification and also only after the authorization is documented. 9. Obtain sample of security violations and verify the process followed for recording, investigating and management reviews done. 10. Conduct walkthrough to verify the PKI servers are monitored continuously 11. Obtain list of users who have access to the system 12. Verify only authorized personnel have access and no generic user is enabled or active						
3.1.3.5	The system administration's password and operation of trusted services must not be written down (in paper or electronic form) or shared with anyone.	1. Obtain the name of system administrator appointed by the CA 2. Check the following: a. Roles and responsibilities of system administrator have been documented. b. a designated System Security Administrator has been appointed for System Administrator with roles and responsibilities documented c. proper training was provided to Security Administrator d. Security Administrator is solely responsible for changes done to information 3. Obtain the password management policy 4. Validate the process implemented in case of forgetting	Mandatory	IT CA Rules SCHEDULE-II 5.1				

		password or changeover to another person 5. Verify if the instance of usage of administrator's passwords is documented securely 6. Obtain the Active Directory of CA employees 7. For sample user accounts verify the System Administrator promptly disabled access if the user is identified as having left the Data Centre, changed assignments, or is no longer requiring system access 8. Validate privileged access is given to users only on a need-to-know and need-to-do basis containing proper business justification and also only after the authorization is documented. 9. Obtain sample of security violations and verify the process followed for recording, investigating and management reviews done. 10. Conduct walkthrough to verify the PKI servers are monitored continuously 11. Obtain list of users who have access to the system 12. Verify only authorized personnel have access and no generic user is enabled or active						
3.1.3.6	A system for password management should be put in place to cover the eventualities such as forgotten password or changeover to another person in case of System Administrator (or System Security Administrator) leaving the organization. Every instance of usage of administrator's passwords must be	1. Obtain the name of system administrator appointed by the CA 2. Check the following: a. Roles and responsibilities of system administrator have been documented. b. a designated System Security Administrator has	Mandatory	IT CA Rules SCHEDULE-II 5.1				

	documented.	been appointed for System Administrator with roles and responsibilities documented c. proper training was provided to Security Administrator d. Security Administrator is solely responsible for changes done to information 3. Obtain the password management policy 4. Validate the process implemented in case of forgetting password or changeover to another person 5. Verify if the instance of usage of administrator's passwords is documented securely 6. Obtain the Active Directory of CA employees 7. For sample user accounts verify the System Administrator promptly disabled access if the user is identified as having left the Data Centre, changed assignments, or is no longer requiring system access 8. Validate privileged access is given to users only on a need-to-know and need-to-do basis containing proper business justification and also only after the authorization is documented. 9. Obtain sample of security violations and verify the process followed for recording, investigating and management reviews done. 10. Conduct walkthrough to verify the PKI servers are monitored continuously 11. Obtain list of users who have access to the system 12. Verify only authorized personnel						
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		have access and no generic user is enabled or active						
3.1.3.7	The System Administrator shall promptly disable access to a user's account if the user is identified as having left the Data Centre, changed assignments, or is no longer requiring system access. Reactivation of the user's account must be authorized in writing by the System Administrator (Digitally signed E-mail may be acceptable)	1. Obtain the name of system administrator appointed by the CA 2. Check the following: a. Roles and responsibilities of system administrator have been documented. b. a designated System Security Administrator has been appointed for System Administrator with roles and responsibilities documented c. proper training was provided to Security Administrator d. Security Administrator is solely responsible for changes done to information 3. Obtain the password management policy 4. Validate the process implemented in case of forgetting password or changeover to another person 5. Verify if the instance of usage of administrator's passwords is documented securely 6. Obtain the Active Directory of CA employees 7. For sample user accounts verify the System Administrator promptly disabled access if the user is identified as having left the Data Centre, changed assignments, or is no longer requiring system access 8. Validate privileged access is given to users only on a need-to-know and need-to-do basis containing proper business justification and also only after the authorization is	Mandatory	IT CA Rules SCHEDULE-II 5.1				

		documented. 9. Obtain sample of security violations and verify the process followed for recording, investigating and management reviews done. 10. Conduct walkthrough to verify the PKI servers are monitored continuously 11. Obtain list of users who have access to the system 12. Verify only authorized personnel have access and no generic user is enabled or active						
3.1.3.8	The System Administrator must take steps to safeguards classified information as prescribed by its owner.	1. Obtain the name of system administrator appointed by the CA 2. Check the following: a. Roles and responsibilities of system administrator have been documented. b. a designated System Security Administrator has been appointed for System Administrator with roles and responsibilities documented c. proper training was provided to Security Administrator d. Security Administrator is solely responsible for changes done to information 3. Obtain the password management policy 4. Validate the process implemented in case of forgetting password or changeover to another person 5. Verify if the instance of usage of administrator's passwords is documented securely 6. Obtain the Active Directory of CA employees 7. For sample user	Mandatory	IT CA Rules SCHEDULE-II 5.1				

		accounts verify the System Administrator promptly disabled access if the user is identified as having left the Data Centre, changed assignments, or is no longer requiring system access 8. Validate privileged access is given to users only on a need-to-know and need-to-do basis containing proper business justification and also only after the authorization is documented. 9. Obtain sample of security violations and verify the process followed for recording, investigating and management reviews done. 10. Conduct walkthrough to verify the PKI servers are monitored continuously 11. Obtain list of users who have access to the system 12. Verify only authorized personnel have access and no generic user is enabled or active					
3.1.3.9	The System Administrator must authorize privileged access to users only on a need-to-know and need-to-do basis and also only after the authorization is documented.	1. Obtain the name of system administrator appointed by the CA 2. Check the following: a. Roles and responsibilities of system administrator have been documented. b. a designated System Security Administrator has been appointed for System Administrator with roles and responsibilities documented c. proper training was provided to Security Administrator d. Security Administrator is solely responsible for changes done to	Mandatory	IT CA Rules SCHEDULE-II 5.1			

		information 3. Obtain the password management policy 4. Validate the process implemented in case of forgetting password or changeover to another person 5. Verify if the instance of usage of administrator's passwords is documented securely 6. Obtain the Active Directory of CA employees 7. For sample user accounts verify the System Administrator promptly disabled access if the user is identified as having left the Data Centre, changed assignments, or is no longer requiring system access 8. Validate privileged access is given to users only on a need-to-know and need-to-do basis containing proper business justification and also only after the authorization is documented. 9. Obtain sample of security violations and verify the process followed for recording, investigating and management reviews done. 10. Conduct walkthrough to verify the PKI servers are monitored continuously 11. Obtain list of users who have access to the system 12. Verify only authorized personnel have access and no generic user is enabled or active					
3.1.3.10	All security violations must be recorded, investigated, and periodic status reports compiled for review by the management	1. Obtain the name of system administrator appointed by the CA 2. Check the following: a. Roles and	Mandatory	IT CA Rules SCHEDULE-II 5.1			

		responsibilities of system administrator have been documented. b. a designated System Security Administrator has been appointed for System Administrator with roles and responsibilities documented c. proper training was provided to Security Administrator d. Security Administrator is solely responsible for changes done to information 3. Obtain the password management policy 4. Validate the process implemented in case of forgetting password or changeover to another person 5. Verify if the instance of usage of administrator's passwords is documented securely 6. Obtain the Active Directory of CA employees 7. For sample user accounts verify the System Administrator promptly disabled access if the user is identified as having left the Data Centre, changed assignments, or is no longer requiring system access 8. Validate privileged access is given to users only on a need-to-know and need-to-do basis containing proper business justification and also only after the authorization is documented. 9. Obtain sample of security violations and verify the process followed for recording, investigating and management reviews done. 10. Conduct walkthrough to verify the PKI servers are						
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		monitored continuously 11. Obtain list of users who have access to the system 12. Verify only authorized personnel have access and no generic user is enabled or active						
3.1.3.11	The System Administrator together with the system support staff, shall conduct a regular analysis of problems reported to and identify any weaknesses in protection of the information	1. Obtain the name of system administrator appointed by the CA 2. Check the following: a. Roles and responsibilities of system administrator have been documented. b. a designated System Security Administrator has been appointed for System Administrator with roles and responsibilities documented c. proper training was provided to Security Administrator d. Security Administrator is solely responsible for changes done to information 3. Obtain the password management policy 4. Validate the process implemented in case of forgetting password or changeover to another person 5. Verify if the instance of usage of administrator's passwords is documented securely 6. Obtain the Active Directory of CA employees 7. For sample user accounts verify the System Administrator promptly disabled access if the user is identified as having left the Data Centre, changed assignments, or is no longer requiring system access 8. Validate privileged access is given to	Mandatory	IT CA Rules SCHEDULE-II 5.1				

		users only on a need-to-know and need-to-do basis containing proper business justification and also only after the authorization is documented. 9. Obtain sample of security violations and verify the process followed for recording, investigating and management reviews done. 10. Conduct walkthrough to verify the PKI servers are monitored continuously 11. Obtain list of users who have access to the system 12. Verify only authorized personnel have access and no generic user is enabled or active						
3.1.3.12	The System Administrator shall ensure that the data, file and Public Key Infrastructure (PKI) servers are not left unmonitored while these systems are powered on	1. Obtain the name of system administrator appointed by the CA 2. Check the following: a. Roles and responsibilities of system administrator have been documented. b. a designated System Security Administrator has been appointed for System Administrator with roles and responsibilities documented c. proper training was provided to Security Administrator d. Security Administrator is solely responsible for changes done to information 3. Obtain the password management policy 4. Validate the process implemented in case of forgetting password or changeover to another person 5. Verify if the instance of usage of	Mandatory	IT CA Rules SCHEDULE-II 5.1				

		administrator's passwords is documented securely 6. Obtain the Active Directory of CA employees 7. For sample user accounts verify the System Administrator promptly disabled access if the user is identified as having left the Data Centre, changed assignments, or is no longer requiring system access 8. Validate privileged access is given to users only on a need-to-know and need-to-do basis containing proper business justification and also only after the authorization is documented. 9. Obtain sample of security violations and verify the process followed for recording, investigating and management reviews done. 10. Conduct walkthrough to verify the PKI servers are monitored continuously 11. Obtain list of users who have access to the system 12. Verify only authorized personnel have access and no generic user is enabled or active					
3.1.3.13	The System Administrator should ensure that no generic user is enabled or active on the system	1. Obtain the name of system administrator appointed by the CA 2. Check the following: a. Roles and responsibilities of system administrator have been documented. b. a designated System Security Administrator has been appointed for System Administrator with roles and responsibilities documented	Mandatory	IT CA Rules SCHEDULE-II 5.1			

		c. proper training was provided to Security Administrator d. Security Administrator is solely responsible for changes done to information 3. Obtain the password management policy 4. Validate the process implemented in case of forgetting password or changeover to another person 5. Verify if the instance of usage of administrator's passwords is documented securely 6. Obtain the Active Directory of CA employees 7. For sample user accounts verify the System Administrator promptly disabled access if the user is identified as having left the Data Centre, changed assignments, or is no longer requiring system access 8. Validate privileged access is given to users only on a need-to-know and need-to-do basis containing proper business justification and also only after the authorization is documented. 9. Obtain sample of security violations and verify the process followed for recording, investigating and management reviews done. 10. Conduct walkthrough to verify the PKI servers are monitored continuously 11. Obtain list of users who have access to the system 12. Verify only authorized personnel have access and no generic user is enabled or active							
Sensitive Information Control									

3.1.3.14	Electronic communication systems, such as router, switches, network device and computers, used for transmission of sensitive information should be equipped or installed with suitable security software and if necessary with an encryptor or encryption software. The appropriate procedure in this regard shall be documented	1. Obtain documented procedure to ensure safety of communication systems 2. Obtain list of communication systems (i.e. router, switches & computers) used for transmission of sensitive information 3. Verify security implementation in devices is done as per the procedure	Mandatory	IT CA Rules SCHEDULE-II 5.2				
3.1.3.15	The Certifying Authority shall manage its functions in accordance with the levels of integrity and security approved by the Controller from time to time	1. Verify the CA manages its functions in accordance with the levels of integrity and security approved by the Controller from time to time. 2. Validate the CA discloses information on the assurance levels of certificates that it issues and the limitations of its liabilities to each of its subscribers and relying parties	Mandatory	IT Regulations 3				
3.1.3.16	The Certifying Authority shall disclose information on the assurance levels of the certificates that it issues and the limitations of its liabilities to each of its subscribers and relying parties	1. Verify the CA manages its functions in accordance with the levels of integrity and security approved by the Controller from time to time. 2. Validate the CA discloses information on the assurance levels of certificates that it issues and the limitations of its liabilities to each of its subscribers and relying parties	Mandatory	IT Regulations 3				
Sensitive Information Security								
3.1.3.17	Highly sensitive information assets shall be stored on secure removable media and should be in an encrypted format to avoid compromise by unauthorized persons.	1. Refer the Process for handling classified data 2. Verify Highly sensitive information assets is stored on secure removable media and is in encrypted format 3. Verify highly sensitive information	Mandatory	T CA Rules SCHEDULE-II 5.3				

		is classified in accordance as per Control No. 3.1.2.1 4. Validate access control mechanism exist for cases where multiple people have access to sensitive information and data 5. Conduct a walkthrough to verify removable electronic storage media is removed from the computer and properly secured at the end of the work session or workday						
3.1.3.18	Highly sensitive information shall be classified in accordance as per Control No. 3.1.2.1	1. Refer the Process for handling classified data 2. Verify Highly sensitive information assets is stored on secure removable media and is in encrypted format 3. Verify highly sensitive information is classified in accordance as per Control No. 3.1.2.1 4. Validate access control mechanism exist for cases where multiple people have access to sensitive information and data 5. Conduct a walkthrough to verify removable electronic storage media is removed from the computer and properly secured at the end of the work session or workday	Mandatory	IT CA Rules SCHEDULE-II 5.3				
3.1.3.19	Sensitive information and data, which are stored on the fixed disk of a computer shared by more than one person, must be protected by access control software (e.g., password). Security packages must be installed which partition or provide authorization to segregated directories/files.	1. Refer the Process for handling classified data 2. Verify Highly sensitive information assets is stored on secure removable media and is in encrypted format 3. Verify highly sensitive information is classified in accordance as per Control No. 3.1.2.1	Mandatory	IT CA Rules SCHEDULE-II 5.3				

		4. Validate access control mechanism exist for cases where multiple people have access to sensitive information and data 5. Conduct a walkthrough to verify removable electronic storage media is removed from the computer and properly secured at the end of the work session or workday						
3.1.3.20	Removable electronic storage media must be removed from the computer and properly secured at the end of the work session or workday.	1. Refer the Process for handling classified data 2. Verify Highly sensitive information assets is stored on secure removable media and is in encrypted format 3. Verify highly sensitive information is classified in accordance as per Control No. 3.1.2.1 4. Validate access control mechanism exist for cases where multiple people have access to sensitive information and data 5. Conduct a walkthrough to verify removable electronic storage media is removed from the computer and properly secured at the end of the work session or workday	Mandatory	IT CA Rules SCHEDULE-II 5.3				
Prevention of Computer Misuse								
3.1.3.21	The computer system shall be configured with minimum of the required accounts and network services.	1. For sample computer system verify configuration is done with minimum of the required accounts and network services. 2. Validate systems are not working on default settings 3. Obtain documented measures for safeguard the security of computers and computer information from misuse 4. Check the	Mandatory	X.509 Policy 6.5.1				

		following: a. Implementation of measures for safeguarding computers and information from misuse b. Breach reporting process and remedial measures taken c. All incidents are reported to System Administrator d. Procedure to prevent future occurrence of an incident						
3.1.3.22	Prevention, detection, and deterrence measures shall be implemented to safeguard the security of computers and computer information from misuse. The measures taken shall be properly documented and reviewed regularly.	1. For sample computer system verify configuration is done with minimum of the required accounts and network services. 2. Validate systems are not working on default settings 3. Obtain documented measures for safeguard the security of computers and computer information from misuse 4. Check the following: a. Implementation of measures for safeguarding computers and information from misuse b. Breach reporting process and remedial measures taken c. All incidents are reported to System Administrator d. Procedure to prevent future occurrence of an incident	Mandatory	IT CA Rules SCHEDULE-II 5.5				
3.1.3.23	CA shall ensure adequate information is provided to all persons, including management, systems developers and programmers, end-users, and third party users warning them against misuse of computers.	1. For sample computer system verify configuration is done with minimum of the required accounts and network services. 2. Validate systems are not working on default settings 3. Obtain documented measures for safeguard the security	Mandatory	IT CA Rules SCHEDULE-II 5.5				

		of computers and computer information from misuse 4. Check the following: a. Implementation of measures for safeguarding computers and information from misuse b. Breach reporting process and remedial measures taken c. All incidents are reported to System Administrator d. Procedure to prevent future occurrence of an incident						
3.1.3.24	Effective measures including reporting of suspected breach, investigation and remedial measures shall be established	1. For sample computer system verify configuration is done with minimum of the required accounts and network services. 2. Validate systems are not working on default settings 3. Obtain documented measures for safeguard the security of computers and computer information from misuse 4. Check the following: a. Implementation of measures for safeguarding computers and information from misuse b. Breach reporting process and remedial measures taken c. All incidents are reported to System Administrator d. Procedure to prevent future occurrence of an incident	Mandatory	IT CA Rules SCHEDULE-II 5.5				
3.1.3.25	All incidents related to breaches shall be reported to the System Administrator or System Security Administrator for appropriate action to prevent future occurrence. These will be responsible for investigation and	1. For sample computer system verify configuration is done with minimum of the required accounts and network services. 2. Validate systems are not working on	Mandatory	IT CA Rules SCHEDULE-II 5.5				

	follow up action	default settings 3. Obtain documented measures for safeguard the security of computers and computer information from misuse 4. Check the following: a. Implementation of measures for safeguarding computers and information from misuse b. Breach reporting process and remedial measures taken c. All incidents are reported to System Administrator d. Procedure to prevent future occurrence of an incident						
3.1.3.26	Procedure shall be set-up to establish the nature of any alleged abuse and determine the subsequent action required to be taken to prevent its future occurrence	1. For sample computer system verify configuration is done with minimum of the required accounts and network services. 2. Validate systems are not working on default settings 3. Obtain documented measures for safeguard the security of computers and computer information from misuse 4. Check the following: a. Implementation of measures for safeguarding computers and information from misuse b. Breach reporting process and remedial measures taken c. All incidents are reported to System Administrator d. Procedure to prevent future occurrence of an incident	Mandatory	IT CA Rules SCHEDULE-II 5.5				

3.1.4. Physical and Operational security

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Site Location & Design							

3.1.4.1	The location and construction of the facility housing CA and Certificate Status Provider (CSP) equipment shall be consistent with facilities used to house high value, sensitive information. The site shall not be in locations that are prone to natural or man-made disasters, like flood, fire, chemical contamination and explosions	1. Verify the CA facility housing CA and CSP equipment is consistent with facilities used to house high value, sensitive information 2. Validate the site is located in secure area immune from natural or man-made disasters 3. Verify applicable building and safety regulations were complied to during construction 4. Conduct walkthrough to check the following: a. External walls are constructed of brick or reinforced concrete of sufficient thickness b. Ground level windows are fortified with sturdy mild steel grills c. All internal walls are tamper- evident d. Air-conditioning system, power supply system and uninterrupted power supply unit are installed e. Proper backup is present in case of power outage f. Ducting holes are designed to prevent intrusion g. Media library, electrical and mechanical control rooms are housed in separate isolated areas h. Access to these areas is granted to specific, named individuals on a need basis i. Media library, electrical and mechanical control rooms is housed in separate isolated areas 5. For mission-critical and sensitive applications verify the facility is located and designed for reparability, relocation and reconfiguration 6. Verify containers are used for storing movable media and papers containing sensitive or plain text information	Mandatory	X.509 Policy 5.1.1, IT CA Rules SCHEDULE-III 3.1				
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		7. Verify measures CA has implemented to keep the location of DSC system hidden						
3.1.4.2	Construction shall be done in compliance with all applicable building and safety regulations as laid down by the relevant Government agencies with use of fire resistant material and non-toxic chemicals.	1. Verify the CA facility housing CA and CSP equipment is consistent with facilities used to house high value, sensitive information 2. Validate the site is located in secure area immune from natural or man-made disasters 3. Verify applicable building and safety regulations were complied to during construction 4. Conduct walkthrough to check the following: a. External walls are constructed of brick or reinforced concrete of sufficient thickness b. Ground level windows are fortified with sturdy mild steel grills c. All internal walls are tamper- evident d. Air-conditioning system, power supply system and uninterrupted power supply unit are installed e. Proper backup is present in case of power outage f. Ducting holes are designed to prevent intrusion g. Media library, electrical and mechanical control rooms are housed in separate isolated areas h. Access to these areas is granted to specific, named individuals on a need basis i. Media library, electrical and mechanical control rooms is housed in separate isolated areas 5. For mission-critical and sensitive applications verify the facility is located and designed for reparability, relocation	Mandatory	IT CA Rules SCHEDULE-II 4.1.3				

		and reconfiguration 6. Verify containers are used for storing movable media and papers containing sensitive or plain text information 7. Verify measures CA has implemented to keep the location of DSC system hidden						
3.1.4.3	CA Shall ensure all external walls shall be constructed of brick or reinforced concrete of sufficient thickness to resist forcible attack. Ground level windows shall be fortified with sturdy mild steel grills or impact-resistant laminated security glass. All internal walls must be from the floor to the ceiling and must be tamper-evident	1. Verify the CA facility housing CA and CSP equipment is consistent with facilities used to house high value, sensitive information 2. Validate the site is located in secure area immune from natural or man-made disasters 3. Verify applicable building and safety regulations were complied to during construction 4. Conduct walkthrough to check the following: a. External walls are constructed of brick or reinforced concrete of sufficient thickness b. Ground level windows are fortified with sturdy mild steel grills c. All internal walls are tamper- evident d. Air-conditioning system, power supply system and uninterrupted power supply unit are installed e. Proper backup is present in case of power outage f. Ducting holes are designed to prevent intrusion g. Media library, electrical and mechanical control rooms are housed in separate isolated areas h. Access to these areas is granted to specific, named individuals on a need basis i. Media library, electrical and mechanical control	Mandatory	IT CA Rules SCHEDULE-II 4.1.5				

		rooms is housed in separate isolated areas 5. For mission-critical and sensitive applications verify the facility is located and designed for reparability, relocation and reconfiguration 6. Verify containers are used for storing movable media and papers containing sensitive or plain text information 7. Verify measures CA has implemented to keep the location of DSC system hidden						
3.1.4.4	Air-conditioning system, power supply system and uninterrupted power supply unit with proper backup shall be installed depending upon the nature of operation. All ducting holes of the air-conditioning system must be designed so as to prevent intrusion of any kind.	1. Verify the CA facility housing CA and CSP equipment is consistent with facilities used to house high value, sensitive information 2. Validate the site is located in secure area immune from natural or man-made disasters 3. Verify applicable building and safety regulations were complied to during construction 4. Conduct walkthrough to check the following: a. External walls are constructed of brick or reinforced concrete of sufficient thickness b. Ground level windows are fortified with sturdy mild steel grills c. All internal walls are tamper- evident d. Air-conditioning system, power supply system and uninterrupted power supply unit are installed e. Proper backup is present in case of power outage f. Ducting holes are designed to prevent intrusion g. Media library, electrical and mechanical control rooms are housed in	Mandatory	IT CA Rules SCHEDULE-II 4.1.6				

		separate isolated areas h. Access to these areas is granted to specific, named individuals on a need basis i. Media library, electrical and mechanical control rooms is housed in separate isolated areas 5. For mission-critical and sensitive applications verify the facility is located and designed for reparability, relocation and reconfiguration 6. Verify containers are used for storing movable media and papers containing sensitive or plain text information 7. Verify measures CA has implemented to keep the location of DSC system hidden						
3.1.4.5	Media library, electrical and mechanical control rooms shall be housed in separate isolated areas, with access granted only to specific, named individuals on a need basis	1. Verify the CA facility housing CA and CSP equipment is consistent with facilities used to house high value, sensitive information 2. Validate the site is located in secure area immune from natural or man-made disasters 3. Verify applicable building and safety regulations were complied to during construction 4. Conduct walkthrough to check the following: a. External walls are constructed of brick or reinforced concrete of sufficient thickness b. Ground level windows are fortified with sturdy mild steel grills c. All internal walls are tamper- evident d. Air-conditioning system, power supply system and uninterrupted power supply unit are installed e. Proper backup is present in case of	Mandatory	IT CA Rules SCHEDULE-II 4.1.8				

		power outage f. Ducting holes are designed to prevent intrusion g. Media library, electrical and mechanical control rooms are housed in separate isolated areas h. Access to these areas is granted to specific, named individuals on a need basis i. Media library, electrical and mechanical control rooms is housed in separate isolated areas 5. For mission-critical and sensitive applications verify the facility is located and designed for reparability, relocation and reconfiguration 6. Verify containers are used for storing movable media and papers containing sensitive or plain text information 7. Verify measures CA has implemented to keep the location of DSC system hidden						
3.1.4.6	Any facility that supports mission-critical and sensitive applications must be located and designed for reparability, relocation and reconfiguration. The ability to relocate, reconstitute and reconfigure these applications must be tested as part of the business continuity/disaster recovery plan.	1. Verify the CA facility housing CA and CSP equipment is consistent with facilities used to house high value, sensitive information 2. Validate the site is located in secure area immune from natural or man-made disasters 3. Verify applicable building and safety regulations were complied to during construction 4. Conduct walkthrough to check the following: a. External walls are constructed of brick or reinforced concrete of sufficient thickness b. Ground level windows are fortified with sturdy mild steel grills c. All internal walls are tamper- evident	Mandatory	IT CA Rules SCHEDULE-II 4.1.9				

		d. Air-conditioning system, power supply system and uninterrupted power supply unit are installed e. Proper backup is present in case of power outage f. Ducting holes are designed to prevent intrusion g. Media library, electrical and mechanical control rooms are housed in separate isolated areas h. Access to these areas is granted to specific, named individuals on a need basis i. Media library, electrical and mechanical control rooms is housed in separate isolated areas 5. For mission-critical and sensitive applications verify the facility is located and designed for reparability, relocation and reconfiguration 6. Verify containers are used for storing movable media and papers containing sensitive or plain text information 7. Verify measures CA has implemented to keep the location of DSC system hidden					
3.1.4.7	All removable media and papers containing sensitive or plain text information shall be listed, documented and stored in a container properly identified in the CA Facilit	1. Verify the CA facility housing CA and CSP equipment is consistent with facilities used to house high value, sensitive information 2. Validate the site is located in secure area immune from natural or man-made disasters 3. Verify applicable building and safety regulations were complied to during construction 4. Conduct walkthrough to check the following: a. External walls are constructed of brick or	Mandatory	IT CA Rules SCHEDULE-III 3.3			

		reinforced concrete of sufficient thickness b. Ground level windows are fortified with sturdy mild steel grills c. All internal walls are tamper- evident d. Air-conditioning system, power supply system and uninterrupted power supply unit are installed e. Proper backup is present in case of power outage f. Ducting holes are designed to prevent intrusion g. Media library, electrical and mechanical control rooms are housed in separate isolated areas h. Access to these areas is granted to specific, named individuals on a need basis i. Media library, electrical and mechanical control rooms is housed in separate isolated areas 5. For mission-critical and sensitive applications verify the facility is located and designed for reparability, relocation and reconfiguration 6. Verify containers are used for storing movable media and papers containing sensitive or plain text information 7. Verify measures CA has implemented to keep the location of DSC system hidden					
3.1.4.8	CA shall ensure exact location of Digital Signature Certification System shall not be publicly identified	1. Verify the CA facility housing CA and CSP equipment is consistent with facilities used to house high value, sensitive information 2. Validate the site is located in secure area immune from natural or man-made disasters 3. Verify applicable building and safety	Mandatory	IT CA Rules SCHEDULE-III 3.3			

		regulations were complied to during construction 4. Conduct walkthrough to check the following: a. External walls are constructed of brick or reinforced concrete of sufficient thickness b. Ground level windows are fortified with sturdy mild steel grills c. All internal walls are tamper- evident d. Air-conditioning system, power supply system and uninterrupted power supply unit are installed e. Proper backup is present in case of power outage f. Ducting holes are designed to prevent intrusion g. Media library, electrical and mechanical control rooms are housed in separate isolated areas h. Access to these areas is granted to specific, named individuals on a need basis i. Media library, electrical and mechanical control rooms is housed in separate isolated areas 5. For mission-critical and sensitive applications verify the facility is located and designed for reparability, relocation and reconfiguration 6. Verify containers are used for storing movable media and papers containing sensitive or plain text information 7. Verify measures CA has implemented to keep the location of DSC system hidden							
Fire Protection									
3.1.4.9	No combustible materials shall be stored within hundred meters of the operational site	1. Conduct walkthrough to check the following: a. No combustible materials is stored	Mandatory	IT CA Rules SCHEDULE-II 4.2					

		within hundred meters of the operational site b. Automatic fire detection, fire suppression systems and audible alarms are installed at the operational site c. Fire extinguishers are installed at operational site d. Location of fire extinguishers is clearly marked with appropriate signs e. Periodic testing, inspection and maintenance of the fire equipment and fire suppression systems are carried out f. Eating, drinking and smoking are prohibited in the operational site g. Operational site is clean h. Procedures for the safe evacuation of personnel in an emergency are visibly pasted/displayed at prominent places i. Fire doors exist on security perimeters 2. Obtain schedule of safety trainings and fire drills and verify they are conducted on periodic basis					
3.1.4.10	Automatic fire detection, fire suppression systems and audible alarms as prescribed by the Fire Brigade or any other agency of the Central or State Government shall be installed at the operational site	1. Conduct walkthrough to check the following: a. No combustible materials is stored within hundred meters of the operational site b. Automatic fire detection, fire suppression systems and audible alarms are installed at the operational site c. Fire extinguishers are installed at operational site d. Location of fire extinguishers is clearly marked with appropriate signs e. Periodic testing, inspection and maintenance of the fire equipment and fire suppression systems are carried out f. Eating, drinking and	Mandatory	IT CA Rules SCHEDULE-II 4.2			

		smoking are prohibited in the operational site g. Operational site is clean h. Procedures for the safe evacuation of personnel in an emergency are visibly pasted/displayed at prominent places i. Fire doors exist on security perimeters 2. Obtain schedule of safety trainings and fire drills and verify they are conducted on periodic basis						
3.1.4.11	Fire extinguishers shall be installed at the operational site and their locations clearly marked with appropriate signs	1. Conduct walkthrough to check the following: a. No combustible materials is stored within hundred meters of the operational site b. Automatic fire detection, fire suppression systems and audible alarms are installed at the operational site c. Fire extinguishers are installed at operational site d. Location of fire extinguishers is clearly marked with appropriate signs e. Periodic testing, inspection and maintenance of the fire equipment and fire suppression systems are carried out f. Eating, drinking and smoking are prohibited in the operational site g. Operational site is clean h. Procedures for the safe evacuation of personnel in an emergency are visibly pasted/displayed at prominent places i. Fire doors exist on security perimeters 2. Obtain schedule of safety trainings and fire drills and verify they are conducted on periodic basis	Mandatory	IT CA Rules SCHEDULE-II 4.2				
3.1.4.12	Periodic testing, inspection and maintenance of the fire	1. Conduct walkthrough to check the following:	Mandatory	IT CA Rules SCHEDULE-II 4.3				

	equipment and fire suppression systems should be carried out.	a. No combustible materials is stored within hundred meters of the operational site b. Automatic fire detection, fire suppression systems and audible alarms are installed at the operational site c. Fire extinguishers are installed at operational site d. Location of fire extinguishers is clearly marked with appropriate signs e. Periodic testing, inspection and maintenance of the fire equipment and fire suppression systems are carried out f. Eating, drinking and smoking are prohibited in the operational site g. Operational site is clean h. Procedures for the safe evacuation of personnel in an emergency are visibly pasted/displayed at prominent places i. Fire doors exist on security perimeters 2. Obtain schedule of safety trainings and fire drills and verify they are conducted on periodic basis						
3.1.4.13	There shall be no eating, drinking or smoking in the operational site	1. Conduct walkthrough to check the following: a. No combustible materials is stored within hundred meters of the operational site b. Automatic fire detection, fire suppression systems and audible alarms are installed at the operational site c. Fire extinguishers are installed at operational site d. Location of fire extinguishers is clearly marked with appropriate signs e. Periodic testing, inspection and maintenance of the fire equipment and fire suppression systems	Mandatory	IT CA Rules SCHEDULE-II 4.2				

		are carried out f. Eating, drinking and smoking are prohibited in the operational site g. Operational site is clean h. Procedures for the safe evacuation of personnel in an emergency are visibly pasted/displayed at prominent places i. Fire doors exist on security perimeters 2. Obtain schedule of safety trainings and fire drills and verify they are conducted on periodic basis						
3.1.4.14	The work areas shall be kept clean at all times.	1. Conduct walkthrough to check the following: a. No combustible materials is stored within hundred meters of the operational site b. Automatic fire detection, fire suppression systems and audible alarms are installed at the operational site c. Fire extinguishers are installed at operational site d. Location of fire extinguishers is clearly marked with appropriate signs e. Periodic testing, inspection and maintenance of the fire equipment and fire suppression systems are carried out f. Eating, drinking and smoking are prohibited in the operational site g. Operational site is clean h. Procedures for the safe evacuation of personnel in an emergency are visibly pasted/displayed at prominent places i. Fire doors exist on security perimeters 2. Obtain schedule of safety trainings and fire drills and verify they are conducted on periodic basis	Mandatory	IT CA Rules SCHEDULE-II 4.2				
3.1.4.15	Procedures for the safe	1. Conduct	Mandatory	IT CA Rules SCHEDULE-				

	evacuation of personnel in an emergency shall be visibly posted/displayed at prominent places at the operational site	walkthrough to check the following: a. No combustible materials is stored within hundred meters of the operational site b. Automatic fire detection, fire suppression systems and audible alarms are installed at the operational site c. Fire extinguishers are installed at operational site d. Location of fire extinguishers is clearly marked with appropriate signs e. Periodic testing, inspection and maintenance of the fire equipment and fire suppression systems are carried out f. Eating, drinking and smoking are prohibited in the operational site g. Operational site is clean h. Procedures for the safe evacuation of personnel in an emergency are visibly posted/displayed at prominent places i. Fire doors exist on security perimeters 2. Obtain schedule of safety trainings and fire drills and verify they are conducted on periodic basis		II 4.2				
3.1.4.16	Periodic training and fire drills shall be conducted	1. Conduct walkthrough to check the following: a. No combustible materials is stored within hundred meters of the operational site b. Automatic fire detection, fire suppression systems and audible alarms are installed at the operational site c. Fire extinguishers are installed at operational site d. Location of fire extinguishers is clearly marked with appropriate signs e. Periodic testing, inspection and maintenance of the fire	Mandatory	IT CA Rules SCHEDULE-II 4.2				

		equipment and fire suppression systems are carried out f. Eating, drinking and smoking are prohibited in the operational site g. Operational site is clean h. Procedures for the safe evacuation of personnel in an emergency are visibly pasted/displayed at prominent places i. Fire doors exist on security perimeters 2. Obtain schedule of safety trainings and fire drills and verify they are conducted on periodic basis						
3.1.4.17	Fire doors exist on security perimeters around CA operational facilities and are alarmed and conform to local fire regulations.	1. Conduct walkthrough to check the following: a. No combustible materials is stored within hundred meters of the operational site b. Automatic fire detection, fire suppression systems and audible alarms are installed at the operational site c. Fire extinguishers are installed at operational site d. Location of fire extinguishers is clearly marked with appropriate signs e. Periodic testing, inspection and maintenance of the fire equipment and fire suppression systems are carried out f. Eating, drinking and smoking are prohibited in the operational site g. Operational site is clean h. Procedures for the safe evacuation of personnel in an emergency are visibly pasted/displayed at prominent places i. Fire doors exist on security perimeters 2. Obtain schedule of safety trainings and fire drills and verify they are conducted on periodic basis	Mandatory	WebTrust 3.4.6				

Physical Access								
3.1.4.18	Responsibilities round the clock, seven days a week, three hundred sixty five days a year for physical security of the systems used for operation and also actual physical layout at the site of operation shall be defined and assigned to named individuals	1. Obtain copy of the responsibilities assigned for physical security and verify individuals have been assigned for the same 2. Conduct walkthrough to check the following a. All CA personnel have identity and authorization verified before they are included in the access list to CAs system b. All personnel wear visible identification c. Entrance to the main building and entrance to each security zone is video recorded round the clock. Verify on sample basis for last one year for any random date of each year d. Check the recording is carefully scrutinized and maintained for at least one year by verifying logs for a sample dates in the past year e. Biometric physical access security systems are installed and functioning in the operational site f. Unauthorized intrusion is manually or electronically monitored at all times at the CAs Site. Verify by checking the register/list maintained for same g. Physical access to CA facilities and equipment is limited to authorized individuals, protected through restricted security perimeters, and is operated under multiple person (at least dual custody) control. Verify by checking the access list / and checking the name of the personnel for dual custody. Physically see also, how the two personnel are involved in the operation h. A manned reception	Mandatory	IT CA Rules SCHEDULE-II 4.4, CA Browser Forum 5.1.2				

		area is present at operational site i. Unescorted access to server is limited to those personnel identified on an access list. Verify by checking list of personnel allowed j. Access security system is installed to control and audit details of personnel using the DSC System. k. Access security system works on the basis of digital signature certification system						
3.1.4.19	All Certifying Authority personnel must have their identity and authorization verified before they are included in the access list for physical access to the Certifying Authority's system. All personnel are required to wear visible identification. Employees are encouraged to challenge anyone not wearing visible identification	1. Obtain copy of the responsibilities assigned for physical security and verify individuals have been assigned for the same 2. Conduct walkthrough to check the following a. All CA personnel have identity and authorization verified before they are included in the access list to CAs system b. All personnel wear visible identification c. Entrance to the main building and entrance to each security zone is video recorded round the clock. Verify on sample basis for last one year for any random date of each year d. Check the recording is carefully scrutinized and maintained for at least one year by verifying logs for a sample dates in the past year e. Biometric physical access security systems are installed and functioning in the operational site f. Unauthorized intrusion is manually or electronically monitored at all times at the CAs Site. Verify by checking the	Mandatory	WebTrust 3.4.9 IT CA Rules SCHEDULE-				

		register/list maintained for same g. Physical access to CA facilities and equipment is limited to authorized individuals, protected through restricted security perimeters, and is operated under multiple person (at least dual custody) control. Verify by checking the access list / and checking the name of the personnel for dual custody. Physically see also, how the two personnel are involved in the operation h. A manned reception area is present at operational site i. Unescorted access to server is limited to those personnel identified on an access list. Verify by checking list of personnel allowed j. Access security system is installed to control and audit details of personnel using the DSC System. k. Access security system works on the basis of digital signature certification system					
3.1.4.20	Dual control over the inventory and issue of access cards/keys during normal business hours to the Data Centre shall be in place.	1. Obtain copy of the responsibilities assigned for physical security and verify individuals have been assigned for the same 2. Conduct walkthrough to check the following a. All CA personnel have identity and authorization verified before they are included in the access list to CAs system b. All personnel wear visible identification c. Entrance to the main building and entrance to each security zone is video recorded round the clock. Verify on	Mandatory	IT CA Rules SCHEDULE-II 4.4			

		sample basis for last one year for any random date of each year d. Check the recording is carefully scrutinized and maintained for at least one year by verifying logs for a sample dates in the past year e. Biometric physical access security systems are installed and functioning in the operational site f. Unauthorized intrusion is manually or electronically monitored at all times at the CAs Site. Verify by checking the register/list maintained for same g. Physical access to CA facilities and equipment is limited to authorized individuals, protected through restricted security perimeters, and is operated under multiple person (at least dual custody) control. Verify by checking the access list / and checking the name of the personnel for dual custody. Physically see also, how the two personnel are involved in the operation h. A manned reception area is present at operational site i. Unescorted access to server is limited to those personnel identified on an access list. Verify by checking list of personnel allowed j. Access security system is installed to control and audit details of personnel using the DSC System. k. Access security system works on the basis of digital signature certification system						
3.1.4.21	Entrance to the main	1. Obtain copy of the	Mandatory	IT CA Rules SCHEDULE-III 3.5				

	building where the Certifying Authority's facilities such as Data Centre, PKI Server and Network devices are housed and entrance to each security zone shall be video recorded round the clock. The recording must be carefully scrutinized and maintained for at least one year	responsibilities assigned for physical security and verify individuals have been assigned for the same 2. Conduct walkthrough to check the following a. All CA personnel have identity and authorization verified before they are included in the access list to CAs system b. All personnel wear visible identification c. Entrance to the main building and entrance to each security zone is video recorded round the clock. Verify on sample basis for last one year for any random date of each year d. Check the recording is carefully scrutinized and maintained for at least one year by verifying logs for a sample dates in the past year e. Biometric physical access security systems are installed and functioning in the operational site f. Unauthorized intrusion is manually or electronically monitored at all times at the CAs Site. Verify by checking the register/list maintained for same g. Physical access to CA facilities and equipment is limited to authorized individuals, protected through restricted security perimeters, and is operated under multiple person (at least dual custody) control. Verify by checking the access list / and checking the name of the personnel for dual custody. Physically see also, how the two personnel are involved in the operation h. A manned reception area is present at operational site i. Unescorted access to						
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		server is limited to those personnel identified on an access list. Verify by checking list of personnel allowed j. Access security system is installed to control and audit details of personnel using the DSC System. k. Access security system works on the basis of digital signature certification system						
3.1.4.22	Biometric physical access security systems shall be installed to control and audit access to the operational site	1. Obtain copy of the responsibilities assigned for physical security and verify individuals have been assigned for the same 2. Conduct walkthrough to check the following a. All CA personnel have identity and authorization verified before they are included in the access list to CAs system b. All personnel wear visible identification c. Entrance to the main building and entrance to each security zone is video recorded round the clock. Verify on sample basis for last one year for any random date of each year d. Check the recording is carefully scrutinized and maintained for at least one year by verifying logs for a sample dates in the past year e. Biometric physical access security systems are installed and functioning in the operational site f. Unauthorized intrusion is manually or electronically monitored at all times at the CAs Site. Verify by checking the register/list maintained for same g. Physical access to	Mandatory	IT CA Rules SCHEDULE-II 4.4				

		CA facilities and equipment is limited to authorized individuals, protected through restricted security perimeters, and is operated under multiple person (at least dual custody) control. Verify by checking the access list / and checking the name of the personnel for dual custody. Physically see also, how the two personnel are involved in the operation h. A manned reception area is present at operational site i. Unescorted access to server is limited to those personnel identified on an access list. Verify by checking list of personnel allowed j. Access security system is installed to control and audit details of personnel using the DSC System. k. Access security system works on the basis of digital signature certification system					
3.1.4.23	Unauthorized intrusion shall be manually or electronically monitored at all times at the CAs Site	1. Obtain copy of the responsibilities assigned for physical security and verify individuals have been assigned for the same 2. Conduct walkthrough to check the following a. All CA personnel have identity and authorization verified before they are included in the access list to CAs system b. All personnel wear visible identification c. Entrance to the main building and entrance to each security zone is video recorded round the clock. Verify on sample basis for last one year for any random date of each	Mandatory	IT CA Rules SCHEDULE-II 3.4.6			

		year d. Check the recording is carefully scrutinized and maintained for at least one year by verifying logs for a sample dates in the past year e. Biometric physical access security systems are installed and functioning in the operational site f. Unauthorized intrusion is manually or electronically monitored at all times at the CAs Site. Verify by checking the register/list maintained for same g. Physical access to CA facilities and equipment is limited to authorized individuals, protected through restricted security perimeters, and is operated under multiple person (at least dual custody) control. Verify by checking the access list / and checking the name of the personnel for dual custody. Physically see also, how the two personnel are involved in the operation h. A manned reception area is present at operational site i. Unescorted access to server is limited to those personnel identified on an access list. Verify by checking list of personnel allowed j. Access security system is installed to control and audit details of personnel using the DSC System. k. Access security system works on the basis of digital signature certification system						
3.1.4.24	Physical access to CA facilities and equipment should be limited to authorized individuals,	1. Obtain copy of the responsibilities assigned for physical security and verify	Mandatory	WebTrust 4.8.3, IT CA Rules SCHEDULE-II 4.4				

	protected through restricted security perimeters, and shall be operated under multiple person (at least dual custody) control. Personnel authorized for limited physical access shall not be allowed to gain unauthorized access to restricted area within operational site	individuals have been assigned for the same 2. Conduct walkthrough to check the following a. All CA personnel have identity and authorization verified before they are included in the access list to CAs system b. All personnel wear visible identification c. Entrance to the main building and entrance to each security zone is video recorded round the clock. Verify on sample basis for last one year for any random date of each year d. Check the recording is carefully scrutinized and maintained for at least one year by verifying logs for a sample dates in the past year e. Biometric physical access security systems are installed and functioning in the operational site f. Unauthorized intrusion is manually or electronically monitored at all times at the CAs Site. Verify by checking the register/list maintained for same g. Physical access to CA facilities and equipment is limited to authorized individuals, protected through restricted security perimeters, and is operated under multiple person (at least dual custody) control. Verify by checking the access list / and checking the name of the personnel for dual custody. Physically see also, how the two personnel are involved in the operation h. A manned reception area is present at operational site i. Unescorted access to server is limited to those personnel identified on an access						
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		list. Verify by checking list of personnel allowed j. Access security system is installed to control and audit details of personnel using the DSC System. k. Access security system works on the basis of digital signature certification system						
3.1.4.25	A manned reception area or other means to control physical access should be present at the CA operations site	1. Obtain copy of the responsibilities assigned for physical security and verify individuals have been assigned for the same 2. Conduct walkthrough to check the following a. All CA personnel have identity and authorization verified before they are included in the access list to CAs system b. All personnel wear visible identification c. Entrance to the main building and entrance to each security zone is video recorded round the clock. Verify on sample basis for last one year for any random date of each year d. Check the recording is carefully scrutinized and maintained for at least one year by verifying logs for a sample dates in the past year e. Biometric physical access security systems are installed and functioning in the operational site f. Unauthorized intrusion is manually or electronically monitored at all times at the CAs Site. Verify by checking the register/list maintained for same g. Physical access to CA facilities and equipment is limited to authorized	Mandatory	WebTrust 3.4.3				

		individuals, protected through restricted security perimeters, and is operated under multiple person (at least dual custody) control. Verify by checking the access list / and checking the name of the personnel for dual custody. Physically see also, how the two personnel are involved in the operation h. A manned reception area is present at operational site i. Unescorted access to server is limited to those personnel identified on an access list. Verify by checking list of personnel allowed j. Access security system is installed to control and audit details of personnel using the DSC System. k. Access security system works on the basis of digital signature certification system					
3.1.4.26	Unescorted access to Certifying Authority's server shall be limited to those personnel identified on an access list	1. Obtain copy of the responsibilities assigned for physical security and verify individuals have been assigned for the same 2. Conduct walkthrough to check the following a. All CA personnel have identity and authorization verified before they are included in the access list to CAs system b. All personnel wear visible identification c. Entrance to the main building and entrance to each security zone is video recorded round the clock. Verify on sample basis for last one year for any random date of each year d. Check the recording is carefully scrutinized	Mandatory	IT CA Rules SCHEDULE-III 3.3			

		and maintained for at least one year by verifying logs for a sample dates in the past year e. Biometric physical access security systems are installed and functioning in the operational site f. Unauthorized intrusion is manually or electronically monitored at all times at the CAs Site. Verify by checking the register/list maintained for same g. Physical access to CA facilities and equipment is limited to authorized individuals, protected through restricted security perimeters, and is operated under multiple person (at least dual custody) control. Verify by checking the access list / and checking the name of the personnel for dual custody. Physically see also, how the two personnel are involved in the operation h. A manned reception area is present at operational site i. Unescorted access to server is limited to those personnel identified on an access list. Verify by checking list of personnel allowed j. Access security system is installed to control and audit details of personnel using the DSC System. k. Access security system works on the basis of digital signature certification system						
3.1.4.27	Access security system shall be installed to control and audit details of personnel using the Digital Signature Certification System.	1. Obtain copy of the responsibilities assigned for physical security and verify individuals have been assigned for the same 2. Conduct	Mandatory	IT CA Rules SCHEDULE-III 3.6				

		walkthrough to check the following a. All CA personnel have identity and authorization verified before they are included in the access list to CAs system b. All personnel wear visible identification c. Entrance to the main building and entrance to each security zone is video recorded round the clock. Verify on sample basis for last one year for any random date of each year d. Check the recording is carefully scrutinized and maintained for at least one year by verifying logs for a sample dates in the past year e. Biometric physical access security systems are installed and functioning in the operational site f. Unauthorized intrusion is manually or electronically monitored at all times at the CAs Site. Verify by checking the register/list maintained for same g. Physical access to CA facilities and equipment is limited to authorized individuals, protected through restricted security perimeters, and is operated under multiple person (at least dual custody) control. Verify by checking the access list / and checking the name of the personnel for dual custody. Physically see also, how the two personnel are involved in the operation h. A manned reception area is present at operational site i. Unescorted access to server is limited to those personnel identified on an access list. Verify by checking list of personnel allowed						
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		j. Access security system is installed to control and audit details of personnel using the DSC System. k. Access security system works on the basis of digital signature certification system						
3.1.4.28	An inventory of access cards shall be maintained by the CA and periodically reviewed	1. Verify the CA maintains and inventory of access cards and reviews it periodically 2. Obtain the up-to-date list of personnel who possess cards/keys 3. Verify the list is updated regularly and archived for a period of three years 4. Validate the process followed in case of loss of access cards, check list of cases where cards were lost and verify if the process was followed. 5. Conduct a walkthrough to check the following: a. Any personnel not on access list is escorted at all times within the operational site b. All individuals, other than operations staff, sign in and sign out of the operational site and are accompanied by operations staff. Verify by checking register/log where entries have been made. c. All personnel entering and leaving CA operational facilities are logged. Verify by checking the physical register or electronic media used for logging. d. CA operational facility has security zones laid out within the facility. Verify the documented proof for zoning performed. e. Access rights to all	Mandatory	IT CA Rules SCHEDULE-III 3.7				

		security zones are laid out. Verify by checking the access list for various security zones f. Security zones are separate from the other by floor to ceiling concrete reinforced walls. Conduct physical walkthrough of the zones to ascertain the same 6. Collect site access log and verify all entries are recorded and inspected periodically						
3.1.4.29	An up-to-date list of personnel who possess the cards/keys shall be regularly maintained and archived for a period of three years. Loss of access cards/keys shall be reported immediately to the Security Administrator; who shall take appropriate actions to prevent unauthorized access.	1. Verify the CA maintains and inventory of access cards and reviews it periodically 2. Obtain the up-to-date list of personnel who possess cards/keys 3. Verify the list is updated regularly and archived for a period of three years 4. Validate the process followed in case of loss of access cards, check list of cases where cards were lost and verify if the process was followed. 5. Conduct a walkthrough to check the following: a. Any personnel not on access list is escorted at all times within the operational site b. All individuals, other than operations staff, sign in and sign out of the operational site and are accompanied by operations staff. Verify by checking register/log where entries have been made. c. All personnel entering and leaving CA operational facilities are logged. Verify by checking the physical register or electronic media used for logging. d. CA operational facility has security	Mandatory	IT CA Rules SCHEDULE-III 3.8				

		zones laid out within the facility. Verify the documented proof for zoning performed. e. Access rights to all security zones are laid out. Verify by checking the access list for various security zones f. Security zones are separate from the other by floor to ceiling concrete reinforced walls. Conduct physical walkthrough of the zones to ascertain the same 6. Collect site access log and verify all entries are recorded and inspected periodically						
3.1.4.30	Any personnel not on the access list shall be properly escorted and supervised at all times within the operational site premises. All individuals, other than operations staff, shall sign in and sign out of the operational site and shall be accompanied by operations staff.	1. Verify the CA maintains and inventory of access cards and reviews it periodically 2. Obtain the up-to-date list of personnel who possess cards/keys 3. Verify the list is updated regularly and archived for a period of three years 4. Validate the process followed in case of loss of access cards, check list of cases where cards were lost and verify if the process was followed. 5. Conduct a walkthrough to check the following: a. Any personnel not on access list is escorted at all times within the operational site b. All individuals, other than operations staff, sign in and sign out of the operational site and are accompanied by operations staff. Verify by checking register/log where entries have been made. c. All personnel entering and leaving CA operational facilities are logged. Verify by checking the	Mandatory	IT CA Rules IT CA Rules SCHEDULE-III 3.9, SCHEDULE II 4.4				

		physical register or electronic media used for logging. d. CA operational facility has security zones laid out within the facility. Verify the documented proof for zoning performed. e. Access rights to all security zones are laid out. Verify by checking the access list for various security zones f. Security zones are separate from the other by floor to ceiling concrete reinforced walls. Conduct physical walkthrough of the zones to ascertain the same 6. Collect site access log and verify all entries are recorded and inspected periodically						
3.1.4.31	A site access log shall be maintained at the Certifying Authority's operational site and inspected periodically. All personnel entering and leaving CA operational facilities shall be logged (i.e., an audit trail of all access is securely maintained).	1. Verify the CA maintains and inventory of access cards and reviews it periodically 2. Obtain the up-to-date list of personnel who possess cards/keys 3. Verify the list is updated regularly and archived for a period of three years 4. Validate the process followed in case of loss of access cards, check list of cases where cards were lost and verify if the process was followed. 5. Conduct a walkthrough to check the following: a. Any personnel not on access list is escorted at all times within the operational site b. All individuals, other than operations staff, sign in and sign out of the operational site and are accompanied by operations staff. Verify by checking register/log where entries have been made.	Mandatory	WebTrust 3.4.11, IT CA Rules SCHEDULE III 3.3				

		c. All personnel entering and leaving CA operational facilities are logged. Verify by checking the physical register or electronic media used for logging. d. CA operational facility has security zones laid out within the facility. Verify the documented proof for zoning performed. e. Access rights to all security zones are laid out. Verify by checking the access list for various security zones f. Security zones are separate from the other by floor to ceiling concrete reinforced walls. Conduct physical walkthrough of the zones to ascertain the same 6. Collect site access log and verify all entries are recorded and inspected periodically						
3.1.4.32	A multi-tiered access mechanism must be installed at the Certifying Authority's operational site. The facility should have clearly laid out security zones within its facility with well-defined access rights to each security zone.	1. Verify the CA maintains and inventory of access cards and reviews it periodically 2. Obtain the up-to-date list of personnel who possess cards/keys 3. Verify the list is updated regularly and archived for a period of three years 4. Validate the process followed in case of loss of access cards, check list of cases where cards were lost and verify if the process was followed. 5. Conduct a walkthrough to check the following: a. Any personnel not on access list is escorted at all times within the operational site b. All individuals, other than operations staff, sign in and sign out of the operational site and are accompanied by operations staff. Verify	Mandatory	IT CA Rules SCHEDULE III 3.4				

		by checking register/log where entries have been made. c. All personnel entering and leaving CA operational facilities are logged. Verify by checking the physical register or electronic media used for logging. d. CA operational facility has security zones laid out within the facility. Verify the documented proof for zoning performed. e. Access rights to all security zones are laid out. Verify by checking the access list for various security zones f. Security zones are separate from the other by floor to ceiling concrete reinforced walls. Conduct physical walkthrough of the zones to ascertain the same 6. Collect site access log and verify all entries are recorded and inspected periodically						
3.1.4.33	Each security zone must be separated from the other by floor to ceiling concrete reinforced walls.	1. Verify the CA maintains and inventory of access cards and reviews it periodically 2. Obtain the up-to-date list of personnel who possess cards/keys 3. Verify the list is updated regularly and archived for a period of three years 4. Validate the process followed in case of loss of access cards, check list of cases where cards were lost and verify if the process was followed. 5. Conduct a walkthrough to check the following: a. Any personnel not on access list is escorted at all times within the operational site b. All individuals, other than operations	Mandatory	IT CA Rules SCHEDULE III 3.4				

		staff, sign in and sign out of the operational site and are accompanied by operations staff. Verify by checking register/log where entries have been made. c. All personnel entering and leaving CA operational facilities are logged. Verify by checking the physical register or electronic media used for logging. d. CA operational facility has security zones laid out within the facility. Verify the documented proof for zoning performed. e. Access rights to all security zones are laid out. Verify by checking the access list for various security zones f. Security zones are separate from the other by floor to ceiling concrete reinforced walls. Conduct physical walkthrough of the zones to ascertain the same 6. Collect site access log and verify all entries are recorded and inspected periodically						
3.1.4.34	Alarm and intrusion detection system must be installed at every stage with adequate power backup capable of continuing operation even in the event of loss of main power.	1. Verify Alarm and intrusion detection system are installed at every stage with adequate power backup 2. Validate Electrical/Electronic circuits to external security alarm monitoring service are supervised 3. Validate access to PKI Server, root keys or any personal computer system or network device is prohibited	Mandatory	IT CA Rules SCHEDULE III 3.4				
3.1.4.35	Electrical/Electronic circuits to external security alarm monitoring service (if used) shall be supervised	1. Verify Alarm and intrusion detection system are installed at every stage with adequate power backup	Mandatory	IT CA Rules SCHEDULE III 3.4				

		2. Validate Electrical/Electronic circuits to external security alarm monitoring service are supervised 3. Validate access to PKI Server, root keys or any personal computer system or network device is prohibited						
3.1.4.36	Access to PKI Server, root keys or any personal computer system or network device shall be prohibited	1. Verify Alarm and intrusion detection system are installed at every stage with adequate power backup 2. Validate Electrical/Electronic circuits to external security alarm monitoring service are supervised 3. Validate access to PKI Server, root keys or any personal computer system or network device is prohibited	Mandatory	IT CA Rules SCHEDULE III 3.4				
3.1.4.37	Access to infrastructure components essential to operation of Certifying Authority such as power control panels, communication infrastructure, Digital Signature Certification system, cabling, etc. shall be restricted to authorized personnel.	1. Verify list of personnel with access to essential components for CA is maintained 2. Validate no unauthorized personnel has access to these components by checking the controls implemented to prevent unauthorized access such as locked door, biometric etc.	Mandatory	IT CA Rules SCHEDULE III 3.8				
3.1.4.38	By-pass or deactivation of normal physical security arrangements shall be authorized and documented by security personnel	1. Verify all By-pass or deactivation of normal physical security arrangements is authorized and documented by security personnel	Mandatory	IT CA Rules SCHEDULE III 3.9				
3.1.4.39	Intrusion detection systems shall be used to monitor and record physical access to the Digital Signature Certification system during and after office hours.	1. Conduct walkthrough to check the following: a. Intrusion detection systems are used to monitor and record physical access to the DSC system during	Mandatory	IT CA Rules SCHEDULE III 3.10				

		and after office hours b. Computer System or PKI Server performing the DSC functions are dedicated to those functions and should not be used for any other purposes c. Emergency exits are tested periodically. Validate by checking proper functioning of emergency exits. d. Unauthorized access to areas storing hardware is prohibited. Verify by checking the access list for areas storing the hardware. e. Two person physical access control are present at both the cryptographic module and computer system for CAs issuing Class 1, Class 2 and Class 3 certificates. Practical demonstration of the two person physical access should be done in the presence of the auditor.						
3.1.4.40	Computer System or PKI Server performing the Digital Signature Certification functions shall be dedicated to those functions and should not be used for any other purposes.	1. Conduct walkthrough to check the following: a. Intrusion detection systems are used to monitor and record physical access to the DSC system during and after office hours b. Computer System or PKI Server performing the DSC functions are dedicated to those functions and should not be used for any other purposes c. Emergency exits are tested periodically. Validate by checking proper functioning of emergency exits. d. Unauthorized access to areas storing hardware is prohibited. Verify by checking the access list for areas storing the hardware. e. Two person physical access control are present at both the cryptographic module and computer system	Mandatory	IT CA Rules SCHEDULE III 3.11				

		for CAs issuing Class 1, Class 2 and Class 3 certificates. Practical demonstration of the two person physical access should be done in the presence of the auditor.						
3.1.4.41	Emergency exits shall be tested periodically to ensure that the access security systems are operational	1. Conduct walkthrough to check the following: a. Intrusion detection systems are used to monitor and record physical access to the DSC system during and after office hours b. Computer System or PKI Server performing the DSC functions are dedicated to those functions and should not be used for any other purposes c. Emergency exits are tested periodically. Validate by checking proper functioning of emergency exits. d. Unauthorized access to areas storing hardware is prohibited. Verify by checking the access list for areas storing the hardware. e. Two person physical access control are present at both the cryptographic module and computer system for CAs issuing Class 1, Class 2 and Class 3 certificates. Practical demonstration of the two person physical access should be done in the presence of the auditor.	Mandatory	IT CA Rules CHEDULE II 4.4				
3.1.4.42	Unauthorized access to areas storing hardware shall be prohibited	1. Conduct walkthrough to check the following: a. Intrusion detection systems are used to monitor and record physical access to the DSC system during and after office hours b. Computer System or PKI Server performing the DSC functions are dedicated to those functions and should not be used for any	Mandatory	X.509 Policy 5.1.2.1				

		other purposes c. Emergency exits are tested periodically. Validate by checking proper functioning of emergency exits. d. Unauthorized access to areas storing hardware is prohibited. Verify by checking the access list for areas storing the hardware. e. Two person physical access control are present at both the cryptographic module and computer system for CAs issuing Class 1, Class 2 and Class 3 certificates. Practical demonstration of the two person physical access should be done in the presence of the auditor.						
3.1.4.43	Two person physical access control shall be present at both the cryptographic module and computer system for CAs issuing Class 1, Class 2 and Class 3 certificates.	1. Conduct walkthrough to check the following: a. Intrusion detection systems are used to monitor and record physical access to the DSC system during and after office hours b. Computer System or PKI Server performing the DSC functions are dedicated to those functions and should not be used for any other purposes c. Emergency exits are tested periodically. Validate by checking proper functioning of emergency exits. d. Unauthorized access to areas storing hardware is prohibited. Verify by checking the access list for areas storing the hardware. e. Two person physical access control are present at both the cryptographic module and computer system for CAs issuing Class 1, Class 2 and Class 3 certificates. Practical demonstration of the two person physical access should be done in the presence	Mandatory	X.509 Policy 5.1.2.				

		of the auditor.						
3.1.4.44	Removable cryptographic modules shall be deactivated prior to storage. When not in use, removable cryptographic modules, activation information used to access or enable cryptographic modules shall be placed in secure containers. Activation data shall either be memorized, or recorded and stored in a manner commensurate with the security afforded the cryptographic module, and shall not be stored with the cryptographic module.	1. Verify removable cryptographic modules are deactivated before storage 2. Validate activation information used to access or enable cryptographic modules is placed in secure containers. 3. Check the mechanism used to store activation data	Mandatory	X.509 Policy 5.1.2.1				
3.1.4.45	The Certifying Authority must ensure that no single individual may gain access to the Digital Signature Certificate server and the computer server maintaining all information associated with generation, issue and management of Digital Signature Certificate and private keys of the Certifying Authority. Minimum two individuals, preferably using a split-knowledge technique, such as twin passwords, must perform any operation associated with generation, issue and management of Digital Signature Certificate and application of private key of the Certifying Authority.	1. Verify the process of security check to be conducted if the facility housing the CA and CSP equipment is left unattended 2. Validate it covers the requirements mentioned in control description 3. Check the list of persons responsible for doing the security checks mentioned in control - 3.1.4.45 4. Verify if a group of people are assigned the security check responsibility a log identifying the person performing a check at each instance is maintained 5. Check if the facility is not continuously attended, the last person to depart initials a sign-out sheet that indicates the date and time, and asserts that all necessary physical protection mechanisms are in place and activated	Mandatory	IT CA Rules SCHEDULE III 12				
Inventory Management								
3.1.4.46	Inventory control processes and procedures shall be documented and implemented to manage	1. Obtain copy of Inventory control processes and procedures and check	Mandatory	WebTrust 4.8.3				

	the origination, arrival, condition, departure and destination of each device.	the following: a. Policy and procedure are implemented to manage the origination, arrival, condition, departure and destination of each device. b. Details of all media are inventoried c. An independent physical inventory check of all media is conducted at least every six months. Verify by checking latest report of physical inventory check d. An up-to-date inventory list of all documentation is maintained. Verify by checking previous versions of inventory list to validate periodicity of update e. All equipment related to CA Operations are inventoried. Verify by conducting a random check on sample basis for the infrastructure and check its entry in to inventory						
3.1.4.47	Details of all media shall be inventoried. An independent physical inventory check of all media shall be conducted at least every six months.	1. Obtain copy of Inventory control processes and procedures and check the following: a. Policy and procedure are implemented to manage the origination, arrival, condition, departure and destination of each device. b. Details of all media are inventoried c. An independent physical inventory check of all media is conducted at least every six months. Verify by checking latest report of physical inventory check d. An up-to-date inventory list of all documentation is maintained. Verify by checking previous versions of inventory list to validate	Mandatory	IT CA Rules SCHEDULE II 8.3				

		periodicity of update e. All equipment related to CA Operations are inventoried. Verify by conducting a random check on sample basis for the infrastructure and check its entry in to inventory						
3.1.4.48	An up-to-date inventory list of all documentation shall be maintained to ensure control and accountability.	1. Obtain copy of Inventory control processes and procedures and check the following: a. Policy and procedure are implemented to manage the origination, arrival, condition, departure and destination of each device. b. Details of all media are inventoried c. An independent physical inventory check of all media is conducted at least every six months. Verify by checking latest report of physical inventory check d. An up-to-date inventory list of all documentation is maintained. Verify by checking previous versions of inventory list to validate periodicity of update e. All equipment related to CA Operations are inventoried. Verify by conducting a random check on sample basis for the infrastructure and check its entry in to inventory	Mandatory	IT CA Rules SCHEDULE II 16				
3.1.4.49	All equipment related to CA Operations shall be inventoried	1. Obtain copy of Inventory control processes and procedures and check the following: a. Policy and procedure are implemented to manage the origination, arrival, condition, departure and destination of each device. b. Details of all media are inventoried	Mandatory	WebTrust 3.4.16				

		c. An independent physical inventory check of all media is conducted at least every six months. Verify by checking latest report of physical inventory check d. An up-to-date inventory list of all documentation is maintained. Verify by checking previous versions of inventory list to validate periodicity of update e. All equipment related to CA Operations are inventoried. Verify by conducting a random check on sample basis for the infrastructure and check its entry in to inventory						
Media Storage								
3.1.4.50	All sensitive information stored in any media shall bear or be assigned an appropriate security classification.	1. Obtain Media Management policy and verify the following: a. All sensitive information stored in any media is assigned an appropriate security classification b. Media library, electrical and mechanical control rooms are housed in separate isolated areas c. Access to these areas is given only to specific, named individuals on a need basis. Verify by checking the access list d. Such area is fire resistant and free of toxic chemicals. e. Storage media containing sensitive information is secured according to their classification f. Responsibilities for media library management and protection are clearly defined and assigned g. Access to the media library is restricted to the authorized persons only (verify by checking list of	Mandatory	IT CA Rules SCHEDULE II 5.2				

		authorized personnel) h. Media containing sensitive and back up data is stored at three different physical locations in the country. Verify by identifying the three different physical locations where backup is stored i. Media that containing audit, archive, or backup information is duplicated and stored in a location separate from the CA location. j. Media management system has been implemented by providing sample of the process followed. k. All incoming/outgoing media transfers are authorized by management and users. Verify by checking on sample basis. l. All media have external volume identification m. Procedures are in place to ensure that only authorized addition/removal of media from the library is allowed n. Media retention procedure is established and approved by management o. Records of all movements of computer tapes/disks between on-site and off-site media library are maintained. p. Procedure are present for secure transfer and disposal of media q. Computer media that are being transported to off-site data backup locations are stored in locked carrying cases that provide magnetic field protection r. All items of equipment containing storage media are checked to ensure that they do not contain sensitive data prior to their disposal.							
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3.1.4.51	Media library, electrical and mechanical control rooms shall be housed in separate isolated areas, with access granted only to specific, named individuals on a need basis. The area must be fire resistant and free of toxic chemicals	1. Obtain Media Management policy and verify the following: a. All sensitive information stored in any media is assigned an appropriate security classification b. Media library, electrical and mechanical control rooms are housed in separate isolated areas c. Access to these areas is given only to specific, named individuals on a need basis. Verify by checking the access list d. Such area is fire resistant and free of toxic chemicals. e. Storage media containing sensitive information is secured according to their classification f. Responsibilities for media library management and protection are clearly defined and assigned g. Access to the media library is restricted to the authorized persons only (verify by checking list of authorized personnel) h. Media containing sensitive and back up data is stored at three different physical locations in the country. Verify by identifying the three different physical locations where backup is stored i. Media that containing audit, archive, or backup information is duplicated and stored in a location separate from the CA location. j. Media management system has been implemented by providing sample of the process followed. k. All incoming/outgoing media transfers are authorized by	Mandatory	IT CA Rules SCHEDULE II 4.1.8				

		management and users. Verify by checking on sample basis. l. All media have external volume identification m. Procedures are in place to ensure that only authorized addition/removal of media from the library is allowed n. Media retention procedure is established and approved by management o. Records of all movements of computer tapes/disks between on-site and off-site media library are maintained. p. Procedure are present for secure transfer and disposal of media q. Computer media that are being transported to off-site data backup locations are stored in locked carrying cases that provide magnetic field protection r. All items of equipment containing storage media are checked to ensure that they do not contain sensitive data prior to their disposal.						
3.1.4.52	Storage media (i.e. floppy diskettes, magnetic tapes, portable hard disks, optical disks, etc.) containing sensitive information shall be secured according to their classification	1. Obtain Media Management policy and verify the following: a. All sensitive information stored in any media is assigned an appropriate security classification b. Media library, electrical and mechanical control rooms are housed in separate isolated areas c. Access to these areas is given only to specific, named individuals on a need basis. Verify by checking the access list d. Such area is fire resistant and free of toxic chemicals.	Mandatory	IT CA Rules SCHEDULE II 5.2				

		e. Storage media containing sensitive information is secured according to their classification f. Responsibilities for media library management and protection are clearly defined and assigned g. Access to the media library is restricted to the authorized persons only (verify by checking list of authorized personnel) h. Media containing sensitive and back up data is stored at three different physical locations in the country. Verify by identifying the three different physical locations where backup is stored i. Media that containing audit, archive, or backup information is duplicated and stored in a location separate from the CA location. j. Media management system has been implemented by providing sample of the process followed. k. All incoming/outgoing media transfers are authorized by management and users. Verify by checking on sample basis. l. All media have external volume identification m. Procedures are in place to ensure that only authorized addition/removal of media from the library is allowed n. Media retention procedure is established and approved by management o. Records of all movements of computer tapes/disks between on-site and off-site media library are maintained. p. Procedure are present for secure transfer and disposal of media						
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		q. Computer media that are being transported to off-site data backup locations are stored in locked carrying cases that provide magnetic field protection r. All items of equipment containing storage media are checked to ensure that they do not contain sensitive data prior to their disposal.						
3.1.4.53	Responsibilities for media library management and protection shall be clearly defined and assigned	1. Obtain Media Management policy and verify the following: a. All sensitive information stored in any media is assigned an appropriate security classification b. Media library, electrical and mechanical control rooms are housed in separate isolated areas c. Access to these areas is given only to specific, named individuals on a need basis. Verify by checking the access list d. Such area is fire resistant and free of toxic chemicals. e. Storage media containing sensitive information is secured according to their classification f. Responsibilities for media library management and protection are clearly defined and assigned g. Access to the media library is restricted to the authorized persons only (verify by checking list of authorized personnel) h. Media containing sensitive and back up data is stored at three different physical locations in the country. Verify by identifying the three different physical locations where backup is stored i. Media that	Mandatory	IT CA Rules SCHEDULE II 8.3				

		containing audit, archive, or backup information is duplicated and stored in a location separate from the CA location. j. Media management system has been implemented by providing sample of the process followed. k. All incoming/outgoing media transfers are authorized by management and users. Verify by checking on sample basis. l. All media have external volume identification m. Procedures are in place to ensure that only authorized addition/removal of media from the library is allowed n. Media retention procedure is established and approved by management o. Records of all movements of computer tapes/disks between on-site and off-site media library are maintained. p. Procedure are present for secure transfer and disposal of media q. Computer media that are being transported to off-site data backup locations are stored in locked carrying cases that provide magnetic field protection r. All items of equipment containing storage media are checked to ensure that they do not contain sensitive data prior to their disposal.						
3.1.4.54	Access to the media library (both on-site and off-site) shall be restricted to the authorized persons only. A list of personnel authorized to enter the library shall be	1. Obtain Media Management policy and verify the following: a. All sensitive information stored in any media is assigned an appropriate security	Mandatory	IT CA Rules SCHEDULE II 8.3				

	maintained	classification b. Media library, electrical and mechanical control rooms are housed in separate isolated areas c. Access to these areas is given only to specific, named individuals on a need basis. Verify by checking the access list d. Such area is fire resistant and free of toxic chemicals. e. Storage media containing sensitive information is secured according to their classification f. Responsibilities for media library management and protection are clearly defined and assigned g. Access to the media library is restricted to the authorized persons only (verify by checking list of authorized personnel) h. Media containing sensitive and back up data is stored at three different physical locations in the country. Verify by identifying the three different physical locations where backup is stored i. Media that containing audit, archive, or backup information is duplicated and stored in a location separate from the CA location. j. Media management system has been implemented by providing sample of the process followed. k. All incoming/outgoing media transfers are authorized by management and users. Verify by checking on sample basis. l. All media have external volume identification m. Procedures are in place to ensure that only authorized addition/removal of media from the library is allowed						
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		n. Media retention procedure is established and approved by management o. Records of all movements of computer tapes/disks between on-site and off-site media library are maintained. p. Procedure are present for secure transfer and disposal of media q. Computer media that are being transported to off-site data backup locations are stored in locked carrying cases that provide magnetic field protection r. All items of equipment containing storage media are checked to ensure that they do not contain sensitive data prior to their disposal.						
3.1.4.55	Media containing sensitive and back up data must be stored at three different physical locations in the country, which can be reached in few hours. Media that contains audit, archive, or backup information shall be duplicated and stored in a	1. Obtain Media Management policy and verify the following: a. All sensitive information stored in any media is assigned an appropriate security classification b. Media library, electrical and mechanical control rooms are housed in separate isolated areas c. Access to these areas is given only to specific, named individuals on a need basis. Verify by checking the access list d. Such area is fire resistant and free of toxic chemicals. e. Storage media containing sensitive information is secured according to their classification f. Responsibilities for media library management and protection are clearly defined and assigned g. Access to the media library is restricted to	Mandatory	X.509 Policy, IT CA Rules				

		the authorized persons only (verify by checking list of authorized personnel) h. Media containing sensitive and back up data is stored at three different physical locations in the country. Verify by identifying the three different physical locations where backup is stored i. Media that containing audit, archive, or backup information is duplicated and stored in a location separate from the CA location. j. Media management system has been implemented by providing sample of the process followed. k. All incoming/outgoing media transfers are authorized by management and users. Verify by checking on sample basis. l. All media have external volume identification m. Procedures are in place to ensure that only authorized addition/removal of media from the library is allowed n. Media retention procedure is established and approved by management o. Records of all movements of computer tapes/disks between on-site and off-site media library are maintained. p. Procedure are present for secure transfer and disposal of media q. Computer media that are being transported to off-site data backup locations are stored in locked carrying cases that provide magnetic field protection r. All items of equipment containing storage media are checked to ensure that they do not contain							
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		sensitive data prior to their disposal.						
3.1.4.56	Media containing sensitive and back up data must be stored at three different physical locations in the country, which can be reached in few hours. Media that contains audit, archive, or backup information shall be duplicated and stored in a location separate from the CA location.	1. Obtain Media Management policy and verify the following: a. All sensitive information stored in any media is assigned an appropriate security classification b. Media library, electrical and mechanical control rooms are housed in separate isolated areas c. Access to these areas is given only to specific, named individuals on a need basis. Verify by checking the access list d. Such area is fire resistant and free of toxic chemicals. e. Storage media containing sensitive information is secured according to their classification f. Responsibilities for media library management and protection are clearly defined and assigned g. Access to the media library is restricted to the authorized persons only (verify by checking list of authorized personnel) h. Media containing sensitive and back up data is stored at three different physical locations in the country. Verify by identifying the three different physical locations where backup is stored i. Media that contains audit, archive, or backup information is duplicated and stored in a location separate from the CA location. j. Media management system has been implemented by providing sample of the process followed. k. All	Mandatory	X.509 Policy, IT CA Rules				

		incoming/outgoing media transfers are authorized by management and users. Verify by checking on sample basis. l. All media have external volume identification m. Procedures are in place to ensure that only authorized addition/removal of media from the library is allowed n. Media retention procedure is established and approved by management o. Records of all movements of computer tapes/disks between on-site and off-site media library are maintained. p. Procedure are present for secure transfer and disposal of media q. Computer media that are being transported to off-site data backup locations are stored in locked carrying cases that provide magnetic field protection r. All items of equipment containing storage media are checked to ensure that they do not contain sensitive data prior to their disposal.						
3.1.4.57	A media management system shall be in place to account for all media stored on-site and off-site	1. Obtain Media Management policy and verify the following: a. All sensitive information stored in any media is assigned an appropriate security classification b. Media library, electrical and mechanical control rooms are housed in separate isolated areas c. Access to these areas is given only to specific, named individuals on a need basis. Verify by checking the access list	Mandatory	IT CA Rules SCHEDULE II 8.3				

		d. Such area is fire resistant and free of toxic chemicals. e. Storage media containing sensitive information is secured according to their classification f. Responsibilities for media library management and protection are clearly defined and assigned g. Access to the media library is restricted to the authorized persons only (verify by checking list of authorized personnel) h. Media containing sensitive and back up data is stored at three different physical locations in the country. Verify by identifying the three different physical locations where backup is stored i. Media that containing audit, archive, or backup information is duplicated and stored in a location separate from the CA location. j. Media management system has been implemented by providing sample of the process followed. k. All incoming/outgoing media transfers are authorized by management and users. Verify by checking on sample basis. l. All media have external volume identification m. Procedures are in place to ensure that only authorized addition/removal of media from the library is allowed n. Media retention procedure is established and approved by management o. Records of all movements of computer tapes/disks between on-site and off-site media library are maintained. p. Procedure are							
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		present for secure transfer and disposal of media q. Computer media that are being transported to off-site data backup locations are stored in locked carrying cases that provide magnetic field protection r. All items of equipment containing storage media are checked to ensure that they do not contain sensitive data prior to their disposal.						
3.1.4.58	All incoming/outgoing media transfers shall be authorized by management and users.	1. Obtain Media Management policy and verify the following: a. All sensitive information stored in any media is assigned an appropriate security classification b. Media library, electrical and mechanical control rooms are housed in separate isolated areas c. Access to these areas is given only to specific, named individuals on a need basis. Verify by checking the access list d. Such area is fire resistant and free of toxic chemicals. e. Storage media containing sensitive information is secured according to their classification f. Responsibilities for media library management and protection are clearly defined and assigned g. Access to the media library is restricted to the authorized persons only (verify by checking list of authorized personnel) h. Media containing sensitive and back up data is stored at three different physical locations in the country. Verify by identifying the three different physical	Mandatory	IT CA Rules SCHEDULE II 8.3				

		locations where backup is stored i. Media that containing audit, archive, or backup information is duplicated and stored in a location separate from the CA location. j. Media management system has been implemented by providing sample of the process followed. k. All incoming/outgoing media transfers are authorized by management and users. Verify by checking on sample basis. l. All media have external volume identification m. Procedures are in place to ensure that only authorized addition/removal of media from the library is allowed n. Media retention procedure is established and approved by management o. Records of all movements of computer tapes/disks between on-site and off-site media library are maintained. p. Procedure are present for secure transfer and disposal of media q. Computer media that are being transported to off-site data backup locations are stored in locked carrying cases that provide magnetic field protection r. All items of equipment containing storage media are checked to ensure that they do not contain sensitive data prior to their disposal.						
3.1.4.59	All media shall have external volume identification. Internal labels shall be fixed, where available	1. Obtain Media Management policy and verify the following: a. All sensitive	Mandatory	IT CA Rules SCHEDULE II 8.3				

		information stored in any media is assigned an appropriate security classification b. Media library, electrical and mechanical control rooms are housed in separate isolated areas c. Access to these areas is given only to specific, named individuals on a need basis. Verify by checking the access list d. Such area is fire resistant and free of toxic chemicals. e. Storage media containing sensitive information is secured according to their classification f. Responsibilities for media library management and protection are clearly defined and assigned g. Access to the media library is restricted to the authorized persons only (verify by checking list of authorized personnel) h. Media containing sensitive and back up data is stored at three different physical locations in the country. Verify by identifying the three different physical locations where backup is stored i. Media that containing audit, archive, or backup information is duplicated and stored in a location separate from the CA location. j. Media management system has been implemented by providing sample of the process followed. k. All incoming/outgoing media transfers are authorized by management and users. Verify by checking on sample basis. l. All media have external volume identification m. Procedures are in place to ensure that only authorized						
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		addition/removal of media from the library is allowed n. Media retention procedure is established and approved by management o. Records of all movements of computer tapes/disks between on-site and off-site media library are maintained. p. Procedure are present for secure transfer and disposal of media q. Computer media that are being transported to off-site data backup locations are stored in locked carrying cases that provide magnetic field protection r. All items of equipment containing storage media are checked to ensure that they do not contain sensitive data prior to their disposal.						
3.1.4.60	Procedures shall be in place to ensure that only authorized addition/removal of media from the library is allowed.	1. Obtain Media Management policy and verify the following: a. All sensitive information stored in any media is assigned an appropriate security classification b. Media library, electrical and mechanical control rooms are housed in separate isolated areas c. Access to these areas is given only to specific, named individuals on a need basis. Verify by checking the access list d. Such area is fire resistant and free of toxic chemicals. e. Storage media containing sensitive information is secured according to their classification f. Responsibilities for media library management and protection are clearly	Mandatory	IT CA Rules SCHEDULE II 8.3				

		defined and assigned g. Access to the media library is restricted to the authorized persons only (verify by checking list of authorized personnel) h. Media containing sensitive and back up data is stored at three different physical locations in the country. Verify by identifying the three different physical locations where backup is stored i. Media that containing audit, archive, or backup information is duplicated and stored in a location separate from the CA location. j. Media management system has been implemented by providing sample of the process followed. k. All incoming/outgoing media transfers are authorized by management and users. Verify by checking on sample basis. l. All media have external volume identification m. Procedures are in place to ensure that only authorized addition/removal of media from the library is allowed n. Media retention procedure is established and approved by management o. Records of all movements of computer tapes/disks between on-site and off-site media library are maintained. p. Procedure are present for secure transfer and disposal of media q. Computer media that are being transported to off-site data backup locations are stored in locked carrying cases that provide magnetic field protection r. All items of equipment containing						
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		storage media are checked to ensure that they do not contain sensitive data prior to their disposal.						
3.1.4.61	Media retention periods shall be established and approved by management in accordance with legal regulatory and user requirements.	1. Obtain Media Management policy and verify the following: a. All sensitive information stored in any media is assigned an appropriate security classification b. Media library, electrical and mechanical control rooms are housed in separate isolated areas c. Access to these areas is given only to specific, named individuals on a need basis. Verify by checking the access list d. Such area is fire resistant and free of toxic chemicals. e. Storage media containing sensitive information is secured according to their classification f. Responsibilities for media library management and protection are clearly defined and assigned g. Access to the media library is restricted to the authorized persons only (verify by checking list of authorized personnel) h. Media containing sensitive and back up data is stored at three different physical locations in the country. Verify by identifying the three different physical locations where backup is stored i. Media that containing audit, archive, or backup information is duplicated and stored in a location separate from the CA location. j. Media management system has been implemented by	Mandatory	IT CA Rules SCHEDULE II 8.3				

		providing sample of the process followed. k. All incoming/outgoing media transfers are authorized by management and users. Verify by checking on sample basis. l. All media have external volume identification m. Procedures are in place to ensure that only authorized addition/removal of media from the library is allowed n. Media retention procedure is established and approved by management o. Records of all movements of computer tapes/disks between on-site and off-site media library are maintained. p. Procedure are present for secure transfer and disposal of media q. Computer media that are being transported to off-site data backup locations are stored in locked carrying cases that provide magnetic field protection r. All items of equipment containing storage media are checked to ensure that they do not contain sensitive data prior to their disposal.						
3.1.4.62	Proper records of all movements of computer tapes/disks between on-site and off-site media library must be maintained.	1. Obtain Media Management policy and verify the following: a. All sensitive information stored in any media is assigned an appropriate security classification b. Media library, electrical and mechanical control rooms are housed in separate isolated areas c. Access to these areas is given only to specific, named	Mandatory	IT CA Rules SCHEDULE II 8.4				

		individuals on a need basis. Verify by checking the access list d. Such area is fire resistant and free of toxic chemicals. e. Storage media containing sensitive information is secured according to their classification f. Responsibilities for media library management and protection are clearly defined and assigned g. Access to the media library is restricted to the authorized persons only (verify by checking list of authorized personnel) h. Media containing sensitive and back up data is stored at three different physical locations in the country. Verify by identifying the three different physical locations where backup is stored i. Media that containing audit, archive, or backup information is duplicated and stored in a location separate from the CA location. j. Media management system has been implemented by providing sample of the process followed. k. All incoming/outgoing media transfers are authorized by management and users. Verify by checking on sample basis. l. All media have external volume identification m. Procedures are in place to ensure that only authorized addition/removal of media from the library is allowed n. Media retention procedure is established and approved by management o. Records of all movements of computer tapes/disks between on-site and							
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		off-site media library are maintained. p. Procedure are present for secure transfer and disposal of media q. Computer media that are being transported to off-site data backup locations are stored in locked carrying cases that provide magnetic field protection r. All items of equipment containing storage media are checked to ensure that they do not contain sensitive data prior to their disposal.						
3.1.4.63	There shall be procedures to ensure the authorized and secure transfer to media to/from external parties and the off-site location including a means to authenticate the receipt shall be in place.	1. Obtain Media Management policy and verify the following: a. All sensitive information stored in any media is assigned an appropriate security classification b. Media library, electrical and mechanical control rooms are housed in separate isolated areas c. Access to these areas is given only to specific, named individuals on a need basis. Verify by checking the access list d. Such area is fire resistant and free of toxic chemicals. e. Storage media containing sensitive information is secured according to their classification f. Responsibilities for media library management and protection are clearly defined and assigned g. Access to the media library is restricted to the authorized persons only (verify by checking list of authorized personnel) h. Media containing sensitive and back up data is stored at three different physical locations in the	Mandatory	IT CA Rules SCHEDULE II 8.4				

		country. Verify by identifying the three different physical locations where backup is stored i. Media that containing audit, archive, or backup information is duplicated and stored in a location separate from the CA location. j. Media management system has been implemented by providing sample of the process followed. k. All incoming/outgoing media transfers are authorized by management and users. Verify by checking on sample basis. l. All media have external volume identification m. Procedures are in place to ensure that only authorized addition/removal of media from the library is allowed n. Media retention procedure is established and approved by management o. Records of all movements of computer tapes/disks between on-site and off-site media library are maintained. p. Procedure are present for secure transfer and disposal of media q. Computer media that are being transported to off-site data backup locations are stored in locked carrying cases that provide magnetic field protection r. All items of equipment containing storage media are checked to ensure that they do not contain sensitive data prior to their disposal.						
3.1.4.64	Computer media that are being transported to off-	1. Obtain Media Management policy	Mandatory	IT CA Rules SCHEDULE II 8.4				

	site data backup locations should be stored in locked carrying cases that provide magnetic field protection and protection from impact while loading and unloading and during transportation	and verify the following: a. All sensitive information stored in any media is assigned an appropriate security classification b. Media library, electrical and mechanical control rooms are housed in separate isolated areas c. Access to these areas is given only to specific, named individuals on a need basis. Verify by checking the access list d. Such area is fire resistant and free of toxic chemicals. e. Storage media containing sensitive information is secured according to their classification f. Responsibilities for media library management and protection are clearly defined and assigned g. Access to the media library is restricted to the authorized persons only (verify by checking list of authorized personnel) h. Media containing sensitive and back up data is stored at three different physical locations in the country. Verify by identifying the three different physical locations where backup is stored i. Media that containing audit, archive, or backup information is duplicated and stored in a location separate from the CA location. j. Media management system has been implemented by providing sample of the process followed. k. All incoming/outgoing media transfers are authorized by management and users. Verify by checking on sample basis. l. All media have external volume identification						
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		m. Procedures are in place to ensure that only authorized addition/removal of media from the library is allowed n. Media retention procedure is established and approved by management o. Records of all movements of computer tapes/disks between on-site and off-site media library are maintained. p. Procedure are present for secure transfer and disposal of media q. Computer media that are being transported to off-site data backup locations are stored in locked carrying cases that provide magnetic field protection r. All items of equipment containing storage media are checked to ensure that they do not contain sensitive data prior to their disposal.						
3.1.4.65	All items of equipment containing storage media (fixed and removable disks) shall be checked to ensure that they do not contain sensitive data prior to their disposal.	1. Obtain Media Management policy and verify the following: a. All sensitive information stored in any media is assigned an appropriate security classification b. Media library, electrical and mechanical control rooms are housed in separate isolated areas c. Access to these areas is given only to specific, named individuals on a need basis. Verify by checking the access list d. Such area is fire resistant and free of toxic chemicals. e. Storage media containing sensitive information is secured according to their classification f. Responsibilities for	Mandatory	WebTrust 3.4.21				

		media library management and protection are clearly defined and assigned g. Access to the media library is restricted to the authorized persons only (verify by checking list of authorized personnel) h. Media containing sensitive and backup data is stored at three different physical locations in the country. Verify by identifying the three different physical locations where backup is stored i. Media that containing audit, archive, or backup information is duplicated and stored in a location separate from the CA location. j. Media management system has been implemented by providing sample of the process followed. k. All incoming/outgoing media transfers are authorized by management and users. Verify by checking on sample basis. l. All media have external volume identification m. Procedures are in place to ensure that only authorized addition/removal of media from the library is allowed n. Media retention procedure is established and approved by management o. Records of all movements of computer tapes/disks between on-site and off-site media library are maintained. p. Procedure are present for secure transfer and disposal of media q. Computer media that are being transported to off-site data backup locations are stored in locked carrying cases that provide magnetic						
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		field protection r. All items of equipment containing storage media are checked to ensure that they do not contain sensitive data prior to their disposal.						
3.1.4.66	Procedures shall be in place to ensure the secure disposal of sensitive information assets on all corrupted/damaged or affected media both internal (e.g. hard disk/optical disk) and external (e.g. diskette, disk drive, tapes etc.) to the system	1. Obtain Media Management policy and verify the following: a. All sensitive information stored in any media is assigned an appropriate security classification b. Media library, electrical and mechanical control rooms are housed in separate isolated areas c. Access to these areas is given only to specific, named individuals on a need basis. Verify by checking the access list d. Such area is fire resistant and free of toxic chemicals. e. Storage media containing sensitive information is secured according to their classification f. Responsibilities for media library management and protection are clearly defined and assigned g. Access to the media library is restricted to the authorized persons only (verify by checking list of authorized personnel) h. Media containing sensitive and back up data is stored at three different physical locations in the country. Verify by identifying the three different physical locations where backup is stored i. Media that containing audit, archive, or backup information is duplicated and stored in a location separate from the CA location.	Mandatory	IT CA Rules SCHEDULE II 5.2				

		j. Media management system has been implemented by providing sample of the process followed. k. All incoming/outgoing media transfers are authorized by management and users. Verify by checking on sample basis. l. All media have external volume identification m. Procedures are in place to ensure that only authorized addition/removal of media from the library is allowed n. Media retention procedure is established and approved by management o. Records of all movements of computer tapes/disks between on-site and off-site media library are maintained. p. Procedure are present for secure transfer and disposal of media q. Computer media that are being transported to off-site data backup locations are stored in locked carrying cases that provide magnetic field protection r. All items of equipment containing storage media are checked to ensure that they do not contain sensitive data prior to their disposal.						
Power and Air Conditioning								
3.1.4.67	Air-conditioning system and power supply system unit shall be installed by the CA	1. Conduct walkthrough of CA facility and check the following: a. existence of Air conditioning and power supply system b. backup power is present to automatically lockout input, finish any pending actions, and	Mandatory	IT CA Rules SCHEDULE II 4.1				

		record the state of the equipment before lack of power or air conditioning causes a shutdown						
3.1.4.68	CAs shall have backup power sufficient to automatically lockout input, finish any pending actions, and record the state of the equipment before lack of power or air conditioning causes a shutdown	1. Conduct walkthrough of CA facility and check the following: a. existence of Air conditioning and power supply system b. backup power is present to automatically lockout input, finish any pending actions, and record the state of the equipment before lack of power or air conditioning causes a shutdown	Mandatory	X.509 Policy 5.1.3				
Waste Disposal								
3.1.4.69	Procedures shall be designed and implemented for disposing off sensitive waste material.	1. Verify procedure is present and implemented for disposing off the sensitive waste material 2. Validate all media used for storage of information pertaining to all functions	Mandatory	X.509 Policy 5.1.7, IT CA Rules SCHEDULE III 5				
3.1.4.70	All media used for storage of information pertaining to all functions associated with generation, production, issue and management of Digital Signature Certificate shall be scrutinized before being destroyed or released for disposal	associated of DSC is scrutinized before being destroyed or released for disposal. Verify by checking on sample basis, media was disposed off based on defined procedure	Mandatory	IT CA Rules SCHEDULE III 5				
Data Backup and Off-site Retention								
3.1.4.71	A Certifying Authority must ensure that facility used for off- site backup shall be within the country and shall have the same level of security as the primary Certifying Authority site	1. Conduct walkthrough of the CA backup site and check the following: a. facility used for off-site backup is within the country and has the same level of security as the primary CA site b. backup procedures are documented and implemented	Mandatory	WebTrust, IT CA Rules SCHEDULE III 6				

		c. Full system backups are made on periodic schedule d. Backups are performed and stored off-site not less than once every 7 days e. At least one full backup copy shall be stored at an offsite location f. One set of the original disks for all operating system and application software is maintained g. Backups of the system, application and data is performed on a regular basis h. Backups for application under development and data conversion efforts is also made i. Data backup is performed j. Critical system data and file server software have full backups taken weekly k. Documented proof of verifications of the integrity of backups every six months l. backups are kept in an area physically separate from the server m. information backups are rotated on a periodic basis to an off-site storage location n. Critical system data and file server software have incremental backups taken daily o. Systems that are completely static are backed up after changes or update in information p. Each LAN/system has a primary and backup operator to ensure continuity of business operations q. Security requirements of backup copies are consistent with control for information backed up						
3.1.4.72	Back-up procedures shall be documented, scheduled and	1. Conduct walkthrough of the CA backup site and check	Mandatory	IT CA Rules SCHEDULE II 9				

	monitored.	the following: a. facility used for off-site backup is within the country and has the same level of security as the primary CA site b. backup procedures are documented and implemented c. Full system backups are made on periodic schedule d. Backups are performed and stored off-site not less than once every 7 days e. At least one full backup copy shall be stored at an offsite location f. One set of the original disks for all operating system and application software is maintained g. Backups of the system, application and data is performed on a regular basis h. Backups for application under development and data conversion efforts is also made i. Data backup is performed j. Critical system data and file server software have full backups taken weekly k. Documented proof of verifications of the integrity of backups every six months l. backups are kept in an area physically separate from the server m. information backups are rotated on a periodic basis to an off-site storage location n. Critical system data and file server software have incremental backups taken daily o. Systems that are completely static are backed up after changes or update in information p. Each LAN/system has a primary and backup operator to ensure continuity of business operations q. Security requirements of backup copies are consistent with control for						
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		information backed up						
3.1.4.73	Full system backups of the CAs, sufficient to recover from system failure, shall be made on a periodic schedule	1. Conduct walkthrough of the CA backup site and check the following: a. facility used for off-site backup is within the country and has the same level of security as the primary CA site b. backup procedures are documented and implemented c. Full system backups are made on periodic schedule d. Backups are performed and stored off-site not less than once every 7 days e. At least one full backup copy shall be stored at an offsite location f. One set of the original disks for all operating system and application software is maintained g. Backups of the system, application and data is performed on a regular basis h. Backups for application under development and data conversion efforts is also made i. Data backup is performed j. Critical system data and file server software have full backups taken weekly k. Documented proof of verifications of the integrity of backups every six months l. backups are kept in an area physically separate from the server m. information backups are rotated on a periodic basis to an off-site storage location n. Critical system data and file server software have incremental backups taken daily o. Systems that are completely static are backed up after changes or update in information	Mandatory	X.509 Policy 5.1.8				

		p. Each LAN/system has a primary and backup operator to ensure continuity of business operations q. Security requirements of backup copies are consistent with control for information backed up						
3.1.4.74	Backups shall be performed and stored off-site not less than once every 7 days. At least one full backup copy shall be stored at an offsite location (at a location separate from the CA equipment). Only the latest full backup need be retained.	1. Conduct walkthrough of the CA backup site and check the following: a. facility used for off-site backup is within the country and has the same level of security as the primary CA site b. backup procedures are documented and implemented c. Full system backups are made on periodic schedule d. Backups are performed and stored off-site not less than once every 7 days e. At least one full backup copy shall be stored at an offsite location f. One set of the original disks for all operating system and application software is maintained g. Backups of the system, application and data is performed on a regular basis h. Backups for application under development and data conversion efforts is also made i. Data backup is performed j. Critical system data and file server software have full backups taken weekly k. Documented proof of verifications of the integrity of backups every six months l. backups are kept in an area physically separate from the server m. information backups are rotated on a periodic basis to an off-site storage location	Mandatory	X.509 Policy 5.1.8				

		n. Critical system data and file server software have incremental backups taken daily o. Systems that are completely static are backed up after changes or update in information p. Each LAN/system has a primary and backup operator to ensure continuity of business operations q. Security requirements of backup copies are consistent with control for information backed up						
3.1.4.75	One set of the original disks for all operating system and application software must be maintained to ensure that a valid, virus-free backup exists and is available for use at any time	1. Conduct walkthrough of the CA backup site and check the following: a. facility used for off-site backup is within the country and has the same level of security as the primary CA site b. backup procedures are documented and implemented c. Full system backups are made on periodic schedule d. Backups are performed and stored off-site not less than once every 7 days e. At least one full backup copy shall be stored at an offsite location f. One set of the original disks for all operating system and application software is maintained g. Backups of the system, application and data is performed on a regular basis h. Backups for application under development and data conversion efforts is also made i. Data backup is performed j. Critical system data and file server software have full backups taken weekly k. Documented proof of verifications of the	Mandatory	IT CA Rules SCHEDULE II 9				

		integrity of backups every six months l. backups are kept in an area physically separate from the server m. information backups are rotated on a periodic basis to an off-site storage location n. Critical system data and file server software have incremental backups taken daily o. Systems that are completely static are backed up after changes or update in information p. Each LAN/system has a primary and backup operator to ensure continuity of business operations q. Security requirements of backup copies are consistent with control for information backed up						
3.1.4.76	Backups of the system, application and data shall be performed on a regular basis. Backups should also be made for application under development and data conversion efforts.	1. Conduct walkthrough of the CA backup site and check the following: a. facility used for off-site backup is within the country and has the same level of security as the primary CA site b. backup procedures are documented and implemented c. Full system backups are made on periodic schedule d. Backups are performed and stored off-site not less than once every 7 days e. At least one full backup copy shall be stored at an offsite location f. One set of the original disks for all operating system and application software is maintained g. Backups of the system, application and data is performed on a regular basis h. Backups for application under development and data conversion efforts is	Mandatory	IT CA Rules SCHEDULE II 9				

		also made i. Data backup is performed j. Critical system data and file server software have full backups taken weekly k. Documented proof of verifications of the integrity of backups every six months l. backups are kept in an area physically separate from the server m. information backups are rotated on a periodic basis to an off-site storage location n. Critical system data and file server software have incremental backups taken daily o. Systems that are completely static are backed up after changes or update in information p. Each LAN/system has a primary and backup operator to ensure continuity of business operations q. Security requirements of backup copies are consistent with control for information backed up					
3.1.4.77	Data backup is required for all systems including personal computers, servers and distributed systems and databases.	1. Conduct walkthrough of the CA backup site and check the following: a. facility used for off-site backup is within the country and has the same level of security as the primary CA site b. backup procedures are documented and implemented c. Full system backups are made on periodic schedule d. Backups are performed and stored off-site not less than once every 7 days e. At least one full backup copy shall be stored at an offsite location f. One set of the original disks for all operating system and application software is	Mandatory	IT CA Rules SCHEDULE II 9			

		maintained g. Backups of the system, application and data is performed on a regular basis h. Backups for application under development and data conversion efforts is also made i. Data backup is performed j. Critical system data and file server software have full backups taken weekly k. Documented proof of verifications of the integrity of backups every six months l. backups are kept in an area physically separate from the server m. information backups are rotated on a periodic basis to an off-site storage location n. Critical system data and file server software have incremental backups taken daily o. Systems that are completely static are backed up after changes or update in information p. Each LAN/system has a primary and backup operator to ensure continuity of business operations q. Security requirements of backup copies are consistent with control for information backed up						
3.1.4.78	Critical system data and file server software shall have full backups taken weekly.	1. Conduct walkthrough of the CA backup site and check the following: a. facility used for off-site backup is within the country and has the same level of security as the primary CA site b. backup procedures are documented and implemented c. Full system backups are made on periodic schedule d. Backups are performed and stored off-site not less than	Mandatory	IT CA Rules SCHEDULE II 9				

		once every 7 days e. At least one full backup copy shall be stored at an offsite location f. One set of the original disks for all operating system and application software is maintained g. Backups of the system, application and data is performed on a regular basis h. Backups for application under development and data conversion efforts is also made i. Data backup is performed j. Critical system data and file server software have full backups taken weekly k. Documented proof of verifications of the integrity of backups every six months l. backups are kept in an area physically separate from the server m. information backups are rotated on a periodic basis to an off-site storage location n. Critical system data and file server software have incremental backups taken daily o. Systems that are completely static are backed up after changes or update in information p. Each LAN/system has a primary and backup operator to ensure continuity of business operations q. Security requirements of backup copies are consistent with control for information backed up						
3.1.4.79	The Certifying Authority should verify the integrity of the backups at least once every six months	1. Conduct walkthrough of the CA backup site and check the following: a. facility used for off-site backup is within the country and has the same level of security as the primary CA site	Mandatory	IT CA Rules SCHEDULE II 9				

		b. backup procedures are documented and implemented c. Full system backups are made on periodic schedule d. Backups are performed and stored off-site not less than once every 7 days e. At least one full backup copy shall be stored at an offsite location f. One set of the original disks for all operating system and application software is maintained g. Backups of the system, application and data is performed on a regular basis h. Backups for application under development and data conversion efforts is also made i. Data backup is performed j. Critical system data and file server software have full backups taken weekly k. Documented proof of verifications of the integrity of backups every six months l. backups are kept in an area physically separate from the server m. information backups are rotated on a periodic basis to an off-site storage location n. Critical system data and file server software have incremental backups taken daily o. Systems that are completely static are backed up after changes or update in information p. Each LAN/system has a primary and backup operator to ensure continuity of business operations q. Security requirements of backup copies are consistent with control for information backed up						
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3.1.4.80	The backups shall be kept in an area physically separate from the server. If critical system data on the LAN represents unique versions of the information assets, then the information backups should be rotated on a periodic basis to an off-site storage location.	1. Conduct walkthrough of the CA backup site and check the following: a. facility used for off-site backup is within the country and has the same level of security as the primary CA site b. backup procedures are documented and implemented c. Full system backups are made on periodic schedule d. Backups are performed and stored off-site not less than once every 7 days e. At least one full backup copy shall be stored at an offsite location f. One set of the original disks for all operating system and application software is maintained g. Backups of the system, application and data is performed on a regular basis h. Backups for application under development and data conversion efforts is also made i. Data backup is performed j. Critical system data and file server software have full backups taken weekly k. Documented proof of verifications of the integrity of backups every six months l. backups are kept in an area physically separate from the server m. information backups are rotated on a periodic basis to an off-site storage location n. Critical system data and file server software have incremental backups taken daily o. Systems that are completely static are backed up after changes or update in information p. Each LAN/system has a primary and backup operator to ensure continuity of business operations q. Security	Mandatory	IT CA Rules SCHEDULE II 9					
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		requirements of backup copies are consistent with control for information backed up						
3.1.4.81	Critical system data and file server software must have incremental backups taken daily.	1. Conduct walkthrough of the CA backup site and check the following: a. facility used for off-site backup is within the country and has the same level of security as the primary CA site b. backup procedures are documented and implemented c. Full system backups are made on periodic schedule d. Backups are performed and stored off-site not less than once every 7 days e. At least one full backup copy shall be stored at an offsite location f. One set of the original disks for all operating system and application software is maintained g. Backups of the system, application and data is performed on a regular basis h. Backups for application under development and data conversion efforts is also made i. Data backup is performed j. Critical system data and file server software have full backups taken weekly k. Documented proof of verifications of the integrity of backups every six months l. backups are kept in an area physically separate from the server m. information backups are rotated on a periodic basis to an off-site storage location n. Critical system data and file server software have incremental backups taken daily o. Systems that are completely static are	Mandatory	IT CA Rules SCHEDULE II 9				

		backed up after changes or update in information p. Each LAN/system has a primary and backup operator to ensure continuity of business operations q. Security requirements of backup copies are consistent with control for information backed up						
3.1.4.82	Systems that are completely static may not require periodic backup, but shall be backed up after changes or updates in the information	1. Conduct walkthrough of the CA backup site and check the following: a. facility used for off-site backup is within the country and has the same level of security as the primary CA site b. backup procedures are documented and implemented c. Full system backups are made on periodic schedule d. Backups are performed and stored off-site not less than once every 7 days e. At least one full backup copy shall be stored at an offsite location f. One set of the original disks for all operating system and application software is maintained g. Backups of the system, application and data is performed on a regular basis h. Backups for application under development and data conversion efforts is also made i. Data backup is performed j. Critical system data and file server software have full backups taken weekly k. Documented proof of verifications of the integrity of backups every six months l. backups are kept in an area physically separate from the server m. information	Mandatory	IT CA Rules SCHEDULE II 9				

		backups are rotated on a periodic basis to an off-site storage location n. Critical system data and file server software have incremental backups taken daily o. Systems that are completely static are backed up after changes or update in information p. Each LAN/system has a primary and backup operator to ensure continuity of business operations q. Security requirements of backup copies are consistent with control for information backed up						
3.1.4.83	Each LAN/system should have a primary and backup operator to ensure continuity of business operations.	1. Conduct walkthrough of the CA backup site and check the following: a. facility used for off-site backup is within the country and has the same level of security as the primary CA site b. backup procedures are documented and implemented c. Full system backups are made on periodic schedule d. Backups are performed and stored off-site not less than once every 7 days e. At least one full backup copy shall be stored at an offsite location f. One set of the original disks for all operating system and application software is maintained g. Backups of the system, application and data is performed on a regular basis h. Backups for application under development and data conversion efforts is also made i. Data backup is performed j. Critical system data and file server software have full backups taken	Mandatory	IT CA Rules SCHEDULE II 9				

		weekly k. Documented proof of verifications of the integrity of backups every six months l. backups are kept in an area physically separate from the server m. information backups are rotated on a periodic basis to an off-site storage location n. Critical system data and file server software have incremental backups taken daily o. Systems that are completely static are backed up after changes or update in information p. Each LAN/system has a primary and backup operator to ensure continuity of business operations q. Security requirements of backup copies are consistent with control for information backed up						
3.1.4.84	The security requirements of backup copies shall be consistent with the controls for the information backed up	1. Conduct walkthrough of the CA backup site and check the following: a. facility used for off-site backup is within the country and has the same level of security as the primary CA site b. backup procedures are documented and implemented c. Full system backups are made on periodic schedule d. Backups are performed and stored off-site not less than once every 7 days e. At least one full backup copy shall be stored at an offsite location f. One set of the original disks for all operating system and application software is maintained g. Backups of the system, application and data is performed on a regular basis h. Backups for	Mandatory	WebTrust 4.2.4				

		application under development and data conversion efforts is also made i. Data backup is performed j. Critical system data and file server software have full backups taken weekly k. Documented proof of verifications of the integrity of backups every six months l. backups are kept in an area physically separate from the server m. information backups are rotated on a periodic basis to an off-site storage location n. Critical system data and file server software have incremental backups taken daily o. Systems that are completely static are backed up after changes or update in information p. Each LAN/system has a primary and backup operator to ensure continuity of business operations q. Security requirements of backup copies are consistent with control for information backed up					
Environmental Protection							
3.1.4.85	Water detectors shall be installed under the raised floors throughout the operational site and shall be connected to audible alarms	1. Conduct walkthrough of the CA facility and check the following: a. Water detectors are installed under the raised floors throughout the operational site and are connected to audible alarms b. temperature and humidity condition in the operational site is monitored and controlled periodically c. periodic inspection, testing and maintenance of equipment is scheduled. Verify by checking previous reports on a sample	Mandatory	IT CA Rules SCHEDULE II 4.3			

		basis based on frequency d. personnel at operational site is trained to monitor and control various devices						
3.1.4.86	The temperature and humidity condition in the operational site shall be monitored and controlled periodically	1. Conduct walkthrough of the CA facility and check the following: a. Water detectors are installed under the raised floors throughout the operational site and are connected to audible alarms b. temperature and humidity condition in the operational site is monitored and controlled periodically c. periodic inspection, testing and maintenance of equipment is scheduled. Verify by checking previous reports on a sample basis based on frequency d. personnel at operational site is trained to monitor and control various devices	Mandatory	IT CA Rules SCHEDULE II 4.3				
3.1.4.87	Periodic inspection, testing and maintenance of the equipment and systems shall be scheduled.	1. Conduct walkthrough of the CA facility and check the following: a. Water detectors are installed under the raised floors throughout the operational site and are connected to audible alarms b. temperature and humidity condition in the operational site is monitored and controlled periodically c. periodic inspection, testing and maintenance of equipment is scheduled. Verify by checking previous reports on a sample basis based on frequency d. personnel at operational site is trained to monitor and control various devices	Mandatory	T CA Rules SCHEDULE II 4.3				

3.1.4.88	Personnel at the operational site shall be trained to monitor and control the various equipment and devices installed at the operational site for the purpose of fire and environment protection	1. Conduct walkthrough of the CA facility and check the following: a. Water detectors are installed under the raised floors throughout the operational site and are connected to audible alarms b. temperature and humidity condition in the operational site is monitored and controlled periodically c. periodic inspection, testing and maintenance of equipment is scheduled. Verify by checking previous reports on a sample basis based on frequency d. personnel at operational site is trained to monitor and control various devices	Mandatory	IT CA Rules SCHEDULE II 4				
Third Party Access								
3.1.4.89	Access to the computer systems by other Organizations shall be subjected to a similar level of security protection and controls as in stated in Information Technology security guidelines under IT CA Rules	1. Verify access to computer systems by other organizations is subjected to a similar level of security protection and controls as stated in IT CA Rules 2. Validate in case Data Centre uses external service/facility provider for any of their operations, all related risks are evaluated before onboarding the outsourced vendor 3. Verify on sample basis the external service or facility provider has signed non-disclosure agreements with the management of the Data Centre/operational site 4. Check the external service/facility provider provides an equivalent level of security controls as required by Information Technology Security Guidelines	Mandatory	IT CA Rules SCHEDULE II 5.4				

3.1.4.90	In case the Data Centre uses the facilities of external service/facility provider (outsourcer) for any of their operations, the use of external service/facility providers (e.g. outsourcer) shall be evaluated in light of the possible security exposures and risks involved and all such agreements shall be approved by the information asset owner. The external service or facility provider shall also sign non-disclosure agreements with the management of the Data Centre/operational site.	1. Verify access to computer systems by other organizations is subjected to a similar level of security protection and controls as stated in IT CA Rules 2. Validate in case Data Centre uses external service/facility provider for any of their operations, all related risks are evaluated before onboarding the outsourced vendor 3. Verify on sample basis the external service or facility provider has signed non-disclosure agreements with the management of the Data Centre/operational site 4. Check the external service/facility provider provides an equivalent level of security controls as required by Information Technology Security Guidelines	Mandatory	IT CA Rules SCHEDULE II 5.4				
3.1.4.91	The external service/facility provider (e.g. outsourcer) shall provide an equivalent level of security controls as required by Information Technology Security Guidelines	1. Verify access to computer systems by other organizations is subjected to a similar level of security protection and controls as stated in IT CA Rules 2. Validate in case Data Centre uses external service/facility provider for any of their operations, all related risks are evaluated before onboarding the outsourced vendor 3. Verify on sample basis the external service or facility provider has signed non-disclosure agreements with the management of the Data Centre/operational site 4. Check the external service/facility provider provides an equivalent level of security controls as	Mandatory	IT CA Rules SCHEDULE II 5.4				

		required by Information Technology Security Guidelines						
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3.1.5. Personnel Security

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Personnel Security							
3.1.5.1	The Certifying Authority must ensure that all personnel performing duties with respect to its operation must: • be appointed in writing; • be bound by contract or statute to the terms and conditions of the position they are to fill; • have received comprehensive training with respect to the duties they are to perform; • be bound by statute or contract not to disclose sensitive Certifying Authority's security related information or subscriber information; • not be assigned duties that may cause a conflict of interest with their Certifying Authority's duties; • be aware and trained in the relevant aspects of the Information Technology Security Policy and Security Guidelines framed for carrying out Certifying Authority's operation	1. For sample CA personnel, perform the following checks: a. personnel received an offer letter with roles and responsibilities appointed in writing. Verify on sample basis for personnel b. personnel is bound by contract or statute to the terms and conditions of the position they are to fill. Verify by checking the contract c. personnel has received comprehensive training via sessions organized specifically for the job. Check training logs d. personnel is bound by statute or contract not to disclose sensitive CA's security related information or subscriber information and has signed NDA for same. e. personnel is not assigned duties that may cause a conflict of interest with their CA's duties f. personnel is aware and trained in the relevant aspects of the Information Technology Security Policy and Security Guidelines	Mandatory	IT CA Rules SCHEDULE III 14			
3.1.5.2	A Certifying Authority must make available to his personnel the Digital Signature Certificate policies it supports, its Certification Practice Statement, Information	1. Validate CA has made available to his personnel the DSC policies, CPS, IT security guideline, Information Technology Security	Mandatory	IT CA Rules SCHEDULE III 17			

	Technology Security Policy and any specific statutes, policies or contracts relevant to their position	Policy and any specific statutes, policies or contracts relevant to their position 2. Verify the CA provides reasonable assurance that personnel and employment practices enhance and support the trustworthiness of the CA's operations and receive regular security/PKI based trainings						
3.1.5.3	The CA shall provide reasonable assurance that personnel and employment practices enhance and support the trustworthiness of the CA's operations	1. Validate CA has made available to his personnel the DSC policies, CPS, IT security guideline, Information Technology Security Policy and any specific statutes, policies or contracts relevant to their position 2. Verify the CA provides reasonable assurance that personnel and employment practices enhance and support the trustworthiness of the CA's operations and receive regular security/PKI based trainings	Mandatory	WebTrust 3.3				
3.1.5.4	Prior to the engagement of any person whether as an employee, agent, or an independent contractor of the CA, the CA shall verify perform a background check. The scope of the background check shall include the following areas covering the past five years: • Employment; • Education (Regardless of the date of award, the highest educational degree shall be verified); • Place of residence (3 years); • Law Enforcement; and • References The background shall be refreshed every three years	1. Check the documented procedure detailing the background check criteria for an employee or an independent contractor of the CA 2. Verify that the scope of background check contains the details covered in control description	Mandatory	WebTrust 3.3.7, X.509 Policy 5.3.2				

3.1.5.5	The CA and CSP shall make available to its personnel this certificate policy, the applicable CPS, and any relevant statutes, policies or contracts	1. Verify the CA and CSP share all relevant statutes, policies or contracts with their personnel	Mandatory	X.509 Policy 5.3.7				
3.1.5.6	Personnel appointed to trusted roles (CA trusted roles) shall: • Have successfully completed an appropriate training program; • Have demonstrated the ability to perform their duties; • Be trustworthy; • Have no other duties that would interfere or conflict with their duties for the trusted role; • Have not been previously relieved of duties for reasons of negligence or nonperformance of duties; • Have not been denied a security clearance, or had a security clearance revoked for cause; • Have not been convicted of a felony offense; and • Be appointed in writing by an approving authority • Be bound by statute or contract not to disclose sensitive Certifying Authority's security related information or subscriber information;	1. Obtain the list of personnel having CA trusted roles 2. Verify for sample personnel, they adhere to the requirements mentioned in control description 3. Validate the personnel in Trusted Roles maintain skill levels consistent with the CA's training and performance programs	Mandatory	X.509 Policy 5.3.1				
3.1.5.7	All personnel in Trusted Roles shall maintain skill levels consistent with the CA's training and performance programs.	1. Obtain the list of personnel having CA trusted roles 2. Verify for sample personnel, they adhere to the requirements mentioned in control description 3. Validate the personnel in Trusted Roles maintain skill levels consistent with the CA's training and performance programs	Mandatory	CA Browser Forum 5.3.4				

3.1.6. System integrity and security measures

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
User's Account Management							

3.1.6.1	Procedures for user account management shall be established to control access to application systems and data	1. For the User Account Management process, check the following on sample basis: a. Procedures have been designed and implemented for user account management b. Users are authorized by the computer system owner to access the computer services c. written statement of access rights is given to all users d. all users sign an undertaking to acknowledge that they understand the conditions of access e. Access is only provided after authorization is complete f. Formal record of all registered users of the computer services is maintained g. Access rights of users who have been transferred, or left the organization are removed immediately h. Periodic checks are carried out for redundant user accounts i. User accounts are suspended if any of the conditions mentioned in control 3.1.6.32 are met	Mandatory	IT CA Rules SCHEDULE II 6.5				
3.1.6.2	Users shall be authorized by the computer system owner to access the computer services	1. For the User Account Management process, check the following on sample basis: a. Procedures have been designed and implemented for user account management b. Users are authorized by the computer system owner to access the computer services c. written statement of access rights is given to all users d. all users sign an undertaking to acknowledge that they understand the conditions of access e. Access is only provided after authorization is complete f. Formal record of all registered users of the	Mandatory	IT CA Rules SCHEDULE II 6.5				

		computer services is maintained g. Access rights of users who have been transferred, or left the organization are removed immediately h. Periodic checks are carried out for redundant user accounts i. User accounts are suspended if any of the conditions mentioned in control 3.1.6.32 are met						
3.1.6.3	A written statement of access rights shall be given to all users.	1. For the User Account Management process, check the following on sample basis: a. Procedures have been designed and implemented for user account management b. Users are authorized by the computer system owner to access the computer services c. written statement of access rights is given to all users d. all users sign an undertaking to acknowledge that they understand the conditions of access e. Access is only provided after authorization is complete f. Formal record of all registered users of the computer services is maintained g. Access rights of users who have been transferred, or left the organization are removed immediately h. Periodic checks are carried out for redundant user accounts i. User accounts are suspended if any of the conditions mentioned in control 3.1.6.32 are met	Mandatory	IT CA Rules SCHEDULE II 6.5				
3.1.6.4	All users shall be required to sign an undertaking to acknowledge that they understand the conditions of access	1. For the User Account Management process, check the following on sample basis: a. Procedures have been designed and implemented for user account management b. Users are authorized by the computer system	Mandatory	IT CA Rules SCHEDULE II 6.5				

		owner to access the computer services c. written statement of access rights is given to all users d. all users sign an undertaking to acknowledge that they understand the conditions of access e. Access is only provided after authorization is complete f. Formal record of all registered users of the computer services is maintained g. Access rights of users who have been transferred, or left the organization are removed immediately h. Periodic checks are carried out for redundant user accounts i. User accounts are suspended if any of the conditions mentioned in control 3.1.6.32 are met						
3.1.6.5	Where access to computer services is administered by service providers, access shall not be provided until authorization is completed. This shall include the acknowledgment of receipt of the accounts by the users	1. For the User Account Management process, check the following on sample basis: a. Procedures have been designed and implemented for user account management b. Users are authorized by the computer system owner to access the computer services c. written statement of access rights is given to all users d. all users sign an undertaking to acknowledge that they understand the conditions of access e. Access is only provided after authorization is complete f. Formal record of all registered users of the computer services is maintained g. Access rights of users who have been transferred, or left the organization are removed immediately h. Periodic checks are carried out for redundant user accounts	Mandatory	IT CA Rules SCHEDULE II 6.5				

		i. User accounts are suspended if any of the conditions mentioned in control 3.1.6.32 are met						
3.1.6.6	A formal record of all registered users of the computer services shall be maintained.	1. For the User Account Management process, check the following on sample basis: a. Procedures have been designed and implemented for user account management b. Users are authorized by the computer system owner to access the computer services c. written statement of access rights is given to all users d. all users sign an undertaking to acknowledge that they understand the conditions of access e. Access is only provided after authorization is complete f. Formal record of all registered users of the computer services is maintained g. Access rights of users who have been transferred, or left the organization are removed immediately h. Periodic checks are carried out for redundant user accounts i. User accounts are suspended if any of the conditions mentioned in control 3.1.6.32 are met	Mandatory	IT CA Rules SCHEDULE II 6.5				
3.1.6.7	Access rights of users who have been transferred, or left the organization shall be removed immediately	1. For the User Account Management process, check the following on sample basis: a. Procedures have been designed and implemented for user account management b. Users are authorized by the computer system owner to access the computer services c. written statement of access rights is given to all users d. all users sign an undertaking to acknowledge that they understand the conditions of access	Mandatory	IT CA Rules SCHEDULE II 6.5				

		e. Access is only provided after authorization is complete f. Formal record of all registered users of the computer services is maintained g. Access rights of users who have been transferred, or left the organization are removed immediately h. Periodic checks are carried out for redundant user accounts i. User accounts are suspended if any of the conditions mentioned in control 3.1.6.32 are met						
3.1.6.8	A periodic check shall be carried out for redundant user accounts and access rights that are no longer required. Redundant user accounts shall not be re-issued to another user	1. For the User Account Management process, check the following on sample basis: a. Procedures have been designed and implemented for user account management b. Users are authorized by the computer system owner to access the computer services c. written statement of access rights is given to all users d. all users sign an undertaking to acknowledge that they understand the conditions of access e. Access is only provided after authorization is complete f. Formal record of all registered users of the computer services is maintained g. Access rights of users who have been transferred, or left the organization are removed immediately h. Periodic checks are carried out for redundant user accounts i. User accounts are suspended if any of the conditions mentioned in control 3.1.6.32 are met	Mandatory	IT CA Rules SCHEDULE II 6.5				
3.1.6.9	User accounts shall be suspended under the following conditions	1. For the User Account Management process, check the following on sample basis:	Mandatory	IT CA Rules SCHEDULE II 6.5				

		a. Procedures have been designed and implemented for user account management b. Users are authorized by the computer system owner to access the computer services c. written statement of access rights is given to all users d. all users sign an undertaking to acknowledge that they understand the conditions of access e. Access is only provided after authorization is complete f. Formal record of all registered users of the computer services is maintained g. Access rights of users who have been transferred, or left the organization are removed immediately h. Periodic checks are carried out for redundant user accounts i. User accounts are suspended if any of the conditions mentioned in control 3.1.6.32 are met						
3.1.6.10	User accounts shall be suspended under the following conditions • When an individual is on extended leave or inactive use of over thirty days. In case of protected computer system, the limit of thirty days may be reduced to fifteen days by the System Administrator. • Immediately upon the termination of the services of an individual. • Suspended or inactive accounts shall be deleted after a two months period. In case of protected computer systems, the limit of two months may be reduced to one month.	1. For the User Account Management process, check the following on sample basis: a. Procedures have been designed and implemented for user account management b. Users are authorized by the computer system owner to access the computer services c. written statement of access rights is given to all users d. all users sign an undertaking to acknowledge that they understand the conditions of access e. Access is only provided after authorization is complete f. Formal record of all registered users of the computer services is maintained g. Access rights of users who have been	Mandatory	IT CA Rules SCHEDULE II 6.5				

		transferred, or left the organization are removed immediately h. Periodic checks are carried out for redundant user accounts i. User accounts are suspended if any of the conditions mentioned in control 3.1.6.32 are met						
Use of Security Systems or Facilities								
3.1.6.11	Security controls shall be installed and maintained on each computer system or computer node to prevent unauthorized users from gaining entry to the information system and to prevent unauthorized access to data.	1. For sample computers, check the minimum baseline security controls are installed and maintained on each computer system 2. Validate system software or resource of the computer system is accessible after being authenticated by access control system	Mandatory	IT CA Rules SCHEDULE II 6.1				
3.1.6.12	Security controls shall be installed and maintained on each computer system or computer node to prevent unauthorized users from gaining entry to the information system and to prevent unauthorized access to data.	1. For sample computers, check the minimum baseline security controls are installed and maintained on each computer system 2. Validate system software or resource of the computer system is accessible after being authenticated by access control system	Mandatory	IT CA Rules SCHEDULE II 6.1				
3.1.6.13	Any system software or resource of the computer system should only be accessible after being authenticated by access control system	1. For sample computers, check the minimum baseline security controls are installed and maintained on each computer system 2. Validate system software or resource of the computer system is accessible after being authenticated by access control system	Mandatory	IT CA Rules SCHEDULE II 6.1				
System Access Control								
3.1.6.14	Access control software and system software security features shall be implemented to protect resources. Management approval	1. For the system access controls implemented in CA facility check on sample basis the following: a. Management approval	Mandatory	IT CA Rules SCHEDULE II 6.2, IT Regulations 3				

	is required to authorize issuance of user identification (ID) and resource privileges. Approved access and integrity controls such as intrusion detection, virus scanning, prevention of denial-of service attacks and physical security measures shall be followed by the Certifying Authority for all its systems that store and process the subscribers' information and certificates.	is required to authorize issuance of user identification (ID) and resource privileges b. Access control software and system software security features are implemented c. Approved access and integrity controls are implemented d. access control software or operating system of the computer system provides features to restrict access to the system and data resources e. All passwords used are resistant to dictionary attacks and contain letters, numerals and special characters. f. A documented Access Control System manual is present g. Each user is assigned a unique user ID h. Sharing IDs is prohibited i. Stored passwords are encrypted j. Stored passwords are protected by access controls k. Automatic time-out for terminal inactivity is implemented l. Activities of all remote users are logged and monitored closely m. facility to login as another user from one user's login is denied n. startup and shutdown procedure of the security software is Automated o. Sensitive Operating System files are protected against all known attacks using proven tools and techniques p. Only System Administrator is allowed to modify systemoperated files						
3.1.6.15	The access control software or operating system of the computer system shall provide features to restrict access to the system and data resources. The use	1. For the system access controls implemented in CA facility check on sample basis the following: a. Management approval is required to authorize	Mandatory	IT CA Rules SCHEDULE II 6.2				

	of common passwords such as "administrator" or "president" or "game" etc. to protect	issuance of user identification (ID) and resource privileges b. Access control software and system software security features are implemented c. Approved access and integrity controls are implemented d. access control software or operating system of the computer system provides features to restrict access to the system and data resources e. All passwords used are resistant to dictionary attacks and contain letters, numerals and special characters. f. A documented Access Control System manual is present g. Each user is assigned a unique user ID h. Sharing IDs is prohibited i. Stored passwords are encrypted j. Stored passwords are protected by access controls k. Automatic time-out for terminal inactivity is implemented l. Activities of all remote users are logged and monitored closely m. facility to login as another user from one user's login is denied n. startup and shutdown procedure of the security software is Automated o. Sensitive Operating System files are protected against all known attacks using proven tools and techniques p. Only System Administrator is allowed to modify systemoperated files						
3.1.6.16	The access control software or operating system of the computer system shall provide features to restrict access to the system and data resources. The use of common passwords	1. For the system access controls implemented in CA facility check on sample basis the following: a. Management approval is required to authorize issuance of user	Mandatory	IT CA Rules SCHEDULE II 6.2				

	such as "administrator" or "president" or "game" etc. to protect access to the system and data resources represent a security exposure and shall be avoided. All passwords used must be resistant to dictionary attacks.	identification (ID) and resource privileges b. Access control software and system software security features are implemented c. Approved access and integrity controls are implemented d. access control software or operating system of the computer system provides features to restrict access to the system and data resources e. All passwords used are resistant to dictionary attacks and contain letters, numerals and special characters. f. A documented Access Control System manual is present g. Each user is assigned a unique user ID h. Sharing IDs is prohibited i. Stored passwords are encrypted j. Stored passwords are protected by access controls k. Automatic time-out for terminal inactivity is implemented l. Activities of all remote users are logged and monitored closely m. facility to login as another user from one user's login is denied n. startup and shutdown procedure of the security software is Automated o. Sensitive Operating System files are protected against all known attacks using proven tools and techniques p. Only System Administrator is allowed to modify systemoperated files					
3.1.6.17	An Access Control System manual documenting the access granted to different level of users shall be prepared to provide guidance to the System Administrator for grant of access.	1. For the system access controls implemented in CA facility check on sample basis the following: a. Management approval is required to authorize issuance of user identification (ID) and	Mandatory	IT CA Rules SCHEDULE II 6.2			

		resource privileges b. Access control software and system software security features are implemented c. Approved access and integrity controls are implemented d. access control software or operating system of the computer system provides features to restrict access to the system and data resources e. All passwords used are resistant to dictionary attacks and contain letters, numerals and special characters. f. A documented Access Control System manual is present g. Each user is assigned a unique user ID h. Sharing IDs is prohibited i. Stored passwords are encrypted j. Stored passwords are protected by access controls k. Automatic time-out for terminal inactivity is implemented l. Activities of all remote users are logged and monitored closely m. facility to login as another user from one user's login is denied n. startup and shutdown procedure of the security software is Automated o. Sensitive Operating System files are protected against all known attacks using proven tools and techniques p. Only System Administrator is allowed to modify systemoperated files						
3.1.6.18	Each user shall be assigned a unique user ID. Adequate user education shall be provided to help users in password choice and password protection. Sharing of user IDs shall not be allowed	1. For the system access controls implemented in CA facility check on sample basis the following: a. Management approval is required to authorize issuance of user identification (ID) and resource privileges	Mandatory	IT CA Rules SCHEDULE II 6.2				

		b. Access control software and system software security features are implemented c. Approved access and integrity controls are implemented d. access control software or operating system of the computer system provides features to restrict access to the system and data resources e. All passwords used are resistant to dictionary attacks and contain letters, numerals and special characters. f. A documented Access Control System manual is present g. Each user is assigned a unique user ID h. Sharing IDs is prohibited i. Stored passwords are encrypted j. Stored passwords are protected by access controls k. Automatic time-out for terminal inactivity is implemented l. Activities of all remote users are logged and monitored closely m. facility to login as another user from one user's login is denied n. startup and shutdown procedure of the security software is Automated o. Sensitive Operating System files are protected against all known attacks using proven tools and techniques p. Only System Administrator is allowed to modify systemoperated files						
3.1.6.19	Stored passwords shall be encrypted using internationally proven encryption techniques to prevent unauthorized disclosure and modification.	1. For the system access controls implemented in CA facility check on sample basis the following: a. Management approval is required to authorize issuance of user identification (ID) and resource privileges b. Access control	Mandatory	IT CA Rules SCHEDULE II 6.2				

		software and system software security features are implemented c. Approved access and integrity controls are implemented d. access control software or operating system of the computer system provides features to restrict access to the system and data resources e. All passwords used are resistant to dictionary attacks and contain letters, numerals and special characters. f. A documented Access Control System manual is present g. Each user is assigned a unique user ID h. Sharing IDs is prohibited i. Stored passwords are encrypted j. Stored passwords are protected by access controls k. Automatic time-out for terminal inactivity is implemented l. Activities of all remote users are logged and monitored closely m. facility to login as another user from one user's login is denied n. startup and shutdown procedure of the security software is Automated o. Sensitive Operating System files are protected against all known attacks using proven tools and techniques p. Only System Administrator is allowed to modify systemoperated files							
3.1.6.20	Stored passwords shall be protected by access controls from unauthorized disclosure and modification	1. For the system access controls implemented in CA facility check on sample basis the following: a. Management approval is required to authorize issuance of user identification (ID) and resource privileges b. Access control software and system	Mandatory	IT CA Rules SCHEDULE II 6.2					

		software security features are implemented c. Approved access and integrity controls are implemented d. access control software or operating system of the computer system provides features to restrict access to the system and data resources e. All passwords used are resistant to dictionary attacks and contain letters, numerals and special characters. f. A documented Access Control System manual is present g. Each user is assigned a unique user ID h. Sharing IDs is prohibited i. Stored passwords are encrypted j. Stored passwords are protected by access controls k. Automatic time-out for terminal inactivity is implemented l. Activities of all remote users are logged and monitored closely m. facility to login as another user from one user's login is denied n. startup and shutdown procedure of the security software is Automated o. Sensitive Operating System files are protected against all known attacks using proven tools and techniques p. Only System Administrator is allowed to modify systemoperated files						
3.1.6.21	Automatic time-out for terminal inactivity should be implemented	1. For the system access controls implemented in CA facility check on sample basis the following: a. Management approval is required to authorize issuance of user identification (ID) and resource privileges b. Access control software and system software security	Mandatory	IT CA Rules SCHEDULE II 6.2				

		features are implemented c. Approved access and integrity controls are implemented d. access control software or operating system of the computer system provides features to restrict access to the system and data resources e. All passwords used are resistant to dictionary attacks and contain letters, numerals and special characters. f. A documented Access Control System manual is present g. Each user is assigned a unique user ID h. Sharing IDs is prohibited i. Stored passwords are encrypted j. Stored passwords are protected by access controls k. Automatic time-out for terminal inactivity is implemented l. Activities of all remote users are logged and monitored closely m. facility to login as another user from one user's login is denied n. startup and shutdown procedure of the security software is Automated o. Sensitive Operating System files are protected against all known attacks using proven tools and techniques p. Only System Administrator is allowed to modify systemoperated files						
3.1.6.22	Where a second level access control is implemented through the application system, password controls similar to those implemented for the computer system shall be in place .	1. For the system access controls implemented in CA facility check on sample basis the following: a. Management approval is required to authorize issuance of user identification (ID) and resource privileges b. Access control software and system software security features are	Mandatory	IT CA Rules SCHEDULE II 6.2				

		implemented c. Approved access and integrity controls are implemented d. access control software or operating system of the computer system provides features to restrict access to the system and data resources e. All passwords used are resistant to dictionary attacks and contain letters, numerals and special characters. f. A documented Access Control System manual is present g. Each user is assigned a unique user ID h. Sharing IDs is prohibited i. Stored passwords are encrypted j. Stored passwords are protected by access controls k. Automatic time-out for terminal inactivity is implemented l. Activities of all remote users are logged and monitored closely m. facility to login as another user from one user's login is denied n. startup and shutdown procedure of the security software is Automated o. Sensitive Operating System files are protected against all known attacks using proven tools and techniques p. Only System Administrator is allowed to modify systemoperated files					
3.1.6.23	Activities of all remote users shall be logged and monitored closely	1. For the system access controls implemented in CA facility check on sample basis the following: a. Management approval is required to authorize issuance of user identification (ID) and resource privileges b. Access control software and system software security features are implemented	Mandatory	IT CA Rules SCHEDULE II 6.2			

		c. Approved access and integrity controls are implemented d. access control software or operating system of the computer system provides features to restrict access to the system and data resources e. All passwords used are resistant to dictionary attacks and contain letters, numerals and special characters. f. A documented Access Control System manual is present g. Each user is assigned a unique user ID h. Sharing IDs is prohibited i. Stored passwords are encrypted j. Stored passwords are protected by access controls k. Automatic time-out for terminal inactivity is implemented l. Activities of all remote users are logged and monitored closely m. facility to login as another user from one user's login is denied n. startup and shutdown procedure of the security software is Automated o. Sensitive Operating System files are protected against all known attacks using proven tools and techniques p. Only System Administrator is allowed to modify systemoperated files					
3.1.6.24	The facility to login as another user from one user's login shall be denied. However, the system should prohibit direct login as a trusted user (e.g. root in Unix, administrator in Windows NT or Windows 2000). This means that there must be a user account configured for the trusted administrator.	1. For the system access controls implemented in CA facility check on sample basis the following: a. Management approval is required to authorize issuance of user identification (ID) and resource privileges b. Access control software and system software security features are implemented c. Approved access and	Mandatory	IT CA Rules SCHEDULE II 6.2			

		integrity controls are implemented d. access control software or operating system of the computer system provides features to restrict access to the system and data resources e. All passwords used are resistant to dictionary attacks and contain letters, numerals and special characters. f. A documented Access Control System manual is present g. Each user is assigned a unique user ID h. Sharing IDs is prohibited i. Stored passwords are encrypted j. Stored passwords are protected by access controls k. Automatic time-out for terminal inactivity is implemented l. Activities of all remote users are logged and monitored closely m. facility to login as another user from one user's login is denied n. startup and shutdown procedure of the security software is Automated o. Sensitive Operating System files are protected against all known attacks using proven tools and techniques p. Only System Administrator is allowed to modify systemoperated files					
3.1.6.25	The startup and shutdown procedure of the security software must be Automated	1. For the system access controls implemented in CA facility check on sample basis the following: a. Management approval is required to authorize issuance of user identification (ID) and resource privileges b. Access control software and system software security features are implemented c. Approved access and integrity controls are	Mandatory	IT CA Rules SCHEDULE II 6.2			

		implemented d. access control software or operating system of the computer system provides features to restrict access to the system and data resources e. All passwords used are resistant to dictionary attacks and contain letters, numerals and special characters. f. A documented Access Control System manual is present g. Each user is assigned a unique user ID h. Sharing IDs is prohibited i. Stored passwords are encrypted j. Stored passwords are protected by access controls k. Automatic time-out for terminal inactivity is implemented l. Activities of all remote users are logged and monitored closely m. facility to login as another user from one user's login is denied n. startup and shutdown procedure of the security software is Automated o. Sensitive Operating System files are protected against all known attacks using proven tools and techniques p. Only System Administrator is allowed to modify systemoperated files						
3.1.6.26	Sensitive Operating System files, which are more prone to hackers must be protected against all known attacks using proven tools and techniques. That is to say no user will be able to modify them except with the permission of System Administrator.	1. For the system access controls implemented in CA facility check on sample basis the following: a. Management approval is required to authorize issuance of user identification (ID) and resource privileges b. Access control software and system software security features are implemented c. Approved access and integrity controls are implemented	Mandatory	IT CA Rules SCHEDULE II 6.2				

		d. access control software or operating system of the computer system provides features to restrict access to the system and data resources e. All passwords used are resistant to dictionary attacks and contain letters, numerals and special characters. f. A documented Access Control System manual is present g. Each user is assigned a unique user ID h. Sharing IDs is prohibited i. Stored passwords are encrypted j. Stored passwords are protected by access controls k. Automatic time-out for terminal inactivity is implemented l. Activities of all remote users are logged and monitored closely m. facility to login as another user from one user's login is denied n. startup and shutdown procedure of the security software is Automated o. Sensitive Operating System files are protected against all known attacks using proven tools and techniques p. Only System Administrator is allowed to modify systemoperated files					
Password Management							
3.1.6.27	Quality standards for password shall be enforced. The quality level shall be increased progressively. The following control features shall be implemented for passwords: • Minimum of eight characters without leading or trailing blanks; • Shall be different from the existing password and the two previous ones;	1. Verify on sample basis the passwords used across CA systems comply with the requirements mentioned in control 2. Verify on sample basis the password retries is limited to maximum three attempts (two for sensitive for sensitive systems) 3. Check the user ID is revoked after the user exhausts permissible limit of password retries on sample basis	Mandatory	IT CA Rules SCHEDULE II 6.3			

	• Shall be changed at least once every ninety days; for sensitive system password shall be changed at least once every thirty days; and • Shall not be shared, displayed or printed. Shall have at least on number, one capital letter, one small case letter and one special character	4. Validate the initial or reset passwords is changed by the user upon first use 5. Check the passwords are encrypted instorage to prevent unauthorized disclosure and protect them from dictionary attacks and all known password cracking algorithms.						
3.1.6.28	Password retries shall be limited to a maximum of three attempted logons after which the user ID shall then be revoked; for sensitive systems, the number of password retries should be limited to a maximum of two.	1. Verify on sample basis the passwords used across CA systems comply with the requirements mentioned in control 2. Verify on sample basis the password retries is limited to maximum three attempts (two for sensitive for sensitive systems) 3. Check the user ID is revoked after the user exhausts permissible limit of password retries on sample basis 4. Validate the initial or reset passwords is changed by the user upon first use 5. Check the passwords are encrypted instorage to prevent unauthorized disclosure and protect them from dictionary attacks and all known password cracking algorithms.	Mandatory	IT CA Rules SCHEDULE II 6.3				
3.1.6.29	Initial or reset passwords must be changed by the user upon first use. Users shall follow defined policies and procedures in the selection and use of passwords.	1. Verify on sample basis the passwords used across CA systems comply with the requirements mentioned in control 2. Verify on sample basis the password retries is limited to maximum three attempts (two for sensitive for sensitive systems) 3. Check the user ID is revoked after the user exhausts permissible limit of password retries on sample basis 4. Validate the initial or reset passwords is changed by the user upon first use 5. Check the passwords	Mandatory	WebTrust 3.6.6, IT CA Rules SCHEDULE II 6.3				

		are encrypted instorage to prevent unauthorized disclosure and protect them from dictionary attacks and all known password cracking algorithms.						
3.1.6.30	Passwords shall always be encrypted in storage to prevent unauthorized disclosure and protect them from dictionary attacks and all known password cracking algorithms.	1. Verify on sample basis the passwords used across CA systems comply with the requirements mentioned in control 2. Verify on sample basis the password retries is limited to maximum three attempts (two for sensitive for sensitive systems) 3. Check the user ID is revoked after the user exhausts permissible limit of password retries on sample basis 4. Validate the initial or reset passwords is changed by the user upon first use 5. Check the passwords are encrypted instorage to prevent unauthorized disclosure and protect them from dictionary attacks and all known password cracking algorithms.	Mandatory	IT CA Rules SCHEDULE II 6.3				
Privileged User`s Management								
3.1.6.31	System privileges shall be granted to users only on a need-to- use basis. Login privileges for highly privileged accounts should be available only from Console and terminals situated within Console room.	1. For the privileged user management process, check the following on sample basis: a. System privileges are granted to users only on a need-to-use basis b. Login privileges for highly privileged accounts are available only from Console and terminals situated within Console room c. An audit trail of activities conducted by highly privileged users is maintained for two years and reviewed periodically at least every week by operator who is independent of System Administrator d. Privileged user is be allowed to log in to the computer system from	Mandatory	IT CA Rules SCHEDULE II 6.4				

		remote terminal e. Separate user IDs are allowed to the user for performing privileged and normal (non-privileged) activities f. user IDs for emergency use are recorded and approved. g. Passwords are reset after emergency use						
3.1.6.32	An audit trail of activities conducted by highly privileged users shall be maintained for two years and reviewed periodically at least every week by operator who is independent of System Administrator	1. For the privileged user management process, check the following on sample basis: a. System privileges are granted to users only on a need-to-use basis b. Login privileges for highly privileged accounts are available only from Console and terminals situated within Console room c. An audit trail of activities conducted by highly privileged users is maintained for two years and reviewed periodically at least every week by operator who is independent of System Administrator d. Privileged user is be allowed to log in to the computer system from remote terminal e. Separate user IDs are allowed to the user for performing privileged and normal (non-privileged) activities f. user IDs for emergency use are recorded and approved. g. Passwords are reset after emergency use	Mandatory	IT CA Rules SCHEDULE II 6.4				
3.1.6.33	Privileged user shall not be allowed to log in to the computer system from remote terminal. The usage of the computer system by the privilege user shall be allowed during a certain time period.	1. For the privileged user management process, check the following on sample basis: a. System privileges are granted to users only on a need-to-use basis b. Login privileges for highly privileged accounts are available only from Console and terminals situated within Console room c. An audit trail of activities conducted by	Mandatory	IT CA Rules SCHEDULE II 6.4				

		highly privileged users is maintained for two years and reviewed periodically at least every week by operator who is independent of System Administrator d. Privileged user is be allowed to log in to the computer system from remote terminal e. Separate user IDs are allowed to the user for performing privileged and normal (non-privileged) activities f. user IDs for emergency use are recorded and approved. g. Passwords are reset after emergency use						
3.1.6.34	Separate user IDs shall be allowed to the user for performing privileged and normal (non-privileged) activities.	1. For the privileged user management process, check the following on sample basis: a. System privileges are granted to users only on a need-to-use basis b. Login privileges for highly privileged accounts are available only from Console and terminals situated within Console room c. An audit trail of activities conducted by highly privileged users is maintained for two years and reviewed periodically at least every week by operator who is independent of System Administrator d. Privileged user is be allowed to log in to the computer system from remote terminal e. Separate user IDs are allowed to the user for performing privileged and normal (non-privileged) activities f. user IDs for emergency use are recorded and approved. g. Passwords are reset after emergency use	Mandatory	IT CA Rules SCHEDULE II 6.4				
3.1.6.35	The use of user IDs for emergency use shall be recorded and approved. The passwords shall be reset after use.	1. For the privileged user management process, check the following on sample basis: a. System privileges are	Mandatory	T CA Rules SCHEDULE II 6.4				

		granted to users only on a need-to-use basis b. Login privileges for highly privileged accounts are available only from Console and terminals situated within Console room c. An audit trail of activities conducted by highly privileged users is maintained for two years and reviewed periodically at least every week by operator who is independent of System Administrator d. Privileged user is be allowed to log in to the computer system from remote terminal e. Separate user IDs are allowed to the user for performing privileged and normal (non-privileged) activities f. user IDs for emergency use are recorded and approved. g. Passwords are reset after emergency use						
Data and Resource Protection								
3.1.6.36	All information assets shall be assigned an "owner" responsible for the integrity of that data/resource. Custodians shall be assigned and shall be jointly responsible for information assets by providing computer controls to assist owners.	1. Check the list of data assets and the corresponding owner/custodian for the same.	Mandatory	IT CA Rules SCHEDULE II 6.6				
3.1.6.37	The operating system or security system of the computer system shall: • Define user authority and enforce access control to data within the computer system; • Be capable of specifying, for each named individual, a list of named data objects (e.g. file, Programme) or groups of named objects, and the type of access allowed.	1. Verify the operating system or security system of the computer system define user authority and enforce access control to data within the computer system Check these are capable of specifying for each named individual, a list of named data objects (e.g. file, Programme) or groups of named objects, and the type of access allowed.	Mandatory	IT CA Rules SCHEDULE II 6.6				

3.1.6.38	For networked or shared computer systems, system users shall be limited to a profile of data objects required to perform their needed tasks	1. Verify for networked or shared computer systems, system users shall be limited to a profile of data objects required to perform their needed tasks	Mandatory	IT CA Rules SCHEDULE II 6.6				
3.1.6.39	Application Programmer shall not be allowed to access the production system	2. Validate that application programmer does not have access to the production system	Mandatory	IT CA Rules SCHEDULE II 6.6				
3.1.6.40	Procedures shall be implemented to ensure the secure storage or distribution of all outputs/reports, in accordance with procedures defined by the owners for each system	1. For the jobs performed in the Data Centre check the following: a. procedures have been established to ensure that all changes to the job schedules are appropriately approved b. list of people who have been assigned authority to approve changes to job schedule c. automated job scheduling is used d. Procedures are established to ensure that only authorized and correct job stream and parameter changes are made. e. Procedures are established to maintain logs of system activities f. Procedures are established to ensure that people other than well-trained computer operators are prohibited from operating the computer equipment g. Procedures are implemented to ensure the secure storage or distribution of all outputs/reports, in accordance with procedures defined by the owners for each system	Mandatory	IT CA Rules SCHEDULE II 8.2				
Sensitive Systems Protection								
3.1.6.41	Security tokens/smart cards/bio-metric technologies such as Iris recognition, finger print verification technologies etc. shall be used to complement the usage of passwords	1. For sensitive system protection, check the following have been implemented: a. Security tokens/smart cards/bio-metric technologies such as Iris recognition etc. are used	Mandatory	IT CA Rules SCHEDULE II 7				

	to access the computer system	to complement the usage of passwords to access the computer system b. For computer system processing sensitive data, identify the list of organizations which have access to sensitive data and check the access control measures implemented to establish its effectiveness c. Data in storage is encrypted						
3.1.6.42	For computer system processing sensitive data, access by other Organizations shall be prohibited or strictly controlled.	1. For sensitive system protection, check the following have been implemented: a. Security tokens/smart cards/bio-metric technologies such as Iris recognition etc. are used to complement the usage of passwords to access the computer system b. For computer system processing sensitive data, identify the list of organizations which have access to sensitive data and check the access control measures implemented to establish its effectiveness c. Data in storage is encrypted	Mandatory	IT CA Rules SCHEDULE II 7				
3.1.6.43	For sensitive data, encryption of data in storage shall be considered to protect its confidentiality and integrity.	1. For sensitive system protection, check the following have been implemented: a. Security tokens/smart cards/bio-metric technologies such as Iris recognition etc. are used to complement the usage of passwords to access the computer system b. For computer system processing sensitive data, identify the list of organizations which have access to sensitive data and check the access control measures implemented to establish its effectiveness c. Data in storage is encrypted	Mandatory	IT CA Rules SCHEDULE II 7				
Data Centre Operations Security								

3.1.6.44	Procedures shall be established to ensure that all changes to the job schedules are appropriately approved. The authority to approve changes to job schedules shall be clearly assigned.	1. For the jobs performed in the Data Centre check the following: a. procedures have been established to ensure that all changes to the job schedules are appropriately approved b. list of people who have been assigned authority to approve changes to job schedule c. automated job scheduling is used d. Procedures are established to ensure that only authorized and correct job stream and parameter changes are made. e. Procedures are established to maintain logs of system activities f. Procedures are established to ensure that people other than well-trained computer operators are prohibited from operating the computer equipment g. Procedures are implemented to ensure the secure storage or distribution of all outputs/reports, in accordance with procedures defined by the owners for each system	Mandatory	IT CA Rules SCHEDULE II 8.1				
3.1.6.45	As far as possible, automated job scheduling should be used. Manual job scheduling should require prior approval from the competent authority	1. For the jobs performed in the Data Centre check the following: a. procedures have been established to ensure that all changes to the job schedules are appropriately approved b. list of people who have been assigned authority to approve changes to job schedule c. automated job scheduling is used d. Procedures are established to ensure that only authorized and correct job stream and parameter changes are made. e. Procedures are established to maintain logs of system activities f. Procedures are established to ensure	Mandatory	IT CA Rules SCHEDULE II 8.1				

		that people other than well-trained computer operators are prohibited from operating the computer equipment g. Procedures are implemented to ensure the secure storage or distribution of all outputs/reports, in accordance with procedures defined by the owners for each system						
3.1.6.46	Procedures shall be established to ensure that only authorized and correct job stream and parameter changes are made.	1. For the jobs performed in the Data Centre check the following: a. procedures have been established to ensure that all changes to the job schedules are appropriately approved b. list of people who have been assigned authority to approve changes to job schedule c. automated job scheduling is used d. Procedures are established to ensure that only authorized and correct job stream and parameter changes are made. e. Procedures are established to maintain logs of system activities f. Procedures are established to ensure that people other than well-trained computer operators are prohibited from operating the computer equipment g. Procedures are implemented to ensure the secure storage or distribution of all outputs/reports, in accordance with procedures defined by the owners for each system	Mandatory	IT CA Rules SCHEDULE II 8.2				
3.1.6.47	Procedures shall be established to maintain logs of system activities. Such logs shall be reviewed by a competent independent party for indications of dubious activities. Appropriate retention periods shall be set for	1. For the jobs performed in the Data Centre check the following: a. procedures have been established to ensure that all changes to the job schedules are appropriately approved b. list of people who	Mandatory	IT CA Rules SCHEDULE II 8.2				

	such logs.	have been assigned authority to approve changes to job schedule c. automated job scheduling is used d. Procedures are established to ensure that only authorized and correct job stream and parameter changes are made. e. Procedures are established to maintain logs of system activities f. Procedures are established to ensure that people other than well-trained computer operators are prohibited from operating the computer equipment g. Procedures are implemented to ensure the secure storage or distribution of all outputs/reports, in accordance with procedures defined by the owners for each system						
3.1.6.48	Procedures shall be established to ensure that people other than well-trained computer operators are prohibited from operating the computer equipment	1. For the jobs performed in the Data Centre check the following: a. procedures have been established to ensure that all changes to the job schedules are appropriately approved b. list of people who have been assigned authority to approve changes to job schedule c. automated job scheduling is used d. Procedures are established to ensure that only authorized and correct job stream and parameter changes are made. e. Procedures are established to maintain logs of system activities f. Procedures are established to ensure that people other than well-trained computer operators are prohibited from operating the computer equipment g. Procedures are implemented to ensure the secure storage or distribution of all outputs/reports, in	Mandatory	IT CA Rules SCHEDULE II 8.2				

		accordance with procedures defined by the owners for each system						
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3.1.7. Disaster Recovery

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Incident and Compromise Handling Procedures							
3.1.7.1	An incident management plan shall be developed and approved by the management. The plan shall include the following areas: • Certifying Authority's certification key compromise; • Hacking of systems and network; • Breach of physical security; • Infrastructure availability; • Fraudulent registration and generation of Digital Signature Certificates; and • Digital Signature Certificate suspension and revocation information.	1. Obtain copy of the incident management plan 2. Verify the plan has been approved by the management 3. Validate the plan includes the areas mentioned in the control description	Mandatory	IT CA Rules SCHEDULE III 11.3			
3.1.7.2	An incident response action plan shall be established to ensure the readiness of the Certifying Authority to respond to incidents. The plan should include the following areas: • Compromise control; • Notification to user community; (if applicable) • Revocation of affected Digital Signature Certificates; (if applicable) • Responsibilities of personnel handling incidents; • Investigation of service disruption; • Service restoration procedure; • Monitoring and audit trail analysis; and • Media and public relation	1. Verify an incident response action plan has been established by the CA 2. Validate the plan includes the areas mentioned in the control description	Mandatory	IT CA Rules SCHEDULE III 11.3			
3.1.7.3	A formal security incident reporting procedure exists setting out the actions to be taken on receipt of an incident report. This	1. Verify the security incident reporting procedure and check the following: a. Procedure details	Mandatory	WebTrust 3.5.9			

	includes a definition and documentation of assigned responsibilities and escalation procedures. Any incidents are reported to PA as a matter of urgency.	actions to be taken on receipt of an incident report b. definition and documentation of assigned responsibilities and escalation procedures exist c. procedures have been established for reporting observed or suspected security weaknesses in, or threats to, systems or services as they are detected. d. Procedure exist and are implemented for reporting hardware and software malfunctions e. Procedures exist and are followed to assess that corrective action is taken for reported incidents f. formal problem management process and procedure exist g. help desk is set up to assist users in the resolution of problems. Check tickets raised on helpdesk and their resolution on sample basis						
3.1.7.4	Users of CA systems are required to note and report observed or suspected security weaknesses in, or threats to, systems or services as they are detected.	1. Verify the security incident reporting procedure and check the following: a. Procedure details actions to be taken on receipt of an incident report b. definition and documentation of assigned responsibilities and escalation procedures exist c. procedures have been established for reporting observed or suspected security weaknesses in, or threats to, systems or services as they are detected. d. Procedure exist and are implemented for reporting hardware and software malfunctions e. Procedures exist	Mandatory	WebTrust 3.5.10				

		and are followed to assess that corrective action is taken for reported incidents f. formal problem management process and procedure exist g. help desk is set up to assist users in the resolution of problems. Check tickets raised on helpdesk and their resolution on sample basis						
3.1.7.5	Procedures exist and are followed for reporting hardware and software malfunctions.	1. Verify the security incident reporting procedure and check the following: a. Procedure details actions to be taken on receipt of an incident report b. definition and documentation of assigned responsibilities and escalation procedures exist c. procedures have been established for reporting observed or suspected security weaknesses in, or threats to, systems or services as they are detected. d. Procedure exist and are implemented for reporting hardware and software malfunctions e. Procedures exist and are followed to assess that corrective action is taken for reported incidents f. formal problem management process and procedure exist g. help desk is set up to assist users in the resolution of problems. Check tickets raised on helpdesk and their resolution on sample basis	Mandatory	WebTrust 3.5.11				
3.1.7.6	Procedures exist and are followed to assess that corrective action is taken for reported incidents	1. Verify the security incident reporting procedure and check the following: a. Procedure details actions to be taken on	Mandatory	WebTrust 3.5.12				

		receipt of an incident report b. definition and documentation of assigned responsibilities and escalation procedures exist c. procedures have been established for reporting observed or suspected security weaknesses in, or threats to, systems or services as they are detected. d. Procedure exist and are implemented for reporting hardware and software malfunctions e. Procedures exist and are followed to assess that corrective action is taken for reported incidents f. formal problem management process and procedure exist g. help desk is set up to assist users in the resolution of problems. Check tickets raised on helpdesk and their resolution on sample basis						
3.1.7.7	A formal problem management process exists that allows the types, volumes and impacts of incidents and malfunctions to be documented, quantified and monitored	1. Verify the security incident reporting procedure and check the following: a. Procedure details actions to be taken on receipt of an incident report b. definition and documentation of assigned responsibilities and escalation procedures exist c. procedures have been established for reporting observed or suspected security weaknesses in, or threats to, systems or services as they are detected. d. Procedure exist and are implemented for reporting hardware and software malfunctions e. Procedures exist and are followed to	Mandatory	WebTrust 3.5.13				

		assess that corrective action is taken for reported incidents f. formal problem management process and procedure exist g. help desk is set up to assist users in the resolution of problems. Check tickets raised on helpdesk and their resolution on sample basis						
3.1.7.8	Procedures for identifying, reporting and resolving problems, such as nonfunctioning of Certifying Authority's system; breaches in Information Technology security; and hacking, shall be established and communicated to all personnel concerned. It shall include emergency procedures. Periodic reports shall be submitted for management review	1. Verify the security incident reporting procedure and check the following: a. Procedure details actions to be taken on receipt of an incident report b. definition and documentation of assigned responsibilities and escalation procedures exist c. procedures have been established for reporting observed or suspected security weaknesses in, or threats to, systems or services as they are detected. d. Procedure exist and are implemented for reporting hardware and software malfunctions e. Procedures exist and are followed to assess that corrective action is taken for reported incidents f. formal problem management process and procedure exist g. help desk is set up to assist users in the resolution of problems. Check tickets raised on helpdesk and their resolution on sample basis	Mandatory	IT CA Rules Section II 22				
3.1.7.9	A help desk shall be set up to assist users in the resolution of problems. A system for recording, tracking and reporting the status of reported problems shall be established to	1. Verify the security incident reporting procedure and check the following: a. Procedure details actions to be taken on receipt of an incident	Mandatory	IT CA Rules Section II 22				

	ensure that they are promptly managed and resolved with minimal impact on the user of the computing resources	report b. definition and documentation of assigned responsibilities and escalation procedures exist c. procedures have been established for reporting observed or suspected security weaknesses in, or threats to, systems or services as they are detected. d. Procedure exist and are implemented for reporting hardware and software malfunctions e. Procedures exist and are followed to assess that corrective action is taken for reported incidents f. formal problem management process and procedure exist g. help desk is set up to assist users in the resolution of problems. Check tickets raised on helpdesk and their resolution on sample basis						
Recovery Procedure								
3.1.7.10	Commitment shall be obtained in writing from computer equipment and supplies vendors to replace critical equipment and supplies within a specified period of time following a destruction of the computing facility.	1. Validate the recover procedure is documented and implemented 2. Verify written commitment in form of a contract has been taken from computer equipment and supplies vendors to replace critical equipment and supplies within a specified period of time following a destruction of the computing facility. 3. Check the business continuity plan includes procedures for emergency ordering of equipment and availability of services 4. Verify need for backup hardware is	Mandatory	IT CA Rules SCHEDULE II 24				

		evaluated in accordance to business needs						
3.1.7.11	The business continuity plan shall be developed which inter alia include the procedures for emergency ordering of the equipment and availability of the services.	1. Validate the recover procedure is documented and implemented 2. Verify written commitment in form of a contract has been taken from computer equipment and supplies vendors to replace critical equipment and supplies within a specified period of time following a destruction of the computing facility. 3. Check the business continuity plan includes procedures for emergency ordering of equipment and availability of services 4. Verify need for backup hardware is evaluated in accordance to business needs	Mandatory	IT CA Rules SCHEDULE II 24				
3.1.7.12	The need for backup hardware and other peripherals should be evaluated in accordance to business needs	1. Validate the recover procedure is documented and implemented 2. Verify written commitment in form of a contract has been taken from computer equipment and supplies vendors to replace critical equipment and supplies within a specified period of time following a destruction of the computing facility. 3. Check the business continuity plan includes procedures for emergency ordering of equipment and availability of services 4. Verify need for backup hardware is evaluated in accordance to	Mandatory	IT CA Rules SCHEDULE II 24				

		business needs						
Emergency Preparedness								
3.1.7.13	Emergency response procedures for all activities connected with computer operation shall be developed and documented. These procedures should be reviewed periodically	1. Verify emergency response procedure has been developed and documented 2. Validate emergency drills are held periodically	Mandatory	IT CA Rules SCHEDULE II 23				
3.1.7.14	Emergency drills should be held periodically to ensure that the documented emergency procedures are effective.	1. Verify emergency response procedure has been developed and documented 2. Validate emergency drills are held periodically	Mandatory	IT CA Rules SCHEDULE II 23				
Disaster Recovery/Management								
3.1.7.15	Disaster recovery plan shall be developed, properly documented, tested and maintained to ensure that in the event of a failure of the information system or destruction of the facility, essential level of service will be provided. The disaster recovery framework should include: • emergency procedures, describing the immediate action to be taken in case of a major incident • fall back procedure, describing the actions to be taken to relocate essential activities or support services to a backup site • restoration procedures, describing the action to be taken to return to normal operation at the original site	1. Obtain copy of the documented disaster recovery plan and check the following: a. Plan has been tested and maintained to ensure service will be provided in case of an emergency b. The DR framework includes requirements covered in control 3.1.7.15 c. The documentation covers definition of disaster, condition for activating plan, stage of crisis, decision making process and requirements covered in control 3.1.7.16 d. Specific disaster management plan is present for critical applications e. Responsibilities and reporting structure is clearly defined and will take effect immediately on the declaration of a disaster f. Each component/aspect of the plan has a person and a backup assigned to its execution g. Periodic training of personnel and users associated with	Mandatory	IT CA Rules SCHEDULE II 26				

		computer system and network is conducted h. Test plan has been developed and documented i. Results of tests are documented for management review j. DR plan is updated regularly						
3.1.7.16	The documentation should include: • definition of a disaster; • condition for activating the plan; • stages of a crisis; • who will make decisions in the crisis • role of individuals for each component of the plan; • composition of the recovery team; and • decision making process for return to normal operation	1. Obtain copy of the documented disaster recovery plan and check the following: a. Plan has been tested and maintained to ensure service will be provided in case of an emergency b. The DR framework includes requirements covered in control 3.1.7.15 c. The documentation covers definition of disaster, condition for activating plan, stage of crisis, decision making process and requirements covered in control 3.1.7.16 d. Specific disaster management plan is present for critical applications e. Responsibilities and reporting structure is clearly defined and will take effect immediately on the declaration of a disaster f. Each component/aspect of the plan has a person and a backup assigned to its execution g. Periodic training of personnel and users associated with computer system and network is conducted h. Test plan has been developed and documented i. Results of tests are documented for management review j. DR plan is updated regularly	Mandatory	IT CA Rules SCHEDULE II 26				
3.1.7.17	Specific disaster	1. Obtain copy of the	Mandatory	IT CA Rules SCHEDULE II 26				

	management plan for critical applications shall be developed, documented, tested and maintained on a regular basis.	documented disaster recovery plan and check the following: a. Plan has been tested and maintained to ensure service will be provided in case of an emergency b. The DR framework includes requirements covered in control 3.1.7.15 c. The documentation covers definition of disaster, condition for activating plan, stage of crisis, decision making process and requirements covered in control 3.1.7.16 d. Specific disaster management plan is present for critical applications e. Responsibilities and reporting structure is clearly defined and will take effect immediately on the declaration of a disaster f. Each component/aspect of the plan has a person and a backup assigned to its execution g. Periodic training of personnel and users associated with computer system and network is conducted h. Test plan has been developed and documented i. Results of tests are documented for management review j. DR plan is updated regularly						
3.1.7.18	Responsibilities and reporting structure shall be clearly defined which will take effect immediately on the declaration of a disaster	1. Obtain copy of the documented disaster recovery plan and check the following: a. Plan has been tested and maintained to ensure service will be provided in case of an emergency b. The DR framework includes requirements covered in control 3.1.7.15 c. The documentation covers definition of	Mandatory	IT CA Rules SCHEDULE II 26				

		disaster, condition for activating plan, stage of crisis, decision making process and requirements covered in control 3.1.7.16 d. Specific disaster management plan is present for critical applications e. Responsibilities and reporting structure is clearly defined and will take effect immediately on the declaration of a disaster f. Each component/aspect of the plan has a person and a backup assigned to its execution g. Periodic training of personnel and users associated with computer system and network is conducted h. Test plan has been developed and documented i. Results of tests are documented for management review j. DR plan is updated regularly					
3.1.7.19	Each component/aspect of the plan should have a person and a backup assigned to its execution	1. Obtain copy of the documented disaster recovery plan and check the following: a. Plan has been tested and maintained to ensure service will be provided in case of an emergency b. The DR framework includes requirements covered in control 3.1.7.15 c. The documentation covers definition of disaster, condition for activating plan, stage of crisis, decision making process and requirements covered in control 3.1.7.16 d. Specific disaster management plan is present for critical applications e. Responsibilities and reporting structure is clearly defined and will take	Mandatory	T CA Rules SCHEDULE II 26			

		effect immediately on the declaration of a disaster f. Each component/aspect of the plan has a person and a backup assigned to its execution g. Periodic training of personnel and users associated with computer system and network is conducted h. Test plan has been developed and documented i. Results of tests are documented for management review j. DR plan is updated regularly						
3.1.7.20	Periodic training of personnel and users associated with computer system and network should be conducted defining their roles and responsibilities in the event of a disaster	1. Obtain copy of the documented disaster recovery plan and check the following: a. Plan has been tested and maintained to ensure service will be provided in case of an emergency b. The DR framework includes requirements covered in control 3.1.7.15 c. The documentation covers definition of disaster, condition for activating plan, stage of crisis, decision making process and requirements covered in control 3.1.7.16 d. Specific disaster management plan is present for critical applications e. Responsibilities and reporting structure is clearly defined and will take effect immediately on the declaration of a disaster f. Each component/aspect of the plan has a person and a backup assigned to its execution g. Periodic training of personnel and users associated with computer system and network is conducted	Mandatory	IT CA Rules SCHEDULE II 26				

		h. Test plan has been developed and documented i. Results of tests are documented for management review j. DR plan is updated regularly						
3.1.7.21	Test plan shall be developed, documented and maintained. Periodic tests shall be carried out to test the effectiveness of the procedures in the plan. The results of the tests shall be documented for management review	1. Obtain copy of the documented disaster recovery plan and check the following: a. Plan has been tested and maintained to ensure service will be provided in case of an emergency b. The DR framework includes requirements covered in control 3.1.7.15 c. The documentation covers definition of disaster, condition for activating plan, stage of crisis, decision making process and requirements covered in control 3.1.7.16 d. Specific disaster management plan is present for critical applications e. Responsibilities and reporting structure is clearly defined and will take effect immediately on the declaration of a disaster f. Each component/aspect of the plan has a person and a backup assigned to its execution g. Periodic training of personnel and users associated with computer system and network is conducted h. Test plan has been developed and documented i. Results of tests are documented for management review j. DR plan is updated regularly	Mandatory	IT CA Rules SCHEDULE II 26				
3.1.7.22	Disaster recovery plan should be updated regularly to ensure its	1. Obtain copy of the documented disaster recovery plan and	Mandatory	IT CA Rules SCHEDULE II 26				

	continuing Effectiveness	check the following: a. Plan has been tested and maintained to ensure service will be provided in case of an emergency b. The DR framework includes requirements covered in control 3.1.7.15 c. The documentation covers definition of disaster, condition for activating plan, stage of crisis, decision making process and requirements covered in control 3.1.7.16 d. Specific disaster management plan is present for critical applications e. Responsibilities and reporting structure is clearly defined and will take effect immediately on the declaration of a disaster f. Each component/aspect of the plan has a person and a backup assigned to its execution g. Periodic training of personnel and users associated with computer system and network is conducted h. Test plan has been developed and documented i. Results of tests are documented for management review j. DR plan is updated regularly						
Compromise and Disaster Recovery								
3.1.7.23	The Certifying Authority must establish business continuity procedures that outline the steps to be taken in the event of the corruption or loss of computing and networking resources, nominated website, repository, software and/or data. Where a repository is not under the control of the Certifying Authority, the Certifying Authority must ensure that	1. Verify the CA has established business continuity procedures 2. Validate the procedures cover the requirements mentioned in control description	Mandatory	IT CA Rules SCHEDULE III 11.1				

	any agreement with the repository provides for business continuity procedures.							
Secure facility after a natural or other type of disaster								
3.1.7.24	The Certifying Authority must establish a disaster recovery plan outlining the steps to be taken to re-establish a secure facility in the event of a natural or other type of disaster. Where a repository is not under the control of the Certifying Authority the Certifying Authority must ensure that any agreement with the repository provides that a disaster recovery plan be established and documented by the repository.	1. Validate the CAs DR plan covers steps to be taken to re-establish a secure facility in the event of a natural or other type of disaster 2. Verify in case the repository is not under CAs control, CA has signed an agreement with the repository provider that a disaster recovery plan be established and documented by the repository	Mandatory	IT CA Rules SCHEDULE III 11.2				

3.1.8. Audit Logging

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Retention Period for Audit Logs and Archive							
3.1.8.1	The Certifying Authority must retain its audit logs onsite for at least twelve months	1. Verify on sample basis CA retains audit logs onsite for at least twelve months 2. Validate process has been established original media cannot retain the data for the required period, a mechanism to periodically transfer the archived data to new media is defined by the archive site	Mandatory	IT CA Rules SCHEDULE III 9.3			
3.1.8.2	If the original media cannot retain the data for the required period, a mechanism to periodically transfer the archived data to new media shall be defined by the archive site.	1. Verify on sample basis CA retains audit logs onsite for at least twelve months 2. Validate process has been established original media cannot retain the data for the required period, a mechanism to periodically transfer	Mandatory	X.509 Policy 5.5.2			

		the archived data to new media is defined by the archive site						
Audit Logging								
3.1.8.3	The CA and each Delegated Third Party shall record details of the actions taken to process a certificate request and to issue a Certificate, including all information generated and documentation received in connection with the certificate request; the time and date; and the personnel involved. The CA shall make these records available to its Qualified Auditor as proof of the CA's compliance with these Requirements.	1. Verify on sample basis the CA records details of the actions taken to process a certificate request and to issue a Certificate 2. Validate security audit logs for activities that are core to CAs operations are automatically collected 3. Check all security audit logs, both electronic and non-electronic, are retained and made available during compliance audits	Mandatory	CA Browser Forum 5.4.1				
3.1.8.4	Where possible, the security audit logs shall be automatically collected. Where this is not possible, a logbook, paper form, or other physical mechanism shall be used	1. Verify on sample basis the CA records details of the actions taken to process a certificate request and to issue a Certificate 2. Validate security audit logs for activities that are core to CAs operations are automatically collected 3. Check all security audit logs, both electronic and non-electronic, are retained and made available during compliance audits	Mandatory	X.509 Policy 5.4				
3.1.8.5	The Certifying Authority must maintain secure and reliable records and logs for activities that are core to its operations and review them regularly to detect any anomaly in the system..All security audit logs, both electronic and non-electronic, shall be retained and made available during compliance audits	1. Verify on sample basis the CA records details of the actions taken to process a certificate request and to issue a Certificate 2. Validate security audit logs for activities that are core to CAs operations are automatically	Mandatory	X.509 Policy 5.4, IT Regulations 3				

		collected 3. Check all security audit logs, both electronic and non-electronic, are retained and made available during compliance audits						
Types of Events Recorded								
3.1.8.6	To facilitate decision-making, all agreements and correspondence relating to services provided by Certifying Authority should be collected and consolidated, either electronically or manually, at a single location.	1. Verify the CA collects and consolidates electronically or manually, security information which may not be generated by his system such as covered in control 3.1.8.10 2. Verify all agreements and correspondence relating to services provided by CA are collected and consolidated, either electronically or manually, at a single location.	Mandatory	IT CA Rules SCHEDULE III 9.1				
3.1.8.7	The Certifying Authority shall maintain record of all events relating to the security of his system. The records should be maintained in audit log file and shall include such events as: • System start-up and shutdown; • Certifying Authority's application start-up and shutdown; • Attempts to create, remove, set passwords or change the system • privileges of the PKI Master Officer, PKI Officer, or PKI Administrator; • Changes to keys of the Certifying Authority or any of his other details; • Changes to Digital Signature Certificate creation policies, e.g. validity period; • Login and logoff attempts; • Unauthorised attempts at network access to the Certifying Authority's system; • Unauthorised attempts to access system files; • Generation of own keys; • Creation and revocation of	1. For sample audit log files, check the following details are recorded: a. System start-up and shutdown; b. CA's application start-up and shutdown; c. Attempts to create, remove, set passwords or change the system d. privileges of the PKI Master Officer, PKI Officer, or PKI Administrator; e. Changes to keys of the CA or any of his other details; f. Changes to DSC creation policies g. Login and logoff attempts; h. Unauthorised attempts at network access i. Unauthorised attempts to access system files; j. Generation of own keys;	Mandatory	IT CA Rules SCHEDULE III 9.1				

	Digital Signature Certificates; • Attempts to initialize remove, enable, and disable subscribers, and update and recover their keys; • Failed read-and-write operations on the Digital Signature Certificate and Certificate Revocation List (CRL) directory	k. Creation and revocation of DSC; l. Attempts to initialize remove, enable, and disable subscribers, and update and recover their keys; m. Failed read-and-write operations on the Digital Signature Certificate and Certificate Revocation List (CRL) directory						
3.1.8.8	CA and Subscriber Certificate lifecycle management events, including the following shall be recorded: • Certificate requests, renewal, and re-key requests, and revocation; • All verification activities stipulated in these Requirements and the CA's Certification Practice Statement; • Date, time, phone number used, persons spoken to, and end results of verification telephone calls; • Acceptance and rejection of certificate requests; Issuance of Certificates; and • Generation of Certificate Revocation Lists and OCSP entries	1. On a sample basis validate the CA and Subscriber Certificate lifecycle management events, including the following are recorded: a. Certificate requests, renewal, and re-key requests, and revocation; b. All verification activities stipulated in these Requirements and the CA's Certification Practice Statement; c. Date, time, phone number used, persons spoken to, and end results of verification telephone calls; d. Acceptance and rejection of certificate requests; e. Issuance of Certificates; and f. Generation of Certificate Revocation Lists and OCSP entries	Mandatory	CA Browser Forum 5.4.1				
3.1.8.9	All security auditing capabilities of the CA, CSP, and RA operating system and the CA, CSP, and RA applications required by this CP shall be enabled	1. Verify all security auditing capabilities of the CA, CSP, and RA operating system and the CA, CSP, and RA applications required by this CP are enabled	Mandatory	X.509 Policy 5.4.1				

3.1.8.10	A Certifying Authority should consider the use of automated security management and monitoring tools providing an integrated view of the security situation at any point in time. Records of the following application transactions shall be maintained: • Registration; • Certification; • Publication; • Suspension; and • Revocation.	1. Verify CA has implemented automated security management and monitoring tools providing an integrated view of the security situation at any point in time 2. Validate records of application transactions as mentioned in control 3.1.8.7 are maintained 3. Check the records and log files are reviewed regularly for the activities mentioned in control 3.1.8.8 4. Verify all logs contain date and time of the event and the identity of subscriber/ subordinate/ entity which caused the event	Mandatory	IT CA Rules SCHEDULE III 9.1				
3.1.8.11	Records and log files shall be reviewed regularly for the following activities: • Misuse; • Errors; • Security violations; • Execution of privileged functions; • Change in access control lists; • Change in system configuration	1. Verify CA has implemented automated security management and monitoring tools providing an integrated view of the security situation at any point in time 2. Validate records of application transactions as mentioned in control 3.1.8.7 are maintained 3. Check the records and log files are reviewed regularly for the activities mentioned in control 3.1.8.8 4. Verify all logs contain date and time of the event and the identity of subscriber/ subordinate/ entity which caused the event	Mandatory	IT CA Rules SCHEDULE III 9.1				
3.1.8.12	All logs, whether maintained through electronic or manual means, should contain the date and time of the event,	1. Verify CA has implemented automated security management and	Mandatory	IT CA Rules SCHEDULE III 9.1				

	and the identity of the subscriber/subordinate/ entity which caused the event	monitoring tools providing an integrated view of the security situation at any point in time 2. Validate records of application transactions as mentioned in control 3.1.8.7 are maintained 3. Check the records and log files are reviewed regularly for the activities mentioned in control 3.1.8.8 4. Verify all logs contain date and time of the event and the identity of subscriber/ subordinate/ entity which caused the event						
3.1.8.13	A Certifying Authority should also collect and consolidate, either electronically or manually, security information which may not be generated by his system, such as: • Physical access logs; • System configuration changes and maintenance; • Personnel changes; • Discrepancy and compromise reports; • Records of the destruction of media containing key material, activation data, or personal subscriber information	1. Verify the CA collects and consolidates electronically or manually, security information which may not be generated by his system such as covered in control 3.1.8.10 2. Verify all agreements and correspondence relating to services provided by CA are collected and consolidated, either electronically or manually, at a single location.	Mandatory	IT CA Rules SCHEDULE III 9.1				
3.1.8.14	Log entries shall include the following elements: • Date and time of entry; • Identity of the person making the journal entry; and • Description of the entry.	1. For sample log entries, check the following are recorded: a. Date and time of entry; b. Identity of the person making the journal entry; and c. Description of the entry.	Mandatory	CA Browser Forum 5.4.1				
3.1.8.15	The audit record shall include the following (either recorded automatically or	1. For sample audit records, check the following are	Mandatory	X.509 Policy 5.4.1				

	manually for each auditable event): • The type of event, • The date and time the event occurred • Success or failure where appropriate, and • The identity of the entity and/or operator that caused the even	included: a. The type of event, b. The date and time the event occurred, c. Success or failure where appropriate, and d. The identity of the entity and/or operator that caused the even						
Frequency of Processing Audit Logs								
3.1.8.16	The Certifying Authority must ensure that its audit logs are reviewed by its personnel at least once every two weeks and all significant events are detailed in an audit log summary. Such reviews should involve verifying that the log has not been tampered with, and then briefly inspecting all log entries, with a more thorough investigation of any alerts or irregularities in the logs. Action taken following these reviews must be documented	1. Verify the CA audit logs are reviewed by its personnel at least once every two weeks and all significant events are detailed in an audit log summary 2. Verify audit logs are reviewed at least every 30 days to examine significant sample of security audit data generated by the CA, CSP since the last review 3. Verify the Audit Administrator explains all significant events in an audit log summary	Mandatory	IT CA Rules SCHEDULE III 9.2				
3.1.8.17	Audit logs shall be reviewed at least once every 30 days. Statistically Significant sample of security audit data generated by the CA, CSP, or RA since the last review shall be examined (where the confidence intervals for each category of security audit data are determined by the security ramifications of the category and the availability of tools to perform such a review), as well as a reasonable search for any evidence of malicious activity.	1. Verify the CA audit logs are reviewed by its personnel at least once every two weeks and all significant events are detailed in an audit log summary 2. Verify audit logs are reviewed at least every 30 days to examine significant sample of security audit data generated by the CA, CSP since the last review 3. Verify the Audit Administrator explains all significant events in an audit log summary	Mandatory	X.509 Policy 5.4.2				
3.1.8.18	The Audit Administrator shall explain all significant	1. Verify the CA audit logs are	Mandatory	X.509 Policy 5.4.2				

	events in an audit log summary. Actions taken as a result of reviews involving verifying that the log has not been tampered with, there is no discontinuity or other loss of audit data, and then briefly inspecting all log entries shall be documented.	reviewed by its personnel at least once every two weeks and all significant events are detailed in an audit log summary 2. Verify audit logs are reviewed at least every 30 days to examine significant sample of security audit data generated by the CA, CSP since the last review 3. Verify the Audit Administrator explains all significant events in an audit log summary						
Protection of Audit Logs								
3.1.8.19	The electronic audit log system must include mechanisms to protect the log files from unauthorized viewing, modification, and deletion. Manual audit information must be protected from unauthorised viewing, modification and destruction	1. Verify the electronic audit log system include mechanisms to protect the log files from unauthorized viewing, modification, and deletion 2. Check the system has been configured to ensure only authorized people read and archive audit logs 3. Validate if it is acceptable for the system to over-write audit logs after they have been backed up and archived by asking the CA for practical demonstration	Mandatory	IT CA Rules SCHEDULE III 9.4				
3.1.8.20	System configuration and procedures shall be implemented together to ensure that: • Only authorized people have read access to the logs; • Only authorized people may archive audit logs; and, • Audit logs are not modified.	1. Verify the electronic audit log system include mechanisms to protect the log files from unauthorized viewing, modification, and deletion 2. Check the system has been configured to ensure only authorized people read and archive audit logs 3. Validate if it is acceptable for the	Mandatory	X.509 Policy 5.4.4				

		system to over-write audit logs after they have been backed up and archived by asking the CA for practical demonstration						
3.1.8.21	It shall be acceptable for the system to over-write audit logs after they have been backed up and archived	1. Verify the electronic audit log system include mechanisms to protect the log files from unauthorized viewing, modification, and deletion 2. Check the system has been configured to ensure only authorized people read and archive audit logs 3. Validate if it is acceptable for the system to over-write audit logs after they have been backed up and archived by asking the CA for practical demonstration	Mandatory	X.509 Policy 5.4.4				
Records Archival								
3.1.8.22	Digital Signature Certificates stored and generated by the Certifying Authority must be retained for at least seven years after the date of its expiration. This requirement does not include the backup of private signature keys. Public Key Certificates and Certificate Revocation Lists must be archived for a minimum period of seven years to enable verification of past transactions	Verify the following are stored for at least seven years a. DSC stored and generated by CA after the date of expiration b. Public Key certificates c. Certificate Revocation List d. Audit information e. subscriber agreements f. verification, identification and authentication information in respect of subscriber 2. Validate a second copy of all information retained or backed up is stored at three locations within the country including the CA site 3. Check the primary and	Mandatory	IT CA Rules SCHEDULE III 10, IT Regulations 3				

		secondary sites have security controls in place 4. Verify all information pertaining to CA’s operation, Subscriber’s application, verification, identification, authentication and Subscriber agreement is stored within the country and taken out of country only with CCA’s permission 5. Verify information stored off-site is periodically verified for data integrity 6. On a sample basis verify archived records are sufficiently detailed to establish the proper operation of the component or the validity of any certificate (including those revoked or expired) issued by the CA					
3.1.8.23	Audit information, subscriber agreements, verification, identification and authentication information in respect of subscriber shall be retained for at least seven years	Verify the following are stored for at least seven years a. DSC stored and generated by CA after the date of expiration b. Public Key certificates c. Certificate Revocation List d. Audit information e. subscriber agreements f. verification, identification and authentication information in respect of subscriber 2. Validate a second copy of all information retained or backed up is stored at three locations within the country including the CA site 3. Check the primary and	Mandatory	IT CA Rules SCHEDULE III 10			

		secondary sites have security controls in place 4. Verify all information pertaining to CA's operation, Subscriber's application, verification, identification, authentication and Subscriber agreement is stored within the country and taken out of country only with CCA's permission 5. Verify information stored off-site is periodically verified for data integrity 6. On a sample basis verify archived records are sufficiently detailed to establish the proper operation of the component or the validity of any certificate (including those revoked or expired) issued by the CA						
3.1.8.24	A second copy of all information retained or backed up must be stored at three locations within the country including the Certifying Authority site and must be protected either by physical security alone, or a combination of physical and cryptographic protection. These secondary sites must provide adequate protection from environmental threats such as temperature, humidity and magnetism. The secondary site should be reachable in few hours	Verify the following are stored for at least seven years a. DSC stored and generated by CA after the date of expiration b. Public Key certificates c. Certificate Revocation List d. Audit information e. subscriber agreements f. verification, identification and authentication information in respect of subscriber 2. Validate a second copy of all information retained or backed up is stored at three locations within the country including the CA site 3. Check the primary and	Mandatory	IT CA Rules SCHEDULE III 10				

		secondary sites have security controls in place 4. Verify all information pertaining to CA's operation, Subscriber's application, verification, identification, authentication and Subscriber agreement is stored within the country and taken out of country only with CCA's permission 5. Verify information stored off-site is periodically verified for data integrity 6. On a sample basis verify archived records are sufficiently detailed to establish the proper operation of the component or the validity of any certificate (including those revoked or expired) issued by the CA						
3.1.8.25	All information pertaining to Certifying Authority's operation, Subscriber's application, verification, identification, authentication and Subscriber agreement shall be stored within the country. This information shall be taken out of the country only with the permission of Controller and where a properly constitutional warrant or such other legally enforceable document is produced	Verify the following are stored for at least seven years a. DSC stored and generated by CA after the date of expiration b. Public Key certificates c. Certificate Revocation List d. Audit information e. subscriber agreements f. verification, identification and authentication information in respect of subscriber 2. Validate a second copy of all information retained or backed up is stored at three locations within the country including the CA site 3. Check the primary and	Mandatory	IT CA Rules SCHEDULE III 10				

		secondary sites have security controls in place 4. Verify all information pertaining to CA's operation, Subscriber's application, verification, identification, authentication and Subscriber agreement is stored within the country and taken out of country only with CCA's permission 5. Verify information stored off-site is periodically verified for data integrity 6. On a sample basis verify archived records are sufficiently detailed to establish the proper operation of the component or the validity of any certificate (including those revoked or expired) issued by the CA						
3.1.8.26	Information stored off-site must be periodically verified for data integrity	Verify the following are stored for at least seven years a. DSC stored and generated by CA after the date of expiration b. Public Key certificates c. Certificate Revocation List d. Audit information e. subscriber agreements f. verification, identification and authentication information in respect of subscriber 2. Validate a second copy of all information retained or backed up is stored at three locations within the country including the CA site 3. Check the primary and	Mandatory	IT CA Rules SCHEDULE III 10				

		secondary sites have security controls in place 4. Verify all information pertaining to CA's operation, Subscriber's application, verification, identification, authentication and Subscriber agreement is stored within the country and taken out of country only with CCA's permission 5. Verify information stored off-site is periodically verified for data integrity 6. On a sample basis verify archived records are sufficiently detailed to establish the proper operation of the component or the validity of any certificate (including those revoked or expired) issued by the CA						
3.1.8.27	CA, CSP, and RA archive records shall be sufficiently detailed to establish the proper operation of the component or the validity of any certificate (including those revoked or expired) issued by the CA	Verify the following are stored for at least seven years a. DSC stored and generated by CA after the date of expiration b. Public Key certificates c. Certificate Revocation List d. Audit information e. subscriber agreements f. verification, identification and authentication information in respect of subscriber 2. Validate a second copy of all information retained or backed up is stored at three locations within the country including the CA site 3. Check the primary and	Mandatory	X.509 Policy 5.5.1				

		secondary sites have security controls in place 4. Verify all information pertaining to CA's operation, Subscriber's application, verification, identification, authentication and Subscriber agreement is stored within the country and taken out of country only with CCA's permission 5. Verify information stored off-site is periodically verified for data integrity 6. On a sample basis verify archived records are sufficiently detailed to establish the proper operation of the component or the validity of any certificate (including those revoked or expired) issued by the CA						
Protection of Archive								
3.1.8.28	No unauthorized user shall be permitted to write to, modify, or delete the archive. For the CA and CSP, the authorized individuals are Audit Administrators. For the RA, authorized individuals are someone other than the RA (e.g., Information Assurance Officer or IAO).	1. Verify only authorized users are permitted to write to, modify, or delete the archive 2. Validate the contents of archives are not released 3. Check the archive media are stored in a safe, secure storage facility separate from the component (CA, CSP) with physical and procedural security controls equivalent or better than those for component. Verify the number of times such request to share individual record was made and the record of subscriber was released to his/ her satisfaction	Mandatory	X.509 Policy 5.5.3				

3.1.8.29	The contents of the archive shall not be released except as determined by the CCA, the Licensed CA, or as required by law. Records of individual may be released upon request of any subscribers involved in the transaction or their legally recognized agents.	1. Verify only authorized users are permitted to write to, modify, or delete the archive 2. Validate the contents of archives are not released 3. Check the archive media are stored in a safe, secure storage facility separate from the component (CA, CSP) with physical and procedural security controls equivalent or better than those for component. Verify the number of times such request to share individual record was made and the record of subscriber was released to his/ her satisfaction	Mandatory	X.509 Policy 5.5.3				
3.1.8.30	Archive media shall be stored in a safe, secure storage facility separate from the component (CA, CSP, or RA) with physical and procedural security controls equivalent or better than those for component	1. Verify only authorized users are permitted to write to, modify, or delete the archive 2. Validate the contents of archives are not released 3. Check the archive media are stored in a safe, secure storage facility separate from the component (CA, CSP) with physical and procedural security controls equivalent or better than those for component. Verify the number of times such request to share individual record was made and the record of subscriber was released to his/ her satisfaction	Mandatory	X.509 Policy 5.5.3				
Requirements for Time-Stamping of Records								
3.1.8.31	The Certifying Authority shall provide Time Stamping Service for its subscribers. Error of the Time Stamping clock shall not be more than 1 in 10 ⁹ . Archived records	1. Verify the VA provides time stamping service for its subscribers with error of the Time Stamping clock less	Mandatory	X.509 Policy 5.5.5, IT Regulations 3				

	shall be time stamped such that order of events can be determined	than 1 in 10 ⁹ . 2. Validate on a sample basis that logs in the archive are time stamped						
Procedures to Obtain & Verify Archive Information								
3.1.8.32	Procedures detailing how to create, verify, package, and transmit archive information shall be published in the applicable CPS.	1. Verify for archived information, procedures have been established to perform following activities: a. Creation b. Verification c. Packaging d. Transmission	Mandatory	X.509 Policy 5.5.7				
Audit Trails and Verification								
3.1.8.33	Transactions that meet exception criteria shall be completely and accurately highlighted and reviewed by personnel independent of those that initiate the transaction	1. Verify on sample basis the transactions that meet exception criteria are completely and accurately highlighted and reviewed by personnel independent of those that initiate the transaction 2. Check audit trails are captured and sensitive information is analyzed to check for indicate possible fraudulent use of the system 3. Verify automated or manual procedures are used to monitor and promptly report all significant security events 4. Validate real time clock of the computer system is set accurately to ensure the accuracy of audit logs, which may be required for investigations or as evidence in legal or disciplinary cases. 5. Check computer system access records are kept for a minimum of two years 6. Verify computer	Mandatory	IT CA Rules SCHEDULE II 10				

		records of applications transactions and significant events are retained for a minimum period of two years or longer depending on specific record retention requirements.						
3.1.8.34	Adequate audit trails shall be captured and certain information needed to determine sensitive events and pattern analysis that would indicate possible fraudulent use of the system (e.g. repeated unsuccessful logons, access attempts over a series of days) shall be analyzed. This information includes such information as who, what, when, where, and any special information such as: • Success or failure of the event • Use of authentication keys, where applicable	1. Verify on sample basis the transactions that meet exception criteria are completely and accurately highlighted and reviewed by personnel independent of those that initiate the transaction 2. Check audit trails are captured and sensitive information is analyzed to check for indicate possible fraudulent use of the system 3. Verify automated or manual procedures are used to monitor and promptly report all significant security events 4. Validate real time clock of the computer system is set accurately to ensure the accuracy of audit logs, which may be required for investigations or as evidence in legal or disciplinary cases. 5. Check computer system access records are kept for a minimum of two years 6. Verify computer records of applications transactions and significant events are retained for a minimum period of two years or longer depending on specific record retention requirements.	Mandatory	IT CA Rules SCHEDULE II 10				

3.1.8.35	Automated or manual procedures shall be used to monitor and promptly report all significant security events, such as accesses, which are out-of pattern relative to time, volume, frequency, type of information asset, and redundancy. Other areas of analysis include: • Significant computer system events (e.g. configuration updates, system crashes) • Security profile changes • Actions taken by computer operations, system administrators, system programmers, and/or security administrators	1. Verify on sample basis the transactions that meet exception criteria are completely and accurately highlighted and reviewed by personnel independent of those that initiate the transaction 2. Check audit trails are captured and sensitive information is analyzed to check for indicate possible fraudulent use of the system 3. Verify automated or manual procedures are used to monitor and promptly report all significant security events 4. Validate real time clock of the computer system is set accurately to ensure the accuracy of audit logs, which may be required for investigations or as evidence in legal or disciplinary cases. 5. Check computer system access records are kept for a minimum of two years 6. Verify computer records of applications transactions and significant events are retained for a minimum period of two years or longer depending on specific record retention requirements.	Mandatory	IT CA Rules SCHEDULE II 10				
3.1.8.36	The real time clock of the computer system shall be set accurately to ensure the accuracy of audit logs, which may be required for investigations or as evidence in legal or disciplinary cases.	1. Verify on sample basis the transactions that meet exception criteria are completely and accurately highlighted and reviewed by personnel independent of	Mandatory	IT CA Rules SCHEDULE II 10				

		those that initiate the transaction 2. Check audit trails are captured and sensitive information is analyzed to check for indicate possible fraudulent use of the system 3. Verify automated or manual procedures are used to monitor and promptly report all significant security events 4. Validate real time clock of the computer system is set accurately to ensure the accuracy of audit logs, which may be required for investigations or as evidence in legal or disciplinary cases. 5. Check computer system access records are kept for a minimum of two years 6. Verify computer records of applications transactions and significant events are retained for a minimum period of two years or longer depending on specific record retention requirements.					
3.1.8.37	The real time clock of the computer or communications device shall be set to Indian Standard Time (IST). Further there shall be a procedure that checks and corrects drift in the real time clock.	1. Verify on sample basis the transactions that meet exception criteria are completely and accurately highlighted and reviewed by personnel independent of those that initiate the transaction 2. Check audit trails are captured and sensitive information is analyzed to check for indicate possible fraudulent use of the system 3. Verify automated	Mandatory	IT CA Rules SCHEDULE II 10			

		or manual procedures are used to monitor and promptly report all significant security events 4. Validate real time clock of the computer system is set accurately to ensure the accuracy of audit logs, which may be required for investigations or as evidence in legal or disciplinary cases. 5. Check computer system access records are kept for a minimum of two years 6. Verify computer records of applications transactions and significant events are retained for a minimum period of two years or longer depending on specific record retention requirements.						
3.1.8.38	Computer system access records shall be kept for a minimum of two years, in either hard copy or electronic form. Records, which are of legal nature and necessary for any legal or regulation requirement or investigation of criminal behavior, shall be retained as per laws of the land	1. Verify on sample basis the transactions that meet exception criteria are completely and accurately highlighted and reviewed by personnel independent of those that initiate the transaction 2. Check audit trails are captured and sensitive information is analyzed to check for indicate possible fraudulent use of the system 3. Verify automated or manual procedures are used to monitor and promptly report all significant security events 4. Validate real time clock of the computer system is set accurately to ensure the accuracy	Mandatory	IT CA Rules SCHEDULE II 10				

		of audit logs, which may be required for investigations or as evidence in legal or disciplinary cases. 5. Check computer system access records are kept for a minimum of two years 6. Verify computer records of applications transactions and significant events are retained for a minimum period of two years or longer depending on specific record retention requirements.						
3.1.8.39	Computer records of applications transactions and significant events must be retained for a minimum period of two years or longer depending on specific record retention requirements.	1. Verify on sample basis the transactions that meet exception criteria are completely and accurately highlighted and reviewed by personnel independent of those that initiate the transaction 2. Check audit trails are captured and sensitive information is analyzed to check for indicate possible fraudulent use of the system 3. Verify automated or manual procedures are used to monitor and promptly report all significant security events 4. Validate real time clock of the computer system is set accurately to ensure the accuracy of audit logs, which may be required for investigations or as evidence in legal or disciplinary cases. 5. Check computer system access records are kept for a minimum of two years 6. Verify computer	Mandatory	IT CA Rules SCHEDULE II 10				

		records of applications transactions and significant events are retained for a minimum period of two years or longer depending on specific record retention requirements.						
Audit Log Backup Procedures								
3.1.8.40	Audit logs and audit summaries must be backed up or copied if in manual form.	1. Validate audit logs and summaries are backed up and copied if in manual form	Mandatory	IT CA Rules SCHEDULE III 9.5				
Vulnerability Assessments								
3.1.8.41	Events in the audit process are logged, in part, to monitor system vulnerabilities. The Certifying Authority must ensure that a vulnerability assessment is performed, reviewed and revised, if necessary, following an examination of these monitored events.	1. Verify the events in the audit process are logged, in part, to monitor system vulnerabilities	Mandatory	IT CA Rules SCHEDULE III 9.6				

3.1.9. Compliance Audit and Other Assessments

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Frequency or circumstances of assessment							
3.1.9.1	The period during which the CA issues Certificates shall be divided into an unbroken sequence of audit periods. An audit period must NOT exceed one year in duration.	1. Verify the annual audits are conducted by the CA 2. Validate CA conducted a point-in-time readiness assessment before issuing Publicly-Trusted Certificates in case a valid audit report did not exist 3. Check the following: a. Point-in-time readiness assessment was completed no earlier than twelve (12) months prior to issuing Publicly-Trusted Certificates b. It was followed	Mandatory	CA Browser Forum 8.1			

		by a complete audit within ninety (90) days of issuing the first Publicly-Trusted Certificate.						
3.1.9.2	If the CA does not have a currently valid Audit Report indicating compliance with the applicable audit schemes, then, before issuing Publicly-Trusted Certificates, the CA shall successfully complete a point-in-time readiness assessment performed in accordance with applicable standards.	1. Verify the annual audits are conducted by the CA 2. Validate CA conducted a point-in-time readiness assessment before issuing Publicly-Trusted Certificates in case a valid audit report did not exist 3. Check the following: a. Point-in-time readiness assessment was completed no earlier than twelve (12) months prior to issuing Publicly-Trusted Certificates b. It was followed by a complete audit within ninety (90) days of issuing the first Publicly-Trusted Certificate.	Mandatory	CA Browser Forum 8.1				
3.1.9.3	The point-in-time readiness assessment shall be completed no earlier than twelve (12) months prior to issuing Publicly-Trusted Certificates and shall be followed by a complete audit under such scheme within ninety (90) days of issuing the first Publicly-Trusted Certificate.	1. Verify the annual audits are conducted by the CA 2. Validate CA conducted a point-in-time readiness assessment before issuing Publicly-Trusted Certificates in case a valid audit report did not exist 3. Check the following: a. Point-in-time readiness assessment was completed no earlier than twelve (12) months prior to issuing Publicly-Trusted Certificates b. It was followed by a complete audit within ninety (90) days of issuing the first Publicly-Trusted Certificate.	Mandatory	CA Browser Forum 8.1				
Identity/qualifications of assessor								

3.1.9.4	The CA’s audit shall be performed by a Qualified Auditor. A Qualified Auditor means a natural person, Legal Entity, or group of natural persons or Legal Entities that collectively possess the following qualifications and skills: • Independence from the subject of the audit • The ability to conduct an audit that addresses the criteria specified in an Eligible Audit Scheme • Employs individuals who have proficiency in examining Public Key Infrastructure technology, information security tools and techniques, information technology and security auditing, and the third party attestation function; • For audits conducted in accordance with any one of the ETSI standard, accredited in accordance with ISO 17065 applying the requirements specified in ETSI EN 319 403; • For audits conducted in accordance with the WebTrust standard, licensed by WebTrust • Bound by law, government regulation, or professional code of ethics; and • Except in the case of an Internal Government Auditing Agency, maintains Professional Liability/Errors	1. Validate that the audit was performed by only by the CCA empaneled agency and auditors. 2. Check the auditor is be independent of the CA being audited	Mandatory	CA Browser Forum 8.2					
Assessor's relationship to assessed entity									
3.1.9.5	The auditor shall be a firm, which is independent from the entity being audited. The Controller shall determine whether an auditor meets this requirement	1. Verify the auditor is a firm, independent from CA 2. Validate the CCA verifies auditor meets above requirement	Mandatory	CA Browser Forum 8.3, X.509 Policy 8.3					
Topics covered by assessment									
3.1.9.6	CAs shall have a compliance audit mechanism in place to ensure that the requirements of this CP and applicable CPS are being implemented and enforced.	1. Obtain details of the compliance audit mechanism to verify requirements of this CP and applicable CPS are being implemented and enforced	Mandatory	CA Browser Forum 8.4, X.509 Policy 8.4					

Actions taken as a result of deficiency								
3.1.9.7	When the auditor finds a discrepancy between how the CA is designed or is being operated or maintained, and the requirements of this CP, or the applicable CPS, the auditor shall take the following actions: • The auditor shall note the discrepancy; • The auditor shall notify the audited CA; and • The auditor shall notify the office of CCA.	1. For sample discrepancies raised by the auditor, verify the following actions were taken by the auditor: a. auditor noted discrepancy; b. The auditor notified the audited CA; and c. The auditor notified the office of CCA.	Mandatory	CA Browser Forum 8.5, X.509 Policy 8.5				
Communication of results								
3.1.9.8	An Audit Report, including identification of corrective measures taken or being taken by the CA, shall be provided to the Controller (in case of grant of license and its renewal) and the audited CA (in case of annual audit). The report shall identify the versions of the CP and CPS used in the assessment	1. Verify the audit reports are submitted to CCA and the audited CA 2. Validate the report identifies the versions of the CP and CPS used in the assessment	Mandatory	CA Browser Forum 8.6, X.509 Policy 8.6				

3.1.10. Licensing of Certifying Authorities

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Licensing of CA Authorities							
3.1.10.1	To ensure the integrity of its digital certificates, the Certifying Authority shall ensure the use of approved security controls in the certificate management processes, i.e. certificate registration, generation, issuance, publication, renewal, suspension, revocation and archival	1. Verify the CA uses approved security controls in the certificate management processes, i.e. certificate registration, generation, issuance, publication, renewal, suspension, revocation and archival 2. Check the method of verification of the identity of the applicant of a Public Key Certificates commensurate with the level of assurance accorded to the certificate	Mandatory	IT Regulations 3 3			
3.1.10.2	The following persons may apply for grant of a license to issue Digital Signature Certificates, namely:- • an individual, being a	1. Verify the following: a. applicant requirements or grant of license mentioned in description of control 3.1.10.1 are met	Mandatory	IT CA Rules 8, IT Regulations 3			

	citizen of India and having a capital of five crores of rupees or more in his business or profession; • a company having— o paid up capital of not less than five crores of rupees; and o net worth of not less than fifty crores of rupees • a firm having – o capital subscribed by all partners of not less than five crores of rupees; and o net worth of not less than fifty crores of rupees • Central Government or a State Government or any of the Ministries or Departments, Agencies or Authorities of such Governments.	b. For licence renewal audit and annual audit, verify the continued maintenance of net worth by CA in the last 5 years(if applicable). c. applicant submitted a performance bond or furnished a banker's guarantee from a scheduled bank in favour of the Controller in such form and in such manner as approved by the Controller for an amount of 50 lakhs/ 01crore rupees and the performance bond or banker's guarantee remains valid for a period of six years from the date of its submission d. validity of license is defined for five years from date of issue e. license is not transferable or heritable f. Controller has the authority to revoke or suspend the license in accordance with the provisions of the Act g. CA is bound to comply with all the parameters against which it was audited prior to issue of license and consistently and continuously complies with those parameters during the period for which the license remains valid. h. CA subjects itself to periodic audits i. CA ensures confidentiality of subscriber information j. CA complies with every order or direction issued by the Controller within the stipulated period.					
3.1.10.3	The applicant being an individual, or a company, or a firm under sub-rule shall submit a performance bond or furnish a banker's guarantee from a scheduled bank in favour of the Controller in such form and in such manner	1. Verify the following: a. applicant requirements or grant of license mentioned in description of control 3.1.10.1 are met b. For licence renewal audit and annual audit, verify the continued maintenance of net	Mandatory	IT CA Rules 8			

	as may be approved by the Controller for an amount of 50 lakhs/ 01 croreof rupees and the performance bond or banker's guarantee shall remain valid for a period of six years from the date of its submission	worth by CA in the last 5 years(if applicable). c. applicant submitted a performance bond or furnished a banker's guarantee from a scheduled bank in favour of the Controller in such form and in such manner as approved by the Controller for an amount of 50 lakhs/ 01crore rupees and the performance bond or banker's guarantee remains valid for a period of six years from the date of its submission d. validity of license is defined for five years from date of issue e. license is not transferable or heritable f. Controller has the authority to revoke or suspend the license in accordance with the provisions of the Act g. CA is bound to comply with all the parameters against which it was audited prior to issue of license and consistently and continuously complies with those parameters during the period for which the license remains valid. h. CA subjects itself to periodic audits i. CA ensures confidentiality of subscriber information j. CA complies with every order or direction issued by the Controller within the stipulated period.					
3.1.10.4	The license shall be valid for a period of five years from the date of issue. The license shall not be transferable or heritable	1. Verify the following: a. applicant requirements or grant of license mentioned in description of control 3.1.10.1 are met b. For licence renewal audit and annual audit, verify the continued maintenance of net worth by CA in the last 5 years(if applicable). c. applicant submitted a performance bond or	Mandatory	IT Regulations 3			

		furnished a banker's guarantee from a scheduled bank in favour of the Controller in such form and in such manner as approved by the Controller for an amount of 50 lakhs/ 01crore rupees and the performance bond or banker's guarantee remains valid for a period of six years from the date of its submission d. validity of license is defined for five years from date of issue e. license is not transferable or heritable f. Controller has the authority to revoke or suspend the license in accordance with the provisions of the Act g. CA is bound to comply with all the parameters against which it was audited prior to issue of license and consistently and continuously complies with those parameters during the period for which the license remains valid. h. CA subjects itself to periodic audits i. CA ensures confidentiality of subscriber information j. CA complies with every order or direction issued by the Controller within the stipulated period.					
3.1.10.5	The Controller can revoke or suspend the licence in accordance with the provisions of the Act	1. Verify the following: a. applicant requirements or grant of license mentioned in description of control 3.1.10.1 are met b. For licence renewal audit and annual audit, verify the continued maintenance of net worth by CA in the last 5 years(if applicable). c. applicant submitted a performance bond or furnished a banker's guarantee from a scheduled bank in favour of the Controller	Mandatory	IT Regulations 3			

		in such form and in such manner as approved by the Controller for an amount of 50 lakhs/ 01crore rupees and the performance bond or banker's guarantee remains valid for a period of six years from the date of its submission d. validity of license is defined for five years from date of issue e. license is not transferable or heritable f. Controller has the authority to revoke or suspend the license in accordance with the provisions of the Act g. CA is bound to comply with all the parameters against which it was audited prior to issue of license and consistently and continuously complies with those parameters during the period for which the license remains valid. h. CA subjects itself to periodic audits i. CA ensures confidentiality of subscriber information j. CA complies with every order or direction issued by the Controller within the stipulated period.						
3.1.10.6	The Certifying Authority shall be bound to comply with all the parameters against which it was audited prior to issue of licence and shall consistently and continuously comply with those parameters during the period for which the licence shall remain valid.	1. Verify the following: a. applicant requirements or grant of license mentioned in description of control 3.1.10.1 are met b. For licence renewal audit and annual audit, verify the continued maintenance of net worth by CA in the last 5 years(if applicable). c. applicant submitted a performance bond or furnished a banker's guarantee from a scheduled bank in favour of the Controller in such form and in such manner as approved by the Controller for an	Mandatory	IT Regulations 3				

		amount of 50 lakhs/ 01crore rupees and the performance bond or banker's guarantee remains valid for a period of six years from the date of its submission d. validity of license is defined for five years from date of issue e. license is not transferable or heritable f. Controller has the authority to revoke or suspend the license in accordance with the provisions of the Act g. CA is bound to comply with all the parameters against which it was audited prior to issue of license and consistently and continuously complies with those parameters during the period for which the license remains valid. h. CA subjects itself to periodic audits i. CA ensures confidentiality of subscriber information j. CA complies with every order or direction issued by the Controller within the stipulated period.						
3.1.10.7	The Certifying Authority shall subject itself to periodic audits to ensure that all conditions of the licence are consistently complied with by it. As the cryptographic components of the Certifying Authority systems are highly sensitive and critical, the components must be subjected to periodic expert review to ensure their integrity and assurance.	1. Verify the following: a. applicant requirements or grant of license mentioned in description of control 3.1.10.1 are met b. For licence renewal audit and annual audit, verify the continued maintenance of net worth by CA in the last 5 years(if applicable). c. applicant submitted a performance bond or furnished a banker's guarantee from a scheduled bank in favour of the Controller in such form and in such manner as approved by the Controller for an amount of 50 lakhs/ 01crore rupees and the performance bond or banker's guarantee	Mandatory	IT Regulations 3				

		remains valid for a period of six years from the date of its submission d. validity of license is defined for five years from date of issue e. license is not transferable or heritable f. Controller has the authority to revoke or suspend the license in accordance with the provisions of the Act g. CA is bound to comply with all the parameters against which it was audited prior to issue of license and consistently and continuously complies with those parameters during the period for which the license remains valid. h. CA subjects itself to periodic audits i. CA ensures confidentiality of subscriber information j. CA complies with every order or direction issued by the Controller within the stipulated period.						
3.1.10.8	The Certifying Authority shall always assure the confidentiality of subscriber information	1. Verify the following: a. applicant requirements or grant of license mentioned in description of control 3.1.10.1 are met b. For licence renewal audit and annual audit, verify the continued maintenance of net worth by CA in the last 5 years(if applicable). c. applicant submitted a performance bond or furnished a banker's guarantee from a scheduled bank in favour of the Controller in such form and in such manner as approved by the Controller for an amount of 50 lakhs/ 01crore rupees and the performance bond or banker's guarantee remains valid for a period of six years from the date of its submission	Mandatory	IT Regulations 3				

		d. validity of license is defined for five years from date of issue e. license is not transferable or heritable f. Controller has the authority to revoke or suspend the license in accordance with the provisions of the Act g. CA is bound to comply with all the parameters against which it was audited prior to issue of license and consistently and continuously complies with those parameters during the period for which the license remains valid. h. CA subjects itself to periodic audits i. CA ensures confidentiality of subscriber information j. CA complies with every order or direction issued by the Controller within the stipulated period.						
3.1.10.9	The Certifying Authority shall comply with every order or direction issued by the Controller within the stipulated period.	1. Verify the following: a. applicant requirements or grant of license mentioned in description of control 3.1.10.1 are met b. For licence renewal audit and annual audit, verify the continued maintenance of net worth by CA in the last 5 years(if applicable). c. applicant submitted a performance bond or furnished a banker's guarantee from a scheduled bank in favour of the Controller in such form and in such manner as approved by the Controller for an amount of 50 lakhs/ 01crore rupees and the performance bond or banker's guarantee remains valid for a period of six years from the date of its submission d. validity of license is defined for five years from date of issue e. license is not	Mandatory	IT Regulations 3				

		transferable or heritable f. Controller has the authority to revoke or suspend the license in accordance with the provisions of the Act g. CA is bound to comply with all the parameters against which it was audited prior to issue of license and consistently and continuously complies with those parameters during the period for which the license remains valid. h. CA subjects itself to periodic audits i. CA ensures confidentiality of subscriber information j. CA complies with every order or direction issued by the Controller within the stipulated period.						
3.1.10.10	The method of verification of the identity of the applicant of a Public Key Certificates shall be commensurate with the level of assurance accorded to the certificate.	1. Verify the CA uses approved security controls in the certificate management processes, i.e. certificate registration, generation, issuance, publication, renewal, suspension, revocation and archival 2. Check the method of verification of the identity of the applicant of a Public Key Certificates commensurate with the level of assurance accorded to the certificate	Mandatory	IT Regulations 3				

3.2. Security Guidelines for Certifying Authorities

3.2.1. CA Business Practices Disclosure

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
CA Business Practices Disclosure							
3.2.1.1	The CA discloses its business practices including but not limited to the topics listed in RFC 3647 in its Certification Practice Statement (CPS)	1. Verify the CA has disclosed its business practices including but not limited to the topics listed in RFC 3647 7 in its CPS and CP	Mandatory	WebTrust 1.1			

3.2.1.2	The CA discloses its business practices including but not limited to the topics listed in RFC 3647 or RFC 2527 in its Certificate Policy (CP)	1. Verify the CA has disclosed its business practices including but not limited to the topics listed in RFC 3647 7 in its CPS and CP	Mandatory	WebTrust 1.2				
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3.2.2. Business Practices Management

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Certification Practice Statement (CPS) Management							
3.2.2.1	The Certification Practice Statement of the Certifying Authority shall comply with, and be governed by, the laws of the country	1. For the CA's CPS check the following: a. CPS complies with and is governed by laws of the country b. PA has approved the CPS c. Responsibilities for maintaining the CPS have been formally assigned to individuals. d. CPS is modified and approved in as per the defined process e. CPS is available to appropriate parties f. Revisions to the CPS are made available to appropriate parties. g. CA updates CPS as and when required	Mandatory	IT CA Rules 18			
3.2.2.2	The Policy Authority (PA) has final authority and responsibility for approving the CA's Certification Practice Statement (CPS).	1. For the CA's CPS check the following: a. CPS complies with and is governed by laws of the country b. PA has approved the CPS c. Responsibilities for maintaining the CPS have been formally assigned to individuals. d. CPS is modified and approved in as per the defined process e. CPS is available	Mandatory	WebTrust 2.1.1			

		to appropriate parties f. Revisions to the CPS are made available to appropriate parties. g. CA updates CPS as and when required						
3.2.2.3	Responsibilities for maintaining the CPS have been formally assigned.	1. For the CA's CPS check the following: a. CPS complies with and is governed by laws of the country b. PA has approved the CPS c. Responsibilities for maintaining the CPS have been formally assigned to individuals. d. CPS is modified and approved in as per the defined process e. CPS is available to appropriate parties f. Revisions to the CPS are made available to appropriate parties. g. CA updates CPS as and when required	Mandatory	WebTrust 2.1.2				
3.2.2.4	The CA's CPS is modified and approved in accordance with a defined review process.	1. For the CA's CPS check the following: a. CPS complies with and is governed by laws of the country b. PA has approved the CPS c. Responsibilities for maintaining the CPS have been formally assigned to individuals. d. CPS is modified and approved in as per the defined process e. CPS is available to appropriate parties f. Revisions to the CPS are made available to appropriate parties.	Mandatory	WebTrust 2.1.3				

		g. CA updates CPS as and when required						
3.2.2.5	The CA makes available its Certification Practice Statement (CPS) to all appropriate parties	1. For the CA's CPS check the following: a. CPS complies with and is governed by laws of the country b. PA has approved the CPS c. Responsibilities for maintaining the CPS have been formally assigned to individuals. d. CPS is modified and approved in as per the defined process e. CPS is available to appropriate parties f. Revisions to the CPS are made available to appropriate parties. g. CA updates CPS as and when required	Mandatory	WebTrust 2.1.4				
3.2.2.6	Revisions to the CA's CPS are made available to appropriate parties.	1. For the CA's CPS check the following: a. CPS complies with and is governed by laws of the country b. PA has approved the CPS c. Responsibilities for maintaining the CPS have been formally assigned to individuals. d. CPS is modified and approved in as per the defined process e. CPS is available to appropriate parties f. Revisions to the CPS are made available to appropriate parties. g. CA updates CPS as and when required	Mandatory	WebTrust 2.1.5				

3.2.2.7	The CA updates its CPS to reflect changes in the environment as they occur	1. For the CA's CPS check the following: a. CPS complies with and is governed by laws of the country b. PA has approved the CPS c. Responsibilities for maintaining the CPS have been formally assigned to individuals. d. CPS is modified and approved in as per the defined process e. CPS is available to appropriate parties f. Revisions to the CPS are made available to appropriate parties. g. CA updates CPS as and when required	Mandatory	WebTrust 2.1.6				
Certificate Policy (CP) Management								
3.2.2.8	The Policy Authority (PA) has the responsibility of defining the business requirements and policies for using digital certificates and specifying them in a Certificate Policy (CP) and supporting agreements	1. No action required by the auditor for checking the controls as CP template is shared by CCA	Mandatory	WebTrust 2.2.1				
3.2.2.9	The PA has final authority and responsibility for specifying and approving Certificate Policy(s).	1. No action required by the auditor for checking the controls as CP template is shared by CCA	Mandatory	WebTrust 2.2.2				
3.2.2.10	Certificate Policy(s) are approved by the Policy Authority in accordance with a defined annual review process, including responsibilities for maintaining and tracking changes to the Certificate Policy(s).	1. No action required by the auditor for checking the controls as CP template is shared by CCA	Mandatory	WebTrust 2.2.3				
3.2.2.11	A defined review process exists to assess that the Certificate Policy(s) are capable of support by the controls specified in the CPS.	1. No action required by the auditor for checking the controls as CP	Mandatory	WebTrust 2.2.4				

		template is shared by CCA						
3.2.2.12	The PA makes available the Certificate Policies supported by the CA to Subscribers and Relying Parties	1. No action required by the auditor for checking the controls as CP template is shared by CCA	Mandatory	WebTrust 2.2.5				
CP and CPS Consistency								
3.2.2.13	The PA is responsible for ensuring that the CA's control processes, as stated in a Certification Practice Statement (CPS) or equivalent, fully comply with the requirements of the CP.	1. No action required by the auditor for checking the controls as CP template is shared by CCA	Mandatory	WebTrust 2.3.1				
3.2.2.14	The CA addresses the requirements of the CP when developing its CPS	1. No action required by the auditor for checking the controls as CP template is shared by CCA	Mandatory	WebTrust 2.3.2				
3.2.2.15	The CA assesses the impact of proposed CPS changes to ensure that they are consistent with the CP.	1. No action required by the auditor for checking the controls as CP template is shared by CCA	Mandatory	WebTrust 2.3.3				
3.2.2.16	A defined review process exists to ensure that Certificate Policy(s) are supported by the CA's CPS.	1. No action required by the auditor for checking the controls as CP template is shared by CCA	Mandatory	WebTrust 2.3.4				
3.2.2.17	All changes in Certificate Policy and certification practice statement shall be published on the web site of the Certifying Authority and brought to the notice of the Controller well in advance of such publication. However any change shall not contravene any provision of the Act, rule or regulation or made there under.	1. No action required by the auditor for checking the controls as CP template is shared by CCA	Mandatory	IT Regulations 3				

3.2.3. Change Management

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Change Management							
3.2.3.1	Software testing and change control procedures exist and are followed for the implementation of software on operational systems including scheduled software releases, modifications, patches, and emergency software fixes.	1. Obtain copy of CAs Change control procedures and check the following: a. Software testing and change control procedures are followed for the implementation of software on operational systems including scheduled software releases, modifications, patches, and emergency software fixes b. Change control procedures are followed for the hardware, network component, and system configuration changes c. Application systems are reviewed and tested when operating system changes occur d. On a sample basis check that the implementation of changes are strictly controlled by the use of formal change control procedures e. responsibilities for the change management process are defined and assigned f. risk and impact analysis, classification and prioritisation process are established as part of the procedure , check on sample basis g. only formally authorized changes are implemented in the production h. Authorisation procedures for change control is defined and documentedi. Owners/Users are notified of all changes made to production system j. Fall-back procedures in the event of a failure in the implementation of the change process are established and documented k. Procedures to protect, control access and changes to production source code,	Mandatory	WebTrust 3.7.2			

		data, execution statements and relevant system documentation are documented and implemented l. Version changes of all application software, system software and communication devices is documented						
3.2.3.2	Change control procedures exist and are followed for the hardware, network component, and system configuration changes.	1. Obtain copy of CAs Change control procedures and check the following: a. Software testing and change control procedures are followed for the implementation of software on operational systems including scheduled software releases, modifications, patches, and emergency software fixes b. Change control procedures are followed for the hardware, network component, and system configuration changes c. Application systems are reviewed and tested when operating system changes occur d. On a sample basis check that the implementation of changes are strictly controlled by the use of formal change control procedures e. responsibilities for the change management process are defined and assigned f. risk and impact analysis, classification and prioritisation process are established as part of the procedure , check on sample basis g. only formally authorized changes are implemented in the production h. Authorisation procedures for change control is defined and documentedi. Owners/Users are notified of all changes made to production system	Mandatory	WebTrust 3.7.3				

		j. Fall-back procedures in the event of a failure in the implementation of the change process are established and documented k. Procedures to protect, control access and changes to production source code, data, execution statements and relevant system documentation are documented and implemented l. Version changes of all application software, system software and communication devices is documented						
3.2.3.3	Application systems are reviewed and tested when operating system changes occur	1. Obtain copy of CAs Change control procedures and check the following: a. Software testing and change control procedures are followed for the implementation of software on operational systems including scheduled software releases, modifications, patches, and emergency software fixes b. Change control procedures are followed for the hardware, network component, and system configuration changes c. Application systems are reviewed and tested when operating system changes occur d. On a sample basis check that the implementation of changes are strictly controlled by the use of formal change control procedures e. responsibilities for the change management process are defined and assigned f. risk and impact analysis, classification and prioritisation process are established as part of the procedure , check on sample basis g. only formally authorized changes are	Mandatory	WebTrust 3.7.6				

		implemented in the production h. Authorisation procedures for change control is defined and documented. Owners/Users are notified of all changes made to production system j. Fall-back procedures in the event of a failure in the implementation of the change process are established and documented k. Procedures to protect, control access and changes to production source code, data, execution statements and relevant system documentation are documented and implemented l. Version changes of all application software, system software and communication devices is documented					
3.2.3.4	The implementation of changes is strictly controlled by the use of formal change control procedures to minimize the risk of corruption of information systems.	1. Obtain copy of CAs Change control procedures and check the following: a. Software testing and change control procedures are followed for the implementation of software on operational systems including scheduled software releases, modifications, patches, and emergency software fixes b. Change control procedures are followed for the hardware, network component, and system configuration changes c. Application systems are reviewed and tested when operating system changes occur d. On a sample basis check that the implementation of changes are strictly controlled by the use of formal change control procedures e. responsibilities for the change management	Mandatory	WebTrust 3.7.7			

		process are defined and assigned f. risk and impact analysis, classification and prioritisation process are established as part of the procedure , check on sample basis g. only formally authorized changes are implemented in the production h. Authorisation procedures for change control is defined and documentedi. Owners/Users are notified of all changes made to production system j. Fall-back procedures in the event of a failure in the implementation of the change process are established and documented k. Procedures to protect, control access and changes to production source code, data, execution statements and relevant system documentation are documented and implemented l. Version changes of all application software, system software and communication devices is documented						
3.2.3.5	Organizational responsibilities for the change management process shall be defined and assigned	1. Obtain copy of CAs Change control procedures and check the following: a. Software testing and change control procedures are followed for the implementation of software on operational systems including scheduled software releases, modifications, patches, and emergency software fixes b. Change control procedures are followed for the hardware, network component, and system configuration changes c. Application systems are reviewed and tested when operating system	Mandatory	IT CA Rules SCHEDULE II 21.1				

		changes occur d. On a sample basis check that the implementation of changes are strictly controlled by the use of formal change control procedures e. responsibilities for the change management process are defined and assigned f. risk and impact analysis, classification and prioritisation process are established as part of the procedure , check on sample basis g. only formally authorized changes are implemented in the production h. Authorisation procedures for change control is defined and documentedi. Owners/Users are notified of all changes made to production system j. Fall-back procedures in the event of a failure in the implementation of the change process are established and documented k. Procedures to protect, control access and changes to production source code, data, execution statements and relevant system documentation are documented and implemented l. Version changes of all application software, system software and communication devices is documented					
3.2.3.6	A risk and impact analysis, classification and prioritisation process shall be established	1. Obtain copy of CAs Change control procedures and check the following: a. Software testing and change control procedures are followed for the implementation of software on operational systems including scheduled software releases, modifications, patches, and emergency	Mandatory	IT CA Rules SCHEDULE II 21.1			

		software fixes b. Change control procedures are followed for the hardware, network component, and system configuration changes c. Application systems are reviewed and tested when operating system changes occur d. On a sample basis check that the implementation of changes are strictly controlled by the use of formal change control procedures e. responsibilities for the change management process are defined and assigned f. risk and impact analysis, classification and prioritisation process are established as part of the procedure , check on sample basis g. only formally authorized changes are implemented in the production h. Authorisation procedures for change control is defined and documentedi. Owners/Users are notified of all changes made to production system j. Fall-back procedures in the event of a failure in the implementation of the change process are established and documented k. Procedures to protect, control access and changes to production source code, data, execution statements and relevant system documentation are documented and implemented l. Version changes of all application software, system software and communication devices is documented						
3.2.3.7	No changes to a production system shall be implemented until such changes have been	1. Obtain copy of CAs Change control procedures and check the following:	Mandatory	IT CA Rules SCHEDULE II 21.1				

	formally authorised. Authorisation procedures for change control shall be defined and documented	a. Software testing and change control procedures are followed for the implementation of software on operational systems including scheduled software releases, modifications, patches, and emergency software fixes b. Change control procedures are followed for the hardware, network component, and system configuration changes c. Application systems are reviewed and tested when operating system changes occur d. On a sample basis check that the implementation of changes are strictly controlled by the use of formal change control procedures e. responsibilities for the change management process are defined and assigned f. risk and impact analysis, classification and prioritisation process are established as part of the procedure , check on sample basis g. only formally authorized changes are implemented in the production h. Authorisation procedures for change control is defined and documentedi. Owners/Users are notified of all changes made to production system j. Fall-back procedures in the event of a failure in the implementation of the change process are established and documented k. Procedures to protect, control access and changes to production source code, data, execution statements and relevant system documentation are documented and implemented l. Version changes of all application software, system software and communication devices						
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		is documented						
3.2.3.8	Owners/Users shall be notified of all changes made to production system which may affect the processing of information on the said production system	1. Obtain copy of CAs Change control procedures and check the following: a. Software testing and change control procedures are followed for the implementation of software on operational systems including scheduled software releases, modifications, patches, and emergency software fixes b. Change control procedures are followed for the hardware, network component, and system configuration changes c. Application systems are reviewed and tested when operating system changes occur d. On a sample basis check that the implementation of changes are strictly controlled by the use of formal change control procedures e. responsibilities for the change management process are defined and assigned f. risk and impact analysis, classification and prioritisation process are established as part of the procedure , check on sample basis g. only formally authorized changes are implemented in the production h. Authorisation procedures for change control is defined and documentedi. Owners/Users are notified of all changes made to production system j. Fall-back procedures in the event of a failure in the implementation of the change process are established and documented k. Procedures to protect, control access and changes to	Mandatory	IT CA Rules SCHEDULE II 21.1				

		production source code, data, execution statements and relevant system documentation are documented and implemented l. Version changes of all application software, system software and communication devices is documented						
3.2.3.9	Fall-back procedures in the event of a failure in the implementation of the change process shall be established and documented.	1. Obtain copy of CAs Change control procedures and check the following: a. Software testing and change control procedures are followed for the implementation of software on operational systems including scheduled software releases, modifications, patches, and emergency software fixes b. Change control procedures are followed for the hardware, network component, and system configuration changes c. Application systems are reviewed and tested when operating system changes occur d. On a sample basis check that the implementation of changes are strictly controlled by the use of formal change control procedures e. responsibilities for the change management process are defined and assigned f. risk and impact analysis, classification and prioritisation process are established as part of the procedure , check on sample basis g. only formally authorized changes are implemented in the production h. Authorisation procedures for change control is defined and documentedi. Owners/Users are notified of all changes made to production	Mandatory	IT CA Rules SCHEDULE II 21.1				

		system j. Fall-back procedures in the event of a failure in the implementation of the change process are established and documented k. Procedures to protect, control access and changes to production source code, data, execution statements and relevant system documentation are documented and implemented l. Version changes of all application software, system software and communication devices is documented						
3.2.3.10	Procedures to protect, control access and changes to production source code, data, execution statements and relevant system documentation shall be documented and implemented	1. Obtain copy of CAs Change control procedures and check the following: a. Software testing and change control procedures are followed for the implementation of software on operational systems including scheduled software releases, modifications, patches, and emergency software fixes b. Change control procedures are followed for the hardware, network component, and system configuration changes c. Application systems are reviewed and tested when operating system changes occur d. On a sample basis check that the implementation of changes are strictly controlled by the use of formal change control procedures e. responsibilities for the change management process are defined and assigned f. risk and impact analysis, classification and prioritisation process are established as part of the procedure , check on sample basis g. only formally	Mandatory	IT CA Rules SCHEDULE II 21.1				

		authorized changes are implemented in the production h. Authorisation procedures for change control is defined and documentedi. Owners/Users are notified of all changes made to production system j. Fall-back procedures in the event of a failure in the implementation of the change process are established and documented k. Procedures to protect, control access and changes to production source code, data, execution statements and relevant system documentation are documented and implemented l. Version changes of all application software, system software and communication devices is documented						
3.2.3.11	Version changes of application software and all system software installed on the computer systems and all communication devices shall be documented. Different versions of application software and system software must be kept in safe custody	1. Obtain copy of CAs Change control procedures and check the following: a. Software testing and change control procedures are followed for the implementation of software on operational systems including scheduled software releases, modifications, patches, and emergency software fixes b. Change control procedures are followed for the hardware, network component, and system configuration changes c. Application systems are reviewed and tested when operating system changes occur d. On a sample basis check that the implementation of changes are strictly controlled by the use of formal change control procedures e. responsibilities for	Mandatory	IT CA Rules SCHEDULE II 21.1				

		the change management process are defined and assigned f. risk and impact analysis, classification and prioritisation process are established as part of the procedure , check on sample basis g. only formally authorized changes are implemented in the production h. Authorisation procedures for change control is defined and documentedi. Owners/Users are notified of all changes made to production system j. Fall-back procedures in the event of a failure in the implementation of the change process are established and documented k. Procedures to protect, control access and changes to production source code, data, execution statements and relevant system documentation are documented and implemented l. Version changes of all application software, system software and communication devices is documented							
Testing Of Changes To Production System									
3.2.3.12	All changes in computer resource proposed in the production system shall be tested and the test results shall be reviewed and accepted by all concerned parties prior to implementation.	1. On a sample basis verify all changes in computer resource proposed in the production system are tested and the test results are reviewed and accepted by all concerned parties prior to implementation. 2. Validate user acceptance tests are performed in a controlled environment which includes requirements mentioned in control 3.2.2.13	Mandatory	IT CA Rules SCHEDULE II 21.2					
3.2.3.13	All user acceptance tests in respect of changes in	1. On a sample basis verify all changes in	Mandatory	IT CA Rules SCHEDULE II 21.2					

	computer resource in production system shall be performed in a controlled environment which includes: (i) Test objectives, (ii) A documented test plan, and (iii) acceptance criteria.	computer resource proposed in the production system are tested and the test results are reviewed and accepted by all concerned parties prior to implementation. 2. Validate user acceptance tests are performed in a controlled environment which includes requirements mentioned in control 3.2.2.13						
Review Of Changes								
3.2.3.14	Procedures shall be established for an independent review of programme changes before they are moved into a production environment to detect unauthorized or malicious codes	1. Verify the change management procedure mentions the following: a. an independent review of programme b. changes before they are moved into a production environment c. schedule and review the implementation of the changes in computer resource in the production system so as to ensure proper functioning 2. Validate on sample basis for latest changes/ fixes in computer resource in the production system are reviewed and approved 3. Verify periodic management reports on the status of the changes implemented in the computer resourced in the production system are submitted for management review 4. Check the software updates and patches are reviewed for security implications before being implemented 5. Verify the software updates and patches to rectify security vulnerability in critical systems used are promptly reviewed and implemented 6. Validate the information on the software updates and patches and their implementation on CA's system shall be clearly and properly	Mandatory	IT CA Rules SCHEDULE II 21.3				

		documented						
3.2.3.15	Procedures shall be established to schedule and review the implementation of the changes in computer resource in the production system so as to ensure proper functioning	1. Verify the change management procedure mentions the following: a. an independent review of programme b. changes before they are moved into a production environment c. schedule and review the implementation of the changes in computer resource in the production system so as to ensure proper functioning 2. Validate on sample basis for latest changes/ fixes in computer resource in the production system are reviewed and approved 3. Verify periodic management reports on the status of the changes implemented in the computer resourced in the production system are submitted for management review 4. Check the software updates and patches are reviewed for security implications before being implemented 5. Verify the software updates and patches to rectify security vulnerability in critical systems used are promptly reviewed and implemented 6. Validate the information on the software updates and patches and their implementation on CA's system shall be clearly and properly documented	Mandatory	IT CA Rules SCHEDULE II 21.3				
3.2.3.16	All emergency changes/fixes in computer resource in the production system shall be reviewed and approved	1. Verify the change management procedure mentions the following: a. an independent review of programme b. changes before they are moved into a production environment c. schedule and review the implementation of the changes in computer resource in the production system	Mandatory	IT CA Rules SCHEDULE II 21.3				

		so as to ensure proper functioning 2. Validate on sample basis for latest changes/ fixes in computer resource in the production system are reviewed and approved 3. Verify periodic management reports on the status of the changes implemented in the computer resourced in the production system are submitted for management review 4. Check the software updates and patches are reviewed for security implications before being implemented 5. Verify the software updates and patches to rectify security vulnerability in critical systems used are promptly reviewed and implemented 6. Validate the information on the software updates and patches and their implementation on CA's system shall be clearly and properly documented					
3.2.3.17	Periodic management reports on the status of the changes implemented in the computer resourced in the production system shall be submitted for management review	1. Verify the change management procedure mentions the following: a. an independent review of programme b. changes before they are moved into a production environment c. schedule and review the implementation of the changes in computer resource in the production system so as to ensure proper functioning 2. Validate on sample basis for latest changes/ fixes in computer resource in the production system are reviewed and approved 3. Verify periodic management reports on the status of the changes implemented in the computer resourced in the production system are submitted for	Mandatory	IT CA Rules SCHEDULE II 21.3			

		management review 4. Check the software updates and patches are reviewed for security implications before being implemented 5. Verify the software updates and patches to rectify security vulnerability in critical systems used are promptly reviewed and implemented 6. Validate the information on the software updates and patches and their implementation on CA's system shall be clearly and properly documented						
3.2.3.18	Software updates and patches shall be reviewed for security implications before being implemented on Certifying Authority's system	1. Verify the change management procedure mentions the following: a. an independent review of programme b. changes before they are moved into a production environment c. schedule and review the implementation of the changes in computer resource in the production system so as to ensure proper functioning 2. Validate on sample basis for latest changes/ fixes in computer resource in the production system are reviewed and approved 3. Verify periodic management reports on the status of the changes implemented in the computer resourced in the production system are submitted for management review 4. Check the software updates and patches are reviewed for security implications before being implemented 5. Verify the software updates and patches to rectify security vulnerability in critical systems used are promptly reviewed and implemented 6. Validate the information on the software updates and	Mandatory	IT CA Rules SCHEDULE III 7				

		patches and their implementation on CA's system shall be clearly and properly documented						
3.2.3.19	Software updates and patches to rectify security vulnerability in critical systems used for Certifying Authority's operation shall be promptly reviewed and implemented	1. Verify the change management procedure mentions the following: a. an independent review of programme b. changes before they are moved into a production environment c. schedule and review the implementation of the changes in computer resource in the production system so as to ensure proper functioning 2. Validate on sample basis for latest changes/ fixes in computer resource in the production system are reviewed and approved 3. Verify periodic management reports on the status of the changes implemented in the computer resourced in the production system are submitted for management review 4. Check the software updates and patches are reviewed for security implications before being implemented 5. Verify the software updates and patches to rectify security vulnerability in critical systems used are promptly reviewed and implemented 6. Validate the information on the software updates and patches and their implementation on CA's system shall be clearly and properly documented	Mandatory	IT CA Rules SCHEDULE III 7				
3.2.3.20	Information on the software updates and patches and their implementation on Certifying Authority's system shall be clearly and properly documented	1. Verify the change management procedure mentions the following: a. an independent review of programme b. changes before they are moved into a production environment c. schedule and review	Mandatory	IT CA Rules SCHEDULE III 7				

		the implementation of the changes in computer resource in the production system so as to ensure proper functioning 2. Validate on sample basis for latest changes/ fixes in computer resource in the production system are reviewed and approved 3. Verify periodic management reports on the status of the changes implemented in the computer resourced in the production system are submitted for management review 4. Check the software updates and patches are reviewed for security implications before being implemented 5. Verify the software updates and patches to rectify security vulnerability in critical systems used are promptly reviewed and implemented 6. Validate the information on the software updates and patches and their implementation on CA's system shall be clearly and properly documented						
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3.2.4. Operations Management

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Operations Management							
3.2.4.1	CA operating procedures are documented and maintained for each functional area.	1. Verify CA has documented operating procedures for functional areas and check the following: a. Formal management responsibilities and procedures exist to control all changes to CA equipment, software and operating procedures b. Duties and areas of responsibility are segregated c. Development and testing facilities are separated from operational facilities	Mandatory	WebTrust 3.5.1			

		d. Prior to using external facilities management services, risks and related controls are identified, agreed upon with the contractor, and incorporated into the contract. e. Capacity demands are monitored and projections of future capacity requirements made f. On a sample basis check that the acceptance criteria for new information systems, upgrades and new versions are established and suitable tests of the system carried out prior to acceptance						
3.2.4.2	Formal management responsibilities and procedures exist to control all changes to CA equipment, software and operating procedures.	1. Verify CA has documented operating procedures for functional areas and check the following: a. Formal management responsibilities and procedures exist to control all changes to CA equipment, software and operating procedures b. Duties and areas of responsibility are segregated c. Development and testing facilities are separated from operational facilities d. Prior to using external facilities management services, risks and related controls are identified, agreed upon with the contractor, and incorporated into the contract. e. Capacity demands are monitored and projections of future capacity requirements made f. On a sample basis check that the acceptance criteria for new information systems, upgrades and new versions are established and suitable tests of the system carried out prior to acceptance	Mandatory	WebTrust 3.5.2				
3.2.4.3	Duties and areas of responsibility are segregated in order to reduce opportunities for	1. Verify CA has documented operating procedures for functional areas and check the	Mandatory	WebTrust 3.5.3				

	unauthorized modification or misuse of information or services	following: a. Formal management responsibilities and procedures exist to control all changes to CA equipment, software and operating procedures b. Duties and areas of responsibility are segregated c. Development and testing facilities are separated from operational facilities d. Prior to using external facilities management services, risks and related controls are identified, agreed upon with the contractor, and incorporated into the contract. e. Capacity demands are monitored and projections of future capacity requirements made f. On a sample basis check that the acceptance criteria for new information systems, upgrades and new versions are established and suitable tests of the system carried out prior to acceptance						
3.2.4.4	Development and testing facilities are separated from operational facilities	1. Verify CA has documented operating procedures for functional areas and check the following: a. Formal management responsibilities and procedures exist to control all changes to CA equipment, software and operating procedures b. Duties and areas of responsibility are segregated c. Development and testing facilities are separated from operational facilities d. Prior to using external facilities management services, risks and related controls are identified, agreed upon with the contractor, and incorporated into the contract. e. Capacity demands are monitored and projections of future capacity requirements made f. On a sample basis	Mandatory	WebTrust 3.5.4				

		check that the acceptance criteria for new information systems, upgrades and new versions are established and suitable tests of the system carried out prior to acceptance						
3.2.4.5	Prior to using external facilities management services, risks and related controls are identified, agreed upon with the contractor, and incorporated into the contract.	1. Verify CA has documented operating procedures for functional areas and check the following: a. Formal management responsibilities and procedures exist to control all changes to CA equipment, software and operating procedures b. Duties and areas of responsibility are segregated c. Development and testing facilities are separated from operational facilities d. Prior to using external facilities management services, risks and related controls are identified, agreed upon with the contractor, and incorporated into the contract. e. Capacity demands are monitored and projections of future capacity requirements made f. On a sample basis check that the acceptance criteria for new information systems, upgrades and new versions are established and suitable tests of the system carried out prior to acceptance	Mandatory	WebTrust 3.5.5				
3.2.4.6	Capacity demands are monitored and projections of future capacity requirements made to ensure that adequate processing power and storage are available.	1. Verify CA has documented operating procedures for functional areas and check the following: a. Formal management responsibilities and procedures exist to control all changes to CA equipment, software and operating procedures b. Duties and areas of responsibility are segregated c. Development and testing facilities are separated from	Mandatory	WebTrust 3.5.6				

		operational facilities d. Prior to using external facilities management services, risks and related controls are identified, agreed upon with the contractor, and incorporated into the contract. e. Capacity demands are monitored and projections of future capacity requirements made f. On a sample basis check that the acceptance criteria for new information systems, upgrades and new versions are established and suitable tests of the system carried out prior to acceptance						
3.2.4.7	Acceptance criteria for new information systems, upgrades and new versions are established and suitable tests of the system carried out prior to acceptance	1. Verify CA has documented operating procedures for functional areas and check the following: a. Formal management responsibilities and procedures exist to control all changes to CA equipment, software and operating procedures b. Duties and areas of responsibility are segregated c. Development and testing facilities are separated from operational facilities d. Prior to using external facilities management services, risks and related controls are identified, agreed upon with the contractor, and incorporated into the contract. e. Capacity demands are monitored and projections of future capacity requirements made f. On a sample basis check that the acceptance criteria for new information systems, upgrades and new versions are established and suitable tests of the system carried out prior to acceptance	Mandatory	WebTrust 3.5.7				

3.2.5. Measures to handle computer virus

Control	Control	Audit Checks	Control	References	Compliance	Document	Remarks
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No.			Type					
Measures to handle computer virus								
3.2.5.1	Detection and prevention controls to protect against viruses and malicious software, including on offline or air gapped systems are implemented. Employee awareness programs are in place.	1. Verify detection and prevention controls to protect against viruses and malicious software, including on offline or air gapped systems are implemented 2. Check an employee awareness program is implemented 3. Validate names of personal responsible for ensuring all file servers and personal computers are equipped with up-to-date virus protection and detection software. 4. Check if virus detection software is being used to check storage drives internal and external to the system on a periodic basis 5. Verify all diskettes and software are screened and verified by virus detection software before being loaded onto the computer system 6. Check magnetic media like tape cartridge, floppies etc. brought from outside are used on the data, file, PKI or computer server or personal computer on Intranet and Internet only after proper screening and verification by virus detection software	Mandatory	WebTrust 3.5.8				
3.2.5.2	Responsibilities and duties shall be assigned to ensure that all file servers and personal computers are equipped with up-to-date virus protection and detection software.	1. Verify detection and prevention controls to protect against viruses and malicious software, including on offline or air gapped systems are implemented 2. Check an employee awareness program is implemented 3. Validate names of personal responsible for ensuring all file servers and personal	Mandatory	IT CA Rules SCHEDULE II 11				

		computers are equipped with up-to-date virus protection and detection software. 4. Check if virus detection software is being used to check storage drives internal and external to the system on a periodic basis 5. Verify all diskettes and software are screened and verified by virus detection software before being loaded onto the computer system 6. Check magnetic media like tape cartridge, floppies etc. brought from outside are used on the data, file, PKI or computer server or personal computer on Intranet and Internet only after proper screening and verification by virus detection software					
3.2.5.3	Virus detection software must be used to check storage drives both internal and external to the system on a periodic basis	1. Verify detection and prevention controls to protect against viruses and malicious software, including on offline or air gapped systems are implemented 2. Check an employee awareness program is implemented 3. Validate names of personal responsible for ensuring all file servers and personal computers are equipped with up-to-date virus protection and detection software. 4. Check if virus detection software is being used to check storage drives internal and external to the system on a periodic basis 5. Verify all diskettes and software are screened and verified by virus detection software before being loaded onto the	Mandatory	IT CA Rules SCHEDULE II 11			

		computer system 6. Check magnetic media like tape cartridge, floppies etc. brought from outside are used on the data, file, PKI or computer server or personal computer on Intranet and Internet only after proper screening and verification by virus detection software						
3.2.5.4	All diskettes and software shall be screened and verified by virus detection software before being loaded onto the computer system. No magnetic media like tape cartridge, floppies etc. brought from outside shall be used on the data, file, PKI or computer server or personal computer on Intranet and Internet without proper screening and verification by virus detection software	1. Verify detection and prevention controls to protect against viruses and malicious software, including on offline or air gapped systems are implemented 2. Check an employee awareness program is implemented 3. Validate names of personal responsible for ensuring all file servers and personal computers are equipped with up-to-date virus protection and detection software. 4. Check if virus detection software is being used to check storage drives internal and external to the system on a periodic basis 5. Verify all diskettes and software are screened and verified by virus detection software before being loaded onto the computer system 6. Check magnetic media like tape cartridge, floppies etc. brought from outside are used on the data, file, PKI or computer server or personal computer on Intranet and Internet only after proper screening and verification by virus detection software	Mandatory	IT CA Rules SCHEDULE II 11				
3.2.5.5	A team shall be designated	1. Verify a team is	Mandatory	IT CA Rules SCHEDULE II 11				

	to deal with reported or suspected incidents of computer virus. The designated team shall ensure that latest version of antivirus software is loaded on all data, file, PKI servers and personal computers.	established for incidents management of computer virus 2. Validate the team ensures latest version of antivirus software is loaded on all data, file, PKI servers and personal computers.						
3.2.5.6	Procedures shall be established to limit the spread of viruses to other organization information assets. Such procedures inter alia shall include: • Communication to other business partners and users who may be at risk from an infected resource • Eradication and recovery procedures • Incident report must be documented and communicated per established procedures	1. Verify by checking procedures are established to limit the spread of viruses to other organization information assets 2. Check awareness and training programs are in place to communicate virus protection practices, available controls, areas of high risk to virus infection and responsibilities	Mandatory	IT CA Rules SCHEDULE II 11				
3.2.5.7	An awareness and training programme shall be established to communicate virus protection practices, available controls, areas of high risk to virus infection and responsibilities	1. Verify by checking procedures are established to limit the spread of viruses to other organization information assets 2. Check awareness and training programs are in place to communicate virus protection practices, available controls, areas of high risk to virus infection and responsibilities	Mandatory	IT CA Rules SCHEDULE II 11				

3.2.6. Relocation of hardware and software

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Relocation of hardware and software							
3.2.6.1	Whenever computers or computer peripherals are relocated (e.g. for maintenance, installation at different sites or storage), the following guidelines shall apply: • All removable media will be removed from the computer system and kept at secure location. • Internal drives will be overwritten, reformatted or removed as the	1. Perform walkthrough and verify Relocation of hardware and software is performed as per requirements mentioned in the control description	Mandatory	IT CA Rules SCHEDULE II 12			

	situation may be. • If applicable, ribbons will be removed from printers. • All paper will be removed from printers.							
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3.2.7. Hardware and software maintenance

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Hardware and software maintenance							
3.2.7.1	Proper placement and installation of Information Technology equipment to reduce the effects of interference due to electromagnetic emanations.	1. For the hardware and software maintenance performed in the CA system check the following: a. secure placement and installation of IT equipment has been done to prevent electromagnetic interference b. inventory and configuration chart of hardware are maintained c. security features within hardware are identified and implemented d. Authorization, documentation, and control of change made to the hardware is defined e. Provision of an uninterruptible power supply is present f. support facilities including power and air conditioning have been installed g. annual maintenance of hardware is in place h. maintenance agreements have been signed with the supplier of computer and communication hardware, software (both system and application) and firmware i. On a sample basis that maintenance personnel have signed non-disclosure agreements	Mandatory	IT CA Rules SCHEDULE II 13			
3.2.7.2	Maintenance of an inventory and configuration chart of	1. For the hardware and software maintenance performed in the CA	Mandatory	IT CA Rules SCHEDULE II 13			

	hardware	system check the following: a. secure placement and installation of IT equipment has been done to prevent electromagnetic interference b. inventory and configuration chart of hardware are maintained c. security features within hardware are identified and implemented d. Authorization, documentation, and control of change made to the hardware is defined e. Provision of an uninterruptible power supply is present f. support facilities including power and air conditioning have been installed g. annual maintenance of hardware is in place h. maintenance agreements have been signed with the supplier of computer and communication hardware, software (both system and application) and firmware i. On a sample basis that maintenance personnel have signed non-disclosure agreements					
3.2.7.3	Identification and use of security features implemented within hardware.	1. For the hardware and software maintenance performed in the CA system check the following: a. secure placement and installation of IT equipment has been done to prevent electromagnetic interference b. inventory and configuration chart of hardware are maintained c. security features within hardware are identified and implemented d. Authorization, documentation, and control of change made	Mandatory	IT CA Rules SCHEDULE II 13			

		to the hardware is defined e. Provision of an uninterruptible power supply is present f. support facilities including power and air conditioning have been installed g. annual maintenance of hardware is in place h. maintenance agreements have been signed with the supplier of computer and communication hardware, software (both system and application) and firmware i. On a sample basis that maintenance personnel have signed non-disclosure agreements						
3.2.7.4	Authorization, documentation, and control of change made to the hardware	1. For the hardware and software maintenance performed in the CA system check the following: a. secure placement and installation of IT equipment has been done to prevent electromagnetic interference b. inventory and configuration chart of hardware are maintained c. security features within hardware are identified and implemented d. Authorization, documentation, and control of change made to the hardware is defined e. Provision of an uninterruptible power supply is present f. support facilities including power and air conditioning have been installed g. annual maintenance of hardware is in place h. maintenance agreements have been signed with the supplier of computer and communication hardware, software (both system and application) and	Mandatory	IT CA Rules SCHEDULE II 13				

		firmware i. On a sample basis that maintenance personnel have signed non-disclosure agreements						
3.2.7.5	Identification of support facilities including power and air conditioning	1. For the hardware and software maintenance performed in the CA system check the following: a. secure placement and installation of IT equipment has been done to prevent electromagnetic interference b. inventory and configuration chart of hardware are maintained c. security features within hardware are identified and implemented d. Authorization, documentation, and control of change made to the hardware is defined e. Provision of an uninterruptible power supply is present f. support facilities including power and air conditioning have been installed g. annual maintenance of hardware is in place h. maintenance agreements have been signed with the supplier of computer and communication hardware, software (both system and application) and firmware i. On a sample basis that maintenance personnel have signed non-disclosure agreements	Mandatory	IT CA Rules SCHEDULE II 13				
3.2.7.6	Provision of an uninterruptible power supply.	1. For the hardware and software maintenance performed in the CA system check the following: a. secure placement and installation of IT equipment has been done to prevent electromagnetic interference	Mandatory	IT CA Rules SCHEDULE II 13				

		b. inventory and configuration chart of hardware are maintained c. security features within hardware are identified and implemented d. Authorization, documentation, and control of change made to the hardware is defined e. Provision of an uninterruptible power supply is present f. support facilities including power and air conditioning have been installed g. annual maintenance of hardware is in place h. maintenance agreements have been signed with the supplier of computer and communication hardware, software (both system and application) and firmware i. On a sample basis that maintenance personnel have signed non-disclosure agreements						
3.2.7.7	Organization must make proper arrangements for maintenance of computer hardware, software (both system and application) and firmware installed and used by them	1. For the hardware and software maintenance performed in the CA system check the following: a. secure placement and installation of IT equipment has been done to prevent electromagnetic interference b. inventory and configuration chart of hardware are maintained c. security features within hardware are identified and implemented d. Authorization, documentation, and control of change made to the hardware is defined e. Provision of an uninterruptible power supply is present f. support facilities including power and air conditioning have been	Mandatory	IT CA Rules SCHEDULE II 13				

		installed g. annual maintenance of hardware is in place h. maintenance agreements have been signed with the supplier of computer and communication hardware, software (both system and application) and firmware i. On a sample basis that maintenance personnel have signed non-disclosure agreements						
3.2.7.8	It shall be the responsibility of the officer in charge of the operational site to ensure that contract for annual maintenance of hardware is always in place.	1. For the hardware and software maintenance performed in the CA system check the following: a. secure placement and installation of IT equipment has been done to prevent electromagnetic interference b. inventory and configuration chart of hardware are maintained c. security features within hardware are identified and implemented d. Authorization, documentation, and control of change made to the hardware is defined e. Provision of an uninterruptible power supply is present f. support facilities including power and air conditioning have been installed g. annual maintenance of hardware is in place h. maintenance agreements have been signed with the supplier of computer and communication hardware, software (both system and application) and firmware i. On a sample basis that maintenance personnel have signed non-disclosure agreements	Mandatory	IT CA Rules SCHEDULE II 13				

3.2.7.9	Organization must enter into maintenance agreements, if necessary, with the supplier of computer and communication hardware, software (both system and application) and firmware.	1. For the hardware and software maintenance performed in the CA system check the following: a. secure placement and installation of IT equipment has been done to prevent electromagnetic interference b. inventory and configuration chart of hardware are maintained c. security features within hardware are identified and implemented d. Authorization, documentation, and control of change made to the hardware is defined e. Provision of an uninterruptible power supply is present f. support facilities including power and air conditioning have been installed g. annual maintenance of hardware is in place h. maintenance agreements have been signed with the supplier of computer and communication hardware, software (both system and application) and firmware i. On a sample basis that maintenance personnel have signed non-disclosure agreements	Mandatory	IT CA Rules SCHEDULE II 13				
3.2.7.10	Maintenance personnel will sign non-disclosure agreements	1. For the hardware and software maintenance performed in the CA system check the following: a. secure placement and installation of IT equipment has been done to prevent electromagnetic interference b. inventory and configuration chart of hardware are maintained c. security features within hardware are identified and implemented	Mandatory	IT CA Rules SCHEDULE II 13				

		d. Authorization, documentation, and control of change made to the hardware is defined e. Provision of an uninterruptible power supply is present f. support facilities including power and air conditioning have been installed g. annual maintenance of hardware is in place h. maintenance agreements have been signed with the supplier of computer and communication hardware, software (both system and application) and firmware i. On a sample basis that maintenance personnel have signed non-disclosure agreements						
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3.2.8. Purchase and Licensing of Hardware and Software

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Purchase and Licensing of Hardware and Software							
3.2.8.1	Hardware and software products that contain or are to be used to enforce security, and intended for use or interface into any organization system or network, must be verified to comply with these Information Technology Security Guidelines prior to the signing of any contract, purchase or lease.	Validate the hardware and software products comply with the Information Technology Security Guidelines	Mandatory	IT CA Rules SCHEDULE II 14			
3.2.8.2	Software, which is capable of bypassing or modifying the security system or operating system, integrity features, must be verified to determine that they conform to these Information Technology Security Guidelines. Where such compliance is not possible, then procedures shall be in place to ensure that the implementation and operation of that software does not compromise the	1. Verify procedures have been implemented to ensure that the implementation and operation of that software does not compromise the security of the system	Mandatory	IT CA Rules SCHEDULE II 14			

	security of the system.							
3.2.8.3	There shall be procedures to identify, select, implement and control software (system and application software) acquisition and installation to ensure compliance with the Indian Copyright Act and Information Technology Security Guidelines.	1. Validate procedures have been implemented to identify, select, implement and control software (system and application software) acquisition and installation 2. Check that it is in compliance with the Indian Copyright Act and Information Technology Security Guidelines	Mandatory	IT CA Rules SCHEDULE II 14				
3.2.8.4	It is prohibited to knowingly install on any system whether test or production, any software which is not licensed for use on the specific systems or networks.	1. For software installed on the CA systems verify the following: a. only licensed software are installed on systems used for CA operations b. licensing agreements exists for software installed and used on CA systems c. Illegally acquired or unauthorized software are not used on any computer, computer network or data communication equipment d. System Administrator or Network Administrator are responsible for removing any illegal software if detected	Mandatory	IT CA Rules SCHEDULE II 14				
3.2.8.5	No software will be installed and used on the system when appropriate licensing agreements do not exist, except during evaluation periods for which the user has documented permission to install and test the software under evaluation.	1. For software installed on the CA systems verify the following: a. only licensed software are installed on systems used for CA operations b. licensing agreements exists for software installed and used on CA systems c. Illegally acquired or unauthorized software are not used on any computer,	Mandatory	IT CA Rules SCHEDULE II 14				

		computer network or data communication equipment d. System Administrator or Network Administrator are responsible for removing any illegal software if detected						
3.2.8.6	Illegally acquired or unauthorized software must not be used on any computer, computer network or data communication equipment. In the event that any illegally acquired or unauthorized software is detected by the System Administrator or Network Administrator, the same must be removed	1. For software installed on the CA systems verify the following: a. only licensed software are installed on systems used for CA operations b. licensing agreements exists for software installed and used on CA systems c. Illegally acquired or unauthorized software are not used on any computer, computer network or data communication equipment d. System Administrator or Network Administrator are responsible for removing any illegal software if detected	Mandatory	IT CA Rules SCHEDULE II 14				

3.2.9. System Software

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
System Software							
3.2.9.1	All system software options and parameters shall be reviewed and approved by the management.	1. For the CAs system software check the following: a. the system software operations and parameters are reviewed and approved by CA management b. System software are tested and its security functionality validated prior to implementation. c. Vendor supplied default IDs are deleted or password changes before users access the computer system	Mandatory	IT CA Rules SCHEDULE II 15			

		d. On sample basis that the versions of system software installed on the computer system and communication devices are regularly updated e. Changes proposed in the system software are appropriately justified and approved by an authorized party (check for recent sample changes performed) f. log of all changes to system software are maintained, completely documented and tested to ensure the desired results g. no standing "Write" access is present for the system libraries h. All "Write" access are logged and reviewed by the System Administrator i. System Programmers are not allowed to have access to the application system's data and programme files in the production environment					
3.2.9.2	System software shall be comprehensively tested and its security functionality validated prior to implementation.	1. For the CAs system software check the following: a. the system software operations and parameters are reviewed and approved by CA management b. System software are tested and its security functionality validated prior to implementation. c. Vendor supplied default IDs are deleted or password changes before users access the computer system d. On sample basis that the versions of system software installed on the computer system and communication devices are regularly updated e. Changes proposed	Mandatory	IT CA Rules SCHEDULE II 15			

		in the system software are appropriately justified and approved by an authorized party (check for recent sample changes performed) f. log of all changes to system software are maintained, completely documented and tested to ensure the desired results g. no standing "Write" access is present for the system libraries h. All "Write" access are logged and reviewed by the System Administrator i. System Programmers are not allowed to have access to the application system's data and programme files in the production environment					
3.2.9.3	All vendor supplied default user IDs shall be deleted or password changed before allowing users to access the computer system	1. For the CAs system software check the following: a. the system software operations and parameters are reviewed and approved by CA management b. System software are tested and its security functionality validated prior to implementation. c. Vendor supplied default IDs are deleted or password changes before users access the computer system d. On sample basis that the versions of system software installed on the computer system and communication devices are regularly updated e. Changes proposed in the system software are appropriately justified and approved by an authorized party (check for recent sample changes performed) f. log of all changes to	Mandatory	IT CA Rules SCHEDULE II 15			

		system software are maintained, completely documented and tested to ensure the desired results g. no standing "Write" access is present for the system libraries h. All "Write" access are logged and reviewed by the System Administrator i. System Programmers are not allowed to have access to the application system's data and programme files in the production environment						
3.2.9.4	Versions of system software installed on the computer system and communication devices shall be regularly updated	1. For the CAs system software check the following: a. the system software operations and parameters are reviewed and approved by CA management b. System software are tested and its security functionality validated prior to implementation. c. Vendor supplied default IDs are deleted or password changes before users access the computer system d. On sample basis that the versions of system software installed on the computer system and communication devices are regularly updated e. Changes proposed in the system software are appropriately justified and approved by an authorized party (check for recent sample changes performed) f. log of all changes to system software are maintained, completely documented and tested to ensure the desired results g. no standing "Write" access is present for	Mandatory	IT CA Rules SCHEDULE II 15				

		the system libraries h. All "Write" access are logged and reviewed by the System Administrator i. System Programmers are not allowed to have access to the application system's data and programme files in the production environment						
3.2.9.5	All changes proposed in the system software must be appropriately justified and approved by an authorized party	1. For the CAs system software check the following: a. the system software operations and parameters are reviewed and approved by CA management b. System software are tested and its security functionality validated prior to implementation. c. Vendor supplied default IDs are deleted or password changes before users access the computer system d. On sample basis that the versions of system software installed on the computer system and communication devices are regularly updated e. Changes proposed in the system software are appropriately justified and approved by an authorized party (check for recent sample changes performed) f. log of all changes to system software are maintained, completely documented and tested to ensure the desired results g. no standing "Write" access is present for the system libraries h. All "Write" access are logged and reviewed by the System Administrator i. System Programmers are not allowed to have access	Mandatory	IT CA Rules SCHEDULE II 15				

		to the application system's data and programme files in the production environment						
3.2.9.6	A log of all changes to system software shall be maintained, completely documented and tested to ensure the desired results.	1. For the CAs system software check the following: a. the system software operations and parameters are reviewed and approved by CA management b. System software are tested and its security functionality validated prior to implementation. c. Vendor supplied default IDs are deleted or password changes before users access the computer system d. On sample basis that the versions of system software installed on the computer system and communication devices are regularly updated e. Changes proposed in the system software are appropriately justified and approved by an authorized party (check for recent sample changes performed) f. log of all changes to system software are maintained, completely documented and tested to ensure the desired results g. no standing "Write" access is present for the system libraries h. All "Write" access are logged and reviewed by the System Administrator i. System Programmers are not allowed to have access to the application system's data and programme files in the production environment	Mandatory	IT CA Rules SCHEDULE II 15				

3.2.9.7	There shall be no standing "Write" access to the system libraries. All "Write" access shall be logged and reviewed by the System Administrator for dubious activities.	1. For the CAs system software check the following: a. the system software operations and parameters are reviewed and approved by CA management b. System software are tested and its security functionality validated prior to implementation. c. Vendor supplied default IDs are deleted or password changes before users access the computer system d. On sample basis that the versions of system software installed on the computer system and communication devices are regularly updated e. Changes proposed in the system software are appropriately justified and approved by an authorized party (check for recent sample changes performed) f. log of all changes to system software are maintained, completely documented and tested to ensure the desired results g. no standing "Write" access is present for the system libraries h. All "Write" access are logged and reviewed by the System Administrator i. System Programmers are not allowed to have access to the application system's data and programme files in the production environment	Mandatory	IT CA Rules SCHEDULE II 15				
3.2.9.8	System Programmers shall not be allowed to have access to the application system's data and programme files in the production environment	1. For the CAs system software check the following: a. the system software operations and parameters are reviewed and approved by CA	Mandatory	IT CA Rules SCHEDULE II 15				

		management b. System software are tested and its security functionality validated prior to implementation. c. Vendor supplied default IDs are deleted or password changes before users access the computer system d. On sample basis that the versions of system software installed on the computer system and communication devices are regularly updated e. Changes proposed in the system software are appropriately justified and approved by an authorized party (check for recent sample changes performed) f. log of all changes to system software are maintained, completely documented and tested to ensure the desired results g. no standing "Write" access is present for the system libraries h. All "Write" access are logged and reviewed by the System Administrator i. System Programmers are not allowed to have access to the application system's data and programme files in the production environment						
3.2.9.9	Procedures to control the use of sensitive system utilities and system programmes that could bypass intended security controls shall be in place and documented. All usage shall be logged and reviewed by the System Administrator and another person independent of System Administrator for dubious activities.	1. Verify documented procedure is present to control the use of sensitive system utilities and system programmes 2. Validate all usage is logged and reviewed by the System Administrator and another person independent of System Administrator for dubious activities	Mandatory	IT CA Rules SCHEDULE II 15				

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3.2.10. Documentation Security

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Documentation Security							
3.2.10.1	All documentation pertaining to application software and sensitive system software and changes made therein shall be updated to the current time, accurately and stored securely. An up-to-date inventory list of all documentation shall be maintained to ensure control and accountability.	1. Conduct a walkthrough of the documentation security process and verify the following: a. Documents pertaining to application software and sensitive system software are updated to the current time, accurately and stored securely. b. up-to-date inventory list of all documentation is maintained c. all documentation and subsequent changes are reviewed and approved by an independent authorized party prior to issue d. Access to application software documentation and sensitive system software documentation is restricted to authorized personnel on a "need-to-use" basis only e. Adequate backups of all documentation are maintained f. Copy of all critical documents and manuals is stored off site g. Documents are classified according to sensitivity h. Clear Desks policy has been implemented	Mandatory	IT CA Rules SCHEDULE II 16			
3.2.10.2	All documentation and subsequent changes shall be reviewed and approved by an independent authorized party prior to issue	1. Conduct a walkthrough of the documentation security process and verify the following: a. Documents	Mandatory	IT CA Rules SCHEDULE II 16			

		pertaining to application software and sensitive system software are updated to the current time, accurately and stored securely. b. up-to-date inventory list of all documentation is maintained c. all documentation and subsequent changes are reviewed and approved by an independent authorized party prior to issue d. Access to application software documentation and sensitive system software documentation is restricted to authorized personnel on a "need-to-use" basis only e. Adequate backups of all documentation are maintained f. Copy of all critical documents and manuals is stored off site g. Documents are classified according to sensitivity h. Clear Desks policy has been implemented						
3.2.10.3	Access to application software documentation and sensitive system software documentation shall be restricted to authorized personnel on a "need-to-use" basis only	1. Conduct a walkthrough of the documentation security process and verify the following: a. Documents pertaining to application software and sensitive system software are updated to the current time, accurately and stored securely. b. up-to-date inventory list of all documentation is maintained c. all documentation and subsequent changes are reviewed and approved by an independent authorized party	Mandatory	IT CA Rules SCHEDULE II 16				

		prior to issue d. Access to application software documentation and sensitive system software documentation is restricted to authorized personnel on a "need-to-use" basis only e. Adequate backups of all documentation are maintained f. Copy of all critical documents and manuals is stored off site g. Documents are classified according to sensitivity h. Clear Desks policy has been implemented						
3.2.10.4	Adequate backups of all documentation shall be maintained and a copy of all critical documentation and manuals shall be stored off-site.	1. Conduct a walkthrough of the documentation security process and verify the following: a. Documents pertaining to application software and sensitive system software are updated to the current time, accurately and stored securely. b. up-to-date inventory list of all documentation is maintained c. all documentation and subsequent changes are reviewed and approved by an independent authorized party prior to issue d. Access to application software documentation and sensitive system software documentation is restricted to authorized personnel on a "need-to-use" basis only e. Adequate backups of all documentation are maintained f. Copy of all critical documents and manuals is stored off	Mandatory	IT CA Rules SCHEDULE II 16				

		site g. Documents are classified according to sensitivity h. Clear Desks policy has been implemented						
3.2.10.5	Documentation shall be classified according to the sensitivity of its contents/ implications	1. Conduct a walkthrough of the documentation security process and verify the following: a. Documents pertaining to application software and sensitive system software are updated to the current time, accurately and stored securely. b. up-to-date inventory list of all documentation is maintained c. all documentation and subsequent changes are reviewed and approved by an independent authorized party prior to issue d. Access to application software documentation and sensitive system software documentation is restricted to authorized personnel on a "need-to-use" basis only e. Adequate backups of all documentation are maintained f. Copy of all critical documents and manuals is stored off site g. Documents are classified according to sensitivity h. Clear Desks policy has been implemented	Mandatory	T CA Rules SCHEDULE II 16				
3.2.10.6	Organizations shall adopt a clean desk policy for papers, diskettes and other documentation in order to reduce the risks of unauthorized access, loss of	1. Conduct a walkthrough of the documentation security process and verify the following: a. Documents	Mandatory	IT CA Rules SCHEDULE II 16				

	and damage to information outside normal working hours.	pertaining to application software and sensitive system software are updated to the current time, accurately and stored securely. b. up-to-date inventory list of all documentation is maintained c. all documentation and subsequent changes are reviewed and approved by an independent authorized party prior to issue d. Access to application software documentation and sensitive system software documentation is restricted to authorized personnel on a "need-to-use" basis only e. Adequate backups of all documentation are maintained f. Copy of all critical documents and manuals is stored off site g. Documents are classified according to sensitivity h. Clear Desks policy has been implemented						
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3.2.11. Firewalls

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Firewalls							
3.2.11.1	All web servers for access by Internet users shall be isolated from other data and host servers.	1. Obtain the network architecture of the CAs system and check the following: a. Networks that operate at varying security levels are isolated from each other by appropriate firewalls b. internal network of the organization is physically and logically isolated from the Internet	Mandatory	IT CA Rules SCHEDULE II 18			

		and any other external connection by a firewall c. Vulnerability assessment is performed on firewalls prior installation and half yearly thereafter d. All web servers for access by Internet users are isolated from other data and host servers.						
3.2.11.2	Intelligent devices generally known as "Firewalls" shall be used to isolate organization's data network with the external network. Firewall device should also be used to limit network connectivity for unauthorized use.	1. Verify firewalls have been installed to protect the data network and to prevent unauthorized use	Mandatory	IT CA Rules SCHEDULE II 18				
3.2.11.3	Networks that operate at varying security levels shall be isolated from each other by appropriate firewalls. The internal network of the organization shall be physically and logically isolated from the Internet and any other external connection by a firewall.	1. Obtain the network architecture of the CAs system and check the following: a. Networks that operate at varying security levels are isolated from each other by appropriate firewalls b. internal network of the organization is physically and logically isolated from the Internet and any other external connection by a firewall c. Vulnerability assessment is performed on firewalls prior installation and half yearly thereafter d. All web servers for access by Internet users are isolated from other data and host servers.	Mandatory	IT CA Rules SCHEDULE II 18				
3.2.11.4	All firewalls shall be subjected to thorough test for vulnerability prior to being put to use and at least half-yearly thereafter.	1. Obtain the network architecture of the CAs system and check the following: a. Networks that	Mandatory	IT CA Rules SCHEDULE II 18				

		operate at varying security levels are isolated from each other by appropriate firewalls b. internal network of the organization is physically and logically isolated from the Internet and any other external connection by a firewall c. Vulnerability assessment is performed on firewalls prior installation and half yearly thereafter d. All web servers for access by Internet users are isolated from other data and host servers.						
3.2.11.5	All web servers for access by Internet users shall be isolated from other data and host servers.	1. Obtain the network architecture of the CAs system and check the following: a. Networks that operate at varying security levels are isolated from each other by appropriate firewalls b. internal network of the organization is physically and logically isolated from the Internet and any other external connection by a firewall c. Vulnerability assessment is performed on firewalls prior installation and half yearly thereafter d. All web servers for access by Internet users are isolated from other data and host servers.	Mandatory	IT CA Rules SCHEDULE II 18				

3.2.12. Connectivity

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Connectivity							

3.2.12.1	Organization shall establish procedure for allowing connectivity of their computer network or computer system to non-organization computer system or networks. The permission to connect other networks and computer system shall be approved by the Network Administrator and documented	1. Conduct a walkthrough to check the implementation of the following: a. procedure is established for allowing connectivity of CA computer network or computer system to non-organization computer system or networks b. permission to connect other networks and computer system are approved by NetworkAdministrator and documented c. All unused connections and network segments are disconnected from active networks d. suitability of new hardware/software particularly the protocol compatibility is assessed before connecting the same to the CA's network e. Internet access is not allowed to database server/ file server or server hosting sensitive data	Mandatory	IT CA Rules SCHEDULE II 19				
3.2.12.2	All unused connections and network segments should be disconnected from active networks. The computer system/personal computer or outside terminal accessing an organization's host system must adhere to the general system security and access control guidelines	1. Conduct a walkthrough to check the implementation of the following: a. procedure is established for allowing connectivity of CA computer network or computer system to non-organization computer system or networks b. permission to connect other networks and computer system are approved by NetworkAdministrator and documented c. All unused connections and network segments are disconnected from active networks d. suitability of new hardware/software particularly the protocol compatibility is assessed before connecting the same to the CA's network e. Internet access is not	Mandatory	IT CA Rules SCHEDULE II 19				

		allowed to database server/ file server or server hosting sensitive data						
3.2.12.3	The suitability of new hardware/software particularly the protocol compatibility should be assessed before connecting the same to the organization's network.	1. Conduct a walkthrough to check the implementation of the following: a. procedure is established for allowing connectivity of CA computer network or computer system to non-organization computer system or networks b. permission to connect other networks and computer system are approved by NetworkAdministrator and documented c. All unused connections and network segments are disconnected from active networks d. suitability of new hardware/software particularly the protocol compatibility is assessed before connecting the same to the CA's network e. Internet access is not allowed to database server/ file server or server hosting sensitive data	Mandatory	IT CA Rules SCHEDULE II 19				
3.2.12.4	No Internet access should be allowed to database server/ file server or server hosting sensitive data.	1. Conduct a walkthrough to check the implementation of the following: a. procedure is established for allowing connectivity of CA computer network or computer system to non-organization computer system or networks b. permission to connect other networks and computer system are approved by NetworkAdministrator and documented c. All unused connections and network segments are disconnected from active networks	Mandatory	IT CA Rules SCHEDULE II 19				

		d. suitability of new hardware/software particularly the protocol compatibility is assessed before connecting the same to the CA's network e. Internet access is not allowed to database server/ file server or server hosting sensitive data						
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3.2.13. Technical Security Controls

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Activation Data							
3.2.13.1	The activation data used to unlock private keys, in conjunction with any other access control, shall have an appropriate level of strength for the keys or data to be protected and shall meet the applicable security policy requirements of the cryptographic module used to store the keys. Activation data may be user selected. If the activation data must be transmitted, it shall be via an appropriately protected channel, and distinct in time and place from the associated cryptographic module	1. Validate the activation data used to unlock private keys have an appropriate level of strength for the keys or data to be protected and meet the applicable security policy requirements of the cryptographic module used to store the keys. 2. Verify data used to unlock private keys is protected from disclosure 3. Check the data used to unlock private keys is protected from disclosure and account gets temporarily locker after a number of failed login attempts 4. Verify activation data is changed whenever the token is re-keyed or returned 5. Validate CA and CSP components are regularly synchronized with IST 6. Check the time derived is used for establishing time of components	Mandatory	X.509 Policy 6.4.1			

		covered in control 3.2.13.4						
3.2.13.2	Data used to unlock private keys shall be protected from disclosure. After a predetermined number of failed login attempts, a facility to temporarily lock the account shall be provided	1. Validate the activation data used to unlock private keys have an appropriate level of strength for the keys or data to be protected and meet the applicable security policy requirements of the cryptographic module used to store the keys. 2. Verify data used to unlock private keys is protected from disclosure 3. Check the data used to unlock private keys is protected from disclosure and account gets temporarily locked after a number of failed login attempts 4. Verify activation data is changed whenever the token is re-keyed or returned 5. Validate CA and CSP components are regularly synchronized with IST 6. Check the time derived is used for establishing time of components covered in control 3.2.13.4	Mandatory	X.509 Policy 6.4.2				
3.2.13.3	CAs, CSPs, shall change the activation data whenever the token is re-keyed or returned from maintenance	1. Validate the activation data used to unlock private keys have an appropriate level of strength for the keys or data to be protected and meet the applicable security policy	Mandatory	X.509 Policy 6.4.3				

		requirements of the cryptographic module used to store the keys. 2. Verify data used to unlock private keys is protected from disclosure 3. Check the data used to unlock private keys is protected from disclosure and account gets temporarily locked after a number of failed login attempts 4. Verify activation data is changed whenever the token is re-keyed or returned 5. Validate CA and CSP components are regularly synchronized with IST 6. Check the time derived is used for establishing time of components covered in control 3.2.13.4						
3.2.13.4	All CA and CSP components shall regularly synchronize with a time service such as Indian Standard Time Service. Time derived from the time service shall be used for establishing the time of: • Initial validity time of a Subscriber's Certificate • Revocation of a Subscriber's Certificate • Posting of CRL updates • OCSP or other CSP responses	1. Validate the activation data used to unlock private keys have an appropriate level of strength for the keys or data to be protected and meet the applicable security policy requirements of the cryptographic module used to store the keys. 2. Verify data used to unlock private keys is protected from disclosure 3. Check the data used to unlock private keys is protected from disclosure and account gets temporarily locked after a number of	Mandatory	X.509 Policy 6.8				

		failed login attempts 4. Verify activation data is changed whenever the token is re-keyed or returned 5. Validate CA and CSP components are regularly synchronized with IST 6. Check the time derived is used for establishing time of components covered in control 3.2.13.4						
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3.2.14. Network Communication Security

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Network Communication Security							
3.2.14.1	All sensitive information on the network shall be protected by using appropriate techniques. The critical network devices such as routers, switches and modems should be protected from physical damage	1. Verify the CA network is protected by using appropriate techniques 2. Check physically critical network devices such as routers, switches and modems are protected from physical damage 3. Check the documented network configuration and inventories and verify these are updated regularly 4. For sample changes in the network check the following: a. Network Administrator's approval was taken b. Changes were documented c. Threats and risk assessment after changes in the network configuration is reviewed d. network operation are monitored for any security irregularity e. formal procedure is in place for identifying and resolving security problems 5. Verify the optical fibre in CAs	Mandatory	IT CA Rules SCHEDULE II 17			

		operational sir is armored cable to ensure security against 6. electromagnetic transmission 7. Validate CA has provisions to use network diagnostic tools on need basis						
3.2.14.2	The network configuration and inventories shall be documented and maintained.	1. Verify the CA network is protected by using appropriate techniques 2. Check physically critical network devices such as routers, switches and modems are protected from physical damage 3. Check the documented network configuration and inventories and verify these are updated regularly 4. For sample changes in the network check the following: a. Network Administrator's approval was taken b. Changes were documented c. Threats and risk assessment after changes in the network configuration is reviewed d. network operation are monitored for any security irregularity e. formal procedure is in place for identifying and resolving security problems 5. Verify the optical fibre in CAs operational sir is armored cable to ensure security against 6. electromagnetic transmission 7. Validate CA has provisions to use network diagnostic tools on need basis	Mandatory	IT CA Rules SCHEDULE II 17				
3.2.14.3	Prior authorization of the Network Administrator shall be obtained for making any changes to network configuration. The changes made in the network configuration shall be documented. The	1. Verify the CA network is protected by using appropriate techniques 2. Check physically critical network devices such as routers, switches and	Mandatory	IT CA Rules SCHEDULE II 17				

	threat and risk assessment of the network after changes in the network configuration shall be reviewed. The network operation shall be monitored for any security irregularity. A formal procedure should be in place for identifying and resolving security problems.	modems are protected from physical damage 3. Check the documented network configuration and inventories and verify these are updated regularly 4. For sample changes in the network check the following: a. Network Administrator's approval was taken b. Changes were documented c. Threats and risk assessment after changes in the network configuration is reviewed d. network operation are monitored for any security irregularity e. formal procedure is in place for identifying and resolving security problems 5. Verify the optical fibre in CAs operational sir is armored cable to ensure security against 6. electromagnetic transmission 7. Validate CA has provisions to use network diagnostic tools on need basis						
3.2.14.4	As far as possible, transmission medium within the Certifying Authority's operational site should be secured against electromagnetic transmission. In this regard, use of Optical Fibre Cable and armored cable may be preferred as transmission media as the case may be.	1. Verify the CA network is protected by using appropriate techniques 2. Check physically critical network devices such as routers, switches and modems are protected from physical damage 3. Check the documented network configuration and inventories and verify these are updated regularly 4. For sample changes in the network check the following: a. Network Administrator's approval was taken b. Changes were documented c. Threats and risk assessment after changes in the network	Mandatory	IT CA Rules SCHEDULE II 17				

		configuration is reviewed d. network operation are monitored for any security irregularity e. formal procedure is in place for identifying and resolving security problems 5. Verify the optical fibre in CAs operational sir is armored cable to ensure security against 6. electromagnetic transmission 7. Validate CA has provisions to use network diagnostic tools on need basis						
3.2.14.5	Network diagnostic tools, e.g., spectrum analyzer, protocol analyzer should be used on a need basis	1. Verify the CA network is protected by using appropriate techniques 2. Check physically critical network devices such as routers, switches and modems are protected from physical damage 3. Check the documented network configuration and inventories and verify these are updated regularly 4. For sample changes in the network check the following: a. Network Administrator's approval was taken b. Changes were documented c. Threats and risk assessment after changes in the network configuration is reviewed d. network operation are monitored for any security irregularity e. formal procedure is in place for identifying and resolving security problems 5. Verify the optical fibre in CAs operational sir is armored cable to ensure security against 6. electromagnetic transmission 7. Validate CA has provisions to use network diagnostic	Mandatory	IT CA Rules SCHEDULE II 17				

		tools on need basis						
Network Administrator								
3.2.14.6	Each organization shall designate a properly trained "Network Administrator" who will be responsible for operation, monitoring security and functioning of the network.	1. Obtain name of the Network administrator and verify the following: a. Network Administrator is properly trained b. Network Administrator regularly undertakes the review of network and also takes adequate measures to provide physical, logical and procedural safeguards for its security c. Network Administrator promptly investigates unusual activity or pattern of access on the computer network d. Network Administrator is alerted in case of a possible breach	Mandatory	IT CA Rules SCHEDULE II 20				
3.2.14.7	Network Administrator shall regularly undertake the review of network and also take adequate measures to provide physical, logical and procedural safeguards for its security. Appropriate follow up of any unusual activity or pattern of access on the computer network shall be investigated promptly by the Network Administrator	1. Obtain name of the Network administrator and verify the following: a. Network Administrator is properly trained b. Network Administrator regularly undertakes the review of network and also takes adequate measures to provide physical, logical and procedural safeguards for its security c. Network Administrator promptly investigates unusual activity or pattern of access on the computer network d. Network Administrator is alerted in case of a possible breach	Mandatory	IT CA Rules SCHEDULE II 20				

3.2.14.8	System must include a mechanism for alerting the Network Administrator of possible breaches in security, e.g., unauthorized access, virus infection and hacking.	1. Obtain name of the Network administrator and verify the following: a. Network Administrator is properly trained b. Network Administrator regularly undertakes the review of network and also takes adequate measures to provide physical, logical and procedural safeguards for its security c. Network Administrator promptly investigates unusual activity or pattern of access on the computer network d. Network Administrator is alerted in case of a possible breach	Mandatory	IT CA Rules SCHEDULE II 20				
3.2.14.9	Secure Network Management System should be implemented to monitor functioning of the computer network. Broadcast of network traffic should be minimized.	1. Verify secure network management system is implemented to monitor functioning of computer network 2. Validate only authorized and legal software are used on the network 3. Check the shared computer systems, network devices used for business applications comply with the requirement established in 3.1.6	Mandatory	IT CA Rules SCHEDULE II 20				
3.2.14.10	Only authorized and legal software shall be used on the network.	1. Verify secure network management system is implemented to monitor functioning of computer network 2. Validate only authorized and legal software are used on the network 3. Check the shared computer systems, network devices used for business applications comply with the requirement established in 3.1.6	Mandatory	IT CA Rules SCHEDULE II 20				

3.2.14.11	Shared computer systems, network devices used for business applications shall comply with the requirement established in System Integrity and Security Measures (Section 3.1.6).	1. Verify secure network management system is implemented to monitor functioning of computer network 2. Validate only authorized and legal software are used on the network 3. Check the shared computer systems, network devices used for business applications comply with the requirement established in 3.1.6	Mandatory	IT CA Rules SCHEDULE II 20				
3.2.14.12	Network connections from the Certifying Authority's system to external networks shall be restricted to only those connections which are essential to facilitate Certifying Authority's functional processes and services. Such network connections to the external network shall be properly secured and monitored regularly.	1. Understand the network connection process and check the following: a. Network connections for CA's system to external networks are restricted to only essential connections b. Network connection are secured and monitored regularly c. Network connections are initiated by the systems performing the functions of generation and management of DSC to connect those systems performing the registration and repository functions but not vice versa	Mandatory	IT CA Rules SCHEDULE III 8				
3.2.14.13	Network connections should be initiated by the systems performing the functions of generation and management of Digital Signature Certificate to connect those systems performing the registration and repository functions but not vice versa. If this is not possible, compensating controls (e.g. use of proxy servers) shall be implemented to protect the systems performing the function of generation and management of Digital Signature Certificate from potential attacks	1. Understand the network connection process and check the following: a. Network connections for CA's system to external networks are restricted to only essential connections b. Network connection are secured and monitored regularly c. Network connections are initiated by the systems performing the functions of generation and management of DSC to connect those systems performing	Mandatory	IT CA Rules SCHEDULE III 8				

		the registration and repository functions but not vice versa						
3.2.14.14	Systems performing the Digital Signature Certification function should be isolated to minimize their exposure to attempts to compromise the confidentiality, integrity and availability of the certification function.	1. Verify Systems performing the DSC function are isolated to minimize their exposure to attempts to compromise the confidentiality, integrity and availability of the certification function 2. communications between the CA's systems connected on a network should be secured (encrypted and digitally signed) 3. Intrusion detection tools are deployed to monitor critical networks and perimeter networks and alert administrators of network intrusions and penetration attempts in a timely manner	Mandatory	T CA Rules SCHEDULE III 8				
3.2.14.15	Communication between the Certifying Authority systems connected on a network shall be secure to ensure confidentiality and integrity of the information. For example, communications between the Certifying Authority's systems connected on a network should be encrypted and digitally signed	1. Verify Systems performing the DSC function are isolated to minimize their exposure to attempts to compromise the confidentiality, integrity and availability of the certification function 2. communications between the CA's systems connected on a network should be secured (encrypted and digitally signed) 3. Intrusion detection tools are deployed to monitor critical networks and perimeter networks and alert administrators of network intrusions and penetration attempts in a timely manner	Mandatory	IT CA Rules SCHEDULE III 8				
3.2.14.16	Intrusion detection tools should be deployed to monitor critical networks and perimeter networks	1. Verify Systems performing the DSC function are isolated to minimize their	Mandatory	IT CA Rules SCHEDULE III 8				

	and alert administrators of network intrusions and penetration attempts in a timely manner	exposure to attempts to compromise the confidentiality, integrity and availability of the certification function 2. communications between the CA's systems connected on a network should be secured (encrypted and digitally signed) 3. Intrusion detection tools are deployed to monitor critical networks and perimeter networks and alert administrators of network intrusions and penetration attempts in a timely manner						
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3.3. Key Management Controls

3.3.1. Key Lifecycle Management Controls

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Key Generation							
3.3.1.1	The Certifying Authority shall use methods, which are approved by the Controller, to verify the identity of a subscriber before issuing or renewing any Public Key Certificate. The subscriber's key pair shall be generated by the subscriber or on a key generation system in the presence of the subscriber.	1. Conduct a walkthrough of the key generation process and check the following a. CA uses methods approved by Controller to verify identity of subscriber before issuing or renewing public key certificate b. Subscriber's key pair is generated by the subscriber or on a key generation system in the presence of the subscriber. c. key generation process generates statistically random key values that are resistant to known attacks d. Generation of CA keys occur within a cryptographic module meeting the applicable requirements of ISO 15782-1/FIPS	Mandatory	IT CA Rules SCHEDULE III 18.1, IT Regulations 3			

		140-2 (or equivalent)/ANSI X9.66 and the business requirements in accordance with the CPS e. CA generates its own key pair in the same cryptographic device in which it will be used f. CA generates keys in compliance with the requirements mentioned in 3.3.1.5 g. CA key generation ceremonies are independently witnessed by internal or external auditors. h. Private key of CA is adequately secured at each phase of its lifecycle 2. Verify the steps by referring a recent key generation activity						
3.3.1.2	The key generation process shall generate statistically random key values that are resistant to known attacks	1. Conduct a walkthrough of the key generation process and check the following a. CA uses methods approved by Controller to verify identity of subscriber before issuing or renewing public key certificate b. Subscriber's key pair is generated by the subscriber or on a key generation system in the presence of the subscriber. c. key generation process generates statistically random key values that are resistant to known attacks d. Generation of CA keys occur within a cryptographic module meeting the applicable	Mandatory	IT CA Rules SCHEDULE III 18.1				

		requirements of ISO 15782-1/FIPS 140-2 (or equivalent)/ANSI X9.66 and the business requirements in accordance with the CPS e. CA generates its own key pair in the same cryptographic device in which it will be used f. CA generates keys in compliance with the requirements mentioned in 3.3.1.5 g. CA key generation ceremonies are independently witnessed by internal or external auditors. h. Private key of CA is adequately secured at each phase of its lifecycle 2. Verify the steps by referring a recent key generation activity						
3.3.1.3	Generation of CA keys shall occur within a cryptographic module meeting the applicable requirements of ISO 15782-1/FIPS 140-2 (or equivalent)/ANSI X9.66 and the business requirements in accordance with the CPS. Such cryptographic devices perform key generation using a random number generator (RNG) or pseudo random number generator (PRNG).	1. Conduct a walkthrough of the key generation process and check the following a. CA uses methods approved by Controller to verify identity of subscriber before issuing or renewing public key certificate b. Subscriber's key pair is generated by the subscriber or on a key generation system in the presence of the subscriber. c. key generation process generates statistically random key values that are resistant to known attacks d. Generation of CA keys occur within a cryptographic	Mandatory	WebTrust 4.1.1. CA Browser Forum 6.1				

		module meeting the applicable requirements of ISO 15782-1/FIPS 140-2 (or equivalent)/ANSI X9.66 and the business requirements in accordance with the CPS e. CA generates its own key pair in the same cryptographic device in which it will be used f. CA generates keys in compliance with the requirements mentioned in 3.3.1.5 g. CA key generation ceremonies are independently witnessed by internal or external auditors. h. Private key of CA is adequately secured at each phase of its lifecycle 2. Verify the steps by referring a recent key generation activity						
3.3.1.4	The CA shall generate its own key pair in the same cryptographic device in which it will be used or the key pair shall be injected directly from the device where it was generated into the device where it will be used.	1. Conduct a walkthrough of the key generation process and check the following a. CA uses methods approved by Controller to verify identity of subscriber before issuing or renewing public key certificate b. Subscriber's key pair is generated by the subscriber or on a key generation system in the presence of the subscriber. c. key generation process generates statistically random key values that are resistant to known attacks d. Generation of CA keys occur	Mandatory	WebTrust 4.1.2				

		within a cryptographic module meeting the applicable requirements of ISO 15782-1/FIPS 140-2 (or equivalent)/ANSI X9.66 and the business requirements in accordance with the CPS e. CA generates its own key pair in the same cryptographic device in which it will be used f. CA generates keys in compliance with the requirements mentioned in 3.3.1.5 g. CA key generation ceremonies are independently witnessed by internal or external auditors. h. Private key of CA is adequately secured at each phase of its lifecycle 2. Verify the steps by referring a recent key generation activity						
3.3.1.5	CA key generation generates keys that: • use a key generation algorithm as disclosed within the CA's CP and/or CPS; • have a key length that is appropriate for the algorithm and for the validity period of the CA certificate as disclosed in the CA's CP and/or CPS. The public key length to be certified by a CA is less than or equal to that of the CA's private signing key; and • take into account requirements on parent and subordinate CA key sizes and have a key size in accordance with the CA's CP and/or CPS.	1. Conduct a walkthrough of the key generation process and check the following a. CA uses methods approved by Controller to verify identity of subscriber before issuing or renewing public key certificate b. Subscriber's key pair is generated by the subscriber or on a key generation system in the presence of the subscriber. c. key generation process generates statistically random key values that are resistant to known attacks	Mandatory	WebTrust 4.1.3				

		d. Generation of CA keys occur within a cryptographic module meeting the applicable requirements of ISO 15782-1/FIPS 140-2 (or equivalent)/ANSI X9.66 and the business requirements in accordance with the CPS e. CA generates its own key pair in the same cryptographic device in which it will be used f. CA generates keys in compliance with the requirements mentioned in 3.3.1.5 g. CA key generation ceremonies are independently witnessed by internal or external auditors. h. Private key of CA is adequately secured at each phase of its lifecycle 2. Verify the steps by referring a recent key generation activity						
3.3.1.6	CA key generation ceremonies are independently witnessed by internal or external auditors. The private key of the Certifying Authority shall be adequately secured at each phase of its life cycle, i.e. key generation, distribution, storage, usage, backup, archival and destruction.	1. Conduct a walkthrough of the key generation process and check the following a. CA uses methods approved by Controller to verify identity of subscriber before issuing or renewing public key certificate b. Subscriber's key pair is generated by the subscriber or on a key generation system in the presence of the subscriber. c. key generation process generates statistically random key values that are	Mandatory	WebTrust 4.1.4, IT Regulations 3				

		resistant to known attacks d. Generation of CA keys occur within a cryptographic module meeting the applicable requirements of ISO 15782-1/FIPS 140-2 (or equivalent)/ANSI X9.66 and the business requirements in accordance with the CPS e. CA generates its own key pair in the same cryptographic device in which it will be used f. CA generates keys in compliance with the requirements mentioned in 3.3.1.5 g. CA key generation ceremonies are independently witnessed by internal or external auditors. h. Private key of CA is adequately secured at each phase of its lifecycle 2. Verify the steps by referring a recent key generation activity						
3.3.1.7	The CA follows a CA key generation script for key generation ceremonies that includes the following: • definition and assignment of participant roles and responsibilities; • management approval for conduct of the key generation ceremony; • specific cryptographic hardware, software and other materials including identifying information, e.g., serial numbers; • specific steps performed during the key generation ceremony • physical security	1. Conduct a walkthrough of the key generation process and check the following a. CA uses methods approved by Controller to verify identity of subscriber before issuing or renewing public key certificate b. Subscriber's key pair is generated by the subscriber or on a key generation system in the presence of the subscriber. c. key generation process generates	Mandatory	WebTrust 4.1.5				

	requirements for the ceremony location (e.g., barriers, access controls and logging controls); • procedures for secure storage of cryptographic hardware and activation materials following the key generation ceremony (e.g., detailing the allocation of materials between storage locations); • sign-off on the script or in a log from participants and witnesses indicating whether key generation ceremony was performed in accordance with the detailed key generation ceremony script; and • notation of any deviations from the key generation ceremony script (e.g., documentation of steps taken to address any technical issues)	statistically random key values that are resistant to known attacks d. Generation of CA keys occur within a cryptographic module meeting the applicable requirements of ISO 15782-1/FIPS 140-2 (or equivalent)/ANSI X9.66 and the business requirements in accordance with the CPS e. CA generates its own key pair in the same cryptographic device in which it will be used f. CA generates keys in compliance with the requirements mentioned in 3.3.1.5 g. CA key generation ceremonies are independently witnessed by internal or external auditors. h. Private key of CA is adequately secured at each phase of its lifecycle 2. Verify the steps by referring a recent key generation activity						
3.3.1.8	The integrity of the hardware/software used for key generation and the interfaces to the hardware/software is tested before production usage.	1. Conduct a walkthrough of the key generation process and check the following a. CA uses methods approved by Controller to verify identity of subscriber before issuing or renewing public key certificate b. Subscriber's key pair is generated by the subscriber or on a key generation system in the presence of the subscriber.	Mandatory	WebTrust 4.1.6				

		c. key generation process generates statistically random key values that are resistant to known attacks d. Generation of CA keys occur within a cryptographic module meeting the applicable requirements of ISO 15782-1/FIPS 140-2 (or equivalent)/ANSI X9.66 and the business requirements in accordance with the CPS e. CA generates its own key pair in the same cryptographic device in which it will be used f. CA generates keys in compliance with the requirements mentioned in 3.3.1.5 g. CA key generation ceremonies are independently witnessed by internal or external auditors. h. Private key of CA is adequately secured at each phase of its lifecycle 2. Verify the steps by referring a recent key generation activity						
Key Storage, Backup, and Recovery								
3.3.1.9	The CA's private (signing and confidentiality) keys are stored and used within a secure cryptographic device meeting the appropriate ISO 15408 protection profile or FIPS 140-1 level 3 recommendations for Cryptographic Modules Validation List based on a risk assessment and the business requirements of the CA and in accordance with the CA's CPS and	1. For the Key Storage, Backup, and Recovery process, verify the following: a. CA's private (signing and confidentiality) keys are stored and used within a secure cryptographic device meeting the appropriate ISO 15408 protection profile or FIPS	Mandatory	WebTrust 4.2.1, IT Regulations 3, CA Browser Forum 6.2.7, X.				

	applicable Certificate Policy(s)	140-1 level 3 recommendations for Cryptographic Modules b. CA private key is generated, stored and used within the same cryptographic module if CA's private keys are not exported from a secure cryptographic module c. When CA's private keys are exported from a secure cryptographic module the requirements mentioned in control 3.3.1.11 d. Backup copies of the CA's private keys are maintained with the same or greater level of security controls as keys currently in use e. recovery of the CA's keys is carried out in as secure a manner as the backup process, using multi-person control.						
3.3.1.10	If the CA's private keys are not exported from a secure cryptographic module, then the CA private key is generated, stored and used within the same cryptographic module.	1. For the Key Storage, Backup, and Recovery process, verify the following: a. CA's private (signing and confidentiality) keys are stored and used within a secure cryptographic device meeting the appropriate ISO 15408 protection profile or or FIPS 140-1 level 3 recommendations for Cryptographic Modules b. CA private key is generated, stored and used within the same cryptographic module if CA's private keys are not	Mandatory	WebTrust 4.2.2				

		exported from a secure cryptographic module c. When CA's private keys are exported from a secure cryptographic module the requirements mentioned in control 3.3.1.11 d. Backup copies of the CA's private keys are maintained with the same or greater level of security controls as keys currently in use e. recovery of the CA's keys is carried out in as secure a manner as the backup process, using multi-person control.						
3.3.1.11	If the CA's private keys are exported from a secure cryptographic module to secure storage for purposes of offline processing or backup and recovery, then they are exported within a secure key management scheme that may include any of the following: • as cipher-text using a key which is appropriately secured; • as encrypted key fragments using multiple control and split knowledge/ownership; or • In another secure cryptographic module such as a key transportation device using multiple control.	1. For the Key Storage, Backup, and Recovery process, verify the following: a. CA's private (signing and confidentiality) keys are stored and used within a secure cryptographic device meeting the appropriate ISO 15408 protection profile or or FIPS 140-1 level 3 recommendations for Cryptographic Modules b. CA private key is generated, stored and used within the same cryptographic module if CA's private keys are not exported from a secure cryptographic module c. When CA's private keys are exported from a secure cryptographic module the requirements	Mandatory	WebTrust 4.2.3				

		mentioned in control 3.3.1.11 d. Backup copies of the CA's private keys are maintained with the same or greater level of security controls as keys currently in use e. recovery of the CA's keys is carried out in as secure a manner as the backup process, using multi-person control.						
3.3.1.12	Backup copies of the CA's private keys are subject to the same or greater level of security controls as keys currently in use. The recovery of the CA's keys is carried out in as secure a manner as the backup process, using multi-person control.	1. For the Key Storage, Backup, and Recovery process, verify the following: a. CA's private (signing and confidentiality) keys are stored and used within a secure cryptographic device meeting the appropriate ISO 15408 protection profile or or FIPS 140-1 level 3 recommendations for Cryptographic Modules b. CA private key is generated, stored and used within the same cryptographic module if CA's private keys are not exported from a secure cryptographic module c. When CA's private keys are exported from a secure cryptographic module the requirements mentioned in control 3.3.1.11 d. Backup copies of the CA's private keys are maintained with the same or greater level of security controls as keys currently in use e. recovery of the	Mandatory	WebTrust 4.2.4				

		CA's keys is carried out in as secure a manner as the backup process, using multi-person control.						
Key Distribution								
3.3.1.13	Keys shall be transferred from the key generation system to the storage device (if the keys are not stored on the key generation system) using a secure mechanism that ensures confidentiality and integrity	1. Verify the key is transferred from the key generation system to the storage device using a secure mechanism that ensures confidentiality and integrity	Mandatory	IT CA Rules SCHEDULE III 18.2				
3.3.1.14	The initial distribution mechanism for the CA's public key is controlled and initially distributed within a Certificate using one of the following methods: • machine readable media (e.g., smart card, flash drive, CD ROM) from an authenticated source; • embedding in an entity's cryptographic module; or • other secure means that ensure authenticity and integrity.	1. Validate the initial distribution mechanism for the CA's public key is controlled and initially distributed within a Certificate using one of the methods mentioned in control description	Mandatory	WebTrust 4.3.2				
3.3.1.15	The CA's public key is changed (rekeyed) periodically according to the requirements of the CPS with advance notice provided to avoid disruption of the CA services	1. Verify by checking documentation of frequency of change of public key and validate CA's public key is changed (rekeyed) periodically according to the requirements	Mandatory	WebTrust 4.3.3				
3.3.1.16	The subsequent distribution mechanism for the CA's public key is controlled in accordance with the CA's disclosed business practices.	1. On a sample basis verify the distribution mechanism for the CA's public key is controlled in accordance with the CA's disclosed business practices	Mandatory	WebTrust 4.3.4				
3.3.1.17	If an entity already has an	1. Validate on a	Mandatory	WebTrust 4.3.5				

	authenticated copy of the CA's public key, a new CA public key is distributed using one of the following methods: • direct electronic transmission from the CA; • placing into a remote cache or directory; • loading into a cryptographic module; or • any of the methods used for initial distribution	sample basis if an entity has an authenticated copy of CAs public key, a new CA public key is distributed using one of the methods mentioned in control description						
3.3.1.18	The CA provides a mechanism for validating the authenticity and integrity of the CA's public keys. The CA provides a mechanism for validating the authenticity and integrity of the CA's public keys.	1. Verify CA has a mechanism in place for validating the authenticity and integrity of the CA's public keys	Mandatory	WebTrust 4.3.6				
Key Usage								
3.3.1.19	A system and software integrity check shall be performed prior to Certifying Authority's key loading. Custody of and access to the Certifying Authority's keys shall be under split control. In particular, Certifying Authority's key loading shall be performed under split control	1. For the CAs key usage process, check the following: - A system and software integrity check is performed prior to Certifying Authority's key loading - Custody of and access to the CA's keys is under split control - activation of the CA private signing key is performed using multi-party control (i.e., m of n) with a minimum value of m (e.g., m greater than 2 for Root CAs) - activation of the CA private key is performed using multi-factor authentication - CA signing key(s) used for generating certificates and/or issuing revocation status information, are not used for any other purpose - CA follows established	Mandatory	IT CA Rules SCHEDULE III 18.4				

		procedures in event of a compromise of private key of the CA for immediate revocation of the affected subscribers' certificates. g. CA ceases to use a key pair at the end of the key pair's defined operational lifetime						
3.3.1.20	The activation of the CA private signing key shall be performed using multi-party control (i.e., m of n) with a minimum value of m (e.g., m greater than 2 for Root CAs).	1. For the CAs key usage process, check the following: a. A system and software integrity check is performed prior to Certifying Authority's key loading b. Custody of and access to the CA's keys is under split control c. activation of the CA private signing key is performed using multi-party control (i.e., m of n) with a minimum value of m (e.g., m greater than 2 for Root CAs) d. activation of the CA private key is performed using multi-factor authentication e. CA signing key(s) used for generating certificates and/or issuing revocation status information, are not used for any other purpose f. CA follows established procedures in event of a compromise of private key of the CA for immediate revocation of the affected subscribers' certificates. g. CA ceases to use a key pair at the end of the key pair's defined operational lifetime	Mandatory	WebTrust 4.4.1				

3.3.1.21	The activation of the CA private key shall be performed using multi-factor authentication (e.g., smart card and password, biometric and password, etc.).	1. For the CAs key usage process, check the following: a. A system and software integrity check is performed prior to Certifying Authority's key loading b. Custody of and access to the CA's keys is under split control c. activation of the CA private signing key is performed using multi-party control (i.e., m of n) with a minimum value of m (e.g., m greater than 2 for Root CAs) d. activation of the CA private key is performed using multi-factor authentication e. CA signing key(s) used for generating certificates and/or issuing revocation status information, are not used for any other purpose f. CA follows established procedures in event of a compromise of private key of the CA for immediate revocation of the affected subscribers' certificates. g. CA ceases to use a key pair at the end of the key pair's defined operational lifetime	Mandatory	WebTrust 4.4.2				
3.3.1.22	CA signing key(s) used for generating certificates and/or issuing revocation status information, shall not be used for any other purpose.	1. For the CAs key usage process, check the following: a. A system and software integrity check is performed prior to Certifying Authority's key loading b. Custody of and access to the CA's keys is under split control	Mandatory	WebTrust 4.4.3				

		c. activation of the CA private signing key is performed using multi-party control (i.e., m of n) with a minimum value of m (e.g., m greater than 2 for Root CAs) d. activation of the CA private key is performed using multi-factor authentication e. CA signing key(s) used for generating certificates and/or issuing revocation status information, are not used for any other purpose f. CA follows established procedures in event of a compromise of private key of the CA for immediate revocation of the affected subscribers' certificates. g. CA ceases to use a key pair at the end of the key pair's defined operational lifetime						
3.3.1.23	In the event of a compromise of the private key the Certifying Authority shall follow the established procedures for immediate revocation of the affected subscribers' certificates. The CA ceases to use a key pair at the end of the key pair's defined operational lifetime or when the compromise of the private key is known or suspected	1. For the CAs key usage process, check the following: a. A system and software integrity check is performed prior to Certifying Authority's key loading b. Custody of and access to the CA's keys is under split control c. activation of the CA private signing key is performed using multi-party control (i.e., m of n) with a minimum value of m (e.g., m greater than 2 for Root CAs) d. activation of the CA private key is performed using multi-factor authentication	Mandatory	IT Regulations 3, WebTrust 4.4.4				

		e. CA signing key(s) used for generating certificates and/or issuing revocation status information, are not used for any other purpose f. CA follows established procedures in event of a compromise of private key of the CA for immediate revocation of the affected subscribers' certificates. g. CA ceases to use a key pair at the end of the key pair's defined operational lifetime						
3.3.1.24	An annual review is required by the PA on key lengths to determine the appropriate key usage period with recommendations acted upon.	1. For the CAs key usage process, check the following: a. A system and software integrity check is performed prior to Certifying Authority's key loading b. Custody of and access to the CA's keys is under split control c. activation of the CA private signing key is performed using multi-party control (i.e., m of n) with a minimum value of m (e.g., m greater than 2 for Root CAs) d. activation of the CA private key is performed using multi-factor authentication e. CA signing key(s) used for generating certificates and/or issuing revocation status information, are not used for any other purpose f. CA follows established procedures in event of a compromise of private key of the	Mandatory	WebTrust 4.4.5				

		CA for immediate revocation of the affected subscribers' certificates. g. CA ceases to use a key pair at the end of the key pair's defined operational lifetime						
3.3.1.25	Private Keys corresponding to Root Certificates MUST NOT be used to sign Certificates except in the following cases: • Self-signed Certificates to represent the Root CA itself; • Certificates for Subordinate CAs and Cross Certificates; • Certificates for infrastructure purposes (administrative role certificates, internal CA operational device certificates); and • Certificates for OCSP Response verification	1. Verify the private keys corresponding to Root Certificate are not used to sign Certificates except in the cases mentioned in control description	Mandatory	CA Browser Forum 6.1.7, X.509 Policy 6.1.7				
Key Archival								
3.3.1.26	Archived CA keys are subject to the same or greater level of security controls as keys currently in use.	1. For the key archival process, check the following: a. Archived CA keys are subject to the same or greater level of security controls as keys currently in use b. For sample archived CA keys, verify, these were destroyed at the end of the archive period using dual control in a physically secure site. c. Subscriber private signature keys is not archived by the CA d. Archived keys are only accessed where historical evidence requires validation e. Archived keys are recovered for the shortest possible time period technically	Mandatory	WebTrust 4.5.1 CA Browser Forum 6.3.1, X.509 Policy 6.3.1				

		permissible to meet business requirements f. Archived keys are periodically verified to ensure that they are properly destroyed at the end of the archive period						
3.3.1.27	All archived CA keys are destroyed at the end of the archive period using dual control in a physically secure site. Subscriber private signature keys shall not be archived by the CA	1. For the key archival process, check the following: a. Archived CA keys are subject to the same or greater level of security controls as keys currently in use b. For sample archived CA keys, verify, these were destroyed at the end of the archive period using dual control in a physically secure site. c. Subscriber private signature keys is not archived by the CA d. Archived keys are only accessed where historical evidence requires validation e. Archived keys are recovered for the shortest possible time period technically permissible to meet business requirements f. Archived keys are periodically verified to ensure that they are properly destroyed at the end of the archive period	Mandatory	WebTrust 4.5.2, X.509 Policy 6.2.5				
3.3.1.28	Archived keys are only accessed where historical evidence requires validation. Control processes are required to ensure the integrity of the CA systems and the key sets.	1. For the key archival process, check the following: a. Archived CA keys are subject to the same or greater level of security controls as keys currently in use b. For sample archived CA keys,	Mandatory	WebTrust 4.5.3				

		verify, these were destroyed at the end of the archive period using dual control in a physically secure site. c. Subscriber private signature keys is not archived by the CA d. Archived keys are only accessed where historical evidence requires validation e. Archived keys are recovered for the shortest possible time period technically permissible to meet business requirements f. Archived keys are periodically verified to ensure that they are properly destroyed at the end of the archive period						
3.3.1.29	Archived keys are recovered for the shortest possible time period technically permissible to meet business requirements.	1. For the key archival process, check the following: a. Archived CA keys are subject to the same or greater level of security controls as keys currently in use b. For sample archived CA keys, verify, these were destroyed at the end of the archive period using dual control in a physically secure site. c. Subscriber private signature keys is not archived by the CA d. Archived keys are only accessed where historical evidence requires validation e. Archived keys are recovered for the shortest possible time period technically permissible to meet business requirements f. Archived keys	Mandatory	WebTrust 4.5.4				

		are periodically verified to ensure that they are properly destroyed at the end of the archive period						
3.3.1.30	Archived keys are periodically verified to ensure that they are properly destroyed at the end of the archive period.	1. For the key archival process, check the following: a. Archived CA keys are subject to the same or greater level of security controls as keys currently in use b. For sample archived CA keys, verify, these were destroyed at the end of the archive period using dual control in a physically secure site. c. Subscriber private signature keys is not archived by the CA d. Archived keys are only accessed where historical evidence requires validation e. Archived keys are recovered for the shortest possible time period technically permissible to meet business requirements f. Archived keys are periodically verified to ensure that they are properly destroyed at the end of the archive period	Mandatory	WebTrust 4.5.5				
Key Destruction								
3.3.1.31	Upon termination of use of a private key, all copies of the private key in computer memory and shared disk space must be securely destroyed by over-writing. Private key destruction procedures must be described in the Certification Practice Statement or other publicly available document.	1. Conduct walkthrough of the key destruction process and check the following: a. Upon termination of use of a private key, all copies of the private key in computer memory and shared disk space are securely destroyed by over-	Mandatory	IT CA Rules SCHEDULE III 20				

		writing b. Private key destruction procedures must be described in the CPS c. CA's private keys is destroyed only if the business purpose or application has ceased to have value or legal obligations have expired d. Authorization to destroy a CA private key and how the CA's private key is destroyed is done in accordance with CA's CPS 2. Validate if in case a secure cryptographic device is accessible and known to be permanently removed from service, all CA private keys stored within the device that have ever been or potentially could be used for any cryptographic purpose are destroyed. 3. Validate if a CA cryptographic device is being permanently removed from service, then any key contained within the device that has been used for any cryptographic purpose is erased from the device. 4. Verify if the CA cryptographic device case is intended to provide tamper-evident characteristics and the device is being permanently removed from service, then the case is destroyed 5. Validate backup or additional copies of CA keys that no longer serve a valid business purpose						
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		are destroyed in accordance with the CA's disclosed business practices 6. Verify the CA follows a CA key destruction script for key destruction ceremonies that includes the requirements covered in control 3.3.1.39 7. For a recent key destruction ceremony, verify name of the auditor who independently witnessed the process						
3.3.1.32	The CA's private keys shall be destroyed only if the business purpose or application has ceased to have value or legal obligations have expired as disclosed within the CA's CPS.	1. Conduct walkthrough of the key destruction process and check the following: a. Upon termination of use of a private key, all copies of the private key in computer memory and shared disk space are securely destroyed by overwriting b. Private key destruction procedures must be described in the CPS c. CA's private keys is destroyed only if the business purpose or application has ceased to have value or legal obligations have expired d. Authorization to destroy a CA private key and how the CA's private key is destroyed is done in accordance with CA's CPS 2. Validate if in case a secure cryptographic device is accessible and known to be permanently removed from service, all CA	Mandatory	WebTrust 4.6.1				

		private keys stored within the device that have ever been or potentially could be used for any cryptographic purpose are destroyed. 3. Validate if a CA cryptographic device is being permanently removed from service, then any key contained within the device that has been used for any cryptographic purpose is erased from the device. 4. Verify if the CA cryptographic device case is intended to provide tamper-evident characteristics and the device is being permanently removed from service, then the case is destroyed 5. Validate backup or additional copies of CA keys that no longer serve a valid business purpose are destroyed in accordance with the CA's disclosed business practices 6. Verify the CA follows a CA key destruction script for key destruction ceremonies that includes the requirements covered in control 3.3.1.39 7. For a recent key destruction ceremony, verify name of the auditor who independently witnessed the process						
3.3.1.33	The CA's private keys shall be destroyed only if the business purpose or application has ceased to have value or legal obligations have expired as disclosed within the	1. Conduct walkthrough of the key destruction process and check the following: a. Upon termination of use of a private	Mandatory	WebTrust 4.6.1				

	CA's CPS.	key, all copies of the private key in computer memory and shared disk space are securely destroyed by over-writing b. Private key destruction procedures must be described in the CPS c. CA's private keys is destroyed only if the business purpose or application has ceased to have value or legal obligations have expired d. Authorization to destroy a CA private key and how the CA's private key is destroyed is done in accordance with CA's CPS 2. Validate if in case a secure cryptographic device is accessible and known to be permanently removed from service, all CA private keys stored within the device that have ever been or potentially could be used for any cryptographic purpose are destroyed. 3. Validate if a CA cryptographic device is being permanently removed from service, then any key contained within the device that has been used for any cryptographic purpose is erased from the device. 4. Verify if the CA cryptographic device case is intended to provide tamper-evident characteristics and the device is being permanently removed from service, then the						
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		case is destroyed 5. Validate backup or additional copies of CA keys that no longer serve a valid business purpose are destroyed in accordance with the CA's disclosed business practices 6. Verify the CA follows a CA key destruction script for key destruction ceremonies that includes the requirements covered in control 3.3.1.39 7. For a recent key destruction ceremony, verify name of the auditor who independently witnessed the process						
3.3.1.34	Authorization to destroy a CA private key and how the CA's private key is destroyed (e.g., token surrender, token destruction, or key overwrite) are limited in accordance with the CA's CPS.	1. Conduct walkthrough of the key destruction process and check the following: a. Upon termination of use of a private key, all copies of the private key in computer memory and shared disk space are securely destroyed by overwriting b. Private key destruction procedures must be described in the CPS c. CA's private keys is destroyed only if the business purpose or application has ceased to have value or legal obligations have expired d. Authorization to destroy a CA private key and how the CA's private key is destroyed is done in accordance with CA's CPS 2. Validate if in	Mandatory	WebTrust 4.6.2				

		case a secure cryptographic device is accessible and known to be permanently removed from service, all CA private keys stored within the device that have ever been or potentially could be used for any cryptographic purpose are destroyed. 3. Validate if a CA cryptographic device is being permanently removed from service, then any key contained within the device that has been used for any cryptographic purpose is erased from the device. 4. Verify if the CA cryptographic device case is intended to provide tamper-evident characteristics and the device is being permanently removed from service, then the case is destroyed 5. Validate backup or additional copies of CA keys that no longer serve a valid business purpose are destroyed in accordance with the CA's disclosed business practices 6. Verify the CA follows a CA key destruction script for key destruction ceremonies that includes the requirements covered in control 3.3.1.39 7. For a recent key destruction ceremony, verify name of the auditor who independently witnessed the process						
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3.3.1.35	All copies and fragments of the CA's private key are destroyed at the end of the key pair life cycle in a manner such that the private key cannot be retrieved.	1. Conduct walkthrough of the key destruction process and check the following: a. Upon termination of use of a private key, all copies of the private key in computer memory and shared disk space are securely destroyed by over-writing b. Private key destruction procedures must be described in the CPS c. CA's private keys is destroyed only if the business purpose or application has ceased to have value or legal obligations have expired d. Authorization to destroy a CA private key and how the CA's private key is destroyed is done in accordance with CA's CPS 2. Validate if in case a secure cryptographic device is accessible and known to be permanently removed from service, all CA private keys stored within the device that have ever been or potentially could be used for any cryptographic purpose are destroyed. 3. Validate if a CA cryptographic device is being permanently removed from service, then any key contained within the device that has been used for any cryptographic purpose is erased from the device. 4. Verify if the CA cryptographic device case is	Mandatory	WebTrust 4.6.3				
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		intended to provide tamper-evident characteristics and the device is being permanently removed from service, then the case is destroyed 5. Validate backup or additional copies of CA keys that no longer serve a valid business purpose are destroyed in accordance with the CA's disclosed business practices 6. Verify the CA follows a CA key destruction script for key destruction ceremonies that includes the requirements covered in control 3.3.1.39 7. For a recent key destruction ceremony, verify name of the auditor who independently witnessed the process						
3.3.1.36	If a secure cryptographic device is accessible and known to be permanently removed from service, all CA private keys stored within the device that have ever been or potentially could be used for any cryptographic purpose are destroyed.	1. Conduct walkthrough of the key destruction process and check the following: a. Upon termination of use of a private key, all copies of the private key in computer memory and shared disk space are securely destroyed by over-writing b. Private key destruction procedures must be described in the CPS c. CA's private keys is destroyed only if the business purpose or application has ceased to have value or legal obligations have expired d. Authorization to destroy a CA	Mandatory	WebTrust 4.6.4				

		private key and how the CA's private key is destroyed is done in accordance with CA's CPS 2. Validate if in case a secure cryptographic device is accessible and known to be permanently removed from service, all CA private keys stored within the device that have ever been or potentially could be used for any cryptographic purpose are destroyed. 3. Validate if a CA cryptographic device is being permanently removed from service, then any key contained within the device that has been used for any cryptographic purpose is erased from the device. 4. Verify if the CA cryptographic device case is intended to provide tamper-evident characteristics and the device is being permanently removed from service, then the case is destroyed 5. Validate backup or additional copies of CA keys that no longer serve a valid business purpose are destroyed in accordance with the CA's disclosed business practices 6. Verify the CA follows a CA key destruction script for key destruction ceremonies that includes the requirements covered in control 3.3.1.39 7. For a recent key destruction ceremony, verify							
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		name of the auditor who independently witnessed the process						
3.3.1.37	If a CA cryptographic device is being permanently removed from service, then any key contained within the device that has been used for any cryptographic purpose is erased from the device.	1. Conduct walkthrough of the key destruction process and check the following: a. Upon termination of use of a private key, all copies of the private key in computer memory and shared disk space are securely destroyed by over-writing b. Private key destruction procedures must be described in the CPS c. CA's private keys is destroyed only if the business purpose or application has ceased to have value or legal obligations have expired d. Authorization to destroy a CA private key and how the CA's private key is destroyed is done in accordance with CA's CPS 2. Validate if in case a secure cryptographic device is accessible and known to be permanently removed from service, all CA private keys stored within the device that have ever been or potentially could be used for any cryptographic purpose are destroyed. 3. Validate if a CA cryptographic device is being permanently removed from service, then any key contained within the device that has been used	Mandatory	WebTrust 4.6.5				

		for any cryptographic purpose is erased from the device. 4. Verify if the CA cryptographic device case is intended to provide tamper-evident characteristics and the device is being permanently removed from service, then the case is destroyed 5. Validate backup or additional copies of CA keys that no longer serve a valid business purpose are destroyed in accordance with the CA's disclosed business practices 6. Verify the CA follows a CA key destruction script for key destruction ceremonies that includes the requirements covered in control 3.3.1.39 7. For a recent key destruction ceremony, verify name of the auditor who independently witnessed the process						
3.3.1.38	If a CA cryptographic device case is intended to provide tamper-evident characteristics and the device is being permanently removed from service, then the case is destroyed	1. Conduct walkthrough of the key destruction process and check the following: a. Upon termination of use of a private key, all copies of the private key in computer memory and shared disk space are securely destroyed by over-writing b. Private key destruction procedures must be described in the CPS c. CA's private keys are destroyed only if the business purpose or	Mandatory	WebTrust 4.6.6				

		application has ceased to have value or legal obligations have expired d. Authorization to destroy a CA private key and how the CA's private key is destroyed is done in accordance with CA's CPS 2. Validate if in case a secure cryptographic device is accessible and known to be permanently removed from service, all CA private keys stored within the device that have ever been or potentially could be used for any cryptographic purpose are destroyed. 3. Validate if a CA cryptographic device is being permanently removed from service, then any key contained within the device that has been used for any cryptographic purpose is erased from the device. 4. Verify if the CA cryptographic device case is intended to provide tamper-evident characteristics and the device is being permanently removed from service, then the case is destroyed 5. Validate backup or additional copies of CA keys that no longer serve a valid business purpose are destroyed in accordance with the CA's disclosed business practices 6. Verify the CA follows a CA key destruction script for key destruction ceremonies that						
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		includes the requirements covered in control 3.3.1.39 7. For a recent key destruction ceremony, verify name of the auditor who independently witnessed the process						
3.3.1.39	Backup or additional copies of CA keys that no longer serve a valid business purpose are destroyed in accordance with the CA's disclosed business practices.	1. Conduct walkthrough of the key destruction process and check the following: a. Upon termination of use of a private key, all copies of the private key in computer memory and shared disk space are securely destroyed by overwriting b. Private key destruction procedures must be described in the CPS c. CA's private keys is destroyed only if the business purpose or application has ceased to have value or legal obligations have expired d. Authorization to destroy a CA private key and how the CA's private key is destroyed is done in accordance with CA's CPS 2. Validate if in case a secure cryptographic device is accessible and known to be permanently removed from service, all CA private keys stored within the device that have ever been or potentially could be used for any cryptographic purpose are destroyed. 3. Validate if a CA cryptographic	Mandatory	WebTrust 4.6.7				

		device is being permanently removed from service, then any key contained within the device that has been used for any cryptographic purpose is erased from the device. 4. Verify if the CA cryptographic device case is intended to provide tamper-evident characteristics and the device is being permanently removed from service, then the case is destroyed 5. Validate backup or additional copies of CA keys that no longer serve a valid business purpose are destroyed in accordance with the CA's disclosed business practices 6. Verify the CA follows a CA key destruction script for key destruction ceremonies that includes the requirements covered in control 3.3.1.39 7. For a recent key destruction ceremony, verify name of the auditor who independently witnessed the process						
3.3.1.40	The CA follows a CA key destruction script for key destruction ceremonies that includes the following: • definition and assignment of participant roles and responsibilities; • management approval for conduct of the key destruction ceremony; • specific cryptographic hardware, software and other materials including identifying information, e.g., serial numbers, that contain the CA key	1. Conduct walkthrough of the key destruction process and check the following: a. Upon termination of use of a private key, all copies of the private key in computer memory and shared disk space are securely destroyed by over-writing b. Private key destruction	Mandatory	WebTrust 4.6.8				

	copies to be destroyed; • specific steps performed during the key destruction ceremony, including; o HSM and/or cryptographic hardware zeroisation/initialisation o HSM and/or cryptographic hardware physical destruction o Deletion of any encrypted files containing the CA key or fragments thereof • physical security requirements for the ceremony location(e.g., barriers, access controls and logging controls); • procedures for secure storage of cryptographic hardware and any associated activation materials following the key destruction ceremony pending their disposal or additional destruction • sign-off on the script or in a log from participants and witnesses indicating whether the key destruction ceremony was performed in accordance with the detailed key destruction ceremony script; and • notation of any deviations from the key destruction ceremony script (e.g., documentation of steps taken to address any technical issues).	procedures must be described in the CPS c. CA's private keys is destroyed only if the business purpose or application has ceased to have value or legal obligations have expired d. Authorization to destroy a CA private key and how the CA's private key is destroyed is done in accordance with CA's CPS 2. Validate if in case a secure cryptographic device is accessible and known to be permanently removed from service, all CA private keys stored within the device that have ever been or potentially could be used for any cryptographic purpose are destroyed. 3. Validate if a CA cryptographic device is being permanently removed from service, then any key contained within the device that has been used for any cryptographic purpose is erased from the device. 4. Verify if the CA cryptographic device case is intended to provide tamper-evident characteristics and the device is being permanently removed from service, then the case is destroyed 5. Validate backup or additional copies of CA keys that no longer serve a valid business purpose are destroyed in accordance with the						
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		CA's disclosed business practices 6. Verify the CA follows a CA key destruction script for key destruction ceremonies that includes the requirements covered in control 3.3.1.39 7. For a recent key destruction ceremony, verify name of the auditor who independently witnessed the process						
3.3.1.41	CA key destruction ceremonies are independently witnessed by internal or external auditors.	1. Conduct walkthrough of the key destruction process and check the following: a. Upon termination of use of a private key, all copies of the private key in computer memory and shared disk space are securely destroyed by overwriting b. Private key destruction procedures must be described in the CPS c. CA's private keys is destroyed only if the business purpose or application has ceased to have value or legal obligations have expired d. Authorization to destroy a CA private key and how the CA's private key is destroyed is done in accordance with CA's CPS 2. Validate if in case a secure cryptographic device is accessible and known to be permanently removed from service, all CA private keys stored within the device that have ever been	Mandatory	WebTrust 4.6.9				

		or potentially could be used for any cryptographic purpose are destroyed. 3. Validate if a CA cryptographic device is being permanently removed from service, then any key contained within the device that has been used for any cryptographic purpose is erased from the device. 4. Verify if the CA cryptographic device case is intended to provide tamper-evident characteristics and the device is being permanently removed from service, then the case is destroyed 5. Validate backup or additional copies of CA keys that no longer serve a valid business purpose are destroyed in accordance with the CA's disclosed business practices 6. Verify the CA follows a CA key destruction script for key destruction ceremonies that includes the requirements covered in control 3.3.1.39 7. For a recent key destruction ceremony, verify name of the auditor who independently witnessed the process						
Key Compromise								
3.3.1.42	A procedure shall be pre-established to handle cases where a compromise of the Certifying Authority's Digital Signature private key has occurred.	1. Conduct walkthrough of the key compromise process and check the following for sample incidents: a. Procedure has been pre-	Mandatory	IT CA Rules SCHEDULE III 21.3, IT Regulations 3, X.509 Polic				

		established to handle cases where a compromise of the CA's Digital Signature private key has occurred. b. CA immediately revokes all affected subscriber DSC c. CAs public keys are archived permanently to facilitate audit d. Archives of CA's publickeys are protected from unauthorized modification						
3.3.1.43	The Certifying Authority should immediately revoke the affected keys and Digital Signature Certificates in the case of Subscriber private key compromise.	1. Conduct walkthrough of the key compromise process and check the following for sample incidents: a. Procedure has been pre-established to handle cases where a compromise of the CA's Digital Signature private key has occurred. b. CA immediately revokes all affected subscriber DSC c. CAs public keys are archived permanently to facilitate audit d. Archives of CA's publickeys are protected from unauthorized modification	Mandatory	IT CA Rules SCHEDULE III 21.3, IT Regulations 3				
3.3.1.44	The Certifying Authority's public keys shall be archived permanently to facilitate audit or investigation requirements	1. Conduct walkthrough of the key compromise process and check the following for sample incidents: a. Procedure has been pre-established to handle cases where a compromise of the CA's Digital Signature private key has occurred. b. CA immediately revokes all affected	Mandatory	IT CA Rules SCHEDULE III 21.3				

		subscriber DSC c. CAs public keys are archived permanently to facilitate audit d. Archives of CA's publickeys are protected from unauthorized modification						
3.3.1.45	Archives of Certifying Authority's public keys shall be protected from unauthorized modification	1. Conduct walkthrough of the key compromise process and check the following for sample incidents: a. Procedure has been pre-established to handle cases where a compromise of the CA's Digital Signature private key has occurred. b. CA immediately revokes all affected subscriber DSC c. CAs public keys are archived permanently to facilitate audit d. Archives of CA's publickeys are protected from unauthorized modification	Mandatory	IT CA Rules SCHEDULE III 21.3				
3.3.1.46	The CA's business continuity plans address the compromise or suspected compromise of a CA's private keys as a disaster.	1. Verify the CA's business continuity plans address the compromise or suspected compromise of a CA's private keys as a disaster 2. Validate disaster recovery procedures include the revocation and reissuance of all certificates that were signed with that CA's private key, in the event of the compromise or suspected compromise 3. Check the recovery procedures used if the CA's private	Mandatory	WebTrust 4.7.1				

		key covers the requirements mentioned in control 3.3.1.47						
3.3.1.47	Disaster recovery procedures include the revocation and reissuance of all certificates that were signed with that CA's private key, in the event of the compromise or suspected compromise of a CA's private signing key.	1. Verify the CA's business continuity plans address the compromise or suspected compromise of a CA's private keys as a disaster 2. Validate disaster recovery procedures include the revocation and reissuance of all certificates that were signed with that CA's private key, in the event of the compromise or suspected compromise 3. Check the recovery procedures used if the CA's private key covers the requirements mentioned in control 3.3.1.47	Mandatory	WebTrust 4.7.2				
3.3.1.48	The recovery procedures used if the CA's private key is compromised include the following actions: • how secure key usage in the environment is re-established; • how the CA's old public key is revoked; • how affected parties are notified (e.g., impacted CAs, Repositories, Subscribers and CVSPs); • how the CA's new public key is provided to the end entities and Relying Parties together with the mechanism for their authentication; and • how the subscriber's public keys are re-certified.	1. Verify the CA's business continuity plans address the compromise or suspected compromise of a CA's private keys as a disaster 2. Validate disaster recovery procedures include the revocation and reissuance of all certificates that were signed with that CA's private key, in the event of the compromise or suspected compromise 3. Check the recovery procedures used if the CA's private key covers the requirements mentioned in control 3.3.1.47	Mandatory	WebTrust 4.7.3				
3.3.1.49	In the event that the CA	1. Validate CA has	Mandatory	WebTrust 4.7.4				

	has to replace its Root CA private key, procedures are in place for the secure and authenticated revocation of the following: • the old CA root public key; • the set of all certificates (including any self-signed) issued by a Root CA or any CA based on the compromised private key; and • any subordinate CA public keys and corresponding certificates that require recertification	a procedure in place in the event that the CA has to replace its private key						
3.3.1.50	The CA's business continuity plan for key compromise addresses who is notified and what actions are taken with system software and hardware, symmetric and asymmetric keys, previously generated signatures and encrypted data.	1. Verify CA's business continuity plan addresses list of people to be notified and what actions are taken with system software and hardware, symmetric and asymmetric keys, previously generated signatures and encrypted data	Mandatory	WebTrust 4.7.5				
Cryptographic Hardware Life Cycle Management								
3.3.1.51	To prevent tampering, CA cryptographic hardware is stored and used in a secure site, with access limited to authorized personnel, having the following characteristics: • inventory control processes and procedures to manage the origination, arrival, condition, departure and destination of each device; • access control processes and procedures to limit physical access to authorized personnel; • recording of all successful or failed access attempts to the CA facility and device storage mechanism (e.g., a safe) in audit logs; • incident handling processes and procedures to handle abnormal	1. Conduct a walkthrough of process of transfer of cryptographic hardware 2. Verify for sample cryptographic hardware which does not contain CA keys is sent from the manufacturer or alternate CA site via registered mail (or equivalent) using tamper evident packaging 3. Validate the cryptographic hardware from the manufacturer, acceptance testing and verification of firmware settings is performed upon the receipt of CA 4. Validate CA takes measure to	Mandatory	WebTrust 4.8.3				

	events, security breaches, and investigation and reports; and • monitoring processes and procedures to verify the ongoing effectiveness of the controls.	prevent tampering of cryptographic hardware covering the characteristics mentioned in control (3.3.1.52) description						
3.3.1.52	Correct processing of CA cryptographic hardware is verified on a periodic basis.	1. Validate the processing of cryptographic hardware is verified on a periodic basis.	Mandatory	WebTrust 4.8.7				
3.3.1.53	Diagnostic support is provided during troubleshooting of CA cryptographic hardware in the presence of no less than two trusted employees	1. Validate diagnostic support is provided and check the names of trusted employees responsible for overlooking support	Mandatory	WebTrust 4.8.8				
3.3.1.54	CA cryptographic hardware which does not contain CA keys is sent from the manufacturer or alternate CA site via registered mail (or equivalent) using tamper evident packaging. Upon the receipt of CA cryptographic hardware from the manufacturer or alternate site, authorized CA personnel inspects the tamper evident packaging to determine whether the seal is intact.	1. Conduct a walkthrough of process of transfer of cryptographic hardware 2. Verify for sample cryptographic hardware which does not contain CA keys is sent from the manufacturer or alternate CA site via registered mail (or equivalent) using tamper evident packaging 3. Validate the cryptographic hardware from the manufacturer, acceptance testing and verification of firmware settings is performed upon the receipt of CA 4. Validate CA takes measure to prevent tampering of cryptographic hardware covering the characteristics mentioned in control (3.3.1.52) description	Mandatory	WebTrust 4.8.1				
3.3.1.55	Upon the receipt of CA cryptographic hardware from the manufacturer,	1. Conduct a walkthrough of process of transfer	Mandatory	WebTrust 4.8.2				

	acceptance testing and verification of firmware settings is performed. Upon the receipt of CA cryptographic hardware that has been serviced or repaired, acceptance testing and verification of firmware settings is performed.	of cryptographic hardware 2. Verify for sample cryptographic hardware which does not contain CA keys is sent from the manufacturer or alternate CA site via registered mail (or equivalent) using tamper evident packaging 3. Validate the cryptographic hardware from the manufacturer, acceptance testing and verification of firmware settings is performed upon the receipt of CA 4. Validate CA takes measure to prevent tampering of cryptographic hardware covering the characteristics mentioned in control (3.3.1.52) description						
3.3.1.56	When not attached to the CA system, the CA cryptographic hardware is stored in a tamper resistant container that is stored securely under multiple controls (i.e., a safe)	1. Validate use of tamper resistant containers to store the CA cryptographic hardware	Mandatory	WebTrust 4.8.4				
3.3.1.57	The handling of CA cryptographic hardware, including the following tasks, is performed in the presence of no less than two trusted employees: • installation of CA cryptographic hardware; • removal of CA cryptographic hardware from production; • servicing or repair of CA cryptographic hardware (including installation of new hardware, firmware, or software); and • disassembly and permanent removal from use	1. Obtain names of employees responsible for handling of CA cryptographic hardware 2. Validate by discussing with them if they were present during installation, removal servicing, repair, disassembly and permanent removal of CA cryptographic hardware	Mandatory	WebTrust 4.8.5				

3.3.1.58	Devices used for private key storage and recovery and the interfaces to these devices are tested before usage for integrity.	1. Verify by checking reports of the testing performed on devices used for private key storage	Mandatory	WebTrust 4.8.6				
Key Escrow								
3.3.1.59	If a third party provides CA private key escrow services, a contract exists that outlines the liabilities and remedies between the parties.	1. Verify if third party provides CA private key escrow services, a contract exists that outlines the liabilities and remedies between the parties 2. Validate if CA private keys are held in escrow, escrowed copies of CA private signing keys have the same or greater level of security controls as keys currently in use	Mandatory	WebTrust 4.9.1				
3.3.1.60	If CA private signing keys are held in escrow, escrowed copies of the CA private signing keys have the same or greater level of security controls as keys currently in use.	1. Verify if third party provides CA private key escrow services, a contract exists that outlines the liabilities and remedies between the parties 2. Validate if CA private keys are held in escrow, escrowed copies of CA private signing keys have the same or greater level of security controls as keys currently in use	Mandatory	WebTrust 4.9.2				
Key Transportation								
3.3.1.61	CA keys are prepared for transport in a physically secure environment by personnel in Trusted Roles and under multi-person control	1. Conduct walkthrough of the key transportation process and check the following: a. CA keys are prepared for transport in a physically secure environment by personnel in Trusted Roles and under multi-person control b. CA has	Mandatory	WebTrust 4.10.1				

		individuals assigned for Trusted Roles and they have received training to perform their responsibilities. c. CA keys remain in a physically secure environment until ready to be transported by CA personnel or common carrier d. CA keys are only transported on hardware devices and in tamper-evident packaging e. For hardware device containing entire CA key, two CA employees will be required to physical transport the key f. For CA key divided into fragments on multiple hardware devices, measures mentioned in control 3.3.1.64 are followed g. Activation materials are transported separately from the CA key in tamper evident packaging h. Packaging for CA keys and activation materials are reviewed for evidence of tampering upon receipt i. CA keys and activation materials are stored in a physically secure environment by personnel in Trusted Roles and under multi-person control upon receipt j. All CA key transportation activities are logged k. For a recent key transportation ceremony, verify name of the auditor who independently witnessed the process							
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3.3.1.62	CA keys remain in a physically secure environment until ready to be transported by CA personnel or common carrier.	1. Conduct walkthrough of the key transportation process and check the following: a. CA keys are prepared for transport in a physically secure environment by personnel in Trusted Roles and under multi-person control b. CA has individuals assigned for Trusted Roles and they have received training to perform their responsibilities. c. CA keys remain in a physically secure environment until ready to be transported by CA personnel or common carrier d. CA keys are only transported on hardware devices and in tamper-evident packaging e. For hardware device containing entire CA key, two CA employees will be required to physical transport the key f. For CA key divided into fragments on multiple hardware devices, measures mentioned in control 3.3.1.64 are followed g. Activation materials are transported separately from the CA key in tamper evident packaging h. Packaging for CA keys and activation materials are reviewed for evidence of tampering upon receipt i. CA keys and activation materials are stored in a physically secure environment by personnel in Trusted Roles and	Mandatory	WebTrust 4.10.2					
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		under multi-person control upon receipt j. All CA key transportation activities are logged k. For a recent key transportation ceremony, verify name of the auditor who independently witnessed the process						
3.3.1.63	CA keys are only transported on hardware devices and in tamper-evident packaging as disclosed in the CA's business practices.	1. Conduct walkthrough of the key transportation process and check the following: a. CA keys are prepared for transport in a physically secure environment by personnel in Trusted Roles and under multi-person control b. CA has individuals assigned for Trusted Roles and they have received training to perform their responsibilities. c. CA keys remain in a physically secure environment until ready to be transported by CA personnel or common carrier d. CA keys are only transported on hardware devices and in tamper-evident packaging e. For hardware device containing entire CA key, two CA employees will be required to physical transport the key f. For CA key divided into fragments on multiple hardware devices, measures mentioned in control 3.3.1.64 are followed g. Activation materials are transported separately from the	Mandatory	WebTrust 4.10.3				

		CA key in tamper evident packaging h. Packaging for CA keys and activation materials are reviewed for evidence of tampering upon receipt i. CA keys and activation materials are stored in a physically secure environment by personnel in Trusted Roles and under multi-person control upon receipt j. All CA key transportation activities are logged k. For a recent key transportation ceremony, verify name of the auditor who independently witnessed the process						
3.3.1.64	If the hardware device contains the entire CA key, it is physically transported by at least two CA employees and remains under multi-person control from origin to destination.	1. Conduct walkthrough of the key transportation process and check the following: a. CA keys are prepared for transport in a physically secure environment by personnel in Trusted Roles and under multi-person control b. CA has individuals assigned for Trusted Roles and they have received training to perform their responsibilities. c. CA keys remain in a physically secure environment until ready to be transported by CA personnel or common carrier d. CA keys are only transported on hardware devices and in tamper-evident packaging e. For hardware device containing entire CA key, two	Mandatory	WebTrust 4.10.4				

		CA employees will be required to physical transport the key f. For CA key divided into fragments on multiple hardware devices, measures mentioned in control 3.3.1.64 are followed g. Activation materials are transported separately from the CA key in tamper evident packaging h. Packaging for CA keys and activation materials are reviewed for evidence of tampering upon receipt i. CA keys and activation materials are stored in a physically secure environment by personnel in Trusted Roles and under multi-person control upon receipt j. All CA key transportation activities are logged k. For a recent key transportation ceremony, verify name of the auditor who independently witnessed the process						
3.3.1.65	If the CA key is divided into fragments on multiple hardware devices: • If transported by CA employees, each fragment is transported separately using different transportation routes, methods, and/or times; or • If transported by common carrier, each fragment is sent using a different common carrier at different times. Shipments require signature service, tracking, are insured.	1. Conduct walkthrough of the key transportation process and check the following: a. CA keys are prepared for transport in a physically secure environment by personnel in Trusted Roles and under multi-person control b. CA has individuals assigned for Trusted Roles and they have received training to perform their	Mandatory	WebTrust 4.10.5				

		responsibilities. c. CA keys remain in a physically secure environment until ready to be transported by CA personnel or common carrier d. CA keys are only transported on hardware devices and in tamper-evident packaging e. For hardware device containing entire CA key, two CA employees will be required to physical transport the key f. For CA key divided into fragments on multiple hardware devices, measures mentioned in control 3.3.1.64 are followed g. Activation materials are transported separately from the CA key in tamper evident packaging h. Packaging for CA keys and activation materials are reviewed for evidence of tampering upon receipt i. CA keys and activation materials are stored in a physically secure environment by personnel in Trusted Roles and under multi-person control upon receipt j. All CA key transportation activities are logged k. For a recent key transportation ceremony, verify name of the auditor who independently witnessed the process						
3.3.1.66	Activation materials are transported separately from the CA key (i.e. by a different method and/or at a different time) in	1. Conduct walkthrough of the key transportation process and check the following:	Mandatory	WebTrust 4.10.6				

	tamper-evident packaging	a. CA keys are prepared for transport in a physically secure environment by personnel in Trusted Roles and under multi-person control b. CA has individuals assigned for Trusted Roles and they have received training to perform their responsibilities. c. CA keys remain in a physically secure environment until ready to be transported by CA personnel or common carrier d. CA keys are only transported on hardware devices and in tamper-evident packaging e. For hardware device containing entire CA key, two CA employees will be required to physical transport the key f. For CA key divided into fragments on multiple hardware devices, measures mentioned in control 3.3.1.64 are followed g. Activation materials are transported separately from the CA key in tamper evident packaging h. Packaging for CA keys and activation materials are reviewed for evidence of tampering upon receipt i. CA keys and activation materials are stored in a physically secure environment by personnel in Trusted Roles and under multi-person control upon receipt j. All CA key transportation						
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		activities are logged k. For a recent key transportation ceremony, verify name of the auditor who independently witnessed the process						
3.3.1.67	Upon receipt at the destination, packaging for CA keys and activation materials are reviewed for evidence of tampering. If evidence of tampering is discovered, the Policy Authority is notified of a possible breach event	1. Conduct walkthrough of the key transportation process and check the following: a. CA keys are prepared for transport in a physically secure environment by personnel in Trusted Roles and under multi-person control b. CA has individuals assigned for Trusted Roles and they have received training to perform their responsibilities. c. CA keys remain in a physically secure environment until ready to be transported by CA personnel or common carrier d. CA keys are only transported on hardware devices and in tamper-evident packaging e. For hardware device containing entire CA key, two CA employees will be required to physical transport the key f. For CA key divided into fragments on multiple hardware devices, measures mentioned in control 3.3.1.64 are followed g. Activation materials are transported separately from the CA key in tamper evident packaging h. Packaging for CA keys and activation materials	Mandatory	WebTrust 4.10.7				

		are reviewed for evidence of tampering upon receipt i. CA keys and activation materials are stored in a physically secure environment by personnel in Trusted Roles and under multi-person control upon receipt j. All CA key transportation activities are logged k. For a recent key transportation ceremony, verify name of the auditor who independently witnessed the process						
3.3.1.68	Upon receipt at the destination, CA keys and activation materials are stored in a physically secure environment by personnel in Trusted Roles and under multi-person control.	1. Conduct walkthrough of the key transportation process and check the following: a. CA keys are prepared for transport in a physically secure environment by personnel in Trusted Roles and under multi-person control b. CA has individuals assigned for Trusted Roles and they have received training to perform their responsibilities. c. CA keys remain in a physically secure environment until ready to be transported by CA personnel or common carrier d. CA keys are only transported on hardware devices and in tamper-evident packaging e. For hardware device containing entire CA key, two CA employees will be required to physical transport the key f. For CA key	Mandatory	WebTrust 4.10.8				

		divided into fragments on multiple hardware devices, measures mentioned in control 3.3.1.64 are followed g. Activation materials are transported separately from the CA key in tamper evident packaging h. Packaging for CA keys and activation materials are reviewed for evidence of tampering upon receipt i. CA keys and activation materials are stored in a physically secure environment by personnel in Trusted Roles and under multi-person control upon receipt j. All CA key transportation activities are logged k. For a recent key transportation ceremony, verify name of the auditor who independently witnessed the process						
3.3.1.69	Personnel involved in a CA key transportation event are in Trusted Roles and have received training in their role and responsibilities.	1. Conduct walkthrough of the key transportation process and check the following: a. CA keys are prepared for transport in a physically secure environment by personnel in Trusted Roles and under multi-person control b. CA has individuals assigned for Trusted Roles and they have received training to perform their responsibilities. c. CA keys remain in a physically secure environment until ready to be	Mandatory	WebTrust 4.10.9				

		transported by CA personnel or common carrier d. CA keys are only transported on hardware devices and in tamper-evident packaging e. For hardware device containing entire CA key, two CA employees will be required to physical transport the key f. For CA key divided into fragments on multiple hardware devices, measures mentioned in control 3.3.1.64 are followed g. Activation materials are transported separately from the CA key in tamper evident packaging h. Packaging for CA keys and activation materials are reviewed for evidence of tampering upon receipt i. CA keys and activation materials are stored in a physically secure environment by personnel in Trusted Roles and under multi-person control upon receipt j. All CA key transportation activities are logged k. For a recent key transportation ceremony, verify name of the auditor who independently witnessed the process					
3.3.1.70	A log is maintained of all actions taken as part of the CA key transportation event and is retained in accordance with the CA’s disclosed business practices.	1. Conduct walkthrough of the key transportation process and check the following: a. CA keys are prepared for transport in a physically secure environment by	Mandatory	WebTrust 4.10.10			

		personnel in Trusted Roles and under multi-person control b. CA has individuals assigned for Trusted Roles and they have received training to perform their responsibilities. c. CA keys remain in a physically secure environment until ready to be transported by CA personnel or common carrier d. CA keys are only transported on hardware devices and in tamper-evident packaging e. For hardware device containing entire CA key, two CA employees will be required to physical transport the key f. For CA key divided into fragments on multiple hardware devices, measures mentioned in control 3.3.1.64 are followed g. Activation materials are transported separately from the CA key in tamper evident packaging h. Packaging for CA keys and activation materials are reviewed for evidence of tampering upon receipt i. CA keys and activation materials are stored in a physically secure environment by personnel in Trusted Roles and under multi-person control upon receipt j. All CA key transportation activities are logged k. For a recent key transportation ceremony, verify name of the auditor							
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		who independently witnessed the process						
3.3.1.71	Internal or external auditors accompany CA personnel during CA key transportation events.	1. Conduct walkthrough of the key transportation process and check the following: a. CA keys are prepared for transport in a physically secure environment by personnel in Trusted Roles and under multi-person control b. CA has individuals assigned for Trusted Roles and they have received training to perform their responsibilities. c. CA keys remain in a physically secure environment until ready to be transported by CA personnel or common carrier d. CA keys are only transported on hardware devices and in tamper-evident packaging e. For hardware device containing entire CA key, two CA employees will be required to physical transport the key f. For CA key divided into fragments on multiple hardware devices, measures mentioned in control 3.3.1.64 are followed g. Activation materials are transported separately from the CA key in tamper evident packaging h. Packaging for CA keys and activation materials are reviewed for evidence of tampering upon receipt i. CA keys and	Mandatory	WebTrust 4.10.11				

		activation materials are stored in a physically secure environment by personnel in Trusted Roles and under multi-person control upon receipt j. All CA key transportation activities are logged k. For a recent key transportation ceremony, verify name of the auditor who independently witnessed the process						
Key Changeover								
3.3.1.72	Certifying Authority and Subscriber keys shall be changed periodically. Key change shall be processed as per Key Generation guidelines.	1. Conduct walkthrough of the key changeover process and check the following: a. The frequency of changing CA and subscriber keys b. Key change is processed as per Key Generation guidelines c. CA provides reasonable notice to subscriber's relying parties of any change to new key pair used by CA d. All keys have validity period less than 5 years e. Validity period of keys is defined as per requirements mentioned in control 3.3.1.75 f. Key is often changed to minimize risk from compromise	Mandatory	IT CA Rules SCHEDULE III 21.1, X.509 Policy 5.6, CA Browser				
3.3.1.73	The Certifying Authority shall provide reasonable notice to the Subscriber's relying parties of any change to a new key pair used by the Certifying Authority to sign Digital Signature Certificates.	1. Conduct walkthrough of the key changeover process and check the following: a. The frequency of changing CA and subscriber keys b. Key change is processed as per Key Generation guidelines c. CA provides	Mandatory	IT CA Rules SCHEDULE III 21.1				

		reasonable notice to subscriber's relying parties of any change to new key pair used by CA d. All keys have validity period less than 5 years e. Validity period of keys is defined as per requirements mentioned in control 3.3.1.75 f. Key is often changed to minimize risk from compromise						
3.3.1.74	The Certifying Authority shall define its key change process that ensures reliability of the process by showing how the generation of key interlocks – such as signing a hash of the new key with the old key.	1. Conduct walkthrough of the key changeover process and check the following: a. The frequency of changing CA and subscriber keys b. Key change is processed as per Key Generation guidelines c. CA provides reasonable notice to subscriber's relying parties of any change to new key pair used by CA d. All keys have validity period less than 5 years e. Validity period of keys is defined as per requirements mentioned in control 3.3.1.75 f. Key is often changed to minimize risk from compromise	Mandatory	IT CA Rules SCHEDULE III 21.1				
3.3.1.75	All CA keys must have validity periods of no more than ten years.	1. Conduct walkthrough of the key changeover process and check the following: a. The frequency of changing CA and subscriber keys b. Key change is processed as per Key Generation guidelines c. CA provides reasonable notice to subscriber's relying parties of any change to new key	Mandatory	IT CA Rules SCHEDULE III 21.1				

		pair used by CA d. All keys have validity period less than 5 years e. Validity period of keys is defined as per requirements mentioned in control 3.3.1.75 f. Key is often changed to minimize risk from compromise						
3.3.1.76	To minimize risk from compromise of a CA's private signing key, that key may be changed often; from that time on, only the new key shall be used for certificate signing purposes. The older, but still valid, certificate will be available to verify old signatures until all of the certificates signed using the associated private key have also expired. If the old private key is used to sign CRLs, then the old key shall be retained and protected.	1. Conduct walkthrough of the key changeover process and check the following: a. The frequency of changing CA and subscriber keys b. Key change is processed as per Key Generation guidelines c. CA provides reasonable notice to subscriber's relying parties of any change to new key pair used by CA d. All keys have validity period less than 5 years e. Validity period of keys is defined as per requirements mentioned in control 3.3.1.75 f. Key is often changed to minimize risk from compromise	Mandatory	X.509 Policy				
Key Migration								
3.3.1.77	The CA follows a CA key migration script for key migration events that includes the following: • definition and assignment of participant roles and responsibilities; • management approval for conduct of the key migration event • specific cryptographic hardware, software and other materials including identifying information, e.g., serial numbers, that contain the CA key copies to be migrated and new hardware where the keys are being migrated	1. Obtain a copy of the CA key migration script and check the following: a. Script covers roles assigned, management approval, serial numbers, procedures for secure storage of cryptographic hardware etc. b. Steps for performing key migration are stated clearly in the script c. physical security	Mandatory	WebTrust 4.11.4				

	to; • specific steps performed during the key migration ceremony, including; • Hardware preparation - o Software tool installation and setup - o Cryptographic hardware setup and initialization - o CA key migration - o CA key verification • physical security requirements for the event location (e.g., barriers, access controls and logging controls); • procedures for secure storage of cryptographic hardware and any associated activation materials following the migration event • sign-off on the script or in a log from participants and witnesses indicating whether the key migration was performed in accordance with the detailed key migration script; and • notation of any deviations from the key migration script (e.g., documentation of steps taken to address any technical issues).	requirements for the ceremony location are met d. sign-off is given for the key migration ceremony e. notation of any deviations from the key generation ceremony script are addressed						
3.3.1.78	CA key migration events occur in a physically secure environment by those in Trusted Roles under multi-person control	1. Validate CA migration is done in a physically secure environment 2. Check vendor-supplied hardware and software tools are tested by the CA prior the key migration event 3. Verify In house developed software tools are developed and tested by the CA prior to the key migration event	Mandatory	WebTrust 4.11.1				
3.3.1.79	Vendor-supplied hardware and software tools are tested by the CA prior the key migration event, and are operated in accordance with vendor-supplied documentation and instructions.	1. Validate CA migration is done in a physically secure environment 2. Check vendor-supplied hardware and software tools are tested by the CA prior the key	Mandatory	WebTrust 4.11.2				

		migration event 3. Verify In house developed software tools are developed and tested by the CA prior to the key migration event						
3.3.1.80	In-house developed software tools are developed and tested by the CA prior to the key migration event in accordance with its standard software development process	1. Validate CA migration is done in a physically secure environment 2. Check vendor-supplied hardware and software tools are tested by the CA prior the key migration event 3. Verify In house developed software tools are developed and tested by the CA prior to the key migration event	Mandatory	WebTrust 4.11.3				
3.3.1.81	The CA follows a CA key migration script for key migration events that includes the following: • definition and assignment of participant roles and responsibilities; • management approval for conduct of the key migration event • specific cryptographic hardware, software and other materials including identifying information, e.g., serial numbers, that contain the CA key copies to be migrated and new hardware where the keys are being migrated to; • specific steps performed during the key migration ceremony, including; • Hardware preparation - o Software tool installation and setup - o Cryptographic hardware setup and initialization - o CA key migration - o CA key verification • physical security requirements for the event location (e.g., barriers, access controls and logging controls); • procedures for secure storage of cryptographic hardware and any associated activation	1. Obtain a copy of the CA key migration script and check the following: a. Script covers roles assigned, management approval, serial numbers, procedures for secure storage of cryptographic hardware etc. b. Steps for performing key migration are stated clearly in the script c. physical security requirements for the ceremony location are met d. sign-off is given for the key migration ceremony e. notation of any deviations from the key generation ceremony script are addressed	Mandatory	WebTrust 4.11.4				

	materials following the migration event • sign-off on the script or in a log from participants and witnesses indicating whether the key migration was performed in accordance with the detailed key migration script; and • notation of any deviations from the key migration script (e.g., documentation of steps taken to address any technical issues).							
3.3.1.82	A log is maintained of all actions taken as part of the CA key migration event and is retained in accordance with the CA's disclosed business practices.	1. Validate logs are maintained for CA key migration event 2. For a recent key migration ceremony, verify name of the auditor who independently witnessed the process 3. Verify upon successful completion of a CA key migration event, remaining copies of the CA keys, and older cryptographic hardware that no longer serve a business purpose shall be securely destroyed	Mandatory	WebTrust 4.11.5				
3.3.1.83	CA key migration events shall be witnessed by internal or external auditors	1. Validate logs are maintained for CA key migration event 2. For a recent key migration ceremony, verify name of the auditor who independently witnessed the process 3. Verify upon successful completion of a CA key migration event, remaining copies of the CA keys, and older cryptographic hardware that no longer serve a business purpose shall be securely destroyed	Mandatory	WebTrust 4.11.6				

3.3.1.84	Upon successful completion of a CA key migration event, remaining copies of the CA keys, and older cryptographic hardware that no longer serve a business purpose shall be securely destroyed in accordance with the CA's disclosed business practices	1. Validate logs are maintained for CA key migration event 2. For a recent key migration ceremony, verify name of the auditor who independently witnessed the process 3. Verify upon successful completion of a CA key migration event, remaining copies of the CA keys, and older cryptographic hardware that no longer serve a business purpose shall be securely destroyed	Mandatory	WebTrust 4.11.7				
Private Key Protection and Cryptographic Module Engineering Controls								
3.3.1.85	The Certifying Authority must protect its private keys from disclosure. The Certifying Authority must back-up its private keys. Backed-up keys must be stored in encrypted form and protected at a level no lower than those followed for storing the primary version of the key.	1. Conduct walkthrough of Private Key Protection and Cryptographic Module Engineering Controls and check the following: a. CA protects private keys from disclosure b. CA backs up its private keys and stores them in encrypted format c. private key backups are stored in a secure storage facility, away from where the original key is stored d. Backup measures are taken to ensure continued availability of private key e. Use of a CA private signing key requires action by at least two persons f. Signature keys are not escrowed by a third party	Mandatory	IT CA Rules SCHEDULE III 19				
3.3.1.86	The Certifying Authority's private key backups should be stored in a secure storage facility, away from where	1. Conduct walkthrough of Private Key Protection and Cryptographic	Mandatory	IT CA Rules SCHEDULE III 19, IT Regulations 3				

	the original key is stored. Continued availability of the private key be ensured through approved backup measures in the event of loss or corruption of its private key.	Module Engineering Controls and check the following: a. CA protects private keys from disclosure b. CA backs up its private keys and stores them in encrypted format c. private key backups are stored in a secure storage facility, away from where the original key is stored d. Backup measures are taken to ensure continued availability of private key e. Use of a CA private signing key requires action by at least two persons f. Signature keys are not escrowed by a third party						
3.3.1.87	Use of a CA private signing key shall require action by at least two persons	1. Conduct walkthrough of Private Key Protection and Cryptographic Module Engineering Controls and check the following: a. CA protects private keys from disclosure b. CA backs up its private keys and stores them in encrypted format c. private key backups are stored in a secure storage facility, away from where the original key is stored d. Backup measures are taken to ensure continued availability of private key e. Use of a CA private signing key requires action by at least two persons f. Signature keys are not escrowed by a third party	Mandatory	X.509 Policy 6.2.2				
3.3.1.88	The cryptographic	1. Conduct	Recommended	X.509 Policy 6.2.7				

	module may store Private Keys in any form as long as the keys are not accessible without authentication mechanism that is in compliance with FIPS 140-2 rating of the cryptographic module.	walkthrough of Private Key Protection and Cryptographic Module Engineering Controls and check the following: a. CA protects private keys from disclosure b. CA backs up its private keys and stores them in encrypted format c. private key backups are stored in a secure storage facility, away from where the original key is stored d. Backup measures are taken to ensure continued availability of private key e. Use of a CA private signing key requires action by at least two persons f. Signature keys are not escrowed by a third party						
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3.3.2. Subscriber Key Lifecycle Controls

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
CA-Provided Subscriber Key Generation Services							
3.3.2.1	The Certifying Authority's public verification key must be delivered to the prospective Digital Signature Certificate holder in an on-line transaction in accordance with PKIX-3 Certificate Management Protocol, or via an equally secure manner. The Certifying Authority shall ensure that the subscriber can verify the Certifying Authority's Public Key Certificate, if he chooses to do so, by having access to the Public Key Certificate of the Controller	1. Conduct walkthrough and check the following: a. CAs public verification key is delivered to the prospective Digital Signature Certificate holder in an on-line transaction in accordance with PKIX-3 Certificate Management Protocol, or via an equally secure manner. b. Certifying Authority ensures that the subscriber can verify the Certifying Authority's Public Key Certificate, if he chooses to do so, by having access to the Public	Optional				

		Key Certificate of the Controller						
3.3.2.2	Subscriber key generation is performed within a secure cryptographic device meeting the applicable ISO 15782-1/FIPS140-2/ANSI X9.66 requirements based on a risk assessment and the business requirements of the CA and in accordance with the applicable CP.	1. No action is required from the auditor to check the controls as Subscriber key generations services are not provided by CAs	Not Applicable	WebTrust 5.1.1, CA Browser Forum 6.1.1.3				
3.3.2.3	Subscriber key generation performed by the CA (or RA or card bureau) uses a key generation algorithm as specified in the CP.	1. No action is required from the auditor to check the controls as Subscriber key generations services are not provided by CAs	Not Applicable	WebTrust 5.1.2				
3.3.2.4	Subscriber key generation performed by the CA (or RA) uses a prime number generator as specified in an ANSI X9 or ISO standard	1. No action is required from the auditor to check the controls as Subscriber key generations services are not provided by CAs	Not Applicable	WebTrust 5.1.3				
3.3.2.5	Subscriber key generation performed by the CA (or RA or card bureau) results in key sizes in accordance with the CP.	1. No action is required from the auditor to check the controls as Subscriber key generations services are not provided by CAs	Not Applicable	WebTrust 5.1.4				
3.3.2.6	Subscriber key generation performed by the CA (or RA) is performed by authorized personnel in accordance with the CA's CPS.	1. No action is required from the auditor to check the controls as Subscriber key generations services are not provided by CAs	Not Applicable	WebTrust 5.1.5				
3.3.2.7	When subscriber key generation is performed by the CA (or RA or card bureau), the CA (or RA or card bureau) securely (confidentially)	1. No action is required from the auditor to check the controls as Subscriber key generations services	Not Applicable	WebTrust 5.1.6				

	delivers the subscriber key pair(s) generated by the CA (or RA or card bureau) to the subscriber in accordance with the CP.	are not provided by CAs						
CA-Provided Subscriber Key Storage and Recovery Services								
3.3.2.8	Certifying Authority's keys shall be stored in tamper-resistant devices and can only be activated under split-control by parties who are not involved in the set-up and maintenance of the systems and operations of the Certifying Authority. The key of the Certifying Authority may be stored in a tamper-resistant cryptographic module or split into sub-keys stored in tamper-resistant devices under the custody of the key custodians	1. Conduct walkthrough of the CA- Provided Subscriber Key Storage and Recovery Services and check the following: a. CAs key is stored in tamper resistant device b. CAs key custodians ensure that the CA's key component or the activation code is always under his sole custody c. Change of key custodian is approved by CAs management and documented d. Subscriber private keys stored by the CA are stored in encrypted form using a cryptographic algorithm and key length e. CA ensures that the subscriber's private keys are not disclosed to any entity other than the owner (i.e., the subscriber) of the keys if Ca generates key pair(s) on behalf of subscriber f. CA does not maintain a copy of any private signing key g. key backup and recovery services are only performed by authorized personnel if the CA provides subscriber (confidentiality) key storage, backup and recovery, subscriber private h. Backup and recovery, controls	Mandatory	IT CA Rules SCHEDULE III 18.3				

		exist for encryptions to ensure that the integrity of the subscriber's private (confidentiality) key is maintained throughout its life cycle.						
3.3.2.9	The Certifying Authority's key custodians shall ensure that the Certifying Authority's key component or the activation code is always under his sole custody. Change of key custodians shall be approved by the Certifying Authority's management and documented	1. Conduct walkthrough of the CA- Provided Subscriber Key Storage and Recovery Services and check the following: a. CAs key is stored in tamper resistant device b. CAs key custodians ensure that the CA's key component or the activation code is always under his sole custody c. Change of key custodian is approved by CAs management and documented d. Subscriber private keys stored by the CA are stored in encrypted form using a cryptographic algorithm and key length e. CA ensures that the subscriber's private keys are not disclosed to any entity other than the owner (i.e., the subscriber) of the keys if Ca generates key pair(s) on behalf of subscriber f. CA does not maintain a copy of any private signing key g. key backup and recovery services are only performed by authorized personnel if the CA provides subscriber (confidentiality) key storage, backup and recovery, subscriber private h. Backup and recovery, controls	Mandatory	IT CA Rules SCHEDULE III 18.3documented				

		exist for encryptions to ensure that the integrity of the subscriber's private (confidentiality) key is maintained throughout its life cycle.						
3.3.2.10	Subscriber private keys stored by the CA (or RA) are stored in encrypted form using a cryptographic algorithm and key length based on a risk assessment and requirements of the CP.	1. Conduct walkthrough of the CA- Provided Subscriber Key Storage and Recovery Services and check the following: a. CAs key is stored in tamper resistant device b. CAs key custodians ensure that the CA's key component or the activation code is always under his sole custody c. Change of key custodian is approved by CAs management and documented d. Subscriber private keys stored by the CA are stored in encrypted form using a cryptographic algorithm and key length e. CA ensures that the subscriber's private keys are not disclosed to any entity other than the owner (i.e., the subscriber) of the keys if Ca generates key pair(s) on behalf of subscriber f. CA does not maintain a copy of any private signing key g. key backup and recovery services are only performed by authorized personnel if the CA provides subscriber (confidentiality) key storage, backup and recovery, subscriber private h. Backup and recovery, controls	Mandatory	WebTrust 5.2.1				

		exist for encryptions to ensure that the integrity of the subscriber's private (confidentiality) key is maintained throughout its life cycle.						
3.3.2.11	If the CA generates key pair(s) on behalf of a Subscriber, the CA (or RA) ensures that the subscriber's private keys are not disclosed to any entity other than the owner (i.e., the subscriber) of the keys.	1. Conduct walkthrough of the CA- Provided Subscriber Key Storage and Recovery Services and check the following: a. CAs key is stored in tamper resistant device b. CAs key custodians ensure that the CA's key component or the activation code is always under his sole custody c. Change of key custodian is approved by CAs management and documented d. Subscriber private keys stored by the CA are stored in encrypted form using a cryptographic algorithm and key length e. CA ensures that the subscriber's private keys are not disclosed to any entity other than the owner (i.e., the subscriber) of the keys if Ca generates key pair(s) on behalf of subscriber f. CA does not maintain a copy of any private signing key g. key backup and recovery services are only performed by authorized personnel if the CA provides subscriber (confidentiality) key storage, backup and recovery, subscriber private h. Backup and recovery, controls	Mandatory	WebTrust 5.2.2				

		exist for encryptions to ensure that the integrity of the subscriber's private (confidentiality) key is maintained throughout its life cycle.						
3.3.2.12	If the CA (or RA) generates public/private signing key pair(s), it does not maintain a copy of any private signing key, once the subscriber confirms receipt of that key.	1. Conduct walkthrough of the CA- Provided Subscriber Key Storage and Recovery Services and check the following: a. CAs key is stored in tamper resistant device b. CAs key custodians ensure that the CA's key component or the activation code is always under his sole custody c. Change of key custodian is approved by CAs management and documented d. Subscriber private keys stored by the CA are stored in encrypted form using a cryptographic algorithm and key length e. CA ensures that the subscriber's private keys are not disclosed to any entity other than the owner (i.e., the subscriber) of the keys if Ca generates key pair(s) on behalf of subscriber f. CA does not maintain a copy of any private signing key g. key backup and recovery services are only performed by authorized personnel if the CA provides subscriber (confidentiality) key storage, backup and recovery, subscriber private h. Backup and recovery, controls	Mandatory	WebTrust 5.2.3				

		exist for encryptions to ensure that the integrity of the subscriber's private (confidentiality) key is maintained throughout its life cycle.						
3.3.2.13	If the CA (or RA) provides subscriber (confidentiality) key storage, backup and recovery, subscriber private (confidentiality) key backup and recovery services are only performed by authorized personnel.	1. Conduct walkthrough of the CA- Provided Subscriber Key Storage and Recovery Services and check the following: a. CAs key is stored in tamper resistant device b. CAs key custodians ensure that the CA's key component or the activation code is always under his sole custody c. Change of key custodian is approved by CAs management and documented d. Subscriber private keys stored by the CA are stored in encrypted form using a cryptographic algorithm and key length e. CA ensures that the subscriber's private keys are not disclosed to any entity other than the owner (i.e., the subscriber) of the keys if Ca generates key pair(s) on behalf of subscriber f. CA does not maintain a copy of any private signing key g. key backup and recovery services are only performed by authorized personnel if the CA provides subscriber (confidentiality) key storage, backup and recovery, subscriber private h. Backup and recovery, controls	Mandatory	WebTrust 5.2.4				

		exist for encryptions to ensure that the integrity of the subscriber's private (confidentiality) key is maintained throughout its life cycle.						
3.3.2.14	If the CA (or RA) provides subscriber key storage, backup and recovery, controls exist to ensure that the integrity of the subscriber's private (confidentiality) key is maintained throughout its life cycle.	1. Conduct walkthrough of the CA- Provided Subscriber Key Storage and Recovery Services and check the following: a. CAs key is stored in tamper resistant device b. CAs key custodians ensure that the CA's key component or the activation code is always under his sole custody c. Change of key custodian is approved by CAs management and documented d. Subscriber private keys stored by the CA are stored in encrypted form using a cryptographic algorithm and key length e. CA ensures that the subscriber's private keys are not disclosed to any entity other than the owner (i.e., the subscriber) of the keys if Ca generates key pair(s) on behalf of subscriber f. CA does not maintain a copy of any private signing key g. key backup and recovery services are only performed by authorized personnel if the CA provides subscriber (confidentiality) key storage, backup and recovery, subscriber private h. Backup and recovery, controls	Mandatory	WebTrust 5.2.5				

		exist for encryptions to ensure that the integrity of the subscriber's private (confidentiality) key is maintained throughout its life cycle.						
Requirements for Subscriber Key Management								
3.3.2.15	The CP specifies the appropriate ISO 15782-1/FIPS 140-2 level requirement for cryptographic modules used for subscriber key generation	1. Conduct walkthrough of the CA- Provided Subscriber Key Management process and check the following: a. CP specifies the appropriate ISO 15782- 1/FIPS 140-2 level requirement for cryptographic modules used for subscriber key generation b. CP specifies the key generation algorithm(s) that is used for subscriber key generation c. CP specifies the acceptable key sizes for subscriber key generation d. CP specifies the private key protection requirements for stored subscriber private keys e. CP states the circumstances and authority of when the subscriber's private key will be restored and the control processes f. CP specifies the private key protection requirements for backup copies of subscriber private keys stored by the subscriber. g. Subscriber Agreements describe the required processes to be followed by the Subscriber h. CP specifies acceptable uses for subscriber key pairs	Mandatory	WebTrust 5.4.1				

		i. CP specifies the requirements for subscriber key usage. j. CP specifies the private key protection requirements for archived subscriber private keys k. CP specifies the requirements for destruction of archived subscriber keys at the end of the archive period l. CP specifies the means through which subscriber key destruction is performed m. CP or CPS specifies the requirements for destruction of all copies and fragments of the subscriber's private key at the end of the key pair life cycle n. CP specifies the requirements for use and handling of cryptographic hardware and subscriber authentication processes o. CP specifies the requirements for notification of the CA in the event of subscriber key compromise.						
3.3.2.16	The CP specifies the key generation algorithm(s) that is used for subscriber key generation.	1. Conduct walkthrough of the CA- Provided Subscriber Key Management process and check the following: a. CP specifies the appropriate ISO 15782- 1/FIPS 140- 2 level requirement for cryptographic modules used for subscriber key generation b. CP specifies the key generation algorithm(s) that is used for subscriber key generation	Mandatory	WebTrust 5.4.2				

		c. CP specifies the acceptable key sizes for subscriber key generation d. CP specifies the private key protection requirements for stored subscriber private keys e. CP states the circumstances and authority of when the subscriber's private key will be restored and the control processes f. CP specifies the private key protection requirements for backup copies of subscriber private keys stored by the subscriber. g. Subscriber Agreements describe the required processes to be followed by the Subscriber h. CP specifies acceptable uses for subscriber key pairs i. CP specifies the requirements for subscriber key usage. j. CP specifies the private key protection requirements for archived subscriber private keys k. CP specifies the requirements for destruction of archived subscriber keys at the end of the archive period l. CP specifies the means through which subscriber key destruction is performed m. CP or CPS specifies the requirements for destruction of all copies and fragments of the subscriber's private key at the end of the key pair life cycle n. CP specifies the requirements for use and handling of cryptographic						
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		hardware and subscriber authentication processes o. CP specifies the requirements for notification of the CA in the event of subscriber key compromise.						
3.3.2.17	The CP specifies the acceptable key sizes for subscriber key generation.	1. Conduct walkthrough of the CA- Provided Subscriber Key Management process and check the following: a. CP specifies the appropriate ISO 15782- 1/FIPS 140- 2 level requirement for cryptographic modules used for subscriber key generation b. CP specifies the key generation algorithm(s) that is used for subscriber key generation c. CP specifies the acceptable key sizes for subscriber key generation d. CP specifies the private key protection requirements for stored subscriber private keys e. CP states the circumstances and authority of when the subscriber's private key will be restored and the control processes f. CP specifies the private key protection requirements for backup copies of subscriber private keys stored by the subscriber. g. Subscriber Agreements describe the required processes to be followed by the Subscriber h. CP specifies acceptable uses for	Mandatory	WebTrust 5.4.3				

		subscriber key pairs i. CP specifies the requirements for subscriber key usage. j. CP specifies the private key protection requirements for archived subscriber private keys k. CP specifies the requirements for destruction of archived subscriber keys at the end of the archive period l. CP specifies the means through which subscriber key destruction is performed m. CP or CPS specifies the requirements for destruction of all copies and fragments of the subscriber's private key at the end of the key pair life cycle n. CP specifies the requirements for use and handling of cryptographic hardware and subscriber authentication processes o. CP specifies the requirements for notification of the CA in the event of subscriber key compromise.						
3.3.2.18	The CA or RA provides or makes available the mechanisms to allow the Subscriber to access (i.e., private key owner verification method), manage and control the usage of their private keys.	1. Conduct walkthrough of the CA- Provided Subscriber Key Management process and check the following: a. CP specifies the appropriate ISO 15782- 1/FIPS 140- 2 level requirement for cryptographic modules used for subscriber key generation b. CP specifies the key generation algorithm(s) that is used for subscriber	Not Applicable	WebTrust 5.4.4				

		key generation c. CP specifies the acceptable key sizes for subscriber key generation d. CP specifies the private key protection requirements for stored subscriber private keys e. CP states the circumstances and authority of when the subscriber's private key will be restored and the control processes f. CP specifies the private key protection requirements for backup copies of subscriber private keys stored by the subscriber. g. Subscriber Agreements describe the required processes to be followed by the Subscriber h. CP specifies acceptable uses for subscriber key pairs i. CP specifies the requirements for subscriber key usage. j. CP specifies the private key protection requirements for archived subscriber private keys k. CP specifies the requirements for destruction of archived subscriber keys at the end of the archive period l. CP specifies the means through which subscriber key destruction is performed m. CP or CPS specifies the requirements for destruction of all copies and fragments of the subscriber's private key at the end of the key pair life cycle n. CP specifies the requirements for use and handling of						
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		cryptographic hardware and subscriber authentication processes o. CP specifies the requirements for notification of the CA in the event of subscriber key compromise.						
3.3.2.19	The CP specifies the private key protection requirements for stored subscriber private keys.	1. Conduct walkthrough of the CA- Provided Subscriber Key Management process and check the following: a. CP specifies the appropriate ISO 15782- 1/FIPS 140- 2 level requirement for cryptographic modules used for subscriber key generation b. CP specifies the key generation algorithm(s) that is used for subscriber key generation c. CP specifies the acceptable key sizes for subscriber key generation d. CP specifies the private key protection requirements for stored subscriber private keys e. CP states the circumstances and authority of when the subscriber's private key will be restored and the control processes f. CP specifies the private key protection requirements for backup copies of subscriber private keys stored by the subscriber. g. Subscriber Agreements describe the required processes to be followed by the Subscriber h. CP specifies	Mandatory	WebTrust 5.4.5				

		acceptable uses for subscriber key pairs i. CP specifies the requirements for subscriber key usage. j. CP specifies the private key protection requirements for archived subscriber private keys k. CP specifies the requirements for destruction of archived subscriber keys at the end of the archive period l. CP specifies the means through which subscriber key destruction is performed m. CP or CPS specifies the requirements for destruction of all copies and fragments of the subscriber's private key at the end of the key pair life cycle n. CP specifies the requirements for use and handling of cryptographic hardware and subscriber authentication processes o. CP specifies the requirements for notification of the CA in the event of subscriber key compromise.						
3.3.2.20	The CP states the circumstances and authority of when the subscriber's private key will be restored and the control processes	1. Conduct walkthrough of the CA- Provided Subscriber Key Management process and check the following: a. CP specifies the appropriate ISO 15782- 1/FIPS 140- 2 level requirement for cryptographic modules used for subscriber key generation b. CP specifies the key generation algorithm(s) that is	Mandatory	WebTrust 5.4.6				

		used for subscriber key generation c. CP specifies the acceptable key sizes for subscriber key generation d. CP specifies the private key protection requirements for stored subscriber private keys e. CP states the circumstances and authority of when the subscriber's private key will be restored and the control processes f. CP specifies the private key protection requirements for backup copies of subscriber private keys stored by the subscriber. g. Subscriber Agreements describe the required processes to be followed by the Subscriber h. CP specifies acceptable uses for subscriber key pairs i. CP specifies the requirements for subscriber key usage. j. CP specifies the private key protection requirements for archived subscriber private keys k. CP specifies the requirements for destruction of archived subscriber keys at the end of the archive period l. CP specifies the means through which subscriber key destruction is performed m. CP or CPS specifies the requirements for destruction of all copies and fragments of the subscriber's private key at the end of the key pair life cycle n. CP specifies the requirements for use						
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		and handling of cryptographic hardware and subscriber authentication processes o. CP specifies the requirements for notification of the CA in the event of subscriber key compromise.						
3.3.2.21	The CP specifies the private key protection requirements for backup copies of subscriber private keys stored by the subscriber.	1. Conduct walkthrough of the CA- Provided Subscriber Key Management process and check the following: a. CP specifies the appropriate ISO 15782- 1/FIPS 140- 2 level requirement for cryptographic modules used for subscriber key generation b. CP specifies the key generation algorithm(s) that is used for subscriber key generation c. CP specifies the acceptable key sizes for subscriber key generation d. CP specifies the private key protection requirements for stored subscriber private keys e. CP states the circumstances and authority of when the subscriber's private key will be restored and the control processes f. CP specifies the private key protection requirements for backup copies of subscriber private keys stored by the subscriber. g. Subscriber Agreements describe the required processes to be followed by the Subscriber	Mandatory	WebTrust 5.4.7				

		h. CP specifies acceptable uses for subscriber key pairs i. CP specifies the requirements for subscriber key usage. j. CP specifies the private key protection requirements for archived subscriber private keys k. CP specifies the requirements for destruction of archived subscriber keys at the end of the archive period l. CP specifies the means through which subscriber key destruction is performed m. CP or CPS specifies the requirements for destruction of all copies and fragments of the subscriber's private key at the end of the key pair life cycle n. CP specifies the requirements for use and handling of cryptographic hardware and subscriber authentication processes o. CP specifies the requirements for notification of the CA in the event of subscriber key compromise.						
3.3.2.22	Subscriber Agreements describe the required processes to be followed by the Subscriber of any use of the cryptographic mechanism (e.g., HSM or ICC and software application).	1. Conduct walkthrough of the CA- Provided Subscriber Key Management process and check the following: a. CP specifies the appropriate ISO 15782- 1/FIPS 140- 2 level requirement for cryptographic modules used for subscriber key generation b. CP specifies the key generation	Mandatory	Subscriber Agreements describe the required processes to be				

		algorithm(s) that is used for subscriber key generation c. CP specifies the acceptable key sizes for subscriber key generation d. CP specifies the private key protection requirements for stored subscriber private keys e. CP states the circumstances and authority of when the subscriber's private key will be restored and the control processes f. CP specifies the private key protection requirements for backup copies of subscriber private keys stored by the subscriber. g. Subscriber Agreements describe the required processes to be followed by the Subscriber h. CP specifies acceptable uses for subscriber key pairs i. CP specifies the requirements for subscriber key usage. j. CP specifies the private key protection requirements for archived subscriber private keys k. CP specifies the requirements for destruction of archived subscriber keys at the end of the archive period l. CP specifies the means through which subscriber key destruction is performed m. CP or CPS specifies the requirements for destruction of all copies and fragments of the subscriber's private key at the end of the key pair life cycle n. CP specifies the						
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		requirements for use and handling of cryptographic hardware and subscriber authentication processes o. CP specifies the requirements for notification of the CA in the event of subscriber key compromise.						
3.3.2.23	The CP specifies the acceptable uses for subscriber key pairs.	1. Conduct walkthrough of the CA- Provided Subscriber Key Management process and check the following: a. CP specifies the appropriate ISO 15782- 1/FIPS 140- 2 level requirement for cryptographic modules used for subscriber key generation b. CP specifies the key generation algorithm(s) that is used for subscriber key generation c. CP specifies the acceptable key sizes for subscriber key generation d. CP specifies the private key protection requirements for stored subscriber private keys e. CP states the circumstances and authority of when the subscriber's private key will be restored and the control processes f. CP specifies the private key protection requirements for backup copies of subscriber private keys stored by the subscriber. g. Subscriber Agreements describe the required processes to be followed by the	Mandatory	WebTrust 5.4.9				

		Subscriber h. CP specifies acceptable uses for subscriber key pairs i. CP specifies the requirements for subscriber key usage. j. CP specifies the private key protection requirements for archived subscriber private keys k. CP specifies the requirements for destruction of archived subscriber keys at the end of the archive period l. CP specifies the means through which subscriber key destruction is performed m. CP or CPS specifies the requirements for destruction of all copies and fragments of the subscriber's private key at the end of the key pair life cycle n. CP specifies the requirements for use and handling of cryptographic hardware and subscriber authentication processes o. CP specifies the requirements for notification of the CA in the event of subscriber key compromise.						
3.3.2.24	The CP specifies the requirements for subscriber key usage.	1. Conduct walkthrough of the CA- Provided Subscriber Key Management process and check the following: a. CP specifies the appropriate ISO 15782- 1/FIPS 140- 2 level requirement for cryptographic modules used for subscriber key generation b. CP specifies the	Mandatory	WebTrust 5.4.10				

		key generation algorithm(s) that is used for subscriber key generation c. CP specifies the acceptable key sizes for subscriber key generation d. CP specifies the private key protection requirements for stored subscriber private keys e. CP states the circumstances and authority of when the subscriber's private key will be restored and the control processes f. CP specifies the private key protection requirements for backup copies of subscriber private keys stored by the subscriber. g. Subscriber Agreements describe the required processes to be followed by the Subscriber h. CP specifies acceptable uses for subscriber key pairs i. CP specifies the requirements for subscriber key usage. j. CP specifies the private key protection requirements for archived subscriber private keys k. CP specifies the requirements for destruction of archived subscriber keys at the end of the archive period l. CP specifies the means through which subscriber key destruction is performed m. CP or CPS specifies the requirements for destruction of all copies and fragments of the subscriber's private key at the end of the key pair life cycle						
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		n. CP specifies the requirements for use and handling of cryptographic hardware and subscriber authentication processes o. CP specifies the requirements for notification of the CA in the event of subscriber key compromise.						
3.3.2.25	The CP specifies the private key protection requirements for archived subscriber private keys.	1. Conduct walkthrough of the CA- Provided Subscriber Key Management process and check the following: a. CP specifies the appropriate ISO 15782- 1/FIPS 140- 2 level requirement for cryptographic modules used for subscriber key generation b. CP specifies the key generation algorithm(s) that is used for subscriber key generation c. CP specifies the acceptable key sizes for subscriber key generation d. CP specifies the private key protection requirements for stored subscriber private keys e. CP states the circumstances and authority of when the subscriber's private key will be restored and the control processes f. CP specifies the private key protection requirements for backup copies of subscriber private keys stored by the subscriber. g. Subscriber Agreements describe the required processes to be	Mandatory	WebTrust 5.4.11				

		followed by the Subscriber h. CP specifies acceptable uses for subscriber key pairs i. CP specifies the requirements for subscriber key usage. j. CP specifies the private key protection requirements for archived subscriber private keys k. CP specifies the requirements for destruction of archived subscriber keys at the end of the archive period l. CP specifies the means through which subscriber key destruction is performed m. CP or CPS specifies the requirements for destruction of all copies and fragments of the subscriber's private key at the end of the key pair life cycle n. CP specifies the requirements for use and handling of cryptographic hardware and subscriber authentication processes o. CP specifies the requirements for notification of the CA in the event of subscriber key compromise.						
3.3.2.26	The CP specifies the requirements for destruction of archived subscriber keys at the end of the archive period.	1. Conduct walkthrough of the CA- Provided Subscriber Key Management process and check the following: a. CP specifies the appropriate ISO 15782- 1/FIPS 140- 2 level requirement for cryptographic modules used for subscriber key generation	Mandatory	WebTrust 5.4.12				

		b. CP specifies the key generation algorithm(s) that is used for subscriber key generation c. CP specifies the acceptable key sizes for subscriber key generation d. CP specifies the private key protection requirements for stored subscriber private keys e. CP states the circumstances and authority of when the subscriber's private key will be restored and the control processes f. CP specifies the private key protection requirements for backup copies of subscriber private keys stored by the subscriber. g. Subscriber Agreements describe the required processes to be followed by the Subscriber h. CP specifies acceptable uses for subscriber key pairs i. CP specifies the requirements for subscriber key usage. j. CP specifies the private key protection requirements for archived subscriber private keys k. CP specifies the requirements for destruction of archived subscriber keys at the end of the archive period l. CP specifies the means through which subscriber key destruction is performed m. CP or CPS specifies the requirements for destruction of all copies and fragments of the subscriber's private key at the end of the						
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		key pair life cycle n. CP specifies the requirements for use and handling of cryptographic hardware and subscriber authentication processes o. CP specifies the requirements for notification of the CA in the event of subscriber key compromise.						
3.3.2.27	The CP specifies the means through which subscriber key destruction is performed.	1. Conduct walkthrough of the CA- Provided Subscriber Key Management process and check the following: a. CP specifies the appropriate ISO 15782- 1/FIPS 140- 2 level requirement for cryptographic modules used for subscriber key generation b. CP specifies the key generation algorithm(s) that is used for subscriber key generation c. CP specifies the acceptable key sizes for subscriber key generation d. CP specifies the private key protection requirements for stored subscriber private keys e. CP states the circumstances and authority of when the subscriber's private key will be restored and the control processes f. CP specifies the private key protection requirements for backup copies of subscriber private keys stored by the subscriber. g. Subscriber Agreements describe the required	Mandatory	WebTrust 5.4.13				

		processes to be followed by the Subscriber h. CP specifies acceptable uses for subscriber key pairs i. CP specifies the requirements for subscriber key usage. j. CP specifies the private key protection requirements for archived subscriber private keys k. CP specifies the requirements for destruction of archived subscriber keys at the end of the archive period l. CP specifies the means through which subscriber key destruction is performed m. CP or CPS specifies the requirements for destruction of all copies and fragments of the subscriber's private key at the end of the key pair life cycle n. CP specifies the requirements for use and handling of cryptographic hardware and subscriber authentication processes o. CP specifies the requirements for notification of the CA in the event of subscriber key compromise.						
3.3.2.28	The CP or CPS specifies the requirements for destruction of all copies and fragments of the subscriber's private key at the end of the key pair life cycle.	1. Conduct walkthrough of the CA- Provided Subscriber Key Management process and check the following: a. CP specifies the appropriate ISO 15782- 1/FIPS 140-2 level requirement for cryptographic modules used for subscriber key	Mandatory	WebTrust 5.4.14				

		generation b. CP specifies the key generation algorithm(s) that is used for subscriber key generation c. CP specifies the acceptable key sizes for subscriber key generation d. CP specifies the private key protection requirements for stored subscriber private keys e. CP states the circumstances and authority of when the subscriber's private key will be restored and the control processes f. CP specifies the private key protection requirements for backup copies of subscriber private keys stored by the subscriber. g. Subscriber Agreements describe the required processes to be followed by the Subscriber h. CP specifies acceptable uses for subscriber key pairs i. CP specifies the requirements for subscriber key usage. j. CP specifies the private key protection requirements for archived subscriber private keys k. CP specifies the requirements for destruction of archived subscriber keys at the end of the archive period l. CP specifies the means through which subscriber key destruction is performed m. CP or CPS specifies the requirements for destruction of all copies and fragments of the subscriber's private						
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		key at the end of the key pair life cycle n. CP specifies the requirements for use and handling of cryptographic hardware and subscriber authentication processes o. CP specifies the requirements for notification of the CA in the event of subscriber key compromise.						
3.3.2.29	CP specifies the requirements for use and handling of cryptographic hardware and subscriber authentication processes (and subsequent actions) where the cryptographic hardware is in other physical locations (i.e., an HSM attached to a mainframe or remote server).	1. Conduct walkthrough of the CA- Provided Subscriber Key Management process and check the following: a. CP specifies the appropriate ISO 15782- 1/FIPS 140- 2 level requirement for cryptographic modules used for subscriber key generation b. CP specifies the key generation algorithm(s) that is used for subscriber key generation c. CP specifies the acceptable key sizes for subscriber key generation d. CP specifies the private key protection requirements for stored subscriber private keys e. CP states the circumstances and authority of when the subscriber's private key will be restored and the control processes f. CP specifies the private key protection requirements for backup copies of subscriber private keys stored by the subscriber. g. Subscriber Agreements describe	Mandatory	WebTrust 5.4.15				

		the required processes to be followed by the Subscriber h. CP specifies acceptable uses for subscriber key pairs i. CP specifies the requirements for subscriber key usage. j. CP specifies the private key protection requirements for archived subscriber private keys k. CP specifies the requirements for destruction of archived subscriber keys at the end of the archive period l. CP specifies the means through which subscriber key destruction is performed m. CP or CPS specifies the requirements for destruction of all copies and fragments of the subscriber's private key at the end of the key pair life cycle n. CP specifies the requirements for use and handling of cryptographic hardware and subscriber authentication processes o. CP specifies the requirements for notification of the CA in the event of subscriber key compromise.					
3.3.2.30	The CP specifies the requirements for notification of the CA or RA in the event of subscriber key compromise.	1. Conduct walkthrough of the CA- Provided Subscriber Key Management process and check the following: a. CP specifies the appropriate ISO 15782- 1/FIPS 140-2 level requirement for cryptographic modules used for	Mandatory	WebTrust 5.4.16			

		subscriber key generation b. CP specifies the key generation algorithm(s) that is used for subscriber key generation c. CP specifies the acceptable key sizes for subscriber key generation d. CP specifies the private key protection requirements for stored subscriber private keys e. CP states the circumstances and authority of when the subscriber's private key will be restored and the control processes f. CP specifies the private key protection requirements for backup copies of subscriber private keys stored by the subscriber. g. Subscriber Agreements describe the required processes to be followed by the Subscriber h. CP specifies acceptable uses for subscriber key pairs i. CP specifies the requirements for subscriber key usage. j. CP specifies the private key protection requirements for archived subscriber private keys k. CP specifies the requirements for destruction of archived subscriber keys at the end of the archive period l. CP specifies the means through which subscriber key destruction is performed m. CP or CPS specifies the requirements for destruction of all copies and fragments of the						
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		subscriber's private key at the end of the key pair life cycle n. CP specifies the requirements for use and handling of cryptographic hardware and subscriber authentication processes o. CP specifies the requirements for notification of the CA in the event of subscriber key compromise.						
Certificate Operational Periods and Key Pair Usage Periods								
3.3.2.31	Subscriber SSL Certificates issued after 1 March 2018 must have a Validity Period no greater than 825 days. Subscriber Certificates issued after 1 July 2016 but prior to 1 March 2018 must have a Validity Period no greater than 39 months.	1. For sample SSL certificates issued after 1 March 2018, verify the validity period is not greater than 825 days 2. For sample certificates issued after 1 July 2016 but prior to 1 March 2018, verify the validity period is not greater than 39 months	Applicable for SSL Certificate	CA Browser Forum 6.3.2				
Confidentiality of Subscriber's Information								
3.3.2.32	Procedures and security controls to protect the privacy and confidentiality of the subscribers' data under the Certifying Authority's custody shall be implemented. Confidential information provided by the subscriber must not be disclosed to a third party without the subscribers' consent, unless the information is required to be disclosed under the law or a court order.	1. Verify CA has implemented procedures and security controls to protect privacy and confidentiality of subscribers' data 2. Validate confidential information provided by the subscriber is not disclosed to a third party without the subscribers' consent, unless the information is required to be disclosed under the law or a court order 3. Check the data on usage of DSC and transactional data is protected to ensure the subscribers' privacy 4. Verify a secure communication	Mandatory	IT CA Rules SCHEDULE III 22				

		channel between the CA and its subscribers is established to ensure the authenticity, integrity and confidentiality of the exchange						
3.3.2.33	Data on the usage of the Digital Signature Certificates by the subscribers and other transactional data relating to the subscribers' activities generated by the Certifying Authority in the course of its operation shall be protected to ensure the subscribers' privacy	1. Verify CA has implemented procedures and security controls to protect privacy and confidentiality of subscribers' data 2. Validate confidential information provided by the subscriber is not disclosed to a third party without the subscribers' consent, unless the information is required to be disclosed under the law or a court order 3. Check the data on usage of DSC and transactional data is protected to ensure the subscribers' privacy 4. Verify a secure communication channel between the CA and its subscribers is established to ensure the authenticity, integrity and confidentiality of the exchange	Mandatory	IT CA Rules SCHEDULE III 22				
3.3.2.34	A secure communication channel between the Certifying Authority and its subscribers shall be established to ensure the authenticity, integrity and confidentiality of the exchanges (e.g. transmission of Digital Signature Certificate, password, private key) during the Digital Signature Certificate issuance process.	1. Verify CA has implemented procedures and security controls to protect privacy and confidentiality of subscribers' data 2. Validate confidential information provided by the subscriber is not disclosed to a third party without the subscribers' consent, unless the information is required to be disclosed under the law or a court order	Mandatory	IT CA Rules SCHEDULE III 22				

		3. Check the data on usage of DSC and transactional data is protected to ensure the subscribers' privacy 4. Verify a secure communication channel between the CA and its subscribers is established to ensure the authenticity, integrity and confidentiality of the exchange						
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3.4. Certificate Management Controls

3.4.1. Certificate Lifecycle Management

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Subscriber Registration							
3.4.1.1	For authenticated certificates, the CA verifies or requires that the RA verify the credentials presented by a subscriber as evidence of identity or authority to perform a specific role in accordance with the requirements of the CP.	1. Conduct a walkthrough of the Subscriber registration process and check the following for a sample of subscriber registrations: a. CA verifies or requires that the RA verify the credentials presented by a subscriber as evidence of identity or authority to perform a specific role in accordance with the requirements of the CP for authenticated certificates b. the CA validates or requires that the RA validate (as determined by the CP) the organization's ownership, control, or right to use the domain name and/or IP address for domain and/or IP address validated certificates c. CA verifies the accuracy of the information included in the requesting entity's certificate request in accordance with the CP d. CA checks the Certificate Request for errors or omissions in accordance with the CP	Mandatory	WebTrust 6.1.1			

		e. CA uses the RA's public key contained in the requesting entity's Certificate Request to verify signature on the Certificate Request submission for end entity certificates f. CA verifies the uniqueness of the subscriber's distinguished name within the boundaries or community defined by the CP						
3.4.1.2	For domain and/or IP address validated certificates, the CA validates or requires that the RA validate (as determined by the CP) the organization's ownership, control, or right to use the domain name and/or IP address	1. Conduct a walkthrough of the Subscriber registration process and check the following for a sample of subscriber registrations: a. CA verifies or requires that the RA verify the credentials presented by a subscriber as evidence of identity or authority to perform a specific role in accordance with the requirements of the CP for authenticated certificates b. the CA validates or requires that the RA validate (as determined by the CP) the organization's ownership, control, or right to use the domain name and/or IP address for domain and/or IP address validated certificates c. CA verifies the accuracy of the information included in the requesting entity's certificate request in accordance with the CP d. CA checks the Certificate Request for errors or omissions in accordance with the CP e. CA uses the RA's public key contained in the requesting entity's Certificate Request to verify signature on the Certificate Request submission for end entity certificates	Mandatory	WebTrust 6.1.2				

		f. CA verifies the uniqueness of the subscriber's distinguished name within the boundaries or community defined by the CP						
3.4.1.3	The CA or RA verifies the accuracy of the information included in the requesting entity's certificate request in accordance with the CP.	1. Conduct a walkthrough of the Subscriber registration process and check the following for a sample of subscriber registrations: a. CA verifies or requires that the RA verify the credentials presented by a subscriber as evidence of identity or authority to perform a specific role in accordance with the requirements of the CP for authenticated certificates b. the CA validates or requires that the RA validate (as determined by the CP) the organization's ownership, control, or right to use the domain name and/or IP address for domain and/or IP address validated certificates c. CA verifies the accuracy of the information included in the requesting entity's certificate request in accordance with the CP d. CA checks the Certificate Request for errors or omissions in accordance with the CP e. CA uses the RA's public key contained in the requesting entity's Certificate Request to verify signature on the Certificate Request submission for end entity certificates f. CA verifies the uniqueness of the subscriber's distinguished name within the boundaries or community defined by the CP	Mandatory	WebTrust 6.1.3				

3.4.1.4	The CA or RA checks the Certificate Request for errors or omissions in accordance with the CP.	1. Conduct a walkthrough of the Subscriber registration process and check the following for a sample of subscriber registrations: a. CA verifies or requires that the RA verify the credentials presented by a subscriber as evidence of identity or authority to perform a specific role in accordance with the requirements of the CP for authenticated certificates b. the CA validates or requires that the RA validate (as determined by the CP) the organization's ownership, control, or right to use the domain name and/or IP address for domain and/or IP address validated certificates c. CA verifies the accuracy of the information included in the requesting entity's certificate request in accordance with the CP d. CA checks the Certificate Request for errors or omissions in accordance with the CP e. CA uses the RA's public key contained in the requesting entity's Certificate Request to verify signature on the Certificate Request submission for end entity certificates f. CA verifies the uniqueness of the subscriber's distinguished name within the boundaries or community defined by the CP	Mandatory	WebTrust 6.1.4				
3.4.1.5	For end entity certificates, the CA uses the RA's public key contained in the requesting entity's Certificate Request to verify signature on the Certificate Request submission.	1. Conduct a walkthrough of the Subscriber registration process and check the following for a sample of subscriber registrations: a. CA verifies or requires that the RA verify the credentials	Mandatory	WebTrust 6.1.5				

		presented by a subscriber as evidence of identity or authority to perform a specific role in accordance with the requirements of the CP for authenticated certificates b. the CA validates or requires that the RA validate (as determined by the CP) the organization's ownership, control, or right to use the domain name and/or IP address for domain and/or IP address validated certificates c. CA verifies the accuracy of the information included in the requesting entity's certificate request in accordance with the CP d. CA checks the Certificate Request for errors or omissions in accordance with the CP e. CA uses the RA's public key contained in the requesting entity's Certificate Request to verify signature on the Certificate Request submission for end entity certificates f. CA verifies the uniqueness of the subscriber's distinguished name within the boundaries or community defined by the CP							
3.4.1.6	The CA verifies the uniqueness of the subscriber's distinguished name within the boundaries or community defined by the CP.	1. Conduct a walkthrough of the Subscriber registration process and check the following for a sample of subscriber registrations: a. CA verifies or requires that the RA verify the credentials presented by a subscriber as evidence of identity or authority to perform a specific role in accordance with the requirements of the CP for authenticated certificates	Mandatory	WebTrust 6.1.6					

		b. the CA validates or requires that the RA validate (as determined by the CP) the organization's ownership, control, or right to use the domain name and/or IP address for domain and/or IP address validated certificates c. CA verifies the accuracy of the information included in the requesting entity's certificate request in accordance with the CP d. CA checks the Certificate Request for errors or omissions in accordance with the CP e. CA uses the RA's public key contained in the requesting entity's Certificate Request to verify signature on the Certificate Request submission for end entity certificates f. CA verifies the uniqueness of the subscriber's distinguished name within the boundaries or community defined by the CP						
3.4.1.7	Encryption and access controls are used to protect the confidentiality and integrity of registration data in transit and in storage.	1. Verify encryption and access controls are used to protect the confidentiality and integrity of registration data in transit and in storage.	Mandatory	WebTrust 6.1.7				
3.4.1.8	At the point of registration (before certificate issuance) the RA or CA informs the Subscriber of the terms and conditions regarding use of the certificate.	1. Conduct walkthrough and check the following: a. Before certificate issuance the CA informs the Subscriber of the terms and conditions regarding use of the certificate. b. The entity requesting a certificate prepares and submits the appropriate certificate request data (Registration Request) to an RA (or the CA) as specified in the CP.	Mandatory	WebTrust 6.1.8				

3.4.1.9	The CA requires that an entity requesting a certificate must prepare and submit the appropriate certificate request data (Registration Request) to an RA (or the CA) as specified in the CP.	1. Conduct walkthrough and check the following: a. Before certificate issuance the CA informs the Subscriber of the terms and conditions regarding use of the certificate. b. The entity requesting a certificate prepares and submits the appropriate certificate request data (Registration Request) to an RA (or the CA) as specified in the CP.	Mandatory	WebTrust 6.1.10				
3.4.1.10	The CA requires that the requesting entity submit its public key in a self-signed message to the CA for certification. The CA requires that the requesting entity digitally sign the Registration Request using the private key that relates to the public key contained in the Registration Request in order to: • allow the detection of errors in the certificate application process; and • Prove possession of the companion private key for the public key being registered.	1. Verify requesting entity submits its public key in a self-signed message to the CA for certification 2. Validate the requesting entity digitally signs the Registration Request using the private key that relates to the public key contained in the Registration Request 3. Validate certificate request is treated as acceptance of the terms of conditions by the requesting entity 4. Check the identity of RA authorized to issue registration requests is validated by the CA	Mandatory	WebTrust 6.1.11				
3.4.1.11	The certificate request is treated as acceptance of the terms of conditions by the requesting entity to use that certificate as described in the Subscriber Agreement	1. Verify requesting entity submits its public key in a self-signed message to the CA for certification 2. Validate the requesting entity digitally signs the Registration Request using the private key that relates to the public key contained in the Registration Request 3. Validate certificate request is treated as acceptance of the terms of conditions by the requesting entity 4. Check the identity of RA authorized to issue registration requests is validated by the CA	Mandatory	WebTrust 6.1.12				
3.4.1.12	The CA validates the identity of the RA authorized to issue	1. Verify requesting entity submits its public key in a self-signed	Mandatory	WebTrust 6.1.13				

	registration requests under a specific CP.	message to the CA for certification 2. Validate the requesting entity digitally signs the Registration Request using the private key that relates to the public key contained in the Registration Request 3. Validate certificate request is treated as acceptance of the terms of conditions by the requesting entity 4. Check the identity of RA authorized to issue registration requests is validated by the CA						
3.4.1.13	The CA requires that RAs submit the requesting entity's certificate request data to the CA in a message (Certificate Request) signed by the RA. The CA verifies the RA's signature on the Certificate Request.	1. Verify the RAs submit the requesting entity's certificate request data to the CA in a message (Certificate Request) signed by the RA	Mandatory	WebTrust 6.1.14				
3.4.1.14	The CA requires that the RA secure that part of the certificate application process for which it (the RA) assumes responsibility in accordance with the CA's CPS.	1. Verify that the RA secures that part of the certificate application process for which it assumes responsibility in accordance with the CA's CPS	Mandatory	WebTrust 6.1.15				
3.4.1.15	The CA requires that RAs record their actions in an audit log.	1. Verify all RA activities are recorded in an audit log	Mandatory	WebTrust 6.1.16				
3.4.1.16	The CA verifies the authenticity of the submission by the RA in accordance with the CA's CPS.	1. Check that the authenticity of the submission by the RA is verified by the CA	Mandatory	WebTrust 6.1.17				
Certificate Issuance								
3.4.1.17	The CA generates certificates using Certificate Request Data and manufactures the certificate as defined by the appropriate Certificate Profile in accordance with ISO 9594/X.509 and ISO 15782-1 formatting rules	1. Conduct walkthrough of the Certificate Issuance process to check the following: a. CA generates certificates using Certificate Request Data and manufactures the certificate as defined by the appropriate	Mandatory	WebTrust 6.4.1, CA Browser Forum 4.3, X.509 Policy 4.3				

	as disclosed within the CP.CA shall verify the source of a certificate request before issuance.	Certificate Profile b. CA verifies source of certificate request before issuance c. Validity periods are set by CA in compliance with the CP d. Extension fields are formatted in accordance with the CP e. CA signs the end entity's public key and other relevant information with the CA's private signing key f. CA publishes the certificate after the certificate has been accepted by the requesting entity as disclosed in the CA's business practices g. CA notifies the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request h. Authorized individual issues direct command for Root CA to perform a certificate signing operation i. Certificates are issued based on approved subscriber registration, certificate renewal or certificate rekey requests in accordance with the CP j. CA issues a signed notification to the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request k. CA issues an out-of-band notification to the Subscriber when a certificate is issued l. copies of certificates are retained for the appropriate period of time specified in the CP						
3.4.1.18	Validity periods are set in the CP and are formatted in accordance with ISO 9594/X.509 and ISO 15782-1 as disclosed within the CP.	1. Conduct walkthrough of the Certificate Issuance process to check the following: a. CA generates certificates using Certificate Request Data and manufactures the certificate as defined by the appropriate Certificate Profile b. CA verifies source of	Mandatory	WebTrust 6.4.2				

		certificate request before issuance c. Validity periods are set by CA in compliance with the CP d. Extension fields are formatted in accordance with with the CP e. CA signs the end entity's public key and other relevant information with the CA's private signing key f. CA publishes the certificate after the certificate has been accepted by the requesting entity as disclosed in the CA's business practices g. CA notifies the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request h. Authorized individual issues direct command for Root CA to perform a certificate signing operation i. Certificates are issued based on approved subscriber registration, certificate renewal or certificate rekey requests in accordance with the CP j. CA issues a signed notification to the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request k. CA issues an out-of-band notification to the Subscriber when a certificate is issued l. copies of certificates are retained for the appropriate period of time specified in the CP						
3.4.1.19	Extension fields are formatted in accordance with ISO 9594/X.509 and ISO 15782-1 as disclosed within the CP.	1. Conduct walkthrough of the Certificate Issuance process to check the following: a. CA generates certificates using Certificate Request Data and manufactures the certificate as defined by the appropriate Certificate Profile b. CA verifies source of certificate request before issuance	Mandatory	WebTrust 6.4.3				

		c. Validity periods are set by CA in compliance with the CP d. Extension fields are formatted in accordance with the CP e. CA signs the end entity's public key and other relevant information with the CA's private signing key f. CA publishes the certificate after the certificate has been accepted by the requesting entity as disclosed in the CA's business practices g. CA notifies the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request h. Authorized individual issues direct command for Root CA to perform a certificate signing operation i. Certificates are issued based on approved subscriber registration, certificate renewal or certificate rekey requests in accordance with the CP j. CA issues a signed notification to the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request k. CA issues an out-of-band notification to the Subscriber when a certificate is issued l. copies of certificates are retained for the appropriate period of time specified in the CP						
3.4.1.20	The CA signs the end entity's public key and other relevant information with the CA's private signing key.	1. Conduct walkthrough of the Certificate Issuance process to check the following: a. CA generates certificates using Certificate Request Data and manufactures the certificate as defined by the appropriate Certificate Profile b. CA verifies source of certificate request before issuance c. Validity periods are set by CA in compliance	Mandatory	WebTrust 6.4.4				

		with the CP d. Extension fields are formatted in accordance with with the CP e. CA signs the end entity’s public key and other relevant information with the CA’s private signing key f. CA publishes the certificate after the certificate has been accepted by the requesting entity as disclosed in the CA’s business practices g. CA notifies the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request h. Authorized individual issues direct command for Root CA to perform a certificate signing operation i. Certificates are issued based on approved subscriber registration, certificate renewal or certificate rekey requests in accordance with the CP j. CA issues a signed notification to the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request k. CA issues an out-of-band notification to the Subscriber when a certificate is issued l. copies of certificates are retained for the appropriate period of time specified in the CP					
3.4.1.21	The CA publishes the certificate after the certificate has been accepted by the requesting entity as disclosed in the CA’s business practices.	1. Conduct walkthrough of the Certificate Issuance process to check the following: a. CA generates certificates using Certificate Request Data and manufactures the certificate as defined by the appropriate Certificate Profile b. CA verifies source of certificate request before issuance c. Validity periods are set by CA in compliance with the CP d. Extension fields are	Mandatory	WebTrust 6.4.5, CA Browser Forum 4.3			

		formatted in accordance with with the CP e. CA signs the end entity's public key and other relevant information with the CA's private signing key f. CA publishes the certificate after the certificate has been accepted by the requesting entity as disclosed in the CA's business practices g. CA notifies the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request h. Authorized individual issues direct command for Root CA to perform a certificate signing operation i. Certificates are issued based on approved subscriber registration, certificate renewal or certificate rekey requests in accordance with the CP j. CA issues a signed notification to the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request k. CA issues an out-of-band notification to the Subscriber when a certificate is issued l. copies of certificates are retained for the appropriate period of time specified in the CP						
3.4.1.22	When an RA is used, the CA notifies the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request. Certificate issuance by the Root CA SHALL require an individual authorized by the CA (i.e. the CA system operator, system officer, or PKI administrator) to deliberately issue a direct command in order for the Root CA to perform a certificate signing operation.	1. Conduct walkthrough of the Certificate Issuance process to check the following: a. CA generates certificates using Certificate Request Data and manufactures the certificate as defined by the appropriate Certificate Profile b. CA verifies source of certificate request before issuance c. Validity periods are set by CA in compliance with the CP d. Extension fields are formatted in accordance with with the CP	Mandatory	WebTrust 6.4.6				

		e. CA signs the end entity's public key and other relevant information with the CA's private signing key f. CA publishes the certificate after the certificate has been accepted by the requesting entity as disclosed in the CA's business practices g. CA notifies the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request h. Authorized individual issues direct command for Root CA to perform a certificate signing operation i. Certificates are issued based on approved subscriber registration, certificate renewal or certificate rekey requests in accordance with the CP j. CA issues a signed notification to the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request k. CA issues an out-of-band notification to the Subscriber when a certificate is issued l. copies of certificates are retained for the appropriate period of time specified in the CP						
3.4.1.23	Certificates are issued based on approved subscriber registration, certificate renewal or certificate rekey requests in accordance with the CP.	1. Conduct walkthrough of the Certificate Issuance process to check the following: a. CA generates certificates using Certificate Request Data and manufactures the certificate as defined by the appropriate Certificate Profile b. CA verifies source of certificate request before issuance c. Validity periods are set by CA in compliance with the CP d. Extension fields are formatted in accordance with the CP e. CA signs the end entity's public key and	Mandatory	WebTrust 6.4.7				

		other relevant information with the CA's private signing key f. CA publishes the certificate after the certificate has been accepted by the requesting entity as disclosed in the CA's business practices g. CA notifies the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request h. Authorized individual issues direct command for Root CA to perform a certificate signing operation i. Certificates are issued based on approved subscriber registration, certificate renewal or certificate rekey requests in accordance with the CP j. CA issues a signed notification to the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request k. CA issues an out-of-band notification to the Subscriber when a certificate is issued l. copies of certificates are retained for the appropriate period of time specified in the CP					
3.4.1.24	The CA issues a signed notification to the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request.	1. Conduct walkthrough of the Certificate Issuance process to check the following: a. CA generates certificates using Certificate Request Data and manufactures the certificate as defined by the appropriate Certificate Profile b. CA verifies source of certificate request before issuance c. Validity periods are set by CA in compliance with the CP d. Extension fields are formatted in accordance with with the CP e. CA signs the end entity's public key and other relevant information with the	Mandatory	WebTrust 6.4.8, X.509 Policy 4.3			

		CA's private signing key f. CA publishes the certificate after the certificate has been accepted by the requesting entity as disclosed in the CA's business practices g. CA notifies the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request h. Authorized individual issues direct command for Root CA to perform a certificate signing operation i. Certificates are issued based on approved subscriber registration, certificate renewal or certificate rekey requests in accordance with the CP j. CA issues a signed notification to the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request k. CA issues an out-of-band notification to the Subscriber when a certificate is issued l. copies of certificates are retained for the appropriate period of time specified in the CP						
3.4.1.25	The CA issues an out-of-band notification to the Subscriber when a certificate is issued. Where this notification includes initial activation data, then control processes ensure safe delivery to the Subscriber.	1. Conduct walkthrough of the Certificate Issuance process to check the following: a. CA generates certificates using Certificate Request Data and manufactures the certificate as defined by the appropriate Certificate Profile b. CA verifies source of certificate request before issuance c. Validity periods are set by CA in compliance with the CP d. Extension fields are formatted in accordance with with the CP e. CA signs the end entity's public key and other relevant information with the CA's private signing key f. CA publishes the	Mandatory	WebTrust 6.4.9				

		certificate after the certificate has been accepted by the requesting entity as disclosed in the CA's business practices g. CA notifies the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request h. Authorized individual issues direct command for Root CA to perform a certificate signing operation i. Certificates are issued based on approved subscriber registration, certificate renewal or certificate rekey requests in accordance with the CP j. CA issues a signed notification to the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request k. CA issues an out-of-band notification to the Subscriber when a certificate is issued l. copies of certificates are retained for the appropriate period of time specified in the CP					
3.4.1.26	Whether certificates expire, are revoked or are suspended, copies of certificates are retained for the appropriate period of time specified in the CP.	1. Conduct walkthrough of the Certificate Issuance process to check the following: a. CA generates certificates using Certificate Request Data and manufactures the certificate as defined by the appropriate Certificate Profile b. CA verifies source of certificate request before issuance c. Validity periods are set by CA in compliance with the CP d. Extension fields are formatted in accordance with with the CP e. CA signs the end entity's public key and other relevant information with the CA's private signing key f. CA publishes the certificate after the certificate has been	Mandatory	WebTrust 6.4.10			

		accepted by the requesting entity as disclosed in the CA's business practices g. CA notifies the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request h. Authorized individual issues direct command for Root CA to perform a certificate signing operation i. Certificates are issued based on approved subscriber registration, certificate renewal or certificate rekey requests in accordance with the CP j. CA issues a signed notification to the RA when a certificate is issued to a subscriber for whom the RA submitted a certificate request k. CA issues an out-of-band notification to the Subscriber when a certificate is issued l. copies of certificates are retained for the appropriate period of time specified in the CP						
Certificate Distribution								
3.4.1.27	Only authorized CA personnel administer the CA's repository or alternative distribution mechanism.	1. Conduct walkthrough of certificate distribution process to check the following: a. Only authorized CA personnel administer the CA's repository or alternative distribution mechanism b. performance of the CA's repository or alternative distribution mechanism is monitored and managed c. integrity of the repository or alternative distribution mechanism is maintained and administered	Mandatory	WebTrust 6.5.2				
3.4.1.28	The performance of the CA's repository or alternative distribution mechanism is monitored and managed.	1. Conduct walkthrough of certificate distribution process to check the following: a. Only authorized CA personnel administer the CA's repository or	Mandatory	WebTrust 6.5.3				

		alternative distribution mechanism b. performance of the CA's repository or alternative distribution mechanism is monitored and managed c. integrity of the repository or alternative distribution mechanism is maintained and administered						
3.4.1.29	The integrity of the repository or alternative distribution mechanism is maintained and administered.	1. Conduct walkthrough of certificate distribution process to check the following: a. Only authorized CA personnel administer the CA's repository or alternative distribution mechanism b. performance of the CA's repository or alternative distribution mechanism is monitored and managed c. integrity of the repository or alternative distribution mechanism is maintained and administered	Mandatory	WebTrust 6.5.4				
Certificate Acceptance								
3.4.1.30	Procedures shall be developed for certificate acceptance and publication of the certificate by the CA. The subject must confirm acceptance of the certificate upon notification of issuance by the CA.	1. Verify procedures have been developed for certificate acceptance and publication and subject confirms acceptance of certificate 2. Verify CA notifies of certificate issuance to other entities	Mandatory	X.509 Policy 4.4, CA Browser Forum 4.4				
3.4.1.31	Notification of certificate issuance by the CA to other entities	1. Verify procedures have been developed for certificate acceptance and publication and subject confirms acceptance of certificate 2. Verify CA notifies of certificate issuance to other entities	Mandatory	X.509 Policy 4.4, CA Browser Forum 4.4				
Key Pair and Certificate Usage								
3.4.1.32	Subscribers and CAs shall protect their private keys from access by any other party. Subscribers and CAs shall use their	1. Check that the CAs protect their private keys from access by any other party. 2. Verify CAs use their	Mandatory	X.509 Policy 4.5.1, CA Browser Forum 4.5				

	private keys for the purposes as constrained by the extensions (such as key usage, extended key usage, certificate policies, etc.) in the certificates issued to them	private keys for the purposes as constrained by the extensions in the certificates issued to them						
Certificate Renewal								
3.4.1.33	A certificate may be renewed if the public key has not reached the end of its validity period, the associated private key has not been revoked or compromised, and the Subscriber name and attributes are unchanged.	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no action is required from the auditor's side.	Not Applicable	X.509 Policy 4.6				
3.4.1.34	In addition, the validity period of the certificate must not exceed the remaining lifetime of the private key	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no action is required from the auditor's side.	Not Applicable	X.509 Policy 4.6				
3.4.1.35	The Certificate Renewal Request includes at least the subscriber's Distinguished Name, the Serial Number of the certificate (or other information that identifies the certificate), and the requested validity period. (The CA will only renew certificates that were issued by itself.)	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no action is required from the auditor's side.	Not Applicable	WebTrust 6.2.1				
3.4.1.36	The CA requires that the requesting entity digitally sign the Certificate Renewal Request using the private key that relates to the public key contained in the requesting entity's existing public key certificate	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no action is required from the auditor's side.	Not Applicable	WebTrust 6.2.2				
3.4.1.37	The CA issues a new certificate using the subscriber's previously certified public key, only if its cryptographic security is still sufficient for the new certificate's	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no action is required from the auditor's side.	Not Applicable	WebTrust 6.2.3				

	intended lifetime and no indications exist that the subscriber's private key has been compromised							
3.4.1.38	For renewal of authenticated certificates, the CA or the RA process the certificate renewal data to verify the identity of the requesting entity and to identify the certificate to be renewed.	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no action is required from the auditor's side.	Not Applicable	WebTrust 6.2.4				
3.4.1.39	For domain validated certificates, the CA or the RA process the certificate renewal data to re-validate the domain in accordance with the requirements of the CP.	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no action is required from the auditor's side.	Not Applicable	WebTrust 6.2.5				
3.4.1.40	The CA or the RA validate the signature on the Certificate Renewal Request	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no action is required from the auditor's side.	Not Applicable	WebTrust 6.2.6				
3.4.1.41	The CA verifies the existence and validity of the certificate The CA verifies the existence and validity of the certificate to be renewed. The CA does not renew certificates that have been revoked, expired or suspended to be renewed. The CA does not renew certificates that have been revoked, expired or suspended	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no action is required from the auditor's side.	Not Applicable	WebTrust 6.2.7				
3.4.1.42	The CA or the RA verifies that the request, including the extension of the validity period, meets the requirements defined in the CP.	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no action is required from the auditor's side.	Not Applicable	WebTrust 6.2.8				
3.4.1.43	The CA requires that RAs submit the Certificate Renewal Data to the CA in a message (Certificate Renewal	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no	Not Applicable	WebTrust 6.2.9				

	Request) signed by the RA.	action is required from the auditor's side.						
3.4.1.44	The CA requires that the RA secures that part of the certificate renewal process for which it (the RA) assumes responsibility in accordance with the CP.	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no action is required from the auditor's side.	Not Applicable	WebTrust 6.2.10				
3.4.1.45	The CA requires that RAs record their actions in an audit log.	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no action is required from the auditor's side.	Not Applicable	WebTrust 6.2.11				
3.4.1.46	The CA verifies the authenticity of the submission by the RA.	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no action is required from the auditor's side.	Not Applicable	WebTrust 6.2.12				
3.4.1.47	The CA verifies the RA's signature on the Certificate Renewal Request	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no action is required from the auditor's side.	Not Applicable	WebTrust 6.2.13				
3.4.1.48	The CA checks the Certificate Renewal Request for errors or omissions. This function may be delegated explicitly to the RA	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no action is required from the auditor's side.	Not Applicable	WebTrust 6.2.14				
3.4.1.49	The CA or RA notifies Subscribers prior to the expiration of their certificate of the need for renewal in accordance with the CP	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no action is required from the auditor's side.	Not Applicable	WebTrust 6.2.15				
3.4.1.50	The CA issues a signed notification indicating the certificate renewal has been successful.	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no action is required from the auditor's side.	Not Applicable	WebTrust 6.2.16				

3.4.1.51	The CA makes the new certificate available to the end entity in accordance with the CP.	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no action is required from the auditor's side.	Not Applicable	WebTrust 6.2.17				
3.4.1.52	A certificate renewal shall be achieved using one of the following processes: • Initial registration process • Identification & Authentication for Re-key, except the old key can also be used as the new key	1. Certificate renewal is not performed as part of current CA systems Hence all the controls are not applicable and no action is required from the auditor's side.	Not Applicable	X.509 Policy 4.6.3				
Certificate Re-Key								
3.4.1.53	A certificate re-key shall be achieved using one of the following processes: • Initial registration process • Identification & Authentication for Re-key	1. Verify Certificate re-key is achieved using one of the following processes: a. Initial registration process	Mandatory	X.509 Policy 4.7.3, CA Browser Forum 4.7, X.509 Policy 4.7				
Certificate Validation								
3.4.1.54	The CA makes certificate status information available to relevant entities (Relying Parties or their agents) using an established mechanism in accordance with the CP. This is achieved using: • Request Response Method – A request signed by the Relying Party to the Certificate Status Provider's responder. In turn, the Certificate Status Provider's responder responds with the certificate status duly signed. (OCSP is an example protocol using this method.) • Delivery Method – A CRL signed by the CA and published within the policy's time frame.	1. Check that the certificate status information is made available to relevant entities by the CA using an established mechanism in accordance with the CP 2. Verify Request Response Method and Delivery Method are used for Validation	Mandatory	WebTrust 6.8.1				
Certificate Revocation List (CRL) Controls								

3.4.1.55	The CA digitally signs each CRL that it issues so that entities can validate the integrity of the CRL and the date and time of issuance.	1. Verify that the CRL issued by the CA is digitally signed by the CA 2. Validate that the CRLs are issued at fixed intervals even if no changes have been made 3. Validate on sample basis a CRL entry identifying a revoked certificate remains on the CRL until the end of the certificate's validity period. 4. CRLs are archived in accordance with the requirements of the CP including the method of retrieval 5. CAs include a monotonically increasing sequence number for each CRL issued by that CA 6. CRL contains entries for all revoked unexpired certificates issued by the CA 7. Old CRLs are retained for the appropriate period of time specified in the CA's CP.	Mandatory	WebTrust 6.8.2				
3.4.1.56	The CA issues CRLs at regular intervals, as specified in the CP, even if no changes have occurred since the last issuance.	1. Verify that the CRL issued by the CA is digitally signed by the CA 2. Validate that the CRLs are issued at fixed intervals even if no changes have been made 3. Validate on sample basis a CRL entry identifying a revoked certificate remains on the CRL until the end of the certificate's validity period. 4. CRLs are archived in accordance with the requirements of the CP including the method of retrieval 5. CAs include a monotonically increasing sequence number for each CRL issued by that CA 6. CRL contains entries for all revoked unexpired certificates issued by the CA 7. Old CRLs are retained for the appropriate period	Mandatory	WebTrust 6.8.3				

		of time specified in the CA's CP.						
3.4.1.57	At a minimum, a CRL entry identifying a revoked certificate remains on the CRL until the end of the certificate's validity period.	1. Verify that the CRL issued by the CA is digitally signed by the CA 2. Validate that the CRLs are issued at fixed intervals even if no changes have been made 3. Validate on sample basis a CRL entry identifying a revoked certificate remains on the CRL until the end of the certificate's validity period. 4. CRLs are archived in accordance with the requirements of the CP including the method of retrieval 5. CAs include a monotonically increasing sequence number for each CRL issued by that CA 6. CRL contains entries for all revoked unexpired certificates issued by the CA 7. Old CRLs are retained for the appropriate period of time specified in the CA's CP.	Mandatory	WebTrust 6.8.4				
3.4.1.58	If certificate suspension is supported, a certificate suspension (hold) entry, with its original action date and expiration date remain on the CRL until the normal expiration of the certificate or until the suspension is lifted	1. Verify that the CRL issued by the CA is digitally signed by the CA 2. Validate that the CRLs are issued at fixed intervals even if no changes have been made 3. Validate on sample basis a CRL entry identifying a revoked certificate remains on the CRL until the end of the certificate's validity period. 4. CRLs are archived in accordance with the requirements of the CP including the method of retrieval 5. CAs include a monotonically increasing sequence number for each CRL issued by that CA 6. CRL contains entries for all revoked unexpired	Mandatory	WebTrust 6.8.5				

		certificates issued by the CA 7. Old CRLs are retained for the appropriate period of time specified in the CA's CP.						
3.4.1.59	CRLs are archived in accordance with the requirements of the CP including the method of retrieval.	1. Verify that the CRL issued by the CA is digitally signed by the CA 2. Validate that the CRLs are issued at fixed intervals even if no changes have been made 3. Validate on sample basis a CRL entry identifying a revoked certificate remains on the CRL until the end of the certificate's validity period. 4. CRLs are archived in accordance with the requirements of the CP including the method of retrieval 5. CAs include a monotonically increasing sequence number for each CRL issued by that CA 6. CRL contains entries for all revoked unexpired certificates issued by the CA 7. Old CRLs are retained for the appropriate period of time specified in the CA's CP.	Mandatory	WebTrust 6.8.6				
3.4.1.60	CAs include a monotonically increasing sequence number for each CRL issued by that CA	1. Verify that the CRL issued by the CA is digitally signed by the CA 2. Validate that the CRLs are issued at fixed intervals even if no changes have been made 3. Validate on sample basis a CRL entry identifying a revoked certificate remains on the CRL until the end of the certificate's validity period. 4. CRLs are archived in accordance with the requirements of the CP including the method of retrieval 5. CAs include a monotonically increasing sequence number for	Mandatory	WebTrust 6.8.7				

		each CRL issued by that CA 6. CRL contains entries for all revoked unexpired certificates issued by the CA 7. Old CRLs are retained for the appropriate period of time specified in the CA's CP.						
3.4.1.61	The CRL contains entries for all revoked unexpired certificates issued by the CA.	1. Verify that the CRL issued by the CA is digitally signed by the CA 2. Validate that the CRLs are issued at fixed intervals even if no changes have been made 3. Validate on sample basis a CRL entry identifying a revoked certificate remains on the CRL until the end of the certificate's validity period. 4. CRLs are archived in accordance with the requirements of the CP including the method of retrieval 5. CAs include a monotonically increasing sequence number for each CRL issued by that CA 6. CRL contains entries for all revoked unexpired certificates issued by the CA 7. Old CRLs are retained for the appropriate period of time specified in the CA's CP.	Mandatory	WebTrust 6.8.8				
3.4.1.62	Old CRLs are retained for the appropriate period of time specified in the CA's CP.	1. Verify that the CRL issued by the CA is digitally signed by the CA 2. Validate that the CRLs are issued at fixed intervals even if no changes have been made 3. Validate on sample basis a CRL entry identifying a revoked certificate remains on the CRL until the end of the certificate's validity period. 4. CRLs are archived in accordance with the requirements of the CP including the method of retrieval	Mandatory	WebTrust 6.8.9				

		5. CAs include a monotonically increasing sequence number for each CRL issued by that CA 6. CRL contains entries for all revoked unexpired certificates issued by the CA 7. Old CRLs are retained for the appropriate period of time specified in the CA's CP.						
Certificate Revocation and Suspension								
3.4.1.63	A certificate shall be revoked when the binding between the subject and the subject's public key defined within a certificate is no longer considered valid. The CA SHALL provide a process for Subscribers to request revocation of their own Certificates. The process MUST be described in the CA's Certificate Policy or Certification Practice Statement. The CA SHALL maintain a continuous 24x7 ability to accept and respond to revocation requests and related inquiries	1. For sample cases, validate certificate is revoked when the binding between the subject and the subject's public key defined within a certificate is no longer considered valid 2. Verify the CA provides a process for Subscribers to request revocation of their own Certificates and the process is described in the CA's Certificate Policy or CCA-CAAC Ver 1.6 122 Certification Practice Statement. 3. Check the CA maintains a continuous 24x7 ability to accept and respond to revocation requests and related inquiries	Mandatory	X.509 Policy 4.9.1, CA Browser Forum 4.9				
3.4.1.64	The CA provides a means of rapid communication to facilitate the secure and authenticated revocation or renovation of the following: • one or more certificates of one or more subscribers; • the set of all certificates issued by a CA based on a single public/private key pair used by a CA to generate certificates; and • all certificates issued by a CA, regardless of the public/private key pair used. The CA SHALL provide Subscribers, Relying Parties,	1. Check that the CA provides a means of rapid communication to facilitate the secure and authenticated revocation or renovation of the elements mentioned in the control 2. Verify the CA provides Subscribers, Relying Parties, Application Software Suppliers, and other third parties with clear instructions for reporting suspected Private Key Compromise, Certificate misuse, or other types of fraud, compromise, misuse, inappropriate conduct, or any other matter related to Certificates. 3. Validate the CA has publicly disclosed the	Mandatory	WebTrust 6.6.1, X.509 Policy 4.9.3, CA Browser Forum 4.9.3				

	Application Software Suppliers, and other third parties with clear instructions for reporting suspected Private Key Compromise, Certificate misuse, or other types of fraud, compromise, misuse, inappropriate conduct, or any other matter related to Certificates. The CA SHALL publicly disclose the instructions through a readily accessible online means.	instructions through a readily accessible online means						
3.4.1.65	The CA provides a means of rapid communication to facilitate the secure and authenticated revocation or renovation of the following: • one or more certificates of one or more subscribers; • the set of all certificates issued by a CA based on a single public/private key pair used by a CA to generate certificates; and • all certificates issued by a CA, regardless of the public/private key pair used. The CA SHALL provide Subscribers, Relying Parties, Application Software Suppliers, and other third parties with clear instructions for reporting suspected Private Key Compromise, Certificate misuse, or other types of fraud, compromise, misuse, inappropriate conduct, or any other matter related to Certificates. The CA SHALL publicly disclose the instructions through a readily accessible online means.	1. Validate that CA verifies the identity and authority of the entity requesting revocation or suspension and reactivation of a certificate in accordance with the CP 2. Verify the Subscriber or Issuing CA can initiate revocation.	Mandatory	WebTrust 6.6.1, X.509 Policy 4.9.3, CA Browser Forum 4.9.3				
3.4.1.66	The CA verifies or requires that the RA verify the identity and authority of the entity requesting revocation or suspension and	1. Validate that CA verifies the identity and authority of the entity requesting revocation or suspension and reactivation of a	Mandatory	WebTrust 6.6.2, X.509 Policy 4.9.2, CA Browser Forum 4.9.2				

	reactivation of a certificate in accordance with the CP. The Subscriber, RA, or Issuing CA can initiate revocation. Additionally, Subscribers, Relying Parties, Application Software Suppliers, and other third parties may submit Certificate Problem Reports informing the issuing CA of reasonable cause to revoke the certificate	certificate in accordance with the CP 2. Verify the Subscriber or Issuing CA can initiate revocation.						
3.4.1.67	If an external RA accepts and forwards revocation requests to the CA, the CA provides a signed acknowledgement of the revocation request and confirmation of actions to the requesting RA.	1. Verify for past instances where the CA provides a signed acknowledgement of the revocation request and confirmation of action	Mandatory	WebTrust 6.6.4				
3.4.1.68	The CA updates the Certificate Revocation List (CRL) and other certificate status mechanisms in the timeframes specified within the CP and in accordance with the format defined in ISO 9594/X.509 and ISO 15782-1.	1. Verify that the CA maintains and updates the Certificate Revocation List (CRL)	Mandatory	WebTrust 6.6.5				
3.4.1.69	The Certifying Authority shall publish a notice of suspension or revocation of any certificate in the Certificate Revocation List in its repository immediately after receiving an authorised request of such suspension or revocation. The CA records all certificate revocation requests and their outcome in an audit log.	1. Verify CA publishes a notice of suspension or revocation of any certificate in the Certificate Revocation List in its repository immediately after receiving an authorised request of such suspension or revocation. 2. Check all certification revocation requests and their outcome are logged	Mandatory	IT Regulation 3, WebTrust 6.6.6				
3.4.1.70	Once a certificate suspension (hold) has been issued, the suspension is handled in one of the following three ways: • an entry for the suspended certificate remains on the CRL with no further action;	1. Check the following on a sample basis: a. Subscriber of a revoked or suspended certificate is informed of the change of status of its certificate. b. Certificate suspension requests are processed and validated in	Mandatory	WebTrust 6.7.8				

	• the CRL entry for the suspended certificate is replaced by a revocation entry for the same certificate; or • the suspended certificate is explicitly released and the entry removed from the CRL.	accordance with the requirements of the CP c. CA notifies the Subscriber in the event of a certificate suspension d. CA updates the Certificate Revocation List (CRL) and other certificate status mechanisms upon certificate suspension e. Suspension is handled in one of the three ways mentioned in the control description (3.4.1.56) f. certificate suspension (hold) entry remains on the CRL until the expiration of the underlying certificate or the expiration of the suspension, whichever is first g. CA verifies or requires that the external RA verify the identity and authority of the entity requesting that the suspension of a certificate be lifted h. Certificate suspensions and the lifting of certificate suspensions are recorded in an audit log					
3.4.1.71	A certificate suspension (hold) entry remains on the CRL until the expiration of the underlying certificate or the expiration of the suspension, whichever is first.	1. Check the following on a sample basis: a. Subscriber of a revoked or suspended certificate is informed of the change of status of its certificate. b. Certificate suspension requests are processed and validated in accordance with the requirements of the CP c. CA notifies the Subscriber in the event of a certificate suspension d. CA updates the Certificate Revocation List (CRL) and other certificate status mechanisms upon certificate suspension e. Suspension is handled in one of the three ways mentioned in the control description (3.4.1.56)	Mandatory	WebTrust 6.7.9			

		f. certificate suspension (hold) entry remains on the CRL until the expiration of the underlying certificate or the expiration of the suspension, whichever is first g. CA verifies or requires that the external RA verify the identity and authority of the entity requesting that the suspension of a certificate be lifted h. Certificate suspensions and the lifting of certificate suspensions are recorded in an audit log						
3.4.1.72	The CA verifies or requires that the external RA verify the identity and authority of the entity requesting that the suspension of a certificate be lifted.	1. Check the following on a sample basis: a. Subscriber of a revoked or suspended certificate is informed of the change of status of its certificate. b. Certificate suspension requests are processed and validated in accordance with the requirements of the CP c. CA notifies the Subscriber in the event of a certificate suspension d. CA updates the Certificate Revocation List (CRL) and other certificate status mechanisms upon certificate suspension e. Suspension is handled in one of the three ways mentioned in the control description (3.4.1.56) f. certificate suspension (hold) entry remains on the CRL until the expiration of the underlying certificate or the expiration of the suspension, whichever is first g. CA verifies or requires that the external RA verify the identity and authority of the entity requesting that the suspension of a certificate be lifted	Mandatory	WebTrust 6.7.11				

		h. Certificate suspensions and the lifting of certificate suspensions are recorded in an audit log						
3.4.1.73	Certificate suspensions and the lifting of certificate suspensions are recorded in an audit log.	1. Check the following on a sample basis: a. Subscriber of a revoked or suspended certificate is informed of the change of status of its certificate. b. Certificate suspension requests are processed and validated in accordance with the requirements of the CP c. CA notifies the Subscriber in the event of a certificate suspension d. CA updates the Certificate Revocation List (CRL) and other certificate status mechanisms upon certificate suspension e. Suspension is handled in one of the three ways mentioned in the control description (3.4.1.56) f. certificate suspension (hold) entry remains on the CRL until the expiration of the underlying certificate or the expiration of the suspension, whichever is first g. CA verifies or requires that the external RA verify the identity and authority of the entity requesting that the suspension of a certificate be lifted h. Certificate suspensions and the lifting of certificate suspensions are recorded in an audit log	Mandatory	WebTrust 6.7.12				
Certificate Status Services								
3.4.1.74	Revocation entries on a CRL or OCSP Response must NOT be removed until after the Expiry Date of the revoked Certificate	1. Check the certificate status services for the following: a. Revocation entries on a CRL or OCSP Response are not removed until after the	Mandatory	CA Browser Forum 4.10.1				

		Expiry Date of the revoked Certificate b. CA operates and maintains its CRL and OCSP capability with resources sufficient to provide a response time of ten seconds or less under normal operating conditions. c. CA maintains an online 24x7 Repository that application software can use to automatically check the current status of all unexpired Certificates d. CA maintains a continuous 24x7 ability to respond internally to a high-priority Certificate Problem Report						
3.4.1.75	The CA shall operate and maintain its CRL and OCSP capability with resources sufficient to provide a response time of ten seconds or less under normal operating conditions.	1. Check the certificate status services for the following: a. Revocation entries on a CRL or OCSP Response are not removed until after the Expiry Date of the revoked Certificate b. CA operates and maintains its CRL and OCSP capability with resources sufficient to provide a response time of ten seconds or less under normal operating conditions. c. CA maintains an online 24x7 Repository that application software can use to automatically check the current status of all unexpired Certificates d. CA maintains a continuous 24x7 ability to respond internally to a high-priority Certificate Problem Report	Mandatory	CA Browser Forum 4.10.2				
3.4.1.76	The CA shall maintain an online 24x7 Repository that application software can use to automatically check the current status of all unexpired Certificates issued by the CA.	1. Check the certificate status services for the following: a. Revocation entries on a CRL or OCSP Response are not removed until after the Expiry Date of the revoked Certificate b. CA operates and maintains its CRL and OCSP capability with resources sufficient to	Mandatory	CA Browser Forum 4.10.2				

		provide a response time of ten seconds or less under normal operating conditions. c. CA maintains an online 24x7 Repository that application software can use to automatically check the current status of all unexpired Certificates d. CA maintains a continuous 24x7 ability to respond internally to a high-priority Certificate Problem Report						
3.4.1.77	The CA shall maintain a continuous 24x7 ability to respond internally to a high-priority Certificate Problem Report, and where appropriate, forward such a complaint to law enforcement authorities, and/or revoke a Certificate that is the subject of such a complaint	1. Check the certificate status services for the following: a. Revocation entries on a CRL or OCSP Response are not removed until after the Expiry Date of the revoked Certificate b. CA operates and maintains its CRL and OCSP capability with resources sufficient to provide a response time of ten seconds or less under normal operating conditions. c. CA maintains an online 24x7 Repository that application software can use to automatically check the current status of all unexpired Certificates d. CA maintains a continuous 24x7 ability to respond internally to a high-priority Certificate Problem Report	Mandatory	CA Browser Forum 4.10.2				

3.4.2. Subordinate CA Certificate and Cross Certificate Lifecycle Management

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Subordinate CA Certificate and Cross Certificate Lifecycle Management							
3.4.2.1	The Parent CA authenticates the Sub-CA certificate rekey request in accordance with the CP.	1. No action required from auditor to check the controls as there is no provision for subordinate CA in the current system. Hence all the controls are not applicable	Not Applicable	WebTrust 7.1.7			

3.4.2.2	The Parent CP specifies the requirements for submission of Sub-CA and cross certification requests.	1. No action required from auditor to check the controls as there is no provision for subordinate CA in the current system. Hence all the controls are not applicable	Not Applicable	WebTrust 7.1.1				
3.4.2.3	The Parent CA authenticates the Sub-CA or cross certificate request in accordance with the Parent's CP.	1. No action required from auditor to check the controls as there is no provision for subordinate CA in the current system. Hence all the controls are not applicable	Not Applicable	WebTrust 7.1.2				
3.4.2.4	The Parent CA performs an assessment of the Sub-CA or cross certificate applicant's compliance with the requirements of the Parent CA's CP before approving a Sub-CA or cross certificate request, or alternatively the Sub-CA or cross certificate applicant presents its CPS for assessment.	1. No action required from auditor to check the controls as there is no provision for subordinate CA in the current system. Hence all the controls are not applicable	Not Applicable	WebTrust 7.1.3				
3.4.2.5	Where Sub-CA and cross certificate renewal is permitted, the Parent CA's CP specifies the requirements for submission of Sub-CA or cross certificate renewal requests	1. No action required from auditor to check the controls as there is no provision for subordinate CA in the current system. Hence all the controls are not applicable	Not Applicable	WebTrust 7.1.4				
3.4.2.6	Where Sub-CA certificate and cross certificate renewal is permitted, the Parent CA authenticates the Sub-CA or cross certificate renewal request in accordance with the CA's CP.	1. No action required from auditor to check the controls as there is no provision for subordinate CA in the current system. Hence all the controls are not applicable	Not Applicable	WebTrust 7.1.5				
3.4.2.7	The Parent CA's CP specifies the requirements for submission of Sub-CA rekey requests	1. No action required from auditor to check the controls as there is no provision for	Not Applicable	WebTrust 7.1.6				

		subordinate CA in the current system. Hence all the controls are not applicable						
3.4.2.8	The Parent CA generates certificates: • using the appropriate certificate profile in accordance with the CP and ISO 9594/X.509 and ISO 15782-1 formatting rules; • with the validity periods formatted in accordance with ISO 9594/X.509, ISO 15782-1 and the CP; and • where extensions are used, with extension fields formatted in accordance with ISO 9594/X.509, ISO 15782-1 and the CP	1. No action required from auditor to check the controls as there is no provision for subordinate CA in the current system. Hence all the controls are not applicable	Not Applicable	WebTrust 7.1.8				
3.4.2.9	The Parent CA signs the Sub-CA or cross certificate with the Parent CA's private signing key	1. No action required from auditor to check the controls as there is no provision for subordinate CA in the current system. Hence all the controls are not applicable	Not Applicable	WebTrust 7.1.9				
3.4.2.10	The Parent CA makes Sub-CA and cross certificates available to relevant entities (e.g., Relying Parties) using an established mechanism (e.g., a repository such as a directory) in accordance with the Parent CA's CP.	1. No action required from auditor to check the controls as there is no provision for subordinate CA in the current system. Hence all the controls are not applicable	Not Applicable	WebTrust 7.1.10				
3.4.2.11	The Parent CA verifies the identity and authority of the entity requesting revocation of a Sub-CA or cross certificate in accordance with the Parent CA's CP.	1. No action required from auditor to check the controls as there is no provision for subordinate CA in the current system. Hence all the controls are not applicable	Not Applicable	WebTrust 7.1.11				
3.4.2.12	The Parent CA updates the Certificate Revocation List (CRL) and other Sub-CA or cross certificate status mechanisms upon certificate	1. No action required from auditor to check the controls as there is no provision for	Not Applicable	WebTrust 7.1.12				

	revocation in accordance with the Parent CA's CP.	subordinate CA in the current system. Hence all the controls are not applicable						
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3.4.3. Publication and repository responsibilities

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Repositories							
3.4.3.1	The CA shall develop, implement, enforce, and annually update a Certificate Policy and/or Certification Practice Statement that describes in detail how the CA implements the latest version of these Requirements	1. Verify the CA develops, implements, enforces, and annually updates a Certificate Policy and/or Certification Practice Statement that describes in detail how the CA implements the latest version of these Requirements 2. Check the CA makes revocation information for Subordinate Certificates and Subscriber Certificates available in accordance with this Policy 3. Validate CA ensures continued accessibility and availability of its Public Key Certificates and Certificate Revocation Lists in its repository to its subscribers and relying parties	Mandatory	CA Browser Forum 2			
3.4.3.2	The CA shall make revocation information for Subordinate Certificates and Subscriber Certificates available in accordance with this Policy. The Certifying Authority shall ensure the continued accessibility and availability of its Public Key Certificates and Certificate Revocation Lists in its repository to its subscribers and relying	1. Verify the CA develops, implements, enforces, and annually updates a Certificate Policy and/or Certification Practice Statement that describes in	Mandatory	CA Browser Forum 2.1, IT Regulations 3			

	parties.	detail how the CA implements the latest version of these Requirements 2. Check the CA makes revocation information for Subordinate Certificates and Subscriber Certificates available in accordance with this Policy 3. Validate CA ensures continued accessibility and availability of its Public Key Certificates and Certificate Revocation Lists in its repository to its subscribers and relying parties						
Publication of information								
3.4.3.3	The CA shall host test Web pages that allow Application Software Suppliers to test their software with Subscriber Certificates that chain up to each publicly trusted Root Certificate. At a minimum, the CA shall host separate Web pages using Subscriber Certificates that are (i) valid, (ii) revoked, and (iii) expired	Check that the CA's repository is publically available in read only format	Mandatory	CA Browser Forum 2.3				
3.4.3.4	The CA shall publicly disclose its Certificate Policy and/or Certification Practice Statement through an appropriate and readily accessible online means that is available on a 24x7 basis.	1. Verify the CA has publicly disclosed its CP and CPS and these are readily accessible online on a 24x7 basis 2. Validate the CP and CPS are structured in accordance with RFC 3647 3. Validate the CA publicly gives effect to the Requirements mentioned in control description (3.4.3.5) and represents that it will adhere to the	Mandatory	CA Browser Forum 2.2				

		latest published version 4. Check that the CA hosts test web pages to allow ASPs to test their software with Subscriber 5. Verify the CA hosts separate Web pages using Subscriber Certificates that are (i) valid, (ii) revoked, and (iii) expired						
3.4.3.5	Effective as of 31 May 2018, the Certificate Policy and/or Certification Practice Statement must be structured in accordance with RFC 3647. Prior to 31 May 2018, the Certificate Policy and/or Certification Practice Statement must be structured in accordance with either RFC 2527 or RFC 3647. The Certificate Policy and/or Certification Practice Statement must include all material required by RFC 3647 or, if structured as such, RFC 2527.	1. Verify the CA has publicly disclosed its CP and CPS and these are readily accessible online on a 24x7 basis 2. Validate the CP and CPS are structured in accordance with RFC 3647 3. Validate the CA publicly gives effect to the Requirements mentioned in control description (3.4.3.5) and represents that it will adhere to the latest published version 4. Check that the CA hosts test web pages to allow ASPs to test their software with Subscriber 5. Verify the CA hosts separate Web pages using Subscriber Certificates that are (i) valid, (ii) revoked, and (iii) expired	Mandatory	CA Browser Forum 2.2				
3.4.3.6	The CA shall publicly give effect to these Requirements and represent that it will adhere to the latest published version. The CA may fulfill this requirement by incorporating these Requirements directly into its Certificate Policy and/or	1. Verify the CA has publicly disclosed its CP and CPS and these are readily accessible online on a 24x7 basis 2. Validate the	Mandatory	CA Browser Forum 2.2				

	Certification Practice Statements or by incorporating them by reference using a clause such as the following (which must include a link to the official version of these Requirements): [Name of CA] conforms to the current version of the Baseline Requirements for the Issuance and Management of Publicly-Trusted Certificates published at http://www.cabforum.org. In the event of any inconsistency between this document and those Requirements, those Requirements take precedence over this document.	CP and CPS are structured in accordance with RFC 3647 3. Validate the CA publicly gives effect to the Requirements mentioned in control description (3.4.3.5) and represents that it will adhere to the latest published version 4. Check that the CA hosts test web pages to allow ASPs to test their software with Subscriber 5. Verify the CA hosts separate Web pages using Subscriber Certificates that are (i) valid, (ii) revoked, and (iii) expired							
Access controls on repositories									

3.4.4. Certificate, crt and oscp profiles

Control No.	Control	Audit Checks	Control Type	References	Compliance	Document	Remarks
Certificate profile							
3.4.4.1	Effective September 30, 2016, CAs shall generate non- sequential Certificate serial numbers greater than zero (0) containing at least 64 bits of output from a CSPRNG.	1. Verify CAs generates non-sequential Certificate serial numbers greater than zero (0) containing at least 64 bits of output from a CSPRNG. 2. For sample Certificates, check the following: a. Certificates is of type X.509 v3 b. basicConstraints extension appears as a critical extension c. basicConstraints extension is set true d. pathLenConstraint field is not present e. keyUsage extension is present and marked critical f. Bit positions for keyCertSign and	Mandatory	CA Browser Forum 7.1			

		cRLSign are set g. digitalSignature bit is set, If the Root CA Private Key is used for signing OCSP responses h. certificatePolicies and extendedKeyUsage are not present						
3.4.4.2	Certificates must be of type X.509 v3.	1. Verify CAs generates non-sequential Certificate serial numbers greater than zero (0) containing at least 64 bits of output from a CSPRNG. 2. For sample Certificates, check the following: a. Certificates is of type X.509 v3 b. basicConstraints extension appears as a critical extension c. basicConstraints extension is set true d. pathLenConstraint field is not present e. keyUsage extension is present and marked critical f. Bit positions for keyCertSign and cRLSign are set g. digitalSignature bit is set, If the Root CA Private Key is used for signing OCSP responses h. certificatePolicies and extendedKeyUsage are not present	Mandatory	CA Browser Forum 7.1.1				
3.4.4.3	basicConstraints - This extension must appear as a critical extension. The CA field must be set true. The pathLenConstraint field should NOT be present	1. Verify CAs generates non-sequential Certificate serial numbers greater than zero (0) containing at least 64 bits of output from a CSPRNG. 2. For sample Certificates, check the following: a. Certificates is of type X.509 v3 b. basicConstraints extension appears as a critical extension c. basicConstraints extension is set true d. pathLenConstraint	Mandatory	CA Browser Forum 7.1.2.1				

		field is not present e. keyUsage extension is present and marked critical f. Bit positions for keyCertSign and cRLSign are set g. digitalSignature bit is set, If the Root CA Private Key is used for signing OCSP responses h. certificatePolicies and extendedKeyUsage are not present						
3.4.4.4	keyUsage - This extension must be present and must be marked critical. Bit positions for keyCertSign and cRLSign must be set. If the Root CA Private Key is used for signing OCSP responses, then the digitalSignature bit must be set.	1. Verify CAs generates non-sequential Certificate serial numbers greater than zero (0) containing at least 64 bits of output from a CSPRNG. 2. For sample Certificates, check the following: a. Certificates is of type X.509 v3 b. basicConstraints extension appears as a critical extension c. basicConstraints extension is set true d. pathLenConstraint field is not present e. keyUsage extension is present and marked critical f. Bit positions for keyCertSign and cRLSign are set g. digitalSignature bit is set, If the Root CA Private Key is used for signing OCSP responses h. certificatePolicies and extendedKeyUsage are not present	Mandatory	CA Browser Forum 7.1.2.1				
3.4.4.5	certificatePolicies - This extension should NOT be present	1. Verify CAs generates non-sequential Certificate serial numbers greater than zero (0) containing at least 64 bits of output from a CSPRNG. 2. For sample Certificates, check the following: a. Certificates is of type X.509 v3 b. basicConstraints	Mandatory	CA Browser Forum 7.1.2.2				

		extension appears as a critical extension c. basicConstraints extension is set true d. pathLenConstraint field is not present e. keyUsage extension is present and marked critical f. Bit positions for keyCertSign and cRLSign are set g. digitalSignature bit is set, If the Root CA Private Key is used for signing OCSP responses h. certificatePolicies and extendedKeyUsage are not present						
3.4.4.6	extendedKeyUsage - This extension must NOT be present	1. Verify CAs generates non-sequential Certificate serial numbers greater than zero (0) containing at least 64 bits of output from a CSPRNG. 2. For sample Certificates, check the following: a. Certificates is of type X.509 v3 b. basicConstraints extension appears as a critical extension c. basicConstraints extension is set true d. pathLenConstraint field is not present e. keyUsage extension is present and marked critical f. Bit positions for keyCertSign and cRLSign are set g. digitalSignature bit is set, If the Root CA Private Key is used for signing OCSP responses h. certificatePolicies and extendedKeyUsage are not present	Mandatory	CA Browser Forum 7.1.2.1				
CRL Profile								
3.4.4.7	CA shall make a full and complete CRL available to the OCSP Responders	1. Check that the CA has made the full and complete CRL available to the OCSP responder	Mandatory	X.504 Policy 7.2.1				

OCSP Profiles								
3.4.4.8	OCSP requests and responses shall be in accordance with RFC 2560	1. Verify the OCSP requests and responses are in accordance with RFC 2560	Mandatory	X.504 Policy 7.3				

Audit Criteria for Certifying Authorities, Ver...

Annexure-A1

Summary of non-compliances observed

S. No.	Control No	Reference(s)	Non-compliance observed	Action taken by CA, if any	Remarks

Project Review Schedules

Project Schedule for PG-DAC & PG-DBDA August 2024 batch

Academic Year: 2024-2025

Task	Date
Guide Allocation & Team Selection (by CDACD Project Coordinator)	Up to Jan 9, 2025
Submission of Project Title & Project Abstract	Up to Jan 10, 2025
Initial Review • Project title Justification • Objective and scope of the project • Time plan of the project	January 14, 2025
Review-1 By External Committee • Explanation of the identified Problem • Basic concepts related to your project • Literature review related to particular problem	January 27, 2025
Status report-1 (Synopsis) (Report after modification based on the suggestions by External Committee in Review 1)	January 31, 2025
Review-2- With Guide • Methodology and the overall proposed approach to solve the particular problem	February 10, 2025
Review-3 With Guide • Algorithms/ Methodology used to overcome the specific problem (general description of algorithms) • Data collection • Experimental setup • 10-30% of initial code implementation	February 14, 2025
Status report-2 With External Committee (Report after modification based on the suggestions in review-2 & 3)	February 17, 2025
Review-4 With Guide • Code implementation • Evaluation and Execution of the project • Result analysis • Performance comparison • 100% of code implementation	February 19, 2025
Final Review with Guide	February 20, 2025
Rough draft of your Project Report to be submitted to Guide	February 20, 2025
Final Project Report to be submitted & Viva Voce- With External Committee	February 20, 2025
• All final reports should be specified in the format given by the institute. • Absenteeism during the review dates will not be entertained. • The schedule is tentative and is subjected to change with prior intimation	

Project Guidelines

A reviewer monitors the running projects on a regular base to ensure the high-level quality projects been developed by Students. Monitoring of running projects is done by means of **Project Progress Reports (PPRs)** and **Project Reviews**.

- PPRs are handled in a separate instruction video
- A Project Review is a face-to-face project evaluation by a multidiscipline review team

The aim of a Project Review is to check if a project is heading towards successful innovative and exploitable results. Advice on corrective measures is provided if necessary.

Choosing a Project:- The idea for your project may be a proposal from a institute, Advisor or your own, or perhaps a combination of the three.

Guides:- The Guides are advised to give projects and suggest project titles focussing more on the current field of research/Problem and ensure the level of innovation. Also, Guides are advised to check for the formatting of the presentation and Project report/ Synopsis.

Following reviews are planned.

The first initial review is planned after roughly selection of the project title with Abstract via Team member and appointed project guide.

The Second Review-1 team members for their final project with their proposed title & project guide and to suggest it to the **external committee** with a brief abstract. Project report/ Synopsis to be prepared as per the guidelines and format given by the institute.

In the Status report-1 (Final Synopsis) In case of re-reviews, any number of re-reviews can happen depending on the discretion of the panel and it should happen in **Status report-1** within the prescribed time.

Guide, can initiate further steps (Review 2 - 5) to ensure the smooth implementation as deems appropriate of guidelines. And submit the status report-2(one- or two-pages max) to the External Committee in terms of presentation. If the work of the candidate is found to be insufficient or with plagiarism, the committee & Advisor will decide the further process.

In the Final Review - The last review will take place shortly before the end date of the project. Make an internal team and review the ready project before final project demonstration to the **external committee**.

Final Project Presentation and Demonstration One of the most important skills which the project aims to assess the student's ability to communicate project ideas and work. As part of the assessment students will be required to give a presentation and demonstration of project to Project Committee.

Project Requirements: Expectations from Students (in each Presentation)

Initial Review	First & Second Review	Third & Fourth Review	Fifth Review
Title Abstract Introduction Literature Survey Proposed System/SDLC Modules Split-up and Gantt Chart References	Title Abstract Architectural Design for Proposed System ER Diagram, DFD, Use case diagram (if necessary) Algorithms / Techniques used with complexity Expected outcomes References 10-30% of initial code implementation	Title Abstract Detailed Design (if any deviation) Contribution of the candidate (if any) Results obtained (intermediate) References 80% of code Implementation	Title Abstract Overall Design Experimental Results Performance Evaluation References Contribution of the Project 100% of code implementation – Demo

Approval Guidelines of Initial Review

- Comparison with the existing systems.
- Deliverables to be mentioned clearly for each review.
- Work distribution among team members. Evaluation based on the complexity of the work.

Note:

- The presentation should have maximum of 12 – 15 slides
- Presentation will be for 10 minutes
- System to be tested using testing software's.
- All Five reviews are compulsory.

External Committee

- The committee is advised to ensure the complexity of the problem statement solved through project work.
- All the three panel members must be presented during the review.
- Reviews to be conducted over online platform.

Guide to check

- Advised to check for the formatting of the presentation and all project documents submitted by students.
- Check for the attendance of the students (Regular meeting for the discussion).
- To ensure that each student of group has contribution in different stages of Project Work.